

Report prepared for the WHO annual consultation on the composition of influenza vaccines for the Northern Hemisphere 2023-2024

Worldwide Influenza Centre, WHO CC for Reference and Research on Influenza, The Francis Crick Institute, 1 Midland Road, NW1 1AT

February 2023



Prof. Nicola Lewis (Director - WIC)

Dr. Rodney Daniels (Deputy Director)

Dr. Ruth Harvey (Assistant Director)

Dr. Monica Galiano (Assistant Director)

Dr. Zheng Xiang, Ms. Lynne Whittaker

Ms. Christine Carr, Ms. Burcu Ermetal

Dr. Karen Cross, Ms. Aine Rattigan

Ms. Alice Lilley, Mr. Michael Bennett

Ms. Chandrika Halai, Ms. Becky Clark, Mr. Lorin Adams

Tel (1): 020 3796 1520 (NL) Tel (2): 020 3796 2444 (RD) Tel (3): 020 3796 0563 (WIC)

Email: Nicola.Lewis@crick.ac.uk; rod.daniels@crick.ac.uk; whooc@crick.ac.uk

Acknowledgements

Laboratories involved in Global Influenza Surveillance

We thank all those who have contributed information, clinical specimens and viruses, and associated data to the WHO Global Influenza Surveillance and Response System (GISRS), which provides the basis for our current understanding of recently circulating influenza viruses included in this summary.

Additional Personnel from the Francis Crick Institute

The work of the WIC is only made possible with the ongoing support and input from the Francis Crick Institute Advance Sequencing Facility (ASF), Biological Research Facility (BRF) and the Bioinformatics Core-Science Technology Platform (BC-STP). We thank all members of staff in these platforms for their assistance.

Influenza Epidemiology in the WHO European Region.

The epidemiological section of this report was prepared using data reported to the European Surveillance System (TESSy) and presented in Flu News Europe (<http://www.flunewseurope.org>) the joint ECDC-WHO Europe weekly influenza update. We acknowledge Cornelia Adlhoch, Eeva Broberg, Annette Kraus, Margaux Mesle, Piers Mook, Karen Nahapetyan, and Richard Pebody for their assistance in preparing the report.

Human serology panels.

We acknowledge Public Health Wales-Microbiology for sharing human serology panels, in particular, Catherine Moore, Laura Gifford, Jonathan Evans and Anastasia Couzens

Contents

Epidemiology	5
Distribution of viruses	5
Influenza virus characterization	5
Susceptibility to Neuraminidase (NA) and Polymerase Acidic protein (PA) inhibitors	6
Timing and spread	6
Hospitalized cases	7
Mortality data	7
Table E1. 2022-2023 Influenza Season in the WHO European Region: weeks 40/2022-05/2023	8
Table E2. Distribution of Influenza Detections in the WHO European Region1, weeks 40/2022-05/2023	8
A. Laboratory confirmed influenza from sentinel and non-sentinel sources by geographic sub-region and influenza type and subtype/lineage	9
B. Laboratory confirmed influenza from sentinel sources by geographic sub-region and influenza type and subtype/lineage	9
Figure E1. Influenza circulation2 in the WHO European Region by (sub)type and Country, territory or area, 2022-2023 season	10
Figure E2 A, B, C. Seasonal influenza activity in the WHO European Region: weeks 40/2022-05/2023	11
Figure E3. Intensity of influenza activity and dominant virus (sub)type in the WHO European Region by week and country, territory or area 2022-2023 season	13
Figure E4. Distribution of virus (sub)type in laboratory-confirmed hospitalized cases in intensive care units by age group in the WHO European Region, 2022-2023 season	14
Table E3. Percentage distribution of severe acute respiratory infection (SARI) cases by age group and country, territory or area, and influenza positives, 2022-2023 season	15
Global influenza activity	16
Microreact: Influenza by type/subtype	16
Submissions	17
Influenza A H1N1	25
Microreact: H1N1 by clade	25
Genetic analyses: H1N1	26
Maximum likelihood phylogenetic trees: H1N1	27
HA-NA tanglegrams	29
Protein alignments HA	30
Protein alignments NA	32
Structure	34
Genotype to Phenotype heatmap: H1N1	35
Antigenic analyses: H1N1	36
Haemagglutination inhibition tables: H1N1	36
Human Serology	48
Panel 1	48
Panel 2	49
Reference reagents and isolates	50
A/H1N1: Egg isolates	50
A/H1N1: HI reagents and references	50
Summary: H1N1	51
Influenza A H3N2	52

Microreact: H3N2 by clade	52
Genetic analyses: H3N2	53
Maximum likelihood phylogenetic tree: H3N2	54
HA-NA tanglegrams	56
Protein alignments HA	57
Protein alignments NA	59
Structure	61
Genotype to Phenotype heatmap: H3N2	63
Antigenic analyses: H3N2	64
Haemagglutination inhibition tables: H3N2	64
Microneutralisation tables: H3N2	80
Human Serology	87
Panel 1	87
Panel 2	88
Reference reagents and isolates	89
A/H3N2: Egg isolates	89
A/H3N2: HI reagents and references	89
A/H3N2: MN reagents and references	90
Summary: H3N2	91
Influenza B	92
Genetic analyses: B/Victoria	93
Maximum likelihood phylogenetic tree: B/Victoria	93
HA-NA tanglegrams	95
Protein alignments HA	96
Protein alignments NA	98
Structure	100
Genotype to Phenotype heatmap: B/Victoria	101
Antigenic analyses: B/Victoria	102
Haemagglutination inhibition tables: B/Victoria	102
B/Victoria: Egg isolates	111
B/Victoria: Reagents and references	111
Summary: B/Victoria	112
Susceptibility to antivirals	113
Annex	114
Tanglegrams with tip labels	114

Epidemiology

Summary of influenza activity in the WHO European Region, October 2022 – February 2023.

Distribution of viruses

Between weeks 40/2022 and 05/2023, a total of 195,984 influenza detections from sentinel and non-sentinel primary care sources were reported to TESSy (**Table E1**). This is compared to 38,832 and 87,955, for the same period in 2021-2022 and 2019-2020 (last pre-pandemic season) respectively, representing 5-fold and 2.2-fold increases, respectively (**Table E1**). The increase in detections for 2022-2023 compared to 2019-20 can be related to some extent to the numbers of specimens tested; 1,420,572 specimens were tested in 2022-2023 compared to 431,066 specimens tested in 2019-2020, a 2.3-fold increase. However, there was a 3% decrease in the number of specimens tested in 2022-2023 compared to 2021-2022 (1,420,572 v 1,471,245) but a 5-fold increase in the number of detections, indicating an early season increase in influenza activity in 2022-2023 compared to 2021-2022 (**Table E1**).

Combined sentinel and non-sentinel data for type A viruses in the Region showed that most viruses detected (90%) were type A and 10% were type B (a ratio of ~ 8.5:1 compared to 25:1 in 2021-2022 and 3.9:1 in 2019-2020) (**Table 1**). This was generally consistent across the Region with all subregions detecting at least 88% type A viruses, except for central Asia where there was a higher proportion of type B viruses (61%). Combined sentinel and non-sentinel data for type A viruses showed a similar proportion of subtypes: 52% were A(H3N2) viruses and 48% were A(H1N1)pdm09 viruses (a ratio of ~ 1.1:1 representing a decrease from 15.3:1 in 2021-2022, but very similar to 1:1 in 2019-2020). This was consistent with the observed distributions in northern Europe (which had the highest number of subtyped influenza A viruses) and west Asia. In central Asia and eastern Europe influenza A(H1N1)pdm09 viruses dominated (97% for central Asia and 90% for eastern Europe) and in south-west Europe and western Europe influenza A(H3N2) viruses dominated (77% and 84%, respectively) (**Table E2-A**). Of 1,894 type B viruses ascribed to a lineage, all were B/Victoria.

When the analysis was restricted to sentinel data only for weeks 40/2022 to 05/2023, the regional distribution of influenza by type was similar: of 17,543 specimens testing positive for influenza, the majority were type A (90%). Of type A viruses, the majority were A(H3N2) subtype (71%) and of the type B viruses ascribed to a lineage, all were B/Victoria (**Table E2-B**). This was consistent with the distribution observed in south-west Europe and western Europe (77% and 84% influenza A(H3N2) viruses, respectively). In central Asia, influenza A(H1N1)pdm09 viruses dominated (97%) and similar proportions of influenza subtypes were observed in eastern Europe, northern Europe and west Asia. Two subregions reported at least 10% type B viruses: central Asia (47%) and western Europe (11%). Of the total influenza detections, 5,364 (31%) were from south-west Europe and 3,960 (23%) were from western Europe (**Table E2-B**).

A dominant virus (sub)type (comprising at least 60% of influenza virus detections from sentinel sources) could be assigned by 47 countries and areas within the WHO European Region, with five (Azerbaijan, Bosnia and Herzegovina, Iceland, Latvia and Turkmenistan) having insufficient numbers of detections (ten or fewer). Forty-three countries and areas reported predominance of influenza type A viruses, while two reported predominance of influenza type B viruses (Kazakhstan and Uzbekistan), and two reported co-dominance of type A and type B viruses (Belgium and Netherlands) (**Figure E1**). Three reported dominance of type A without specifying a subtype (Croatia, Tajikistan and Kosovo), and 34 were able to ascribe dominance (60% or greater) or co-dominance (between 41% and 59%) of influenza A virus subtypes for the season: 19 reported A(H3N2) dominance, 15 reported A(H1N1)pdm09 dominance (mainly in central and eastern Europe) and six reported co-dominance of A(H1N1)pdm09 and A(H3N2) viruses (Denmark, Hungary, Ireland, Poland, Republic of Moldova and Slovenia). Only two countries reported dominance of type B viruses (Kazakhstan and Uzbekistan).

Influenza virus characterization

Of 180,251 influenza virus detections between weeks 40/2022 and 04/2023, genetic characterization of 3,301 viruses was reported to TESSy.

Of the 1,439 A(H3N2) viruses genetically characterized (48% of influenza type A viruses), the majority (1,385; 96%) were attributed to subclade 3C.2a1b.2a.2, represented by A/Darwin/9/2021, A/Bangladesh/4005/2020 and A/Slovenia/8720/2022, which are in the same clade as the recommended vaccine virus strain (A/Darwin/9/2021) for use in the 2022-2023 northern hemisphere (NH) and 2023 southern hemisphere (SH) influenza seasons. Only three viruses (<1%) belonged to the subclade 3C.2a1b.1a, represented by A/Denmark/3264/2019. In addition, two A(H3N2) viruses (<1% of all influenza A viruses), originally not attributed to any predefined genetic clade, fell in the same clade as the recommended vaccine virus, A/Darwin/9/2021, based on the HA sequences provided. Further, 49 A(H3N2) viruses (2% of all influenza A viruses genetically characterized) were reported as belonging to a subgroup not listed in the current guidance. The corresponding HA sequences were reported, and all 49 viruses fell in clade 3C.2a1b.2a.2.

The genetic clade was reported for 1,544 characterized A(H1N1)pdm09 viruses, 697 (45%) were represented by clade 6B.1A.5a.2 viruses like A/Victoria/2570/2019, the virus component for 2022-2023 NH vaccines, and A/Sydney/5/2021 the virus component for 2023 SH vaccines. Four viruses belonged to the 6B.1A.5a.1 group, represented by A/Guangdong-Maonan/SWL1536/2019. Additionally, five A(H1N1)pdm09 viruses originally not attributed to any predefined genetic clade, fell in clade 6B.1A.5a.2 based on the HA sequences provided. For a further 838 A(H1N1)pdm09 viruses, no clade/subclade attribution was made and all but one of these viruses (837, >99%) fell in clade 6B.1A.5a.2 based on the HA sequences, while one virus fell in clade 6B.1A.5a.1.

Of the 318 B/Victoria-lineage viruses characterized, 155 (49%) belonged to subgroup V1A.3a.2, represented by B/Austria/1359417/2021, the recommended vaccine virus for the 2022-2023 NH and 2023 SH influenza seasons. Of the 163 viruses not attributed to a clade/group, phylogenetic analysis showed all these viruses to fall in clade V1A.3a.2.

There were no confirmed detections of B/Yamagata-lineage viruses associated with naturally circulating viruses.

Susceptibility to Neuraminidase (NA) and Polymerase Acidic protein (PA) inhibitors

For weeks 40/2022 to 04/2023 viruses from the 2022-2023 season were assessed phenotypically and/or genotypically (for markers of) reduced susceptibility NA inhibitors: 2,248 to oseltamivir (1,139 A(H3N2), 836 A(H1N1)pdm09 and 273 B viruses) and 1,840 to zanamivir (1,015 A(H3N2), 574 A(H1N1)pdm09 and 251 B viruses).

During this period, 1,695 viruses were assessed genotypically for markers of reduced susceptibility to the PA inhibitor baloxavir marboxil (BM:1,017 A(H3), 443 A(H1N1)pdm09, and 235 B viruses).

Phenotypically no viruses with reduced susceptibility to NA inhibitors were identified. Genotypically, no viruses with potentially reduced susceptibility to BM were identified. However, two A(H1N1)pdm09 viruses carrying NA H275Y amino acid substitutions associated with highly reduced susceptibility to oseltamivir were identified. These two viruses were reported from different countries in consecutive weeks and fell in clade 6B.1A.5a.2 with the closest reference virus being A/Norway/25089/2022.

Timing and spread

Historically, a threshold of 10% of influenza-like illness (ILI) or acute respiratory infection (ARI) specimens from sentinel primary care sites testing positive for influenza viruses for two consecutive weeks has been used to indicate the start of the influenza season in the Region.

The 2022-2023 influenza epidemic in the WHO European Region started in week 45/2022, an earlier start compared to the four previous epidemics: ranging from week 47 (2019-2020 season) to 49 (2021-2022 season) (**Figures E1-A and B**). Between weeks 40/2022 and 05/2023, the proportion of specimens testing positive for influenza on a weekly basis reached a peak of 39% for the season so far in week 51/2022, which was lower than peaks in recent pre-COVID-19 seasons (54-59%; 2015-2016 to 2018-2019 seasons) but higher than in the 2020-2021 influenza surveillance season, the first with dominant co-circulation of SARS-CoV-2,

when influenza-positivity did not exceed 2%. To date, influenza activity where >10% of sentinel specimens have tested positive for influenza has lasted for 13 consecutive weeks (since week 45/2022), compared to seasonal averages of ~10.5 weeks (range: 9-14 weeks) for recent epidemics (2015-2016 to 2018-2019 seasons) (**Figures E2-A/B**). The proportion of subtyped influenza viruses that were A(H1N1)pdm09 has increased over the course of season and there has also been an increase in numbers of influenza type B virus detections. A similar pattern of overall influenza activity and influenza virus type was observed for non-sentinel data (**Figure E2-C**).

Over the season, a total of 49 countries and areas reported influenza activity above their calculated baselines, with 37 having reported medium or higher activity in some weeks. A total of eight countries reported very high intensity in some weeks (Austria, Finland, France, Latvia, Lithuania, Luxembourg, Republic of Moldova and Russian Federation). Activity has reached peaks in many, but not all countries and areas, with no apparent geographic distribution in terms of the start of the season or timing of peak activity. (Note: this indicator is typically based on ILI and/or ARI activity but at least one virus detection per week is required to meet the definition for an intensity level above baseline. Not all countries or areas reporting intensity above baseline have reported detections)(**Figure E3**).

Hospitalized cases

Between weeks 40/2022 and 05/2023, five countries and areas (Czech Republic, France, Ireland, Sweden and United Kingdom (England)) reported a total of 6,556 laboratory-confirmed influenza cases from hospital settings to TESSy. Of these, 2,609 cases were reported from intensive care units (ICUs) in United Kingdom (England) (n=1,457, 56%), France (n= 697, 27%), Sweden (n= 182, 7%), Ireland (n= 151, 6%) and Czech Republic (n= 122, 5%). The remaining cases were reported from non-ICU hospital wards in Ireland (n=3,817, 97%) and Czech Republic (n= 130, 3%). Among the 2,609 cases admitted to ICUs, 2,464 (94%) were infected with type A viruses (of those subtyped, 252 A(H3N2) and 207 A(H1N1)pdm09) and 145 (6%) with type B viruses (none ascribed to a lineage). A total of 1,144 cases had a recorded age, falling within different age groups: 515 were 15-64 years (45%), 499 were 65 years or older (44%), 85 were 0-4 years (7%) and 45 were 5-14 years (4%) (**Figure E4**).

A total of 27 countries and areas (Albania, Armenia, Belarus, Belgium, Bosnia and Herzegovina, Croatia, Georgia, Germany, Ireland, Kazakhstan, Kyrgyzstan, Lithuania, Malta, Montenegro, North Macedonia, Republic of Moldova, Romania, Russian Federation, Serbia, Slovakia, Spain, Tajikistan, Türkiye, Turkmenistan, Ukraine, Uzbekistan and Kosovo) with surveillance systems for severe acute respiratory infection (SARI) reported a total of 100,135 hospitalized patients between weeks 40/2022 and 05/2023. Data on age was available for 60,037 of these patients. Overall, children aged 0-4 years accounted for the largest proportion of cases (55%), followed by those aged 30-64 years (15%), those aged 5 to 14 years (14%), those aged 65 years or older (9%) and those aged 15 to 29 years (6%). SARI cases were predominantly among children aged less than 5 years in 14 countries (Armenia, Belgium, Croatia, Georgia, Germany, Kazakhstan, Kyrgyzstan, Montenegro, North Macedonia, Slovakia, Spain, Tajikistan, Turkmenistan and Uzbekistan). Türkiye had the largest proportion (43%) of reported cases predominantly among those aged 65 years or older (**Table E3**). Of 38,092 SARI cases tested for influenza virus over the period, 9% (n=3,464) were positive. The majority (n=2,691, 78%) were infected with type A viruses and, of 2,270 subtyped viruses, 1,672 (74%) were A(H1N1)pdm09 and 598 (26%) were A(H3N2). A total of 773 infections by influenza type B viruses were reported from Kazakhstan (n=433), Kyrgyzstan (n=180), Russian Federation (n=49), Türkiye (n=23), Uzbekistan (n=19), Albania (n=13), Belgium (n=11), Lithuania (n=10), Serbia (n=8), Armenia (n=4), Georgia (n=4), Ukraine (n=4), Republic of Moldova (n=3), Romania (n=3), Turkmenistan (n=3), Belarus (n=1), Bosnia and Herzegovina (n=1), Croatia (n=1), Ireland (n=1), Malta (n=1) and North Macedonia (n=1). Of the 174 type B viruses ascribed to a lineage, all were B/Victoria.

Mortality data

Pooled analyses of all-cause excess and influenza-attributable mortality from 27 European countries or sub-national regions participating in the EuroMOMO network (Austria, Belgium, Cyprus, Denmark, Estonia, Finland, France, Germany, Germany (Berlin), Germany (Hesse), Greece, Hungary, Ireland, Israel, Italy, Lux-

embourg, Malta, Netherlands, Portugal, Slovenia, Spain, Sweden, Switzerland, UK (England), UK (Northern Ireland), UK (Scotland), and UK (Wales)) showed an overall substantially increased excess all-cause mortality for the participating European countries during the 2022-2023 season which is now declining. This excess mortality was driven by a substantial excess in a few countries. Please refer to the EuroMOMO website for a cautionary note relating to interpretation of these data.

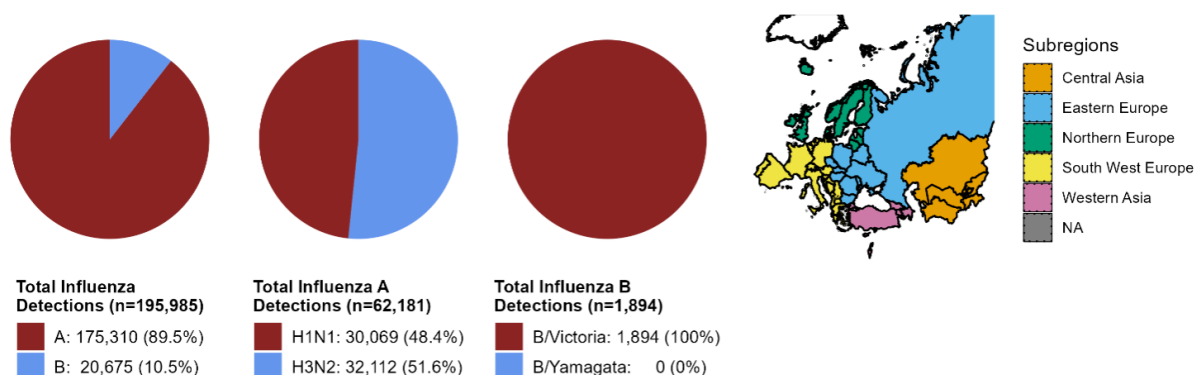
Table E1. 2022-2023 Influenza Season in the WHO European Region: weeks 40/2022-05/2023

Virus type subtype/lineage	Cumulative number of detections for weeks 40/2022-05/2023			Totals*		Cumulative number of detections for weeks 40/2021-05/2022			Totals*		Cumulative number of detections for weeks 40/2019-05/2020			Totals*	
	Sentinel	Non-sentinel	Totals	%	Ratios	Sentinel	Non-sentinel	Totals	%	Ratios	Sentinel	Non-sentinel	Totals	%	Ratios
	sources	sources				sources	sources				sources	sources			
Influenza A	15,738	159,572	175,310	89.5	8.5:1	2,398	34,943	37,341	96.2	25.0:1	7,010	62,896	69,906	79.5	3.9:1
A(H1N1)pdm09	3,736	26,333	30,069	48.4		126	783	909	6.1		4,039	8,979	13,018	49.0	
A(H3N2)	9,285	22,827	32,112	51.6	1.1:1	1,627	12,279	13,906	93.9	15.3:1	2,620	10,948	13,568	51.0	1.0:1
A not subtyped	2,717	110,412				645	21,881	22,526			351	42,969	43,320		
Influenza B	1,806	18,869	20,675	10.5		43	1,448	1,491	3.8		3,219	14,830	18,049	20.5	
Victoria lineage	558	1,336	1,894	99.9	x	13	24	37	100.0	x	1,109	923	2,032	96.8	28.6:1
Yamagata lineage	1	0	1	0.1		0	0	0			15	52	67	3.2	
Lineage not ascribed	1,247	17,533	18,780			30	1,424	1,454			2,095	13,855	15,950		
Total detections	17,544	178,441	195,985			2,441	36,391	38,832			10,229	77,726	87,955		
Total tested	75,665	1,344,907	1,420,572			39,956	1,434,289	1,474,245			30,825	400,241	431,066		

* Percentages are shown for total detections (types A & B [in bold type], and for viruses ascribed to influenza A subtype and influenza B lineage). Ratios are given for type A:B [in bold type], A(H3N2):A(H1N1)pdm09 and Victoria:Yamagata lineages.

Table E2. Distribution of Influenza Detections in the WHO European Region1, weeks 40/2022-05/2023

Laboratory confirmed influenza from sentinel and non-sentinel sources by type and subtype/lineage



WHO European Region Member States grouped into sub-regions according to United Nations geographic composition. Available from: (<http://unstats.un.org/UNSD/METHODS/M49/M49REGIN.HTM>)

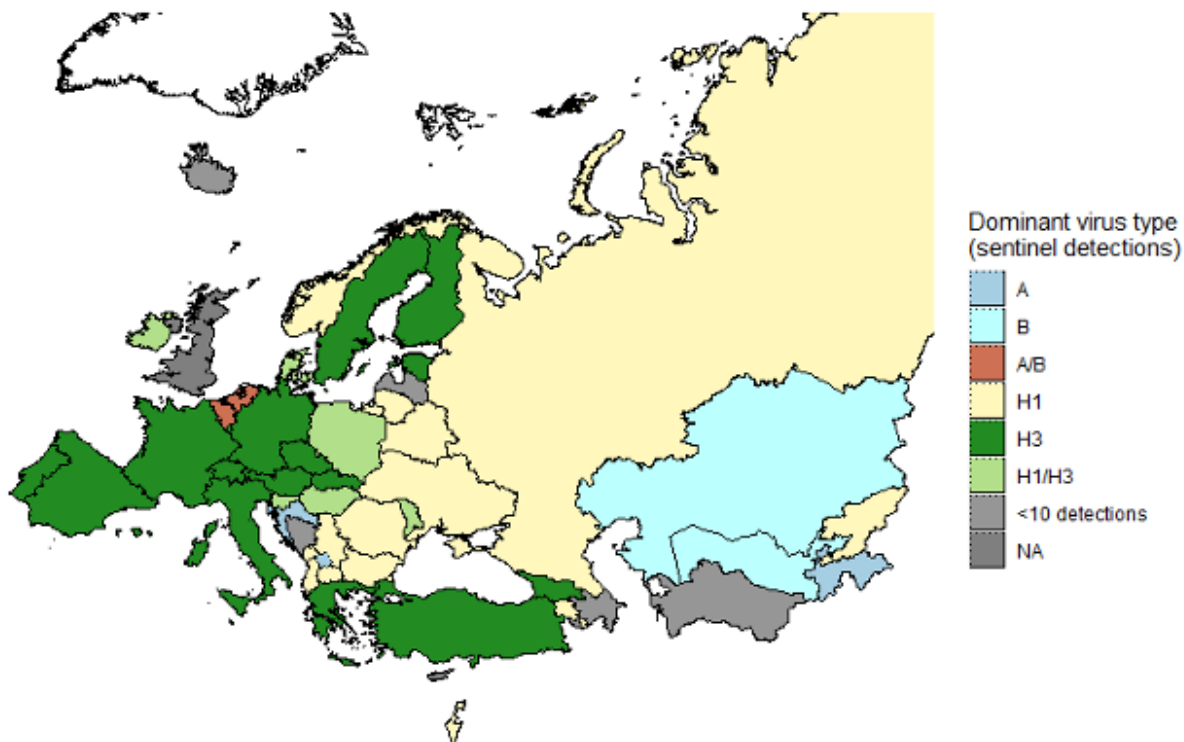
A. Laboratory confirmed influenza from sentinel and non-sentinel sources by geographic sub-region and influenza type and subtype/lineage

Influenza type	South West Europe	Western Europe	Eastern Europe	Northern Europe	West Asia	Central Asia	WHO European Region
Influenza A	27,653 (96%)	50,639 (88%)	33,664 (89%)	60,516 (89%)	2,028 (94%)	810 (39%)	175,310 (89%)
Influenza A subtyped	14,025	10,467	17,303	18,114	1,719	553	62,181
A(H1N1)pdm09	3,174 (23%)	1,707 (16%)	15,524 (90%)	8,224 (45%)	904 (53%)	536 (97%)	30,069 (48%)
A(H3N2)	10,851 (77%)	8,760 (84%)	1,779 (10%)	9,890 (55%)	815 (47%)	17 (3%)	32,112 (52%)
Influenza B	1,193 (4%)	6,599 (12%)	4,296 (11%)	7,206 (11%)	138 (6%)	1,242 (61%)	20,674 (11%)
Influenza B lineage determined	154	548	34	860	14	284	1,895
B/Yamagata-lineage	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
B/Victoria-lineage	154 (100%)	548 (100%)	34 (100%)	860 (100%)	14 (100%)	284 (100%)	1,894 (100%)
Total	28,846	57,238	37,960	67,722	2,166	2,052	195,984

B. Laboratory confirmed influenza from sentinel sources by geographic sub-region and influenza type and subtype/lineage

Influenza type	South West Europe	Western Europe	Eastern Europe	Northern Europe	West Asia	Central Asia	WHO European Region
Influenza A	4,888 (91%)	3,519 (89%)	2,448 (93%)	3,091 (93%)	1,362 (95%)	430 (53%)	15,738 (90%)
Influenza A subtyped	4,144	3,449	1,189	2,728	1,261	250	13,021
A(H1N1)pdm09	768 (19%)	353 (10%)	613 (52%)	1,187 (44%)	577 (46%)	238 (95%)	3,736 (29%)
A(H3N2)	3,376 (81%)	3,096 (90%)	576 (48%)	1,541 (56%)	684 (54%)	12 (5%)	9,285 (71%)
Influenza B	476 (9%)	441 (11%)	194 (7%)	229 (7%)	78 (5%)	387 (47%)	1,805 (10%)
Influenza B lineage determined	75	282	23	32	9	138	559
B/Yamagata-lineage	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0	0 (0%)	0 (0%)
B/Victoria-lineage	75 (100%)	281 (100%)	23 (100%)	32 (100%)	9	138 (100%)	558 (100%)
Total	5,364	3,960	2,642	3,320	1,440	817	17,543

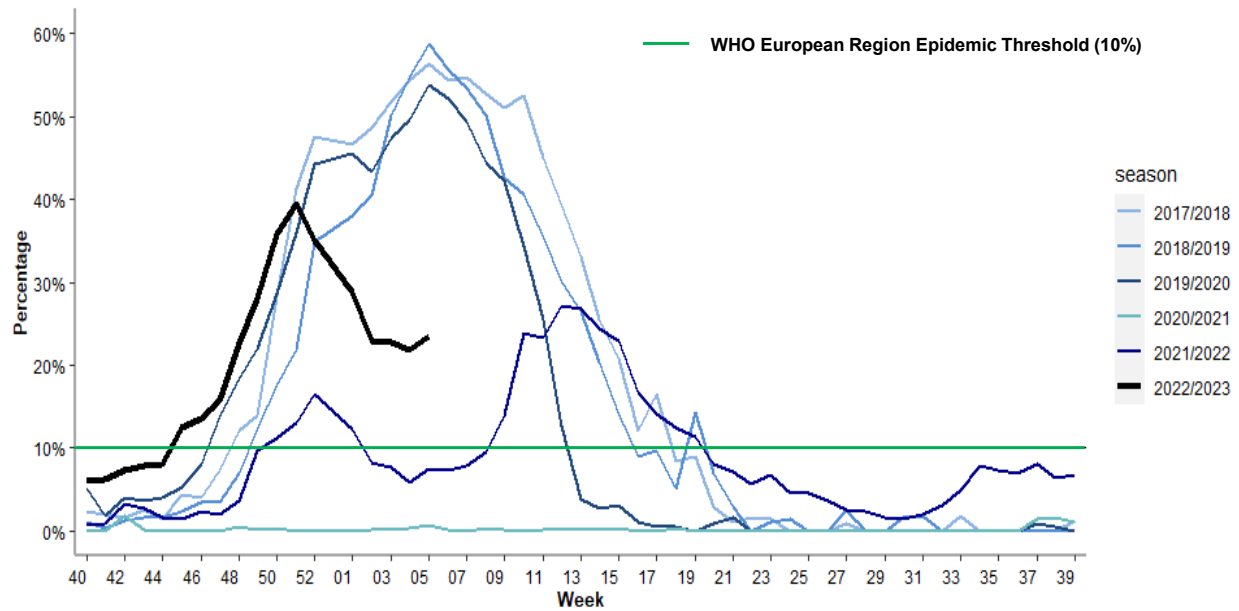
Figure E1. Influenza circulation² in the WHO European Region by (sub)type and Country, territory or area, 2022-2023 season



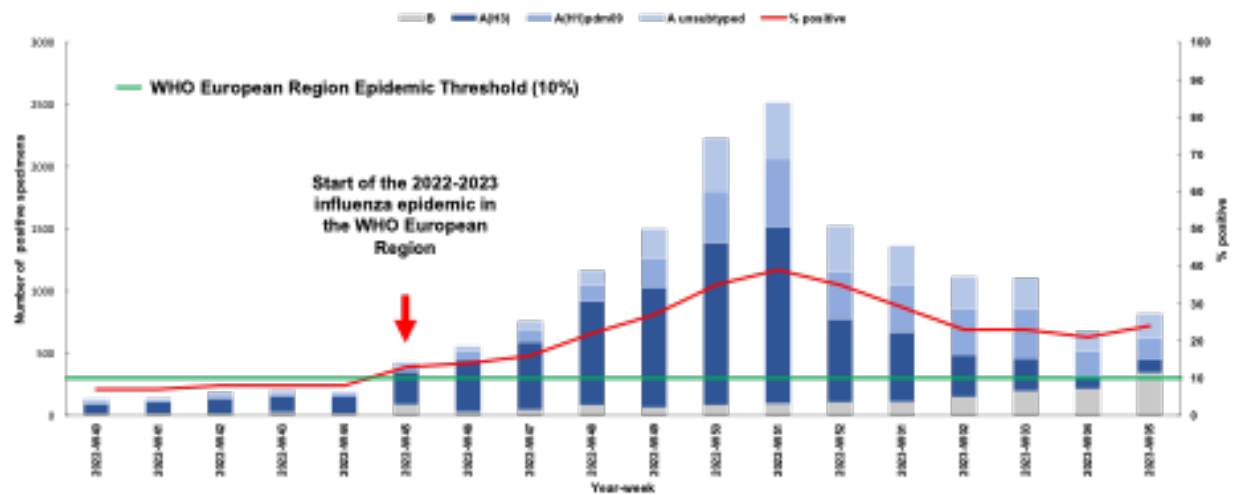
² Dominance of (sub)type is based on at least 60% of detections and co-dominance is based on between 40% and 60% of detections. Calculations are only considered valid if at least 10 viruses were detected.

Figure E2 A, B, C. Seasonal influenza activity in the WHO European Region: weeks 40/2022-05/2023

A. Percentage of sentinel ILI/ARI specimens testing positive for influenza by season



B. Sentinel specimens testing positive for influenza by week, (sub)type and percentage positive, 2022-2023 season



C. Non-sentinel specimens testing positive for influenza by week and (sub)type, 2022-2023 season

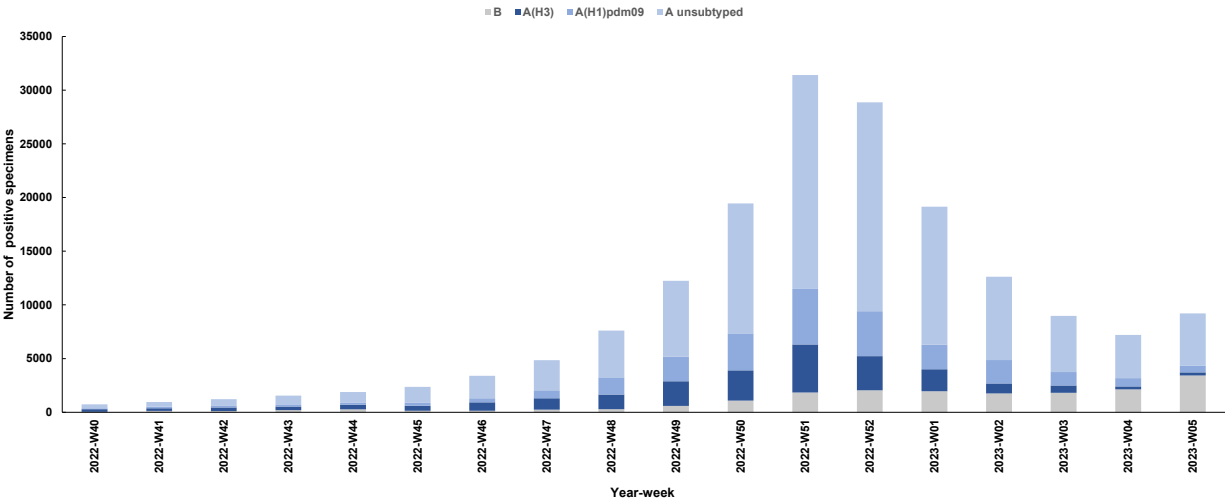
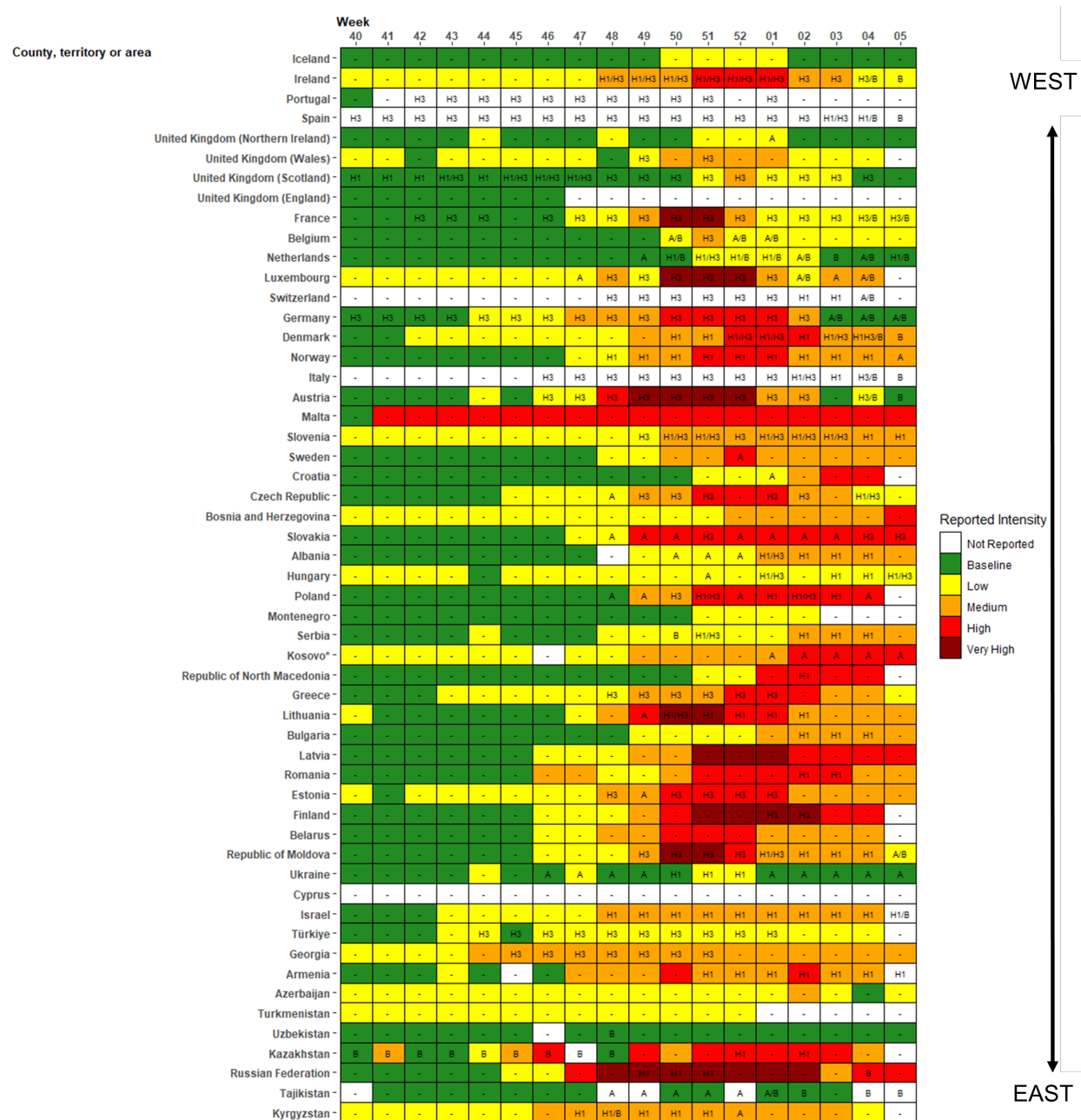


Figure E3. Intensity of influenza activity and dominant virus (sub)type in the WHO European Region by week and country, territory or area 2022-2023 season



*In accordance with United Nations Security Council Resolution 1244 (1999).

Figure E4. Distribution of virus (sub)type in laboratory-confirmed hospitalized cases in intensive care units by age group in the WHO European Region, 2022-2023 season

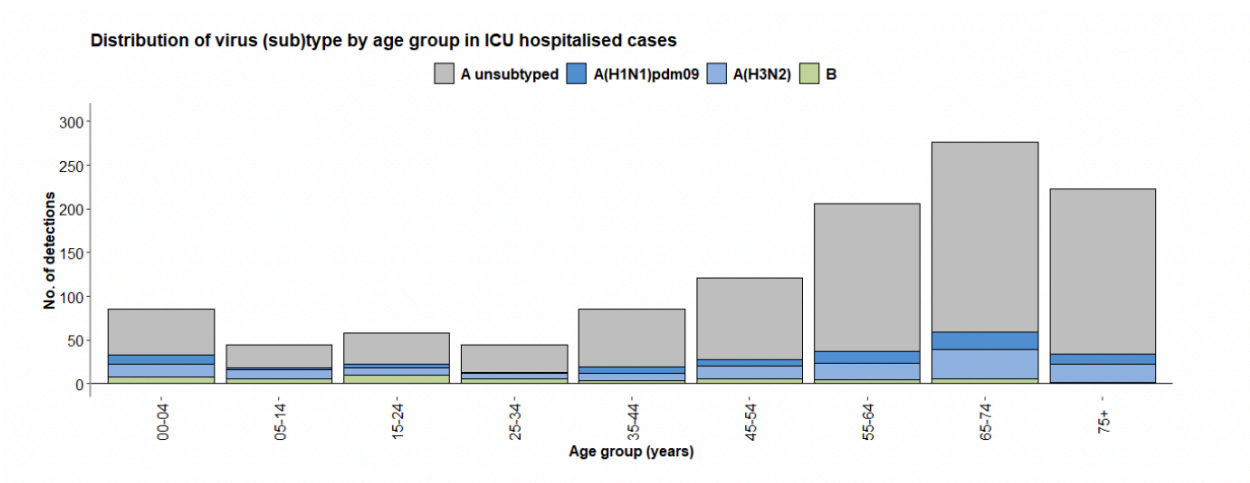


Table E3. Percentage distribution of severe acute respiratory infection (SARI) cases by age group and country, territory or area, and influenza positives, 2022-2023 season

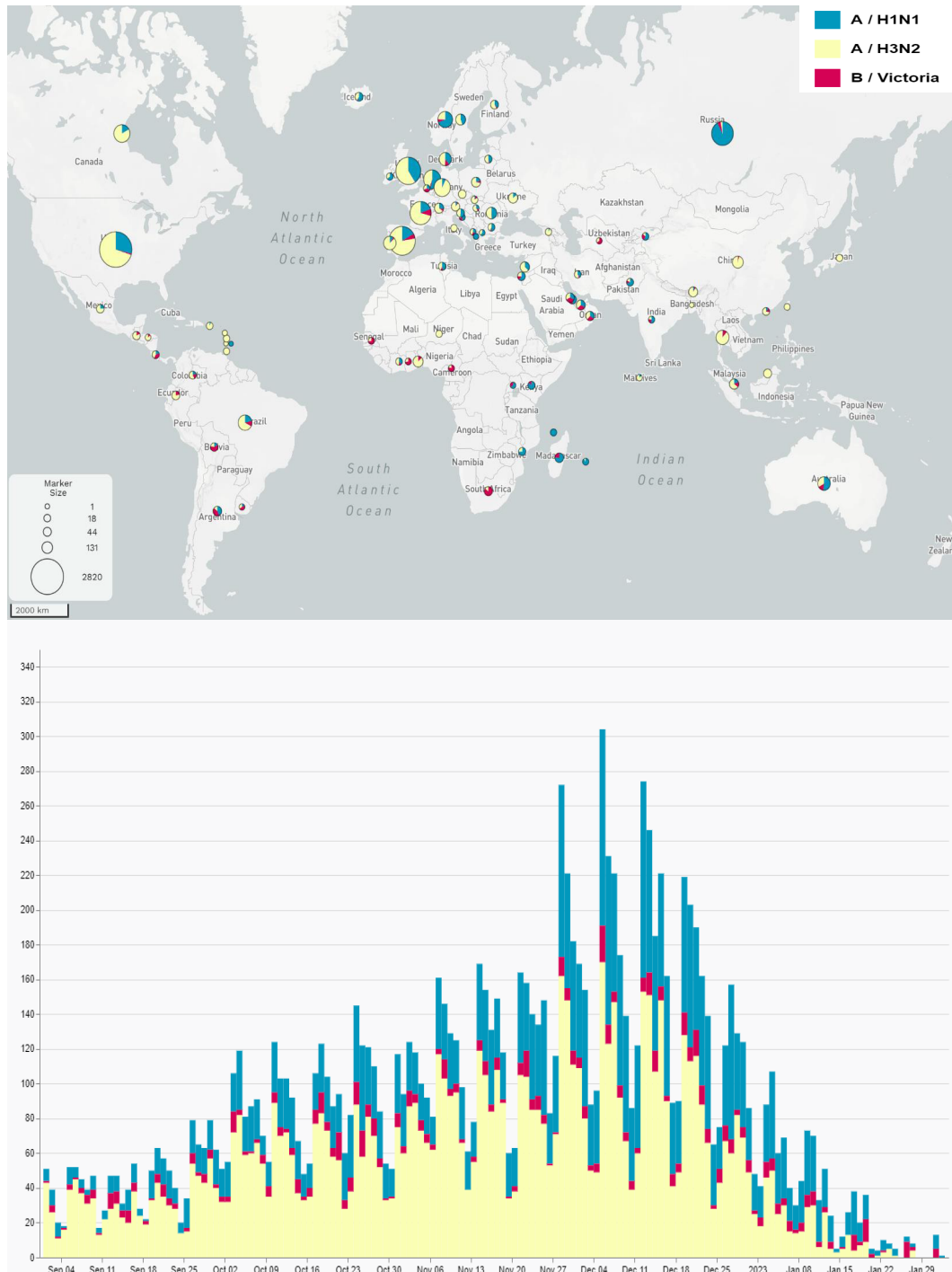
Country, territory or area	Total SARI cases	SARI cases with age data	Percentage (%) of total cases					SARI specimens	Positive detections (%)
			0-4 years	5-14 years	15-29 years	30-64 years	>=65 years		
Albania	1,257	877	34	14	11	41	0	1,267	216 (17%)
Armenia	1,832	1,832	56	13	7	13	11	1,299	276 (21%)
Belarus	928	925	19	8	8	35	30	345	4 (1%)
Belgium	157	139	55	5	3	37	0	269	35 (13%)
Bosnia and Herzegovina	682	682	24	11	9	33	24	191	25 (13%)
Croatia	556	302	62	10	3	25	0	556	19 (3%)
Georgia	3,202	3,198	70	18	3	5	4	1,742	269 (15%)
Germany	17,743	8,218	57	10	5	28	0	17,743	0 (0%)
Ireland	94	94	0	0	14	86	0	345	65 (19%)
Kazakhstan	13,878	13,874	53	19	12	12	3	2,715	801 (30%)
Kyrgyzstan	4,474	4,474	81	11	2	4	2	1,846	488 (26%)
Lithuania	988							218	100 (46%)
Malta	128	128	0	0	12	88	0	471	77 (16%)
Montenegro	250	250	54	5	5	13	23	98	11 (11%)
North Macedonia	372	330	79	12	1	8	0	116	5 (4%)
Republic of Moldova	3,409	3,409	46	7	2	19	27	152	28 (18%)
Romania	202	202	36	9	4	50	0	366	171 (47%)
Russian Federation	1,406	1,406	31	21	23	16	10	1,046	183 (17%)
Serbia	2,277	2,204	10	5	7	45	33	192	66 (34%)
Slovakia	52	52	65	19	6	10	0	92	37 (40%)
Spain	33,674	7,120	87	13	0	0	0	0	0 (-%)
Tajikistan	80	80	85	15	0	0	0	0	0 (-%)
Türkiye	5,189	3,361	42	15	0	0	43	2,394	247 (10%)
Turkmenistan	3,100	2,705	71	24	0	0	5	3,100	9 (0%)
Ukraine	3,705	3,705	15	13	16	33	23	1,114	302 (27%)
Uzbekistan	406	406	59	13	11	14	3	334	30 (9%)
Kosovo*	94	64	39	36	8	17	0	81	0 (0%)
Total	100,135	60,037	32,964 (55%)	8,520 (14%)	3,815 (6%)	9,129 (15%)	5,609 (9%)	38,092	3,464 (9%)

*In accordance with United Nations Security Council Resolution 1244 (1999).

Global influenza activity

Microreact: Influenza by type/subtype

MicroReact map for influenza viruses with collection dates after 31st August 2022 through 7th February 2023 as deposited in GISAID, coloured by Type/subtype. Geographic markers scaled to detection proportions. Full length HA.



Submissions

Microreact map showing the geographic source of influenza virus specimens submitted to the Worldwide Influenza Centre in this reporting period.



Table 1a. Summary for influenza samples received (with collection dates after 2022-08-31) since the September 2022 VCM, by country*

MONTH	TOTAL RECEIVED Seasonal	A		H1N1pdm09		H3N2		B		B Victoria lineage	
Country		Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ²	Number received	Number propagated ¹	Number received	Number propagated ¹
2022											
SEPTEMBER											
Albania	3					3	in process				
Argentina	30			12	in process	3	in process			15	in process
Belgium	1									1	in process
Cameroon	3					1	0			2	in process
Croatia	2					2	1				
Cyprus	1					1	in process				
France	15			3	in process	10	in process			2	2
Germany	6					5	5			1	1
Iran	4			3	3	1	1	5	in process		
Madagascar	5										
Netherlands	9			7	in process	2	in process				
Norway	14			12	1	1	1			1	0
Oman	19			6	in process	8	in process			5	in process
Portugal	2					2	2				
Qatar	14			4	in process	5	3	1	0	4	0
Saudia Arabia	11			2	in process	1	in process			8	in process
Senegal	1									1	in process
South Africa	16									16	16
Spain	38			11	in process	27	in process				
Sweden	5			1	in process	4	in process				
UK (England)	7			4	4	3	in process				
OCTOBER											
Austria	4					4	in process				
Argentina	21			8	in process	4	in process			9	in process
Belgium	1									1	in process
Cameroon	17					7	in process			10	in process
Croatia	10			3	3	5	in process			2	2
Denmark	11			4	3	5	5			2	2
France	28			9	9	12	11			7	in process
Germany	19			1	in process	18	18			8	0
Ghana	15					7	0				
Greece	1					1	in process				
Hong Kong	2					2	2				
Iceland	2					2	in process				
Iran	14			4	4	9	8			1	1
Ireland	13	1	0	6	in process	6	in process	10	in process		
Kazakhstan	10									4	4
Kyrgyzstan	4										
Lithuania	4	1	in process			3	in process				
Madagascar	9	1	0	2	in process			1	0	5	in process
Mauritius	2			2	in process						
Netherlands	13			6	in process	4	in process			3	in process
Norway	35			18	in process	12	11			5	4
Oman	81			25	in process	29	in process	1	0	26	in process
Portugal	33			8	in process	25	25				
Qatar	55			30	in process	1	0	2	0	22	in process
Romania	2			2	in process						
Saudia Arabia	2			1	in process	1	in process				
Senegal	22					1	0			21	in process
Slovenia	3					3	in process				
South Africa	1									1	1
Sweden	8			3	in process	4	in process	1	in process		
Spain	194	3	0	22	in process	167				2	in process
Switzerland	3			1	1	2	2				
Togo	12					6	in process			6	in process
Tunisia	1					1	in process				
UK (England)	6			1	1	5	5				
UK (N. Ireland)	14			13	0	1	0				
NOVEMBER											
Austria	2			1	in process					1	in process
Argentina	3			1	1					2	2
Belgium	8	1	in process	3	3	1	in process			3	in process
Bosnia and Herzegovina	2							2	in process		
Cameroon	9					4	1			5	in process
Croatia	6			3	3					3	in process
Denmark	3					2	2			1	1
Estonia	15			1	in process	14	in process				
France	9					7	in process			2	1
Georgia	15	9	0	2	0	4	in process				
Germany	15			6	in process	8	in process			1	1
Ghana	5					3	0			2	0
Greece	7	3	in process			3	in process			1	in process
Hong Kong	11			1	1	8	8			2	2
Hungary	7			2	2	4	4			1	1
Iceland	14			7	in process	7	in process				
Iran	2					2	1				
Ireland	24	1	0	12	in process	10	in process			1	in process
Israel	8			4	in process	2	2			2	in process
Italy	26			4	4	22	in process				
Kazakhstan	11			1	in process	4	in process	10	in process		
Kyrgyzstan	14			3	in process	1	in process	1	0	3	in process
Lebanon	1										
Lithuania	29			13	13	14	in process	1	0	1	1
Macedonia	1					1	in process				
Madagascar	25	1	0	19	in process			2	in process	3	in process
Mauritius	3			3	in process						
Montenegro	1					1	in process				
Morocco	6									6	in process
Netherlands	6			3	3	3	in process				

1. Propagated to sufficient titre to perform HI assay (the totalled number does not include any from batches that are in process)

2. Propagated to sufficient titre to perform HI assay in the presence of 20nM oseltamivir (the totalled number does not include any from batches that are in process)

Samples received were in lysis buffer - only genetic characterisation possible

Some samples were sent as RNA - only genetic characterisation possible

Some samples not cultured because Ct value high (>30), failed sequence, identical sequence, mixed sequence or SARS-COV-2 positive

* As of 2023-02-14

Table 1b. Summary for influenza samples received (with collection dates after 2022-08-31) since the September 2022 VCM, by country*

MONTH	TOTAL RECEIVED Seasonal	A		H1N1pdm09		H3N2		B		B Victoria lineage	
Country		Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ²	Number received	Number propagated ¹	Number received	Number propagated ¹
Norway	13			9	6	4	2				
Poland	10			3	in process	7	in process				
Portugal	30			5	in process	23	in process			2	2
Romania	8			3	in process	5	in process				
Saudia Arabia	7			3	in process	1	in process			3	in process
Slovakia	7	1	in process			6	in process				
Slovenia	9			6	in process	3	in process				
Spain	101	3	in process	8	in process	89	in process			1	0
Sweden	8			4	in process	3	in process			1	in process
Switzerland	11			3	3	7	7			1	1
Tajikistan	7			7	in process						
Togo	29					2	in process			27	in process
Turkmenistan	1					1	1				in process
Tunisia	16			4	in process	6	in process			6	0
Ukraine	22			3	2	19	in process				
UK (England)	1			1	1						
UK (Wales)	2							2	in process		
DECEMBER											
Albania	43	1	in process	24	in process	18	in process				
Austria	35			5	in process	27	in process	1	0	2	in process
Belgium	26			3	2	12	in process			11	in process
Bosnia and Herzegovina	1							1	in process		
Bulgaria	9			4	in process	5	in process				
Cameroon	8					5	1			3	1
Cyprus	25			4	in process	21	in process				
Estonia	55			12	in process	41	in process			2	in process
Georgia	23			3	1	17	in process	2	0	1	0
Germany	13			8	in process	3	3			2	in process
Greece	2			1	in process					1	in process
Hong Kong	1					1	1				
Hungary	5			2	2	3	3				
Iceland	29			19	in process	9	in process			1	0
Ireland	28			19	in process	9	in process				
Israel	42			28	in process	7	in process	1	in process	6	in process
Italy	5			2	2	3	in process				
Kazakhstan	6			6	in process						
Kyrgyzstan	22	9	in process	11	in process	1	in process	1	0		
Latvia	11			5	in process	4	in process	2	in process		
Lebanon	28			14	in process	14	in process				
Macedonia	2	1	in process			1	in process				
Madagascar	31			29	in process					2	in process
Mauritius	8			7	in process	1	in process				
Montenegro	6			2	in process	3	in process			1	in process
Morocco	24			9	in process	3	in process			12	in process
Netherlands	19			8	in process	6	in process			5	5
Poland	83			14	in process	62	in process			7	in process
Portugal	12			4	in process	6	in process			2	2
Romania	19			9	in process	8	in process			2	1
Saudia Arabia	2			2	in process						
Serbia	12			1	in process	4	in process	7	in process		
Slovakia	28			1	in process	26	in process			1	in process
Slovenia	59	6	in process	17	in process	35	in process			1	in process
Spain	27			9	in process	11	in process	7	in process		
Switzerland	26			8	8	15	15			3	3
Tajikistan	18			13	in process			5	in process		
Togo	8					2	in process			6	in process
Turkmenistan	5					1	1			4	4
Tunisia	79			34	in process	27	in process			18	in process
Ukraine	76	5	0	10	5	61	in process				
UK (Wales)	11							10	in process	1	in process
2023											
JANUARY											
Albania	50			36	in process	14	in process				
Bosnia and Herzegovina	4			1	in process	2	in process	1	in process		
Bulgaria	27			15	in process	8	in process			4	in process
Cyprus	3			2	in process	1	in process				
Estonia	26			12	in process	12	in process			2	in process
Georgia	1			1	0						
Germany	15			4	in process	4	in process			7	in process
Greece	19	2	in process	4	in process	12	in process			1	in process
Israel	15			12	in process	3	in process				
Latvia	9			2	in process	6	in process	1	in process		
Lebanon	21			9	in process	11	in process			1	in process
Macedonia	17			7	in process	9	in process	1	in process		
Montenegro	13			6	in process	5	in process			2	in process
Mauritius	24			21	in process	2	in process	1	in process		
Morocco	9			7	in process	1	in process			1	in process
Netherlands	2									2	2
Portugal	2									2	2
Romania	11			6	6	3	3	1	0	1	in process
Serbia	18			9	in process	6	in process				
Slovakia	4	2	in process			1	in process			1	in process
Slovenia	12			6	in process	4	in process	2	in process		
Spain	18			10	in process	7	in process			1	in process
Tajikistan	5			1	in process			4	in process		
UK (N. Ireland)	30			15	0	13	0	2	0		
UK (Wales)	11							10	in process	1	in process
FEBRUARY											
Albania	2			2	in process						
55 Countries	2788	54	1.9%	917	32.3%	1323	47.5%	103	3.7%	391	14.0%
										17.7%	

1. Propagated to sufficient titre to perform HI assay (the totalled number does not include any from batches that are in process)
2. Propagated to sufficient titre to perform HI assay in the presence of 20mM oseltamivir (the totalled number does not include any from batches that are in process)
Samples received were in lysis buffer - only genetic characterisation possible
Some samples were sent as RNA - only genetic characterisation possible
Some samples not cultured because Ct value high (>30), failed sequence, identical sequence, mixed sequence or SARS-COV-2 positive

* As of 2023-02-14

Table 2. Summary of influenza A(H1N1)pdm09 viruses collected after 2022-08-31, analysed by HI

MONTH	Country	Number received	Number propagated	Post-infection faecal culture spread against																	
				A/Denmark/328/2019 (cell)			108/215 (A/Victoria/207/2018)			A/Sydney/05/2011 (cell)			A/Sydney/05/2011 (egg)			A/Norway/2089/2022 (cell)			A/Norway/31684/2022 (egg)		
				Clade 6a.1			Clade 6a.1			Clade 6a.1			Clade 6a.1			Clade 6a.1			Clade 6a.1		
				Reductions in 10 titre	Reductions in 10 titre	Reductions in 10 titre	Reductions in 10 titre	Reductions in 10 titre	Reductions in 10 titre	Reductions in 10 titre	Reductions in 10 titre	Reductions in 10 titre	Reductions in 10 titre	Reductions in 10 titre	Reductions in 10 titre	Reductions in 10 titre	Reductions in 10 titre	Reductions in 10 titre	Reductions in 10 titre	Reductions in 10 titre	
2022																					
SEPTEMBER																					
	Argentina	12	0																		
	France	3	2	1		1	1	1				1	1		1		1				1
	Iran	3	3			3		3				3			3						
	Netherlands	7	3	2	1		3		1			3			3			1			1
	Norway	12	1			1		1				1			1			1			1
	Oman	6	0																		
	Qatar	4	0																		
	Saudi Arabia	2	0																		
	Spain	11	1			1		1					1			1					
	Sweden	1	0																		
	UK (England)	4	3	2		1	2		1	1	1	1	2		1	2		1	2		1
OCTOBER																					
	Argentina	8	3			3		3				3		3		3		3		3	
	France	3	2	1	1		3	2		1	1	3	1	1	3	2		3	2		3
	Denmark	4	3				1	2				1	2		1	2		1	2		1
	France	9	9	4	1	4	7	1	1		3	2	5	2	2	7	1	1	4		1
	Germany	1	0																		
	Iran	4	4	2	2								4			3	1				
	Ireland	6	0																		
	Madagascar	2	0																		
	Mauritius	2	0																		
	Netherlands	6	0																		
	Norway	18	2			2		2				2			2						
	Oman	25	4	4		4		4		2	2	3	1		2	2			2		4
	Portugal	18	2																		
	Qatar	36	5	5		5		5				5			2	3					
	Romania	2	0																		
	Saudi Arabia	1	0																		
	Sweden	3	0																		
	Spain	22	9	8	1		9		8	1		8	1		9				9		
	Switzerland	1	1			1		1		1		1			1				1		
	UK (England)	1	1	1		1		1		1		1			1				1		
	UK (Ireland)	13	0																		
NOVEMBER																					
	Austria	1	0																		
	Argentina	1	1			1		1				1		1		1		1			1
	Belgium	3	3	3		3		3		3		3		3		3		3		3	
	Croatia	3	3	2	1		3		2	1		2	1		2		3		3		
	Estonia	1	0																		
	Georgia	2	0																		
	Germany	6	2	1	1		1	1		1		1	1	1		1	1		2		
	Hong Kong	1	1									1	1	1		1			1		
	Hungary	2	2			2		2				1	1	1		2			2		
	Ireland	7	0																		
	Israel	12	6	2	4		6		1	5		4	2		6				6		
	Italy	4	4	3	1		3	1		3		1	3		1	3		1	4		
	Kazakhstan	1	0																		
	Kyrgyzstan	3	0																		
	Lithuania	13	13			10	3	13		2	8	3	7	5	1	13			12	1	
	Madagascar	18	0																		
	Mauritius	3	0																		
	Netherlands	3	3	2		3		3		1	1	1	2		1	2		1	2		1
	Norway	9	6	4	2	1	2	6		2	4	1	4	1	1	6		1	2	6	
	Poland	3	0																		
	Portugal	3	0																		
	Romania	3	0																		
	Saudi Arabia	3	0																		
	Russia	6	0																		
	Spain	9	1			1		1		1		1			1			1			
	Sweden	4	0																		
	Switzerland	3	3			3		3		3		3		3		3			2	1	
	Tajikistan	7	0																		
	Tunisia	4	0																		
	Ukraine	3	2	1	1		2			1	1	2			2			2			
	UK (England)	1	0																		
DECEMBER																					
	Albania	24	0																		
	Austria	5	0																		
	Belgium	3	2	1	1		2		2			1	1		2			2			
	Bulgaria	4	0																		
	Cyprus	4	0																		
	Estonia	12	0																		
	Georgia	3	1			1		1		1		1		1		1		1			
	Germany	8	4	3	1		4		3	1		4		3	1	4		3	1	4	
	Greece	1	0																		
	Hungary	2	1			1	1					1		1					1		
	Ireland	19	0																		
	Israel	19	0																		
	Italy	28	2	1		1	1	1		1	1		1	1		1	1		1	1	
	Japan	2	2	2		2		2		2		2		2		2		2		2	
	Kazakhstan	6	0																		
	Kyrgyzstan	11	0																		
	Lithuania	5	0																		
	Madagascar	14	0																		
	Mauritius	29	0																		
	Montenegro	7	0																		
	Morocco	2	0																		
	Netherlands	8	4	4		4		4		4		4		4		4		4			
	Poland	14	0																		
	Portugal	4	0																		
	Romania	9	0																		
	Saudi Arabia	2	0																		
	Slovakia	1	0																		
	Slovenia	17	10	8	2		10		10			10		10		10			10		
	Spain	9	0																		
	Switzerland	8	8	2	2	4	8		3	4	1	6	2		8			5	3		
	Tajikistan	13	0																		
	Tunisia	34	2																		
	Ukraine	10	5	2	3	4	1		2	2	3	4	1		5			2	5		
2023																					
JANUARY																					
	Albania	36	0																		
	Bosnia and Herzegovina	1	0																		
	Bulgaria	15	0																		
	Cyprus	2	0																		
	Estonia	12	0																		
	Georgia	1	0																		
	Germany	4	0																		
	Greece	4	0																		
	Iran	12	2	2		1	1	2		1	1		2		2			2			
	Latvia	2	0																		
	Lebanon	9	0																		
	Madagascar	7	0																		
	Montenegro	6	0																		
	Mauritius	21	0																		
	Morocco	7	0																		

Table 3a. Summary of influenza A(H3N2) viruses collected after 2022-08-31, analysed by HI

MONTH		Number received ¹	Number propagated ²	Post-infection ferret antisera raised against																							
				A/Darwin/9/2021 (Egg)			A/Stockholm/5/2021 (Cell)			A/Slovenia/87/20/2022 (Cell)			A/Catalonia/NSVH1615120/27/2022 (Cell)			A/Poland/97/2022 (Cell)			A/Norway/24873/2021 (Cell)			A/Thuringen/10/2022 (Cell)					
				Clade 2a			Clade 2a			Clade 2a.1			Clade 2a.1b			Clade 2a.2			Clade 2a.3			Clade 2b					
				Reductions in HI titre			Reductions in HI titre			Reductions in HI titre			Reductions in HI titre			Reductions in HI titre			Reductions in HI titre			Reductions in HI titre					
Country			S2 fold	4 fold	≥8 fold	S2 fold	4 fold	≥8 fold	S2 fold	4 fold	≥8 fold	S2 fold	4 fold	≥8 fold	S2 fold	4 fold	≥8 fold	S2 fold	4 fold	≥8 fold	S2 fold	4 fold	≥8 fold				
2022																											
SEPTEMBER																											
Albania	3	0																									
Argentina	3	2				2		1	1			2		1	1			2				1	1				
Cameroun	1	0																									
Croatia	2	1						1	1					1				1			1						
Cyprus	1	0																									
France	10	7	2	5		6	1		4	2	1	7			3	2	2	6	1		2	3	2				
Germany	5	5	5					3	2		2	2	1	2	1	1	3	4	1		5						
Iran	1	1																									
Netherlands	2	1																			1	1					
Norway	1	1																									
Oman	8	0																									
Portugal	2	2																									
Qatar	5	3	2	1		1	2		2	1		3			2	1	3				1	2					
Saudi Arabia	1	0																									
Spain	27	6	3	3		3	3		5			5			3	2		5			1	5					
Sweden	4	0																									
UK (England)	3	1							1			1		1		1		1					1				
OCTOBER																											
Austria	4	0																									
Argentina	4	2				1	1		1	1		1	1		2		2					2					
Cameroun	7	0																									
Croatia	5	2				2			2			2			1	1		1	1			1	1				
Denmark	5	5				1	4	5			5	1		4		5	5				5	1					
France	12	11	2	7	2	11			3	3	5	5	1	5		3	8	8	3		9	2					
Germany	18	18	15	3		11	6	1	8	6	4	7	1	10	4	4	10	11	7		18						
Ghana	7	0																									
Greece	1	0																									
Hong Kong	2	2				1	1		1			2		2		1	1	1	1			2					
Iceland	2	0																									
Iran	9	8	2	6		8				3	5	3	5		3	5	4	4			8						
Ireland	6	0																									
Lithuania	3	0																									
Netherlands	4	2				1	1		2		1	1	1		1		1	1			1	1					
Norway	12	11	1	6	4	6	3	2	3	4	4	5	2	4	3	2	6	3	5	3	5	3	3				
Oman	29	3	1	2		2	3					2	1			3	3				3						
Portugal	25	25	5	4	16	19	6		1	3	21		1	24		1	24	10	15		24		1				
Qatar	1	0																									
Saudi Arabia	1	0																									
Senegal	1	0																									
Slovenia	3	0																									
Sweden	4	0																									
Spain	167	30	1	13	16	17	10	3	2	7	21	5	7	18	2	4	24	13	8	9	27	1	2				
Switzerland	2	1																									
Togo	6	0																									
Tunisia	1	0																									
UK (England)	5	5				5					5		2	3		5	2	1	2		1	2	2				
UK (N. Ireland)	1	0																									
NOVEMBER																											
Belgium	1	0																									
Cameroun	4	1				1	1		1		1	1		1		1	1				1						
Denmark	2	2				1	1	2		1		1	1		1	1	2				1	1					
Estonia	14	0																									
France	7	2				2		2			2		2			2	2				2	1					
Georgia	4	1							1																		
Germany	8	3				1	2	1	2		3	1		3	1		3	1	1	1	1	1	1				
Ghana	3	0																									
Greece	3	0																									
Hong Kong	8	8	6	2		7	1		4	3	1	5	3		3	4	1	8			7	1					
Hungary	4	4	1	1	2	1	3		3		1	3		1	2	1	1		4		1	3					
Iceland	7	0																									
Iran	2	1																									
Ireland	10	5				4	1	3	2		2	1		2	3		2					2	3				
Israel	2	2				1	1	2			2		2		2		2				2	1	1				
Italy	22	2				1	1		1		2		2		2		2										
Kyrgyzstan	4	0																									
Lebanon	1	0																									
Lithuania	14	1	1				1			1		1			1			1				1					
Macedonia	1	0																									
Montenegro	1	0																									
Netherlands	3	0																									
Norway	4	2				1	1	1	1		1	1	1	1	1			1	1	1	1	1	1				
Poland	7	0				1	3	3	2	1	4																
Portugal	23	7	1	3	3	2	1	4																			
Romania	5	0																									
Saudi Arabia	1	0																									
Slovakia	6	0																									
Slovenia	3	0																									
Spain	87	8				2	6	2	5	1	1	7	1	7		1	7	2	4	2	7		1				
Sweden	3	0																									
Switzerland	7	7				6	1	4	3		4	3		4	3		3	4	3	3	1	4	2				
Togo	2	0																									
Turkmenistan	1	1				1		1			1		1			1		1									
Tunisia	6	0																									
Ukraine	19	0																									

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared
2. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir

Numbers highlighted in yellow show the number of viruses subjected to HI assay for 'completed' sample sets. Under a 'sequence first' virus characterisation scheme: where multiple viruses shared the same HA sequence only a selection were propagated to allow assay by HI

Samples received were in lysis buffer - only genetic characterisation possible

Sample sent are RNA, so only genetic characterisation possible

As of 2023-05-16

Table 3b. Summary of influenza A(H3N2) viruses collected after 2022-08-31, analysed by HI

MONTH	Number received ¹	Number propagated ²	Post-infection ferret antisera raised against																							
			A/Darwin/9/2021 (Egg)			A/Stockholm/5/2021 (Cell)			A/Slovenia/8720/2022 (Cell)			A/Catalonia/NSVH1615120/67/2022 (Cell)			A/Poland/97/2022 (Cell)			A/Norway/24873/2021 (Cell)			A/Thuringen/10/2022 (Cell)					
			Clade 2a			Clade 2a			Clade 2a.1			Clade 2a.1b			Clade 2a.2			Clade 2a.3			Clade 2b					
			Reductions in HI titre			Reductions in HI titre			Reductions in HI titre			Reductions in HI titre			Reductions in HI titre			Reductions in HI titre			Reductions in HI titre					
Country			S2 fold	4 fold	28 fold	S2 fold	4 fold	28 fold	S2 fold	4 fold	28 fold	S2 fold	4 fold	28 fold	S2 fold	4 fold	28 fold	S2 fold	4 fold	28 fold	S2 fold	4 fold	28 fold			
DECEMBER																										
Albania	18	0																								
Austria	27	0																								
Belgium	12	6		4	2		2	4		2	4	1	1	4			6		1	5	1	3	2			
Bulgaria	5	0																								
Cameroon	5	1		1				1			1		1				1	1				1				
Cyprus	21	0																								
Estonia	41	0																								
Georgia	17	7		1	2	4		2	5	1	2	4	3	2	2	1	3	3		1	6	1	1	5		
Germany	3	3		1	1	1		3		2	1	1	1	1		2	1	1		2	1	1	2			
Hong Kong	1	1																								
Hungary	3	3		1	2		3		1	1	2	1	1	2	1		2	1		2	1	2	1			
Iceland	9	0																								
Ireland	9	0																								
Israel	7	3		2	1		1	2			3			3			3	1	1	1	1	1	1			
Italy	3	2		1	1						2			2						2	2					
Kyrgyzstan	1	0																								
Latvia	4	0																								
Lebanon	14	0																								
Macedonia	1	0																								
Mauritius	1	0																								
Montenegro	3	0																								
Morocco	3	0																								
Netherlands	6	5		2	3		3	2	2		3	2		3			5			5	1	3	1			
Poland	62	0																								
Portugal	6	1			1		1				1			1			1			1	1					
Romania	8	3		1	2		2	1			3	1		2			3			3	1	1	1			
Serbia	4	0																								
Slovakia	26	0																								
Slovenia	38	6		2	6	3	5		1	1	6	1		7	1		7		8		7	1				
Spain	11	0																								
Switzerland	15	15	1	5	9	4	11		3	4	8	5	3	7	3	3	9	4	8	3	7	7	1			
Togo	2	0																								
Turkmenistan	1	1		1		1				1		1				1		1			1					
Tunisia	27	2	1	1		1	1		1	1	1	2			1	1	2				1	1				
Ukraine	61	5		2	3		5		1	4		5			2	3		4	1		3	2				
JANUARY																										
Albania	14	0																								
Bosnia and Herzegovina	2	0																								
Bulgaria	9	0																								
Cyprus	1	0																								
Estonia	12	0																								
Germany	4	0																								
Greece	12	0																								
Israel	3	1		1			1				1			1			1			1			1			
Latvia	6	0																								
Lebanon	11	0																								
Macedonia	9	0																								
Montenegro	5	0																								
Mauritius	2	0																								
Morocco	1	0																								
Romania	3	3			3		2	1			3			3			3			3	3					
Serbia	6	0																								
Slovakia	1	0																								
Slovenia	4	1		1		1					1			1			1			1						
Spain	9	0																								
UK (N. Ireland)	13	0																								
	1323	283	56	118	109	135	107	41	51	66	165	81	47	154	37	55	190	111	109	62	184	67	32			
51 Countries			19.8%	41.7%	36.5%	47.7%	37.8%	14.5%	18.1%	23.4%	58.5%	28.7%	16.7%	54.6%	13.1%	19.5%	67.4%	39.4%	38.7%	22.0%	65.0%	23.7%	11.3%			

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared
2. Propagated to sufficient titre to perform HI assay in presence of 20mM oseltamivir

Numbers highlighted in yellow show the number of viruses subjected to HI assay for 'completed' sample sets. Under a 'sequence first' virus characterisation scheme: where multiple viruses shared the same HA sequence only a selection were propagated to allow assay by HI

Samples received were in lysis buffer - only genetic characterisation possible

Sample sent as RNA, so only genetic characterisation possible

As of 2023-02-16

Table 4. Summary of influenza B (Victoria-lineage) viruses collected after 2022-08-31 and analysed by HI

MONTH	Country	Number received	Number propagated ^{1,2}	B Victoria lineage											
				Post infection ferret antisera raised against											
				B/Austria/1359417/2021 (cell)			B/Austria/1359417/2021 (egg)			B/Austria/1359417/2021 (egg G141R)			B/Washington/02/2019 (egg)		
				Code V1A.3a.2			Code V1A.3a.2			Code V1A.3a.3			Code V1A.3		
				≥2 fold	4 fold	≥8 fold	≥2 fold	4 fold	≥8 fold	≥2 fold	4 fold	≥8 fold	≥2 fold	4 fold	≥8 fold
2022															
SEPTEMBER															
	Argentina	15	0												
	Belgium	1	0												
	Cameroon	2	0												
	France	2	2	2			2					2			2
	Germany	1	1	1			1					1			1
	Oman	5	0												
	Qatar	4	0												
	Saudi Arabia	8	0												
	Senegal	1	0												
	South Africa	16	16	16			12	4				16			16
OCTOBER															
	Argentina	9	2	2			2					2			2
	Belgium	1	0												
	Cameroon	10	0												
	Croatia	2	1	1			1				1				1
	Denmark	2	2	2			2					2			2
	France	7	6	6			6			1		5			6
	Ghana	8	0												
	Iran	1	1	1			1					1			1
	Kyrgyzstan	4	4	4			4					4			4
	Madagascar	5	0												
	Netherlands	3	1	1			1					1			1
	Norway	5	4	4			4				3	1			4
	Oman	26	2	2			2					2		1	1
	Qatar	22	4	4			4					4			4
	Senegal	21	7	7			7					7			7
	South Africa	1	1	1			1					1			1
	Spain	2	1	1			1					1			1
	Togo	6	0												
NOVEMBER															
	Austria	1	0												
	Argentina	2	2	2			2					2			2
	Belgium	3	0												
	Cameroon	5	0												
	Croatia	3	1	1			1					1			1
	Denmark	1	1	1			1					1			1
	France	2	1	1			1					1			1
	Germany	1	1	1			1					1			1
	Ghana	2	0												
	Greece	1	0												
	Hong Kong	2	2	2			2					2			2
	Hungary	1	1	1			1					1			1
	Ireland	1	0												
	Israel	2	1	1			1				1				1
	Kyrgyzstan	3	0												
	Lithuania	1	1	1			1					1			1
	Madagascar	3	0												
	Morocco	6	0												
	Portugal	2	2	2			2				1	1			2
	Saudi Arabia	3	0												
	Spain	1	0												
	Sweden	1	0												
	Switzerland	1	1	1			1					1			1
	Togo	27	0												
	Tunisia	6	0												
DECEMBER															
	Austria	2	0												
	Belgium	11	6	6			6				3	3			6
	Cameroon	3	1	1			1				1				1
	Estonia	2	0												
	Georgia	1	0												
	Germany	2	0												
	Greece	1	0												
	Iceland	1	0												
	Israel	6	4	4			4				2	2			4
	Madagascar	2	0												
	Montenegro	1	0												
	Morocco	12	0												
	Netherlands	5	5	5			5					5			5
	Poland	7	0												
	Portugal	2	2	2			2				1	1			2
	Romania	1	1	1			1					1			1
	Slovakia	1	0												
	Slovenia	1	0												
	Switzerland	3	3	3			3					3			3
	Togo	6	0												
	Tunisia	18	1	1			1				1				1
	Turkmenistan	4	4	4			4					4			4
	UK (Wales)	1	0												
2023															
JANUARY															
	Bulgaria	4	0												
	Estonia	2	0												
	Germany	7	0												
	Greece	1	0												
	Lebanon	1	0												
	Montenegro	2	0												
	Morocco	1	0												
	Netherlands	2	2	2			2				1	1			2
	Portugal	2	2	2			2					2			2
	Romania	1	0												
	Slovakia	1	0												
	UK (Wales)	1	0												
		390	100	100	0	0	96	4	0	1	15	84	0	1	99
	44 Countries			100.0%	0.0%	0.0%	96.0%	4.0%	0.0%	1.0%	15.0%	84.0%	0.0%	1.0%	99.0%

¹ Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

² Propagated to sufficient titre to perform HI assay

Some samples not cultured because Ct value high (>30), failed sequence, identical sequence, mixed sequence or SARS-CoV-2 positive

As of 2023-02-12

Table 5. Isolation rates for specimens collected after 2022-08-31

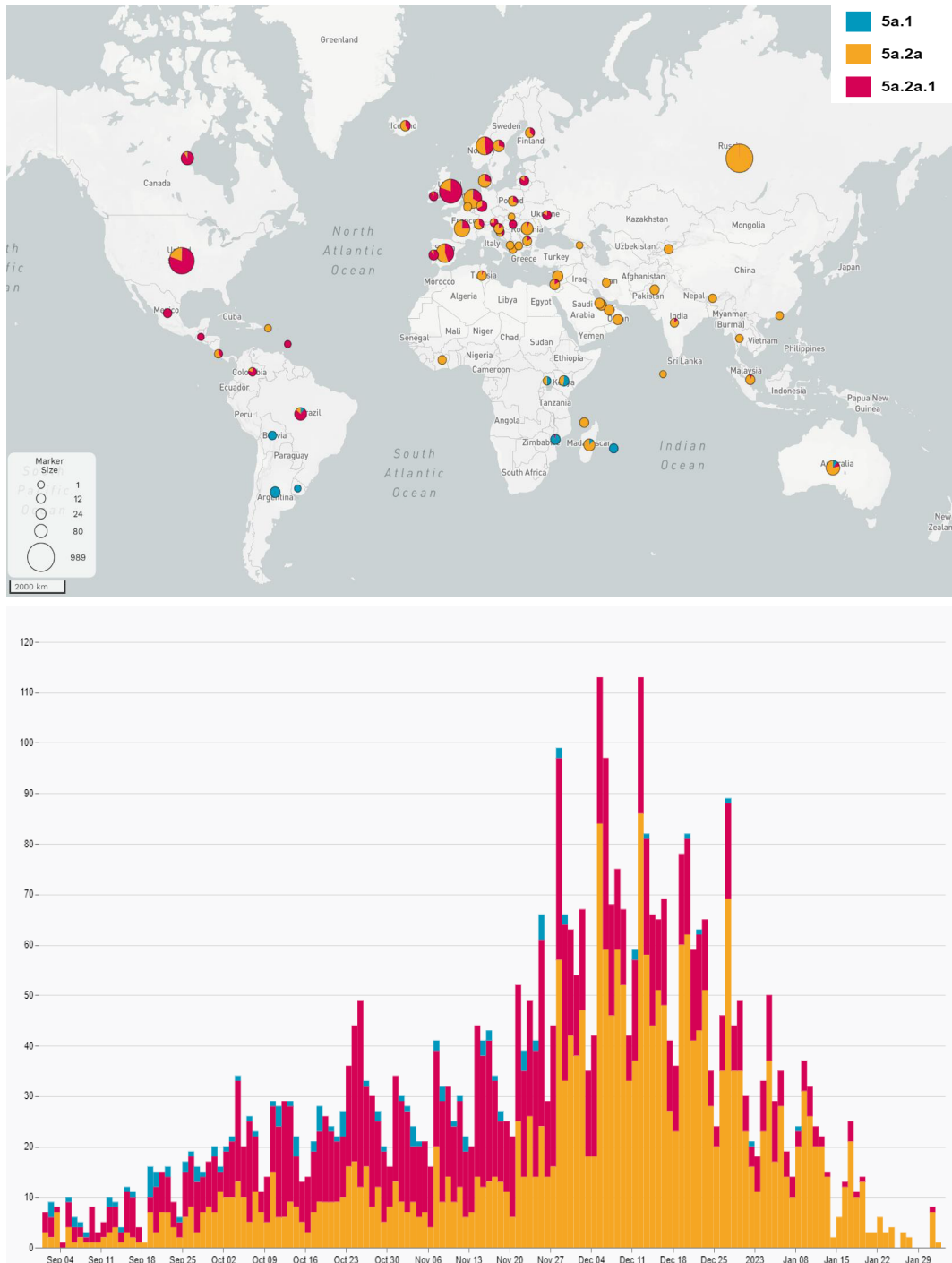
Seasonal Virus	Clinical sample			Virus isolate		
	Number processed	Number propagated¹	Percent recovered	Number processed	Number propagated¹	Percent recovered
H1	107	60	56%	99	96	97%
H3	151	93	62%	199	191	96%
B Victoria	58	20	35%	83	80	96%

1. Propagated to sufficient HA titre to perform HI assay

Influenza A H1N1

Microreact: H1N1 by clade

MicroReact map for influenza viruses with collection dates after 31st August 2022 through to 7th February 2023 as deposited in GISAID, coloured by H1 clade. Geographic markers scaled to detection proportions. Full length HA.



Genetic analyses: H1N1

Clade 5a.1.

Signature amino acid substitutions: D187A, Q189E.

Vaccine viruses: A/Guangdong-Maonan/SWL1536/2019 (egg) and A/Hawaii/70/2019 (cell)

Clade 5a.2.

Signature amino acid substitutions: N156K, L161I, V250A.

Vaccine viruses: A/Victoria/2570/2019 (egg) and A/Wisconsin/588/2019 (cell)

Clade 5a.2a.

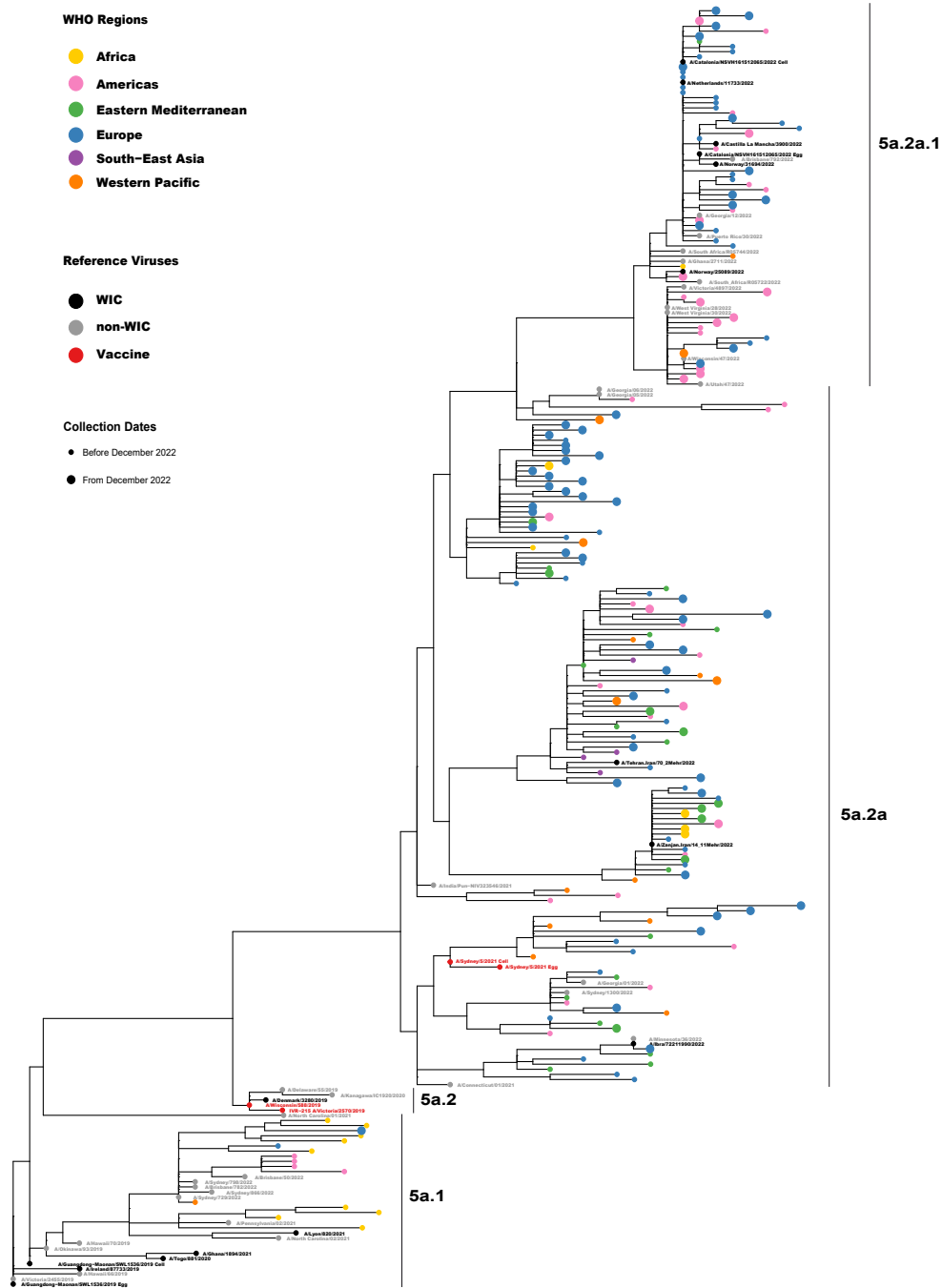
Signature amino acid substitutions: K54Q, A186T, Q189E, E224A, R259K, K308R

Vaccine viruses: A/Sydney/5/2021 (cell and egg) which also contain D94N and T216A

Clade 5a.2a.1. Signature amino acid substitutions: P137S, K142R, D260E

Maximum likelihood phylogenetic trees: H1N1

Maximum likelihood phylogenetic trees inferred from full-length HA sequence data obtained from GISAID for viruses with collection dates after 31st August 2022 and submitted to GISAID up to 7th February 2023, manually curated and then downsampled using Treemmer to retain a representative tree topology of 250 sequences.



Vaccine viruses

Reference viruses (WIC)

Reference viruses (non-WIC)

Collection dates

Sep 2022

Oct 2022

Nov 2022

Dec 2022

Jan 2023



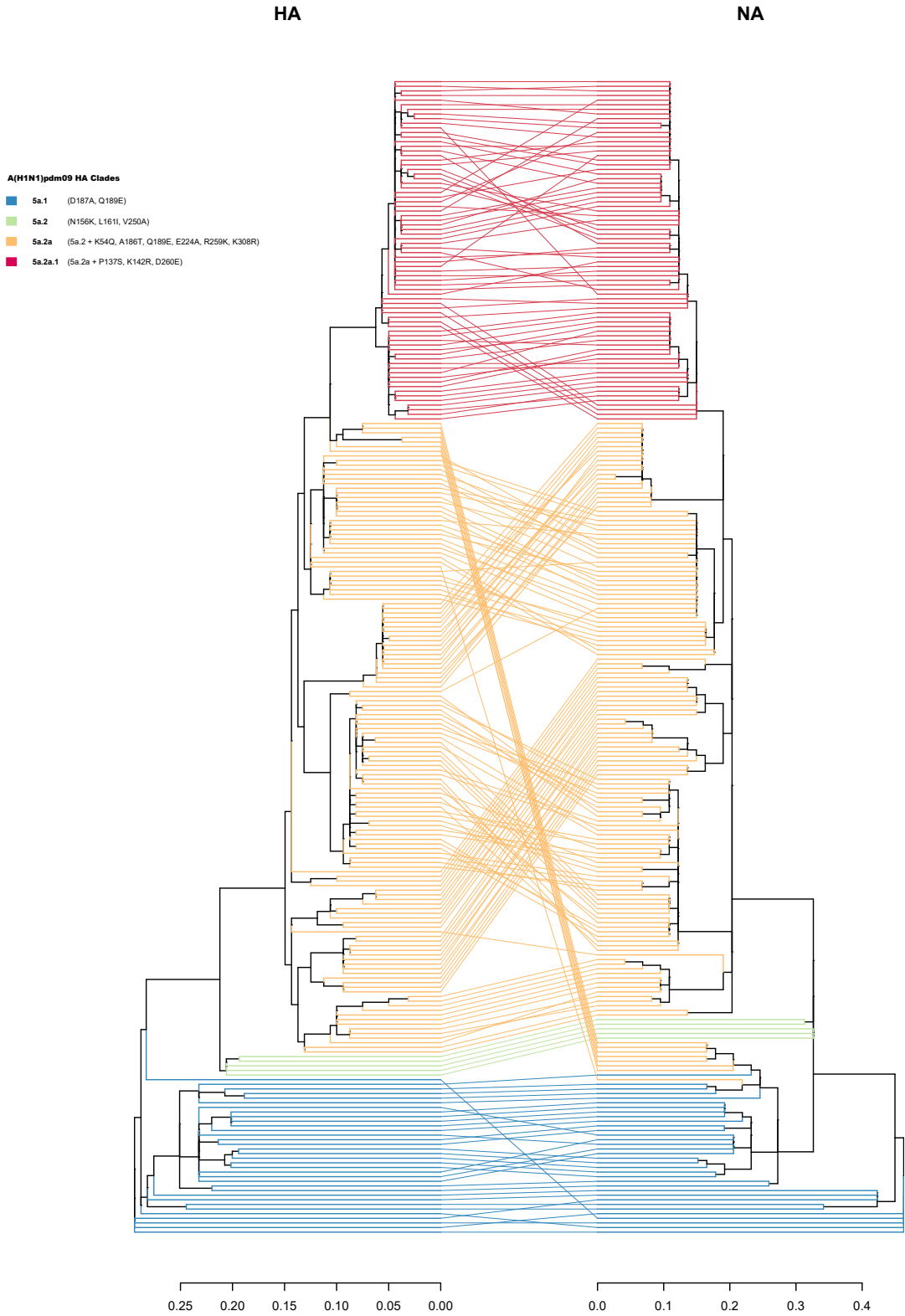
5a.2a.1

5a.2a

5a.2

5a.1

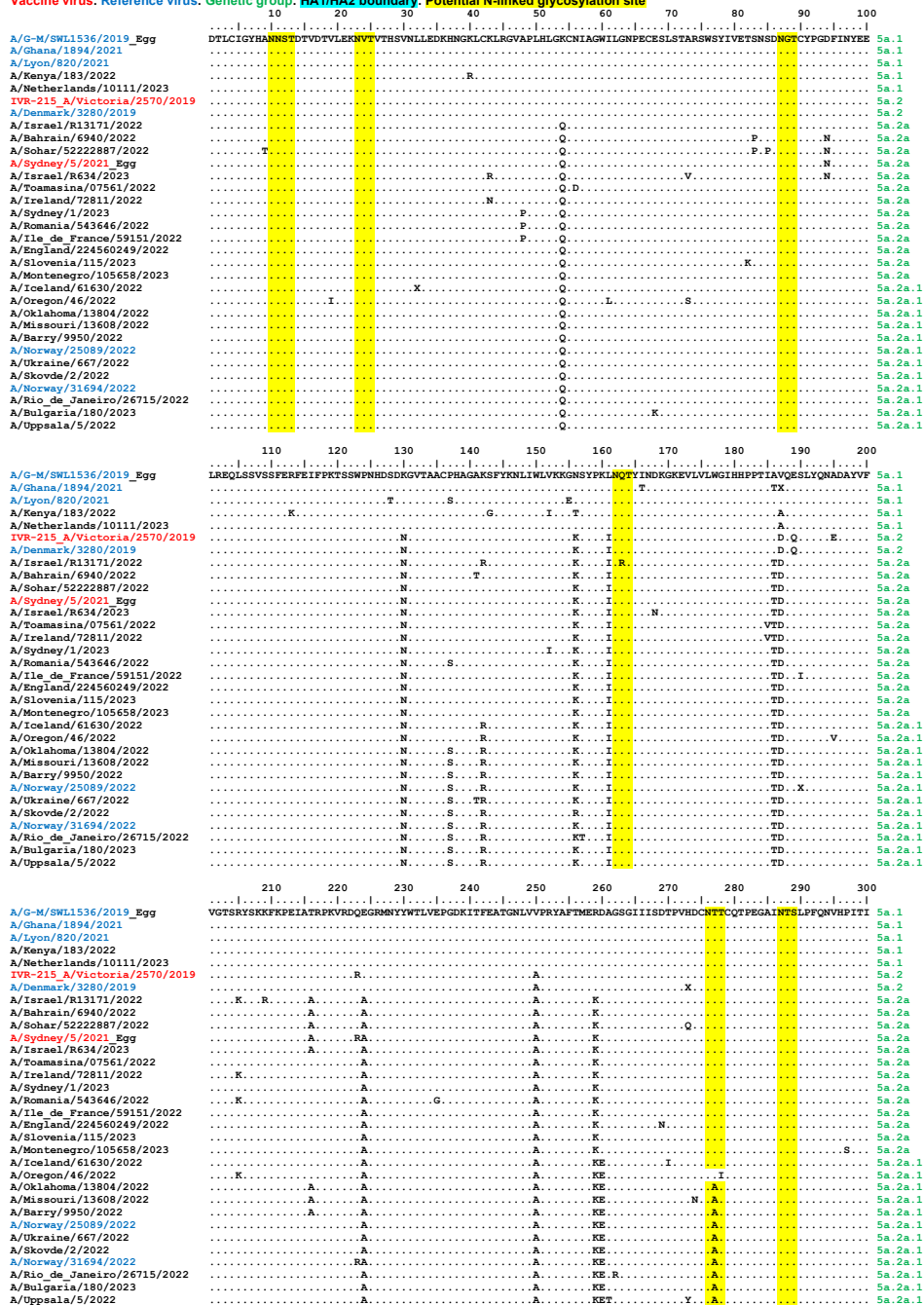
HA-NA tanglegams



Protein alignments HA

Figure H1-P1. HA protein alignment for A(H1N1)pdm09 viruses

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

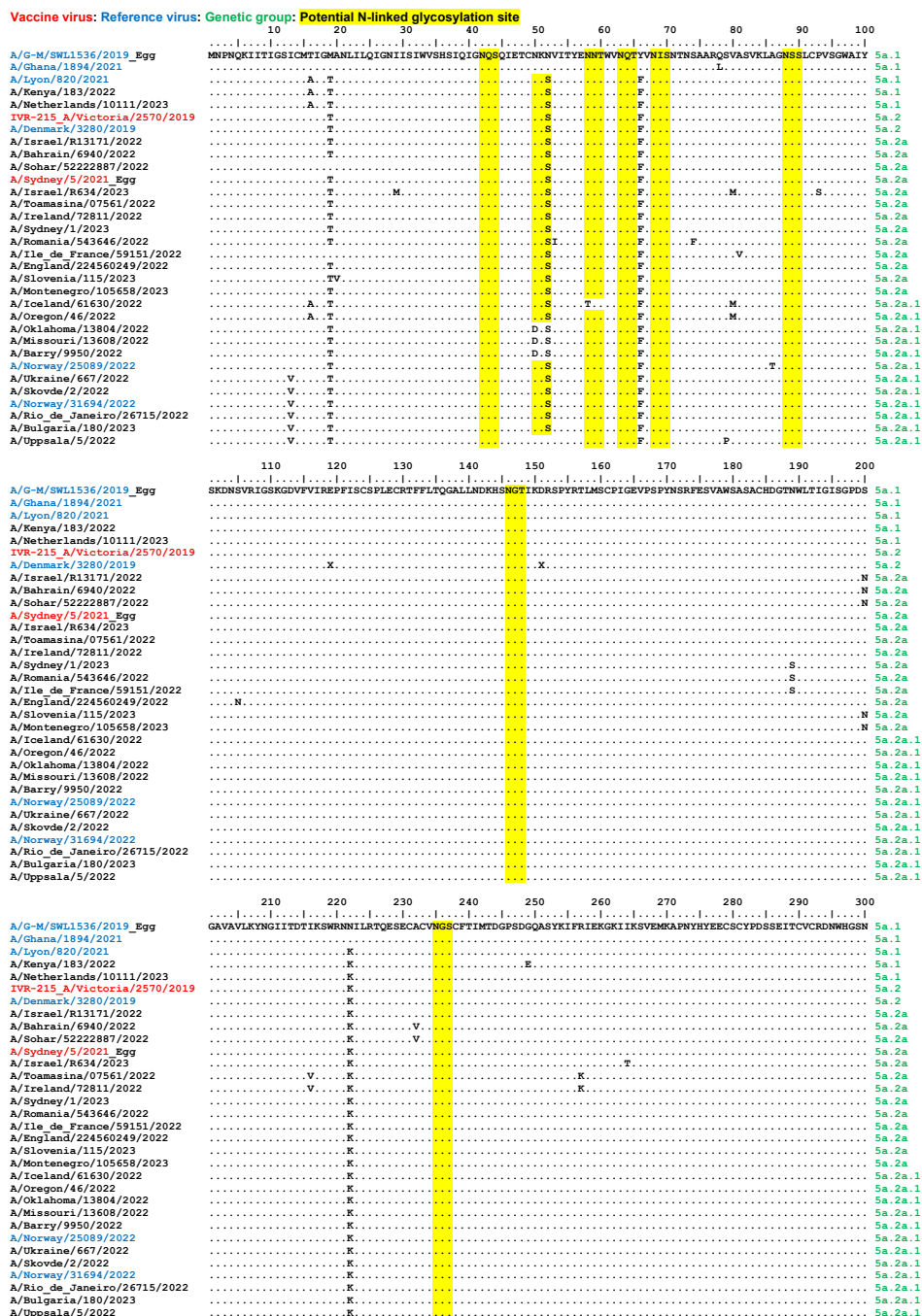


	310	320	330	340	350	360	370	380	390	400	
A/G-M/SWL1536/2019_Egg	GKCPKYVSTKLRLATGLRNVPSIQS	GLGFAIAGFIEGGWTGMVDGWYGYH	HNQNEQGS	GYAADI	KSTQNAIDKITNKVNSVIEKMN	QPTAVGKEFNH					5a.1
A/Ghana/1894/2021											5a.1
A/Lyon/820/2021											5a.1
A/Kenya/183/2022											5a.1
A/Netherlands/10111/2023			I								5a.1
IVR-215_A/Victoria/2570/2019											5a.2
A/Denmark/3280/2019											5a.2
A/Israel/R13171/2022		R									5a.2a
A/Bahrain/6940/2022		R							S		5a.2a
A/Sohar/52222887/2022			I								5a.2a
A/Sydney/5/2021_Egg		R									5a.2a
A/Israel/R634/2023		R									5a.2a
A/Toamasina/07561/2022		R									5a.2a
A/Ireland/72811/2022		R									5a.2a
A/Sydney/1/2023		R							Q		5a.2a
A/Romania/543646/2022		R									5a.2a
A/Ile de France/59151/2022		R									5a.2a
A/England/224560249/2022		R									5a.2a
A/Slovenia/115/2023		R									5a.2a
A/Montenegro/105658/2023		R									5a.2a
A/Iceland/61630/2022		R									5a.2a.1
A/Oregon/46/2022		R					V				5a.2a.1
A/Oklahoma/13804/2022		R								Q	5a.2a.1
A/Missouri/13608/2022		R									5a.2a.1
A/Barry/9950/2022		R									5a.2a.1
A/Norway/25089/2022		R									5a.2a.1
A/Ukraine/667/2022		R		I							5a.2a.1
A/Skovde/2/2022		R									5a.2a.1
A/Norway/31694/2022		R									5a.2a.1
A/Rio de Janeiro/26715/2022		R									5a.2a.1
A/Bulgaria/180/2023		R									5a.2a.1
A/Uppsala/5/2022		R									5a.2a.1
	410	420	430	440	450	460	470	480	490	500	
A/G-M/SWL1536/2019_Egg	EKRLENLNKKVDDGFLDIWTYNAAELV	LLNERLTDYHDSNVKNLYEKVRNQL	KNNAKEI	NGGCFEYHKCDNTCMESVKN	NGTYDYPKYSEAKINREKI						5a.1
A/Ghana/1894/2021											5a.1
A/Lyon/820/2021											5a.1
A/Kenya/183/2022											5a.1
A/Netherlands/10111/2023											5a.1
IVR-215_A/Victoria/2570/2019											5a.2
A/Denmark/3280/2019											5a.2
A/Israel/R13171/2022											5a.2a
A/Bahrain/6940/2022											5a.2a
A/Sohar/52222887/2022											5a.2a
A/Sydney/5/2021_Egg											5a.2a
A/Israel/R634/2023		V									5a.2a
A/Toamasina/07561/2022											5a.2a
A/Ireland/72811/2022											5a.2a
A/Sydney/1/2023											5a.2a
A/Romania/543646/2022											5a.2a
A/Ile de France/59151/2022											5a.2a
A/England/224560249/2022			V								5a.2a
A/Slovenia/115/2023			V								5a.2a
A/Montenegro/105658/2023			V								5a.2a
A/Iceland/61630/2022			V				H				5a.2a.1
A/Oregon/46/2022			V				H	T			5a.2a.1
A/Oklahoma/13804/2022			V				H				5a.2a.1
A/Missouri/13608/2022			V				H				5a.2a.1
A/Barry/9950/2022			V				H				5a.2a.1
A/Norway/25089/2022				X			H				5a.2a.1
A/Ukraine/667/2022			V				H				5a.2a.1
A/Skovde/2/2022			V				H				5a.2a.1
A/Norway/31694/2022			V				H				5a.2a.1
A/Rio de Janeiro/26715/2022			V				H				5a.2a.1
A/Bulgaria/180/2023			V				H				5a.2a.1
A/Uppsala/5/2022			V				H				5a.2a.1
	510	520	530	540							
A/G-M/SWL1536/2019_Egg	DGVKLESTRIYQILAIYSTVASSILV	LSLGAISFWMCSNGSLQCRICI									5a.1
A/Ghana/1894/2021											5a.1
A/Lyon/820/2021	N										5a.1
A/Kenya/183/2022											5a.1
A/Netherlands/10111/2023		T									5a.1
IVR-215_A/Victoria/2570/2019		D									5a.2
A/Denmark/3280/2019		D									5a.2
A/Israel/R13171/2022		D									5a.2a
A/Bahrain/6940/2022		D									5a.2a
A/Sohar/52222887/2022		D									5a.2a
A/Sydney/5/2021_Egg		D									5a.2a
A/Israel/R634/2023		I	D								5a.2a
A/Toamasina/07561/2022		D									5a.2a
A/Ireland/72811/2022		D									5a.2a
A/Sydney/1/2023		D									5a.2a
A/Romania/543646/2022		D									5a.2a
A/Ile de France/59151/2022		D									5a.2a
A/England/224560249/2022		D									5a.2a
A/Slovenia/115/2023		D									5a.2a
A/Montenegro/105658/2023		D									5a.2a
A/Iceland/61630/2022		D	T								5a.2a.1
A/Oregon/46/2022		D		V							5a.2a.1
A/Oklahoma/13804/2022		D									5a.2a.1
A/Missouri/13608/2022		D									5a.2a.1
A/Barry/9950/2022		D									5a.2a.1
A/Norway/25089/2022		D									5a.2a.1
A/Ukraine/667/2022		D									5a.2a.1
A/Skovde/2/2022		D									5a.2a.1
A/Norway/31694/2022		D									5a.2a.1
A/Rio de Janeiro/26715/2022		D									5a.2a.1
A/Bulgaria/180/2023		D									5a.2a.1
A/Uppsala/5/2022		D									5a.2a.1

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Protein alignments NA

Figure H1-P2. NA protein alignment for A(H1N1)pdm09 viruses



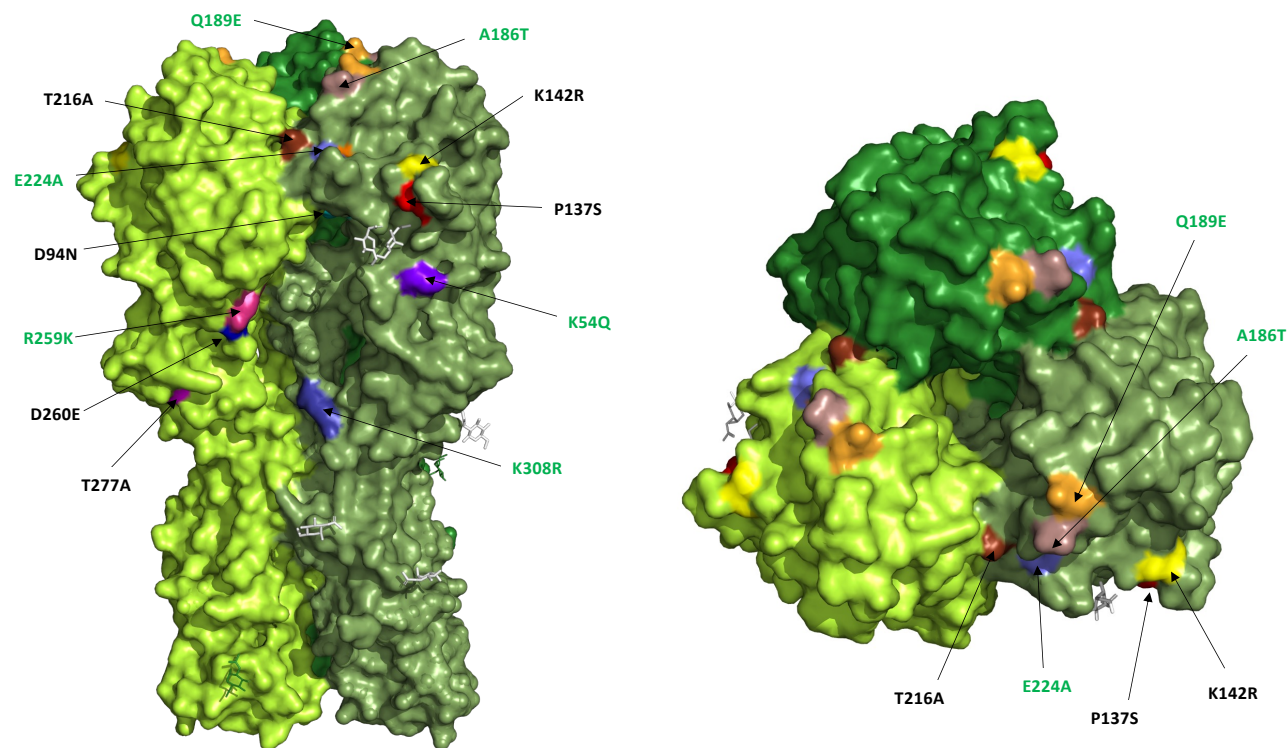
	310	320	330	340	350	360	370	380	390	400						
A/G-M/SWL1536/2019_Egg	RPWVSFNQ	LEYQMG	YICSGV	FGDNPR	PNKDTG	SCGPGV	SSNGANGV	KGFSTKY	GNGVWIG	RKTSISSR	KGFEMIDP	NGWTGT	DNKFSK	KQDIVG	INews	5a.1
A/Ghana/1894/2021	D.....															5a.1
A/Lyon/820/2021	P.....															5a.1
A/Kenya/183/2022	I.....															5a.1
A/Netherlands/10111/2023	I.....						N.....					I.....	T.....			5a.1
IVR-215_A/Victoria/2570/2019																5a.2
A/Denmark/3280/2019																5a.2
A/Israel/R13171/2022																5a.2a
A/Bahrain/6940/2022	T.....															5a.2a
A/Sohar/5222887/2022								T.....								5a.2a
A/Sydney/5/2021_Egg																5a.2a
A/Israel/R634/2023																5a.2a
A/Toamasina/07561/2022																5a.2a
A/Ireland/72811/2022	N.....															5a.2a
A/Sydney/1/2023	D.....															5a.2a
A/Romania/543646/2022																5a.2a
A/Ile_de_France/59151/2022																5a.2a
A/England/224560249/2022																5a.2a
A/Slovenia/115/2023																5a.2a
A/Montenegro/105658/2023																5a.2a
A/Iceland/61630/2022	I.....															5a.2a.1
A/Oregon/46/2022	I.....							N.....				E.....	D.....			5a.2a.1
A/Oklahoma/13804/2022																5a.2a.1
A/Missouri/13608/2022																5a.2a.1
A/Barry/9950/2022																5a.2a.1
A/Norway/25089/2022																5a.2a.1
A/Ukraine/667/2022	X.....			L.....												5a.2a.1
A/Skovde/2/2022				L.....												5a.2a.1
A/Norway/31694/2022				L.....												5a.2a.1
A/Rio_de_Janeiro/26715/2022				L.....												5a.2a.1
A/Bulgaria/180/2023				L.....												5a.2a.1
A/Uppsala/5/2022				L.....												5a.2a.1

	410	420	430	440	450	460					
A/G-M/SWL1536/2019_Egg	GYSGSFVQH	PELTGL	NCIRPC	FWVELIR	GRPEENT	IWTSGSSIS	FCGVSD	IVGMSW	PDGAEL	PFTIDK	5a.1
A/Ghana/1894/2021											5a.1
A/Lyon/820/2021											5a.1
A/Kenya/183/2022	N.....										5a.1
A/Netherlands/10111/2023											5a.1
IVR-215_A/Victoria/2570/2019											5a.2
A/Denmark/3280/2019											5a.2
A/Israel/R13171/2022	M.....										5a.2a
A/Bahrain/6940/2022	M.....										5a.2a
A/Sohar/5222887/2022	M.....										5a.2a
A/Sydney/5/2021_Egg	M.....										5a.2a
A/Israel/R634/2023	A.....										5a.2a
A/Toamasina/07561/2022	M.....										5a.2a
A/Ireland/72811/2022	M.....										5a.2a
A/Sydney/1/2023	M.....										5a.2a
A/Romania/543646/2022	M.....										5a.2a
A/Ile_de_France/59151/2022	Q.....										5a.2a
A/England/224560249/2022	M.....										5a.2a
A/Slovenia/115/2023	M.....										5a.2a
A/Montenegro/105658/2023	M.....										5a.2a
A/Iceland/61630/2022											5a.2a.1
A/Oregon/46/2022											5a.2a.1
A/Oklahoma/13804/2022	M.....										5a.2a.1
A/Missouri/13608/2022	M.....										5a.2a.1
A/Barry/9950/2022	M.....							N.....			5a.2a.1
A/Norway/25089/2022	M.....										5a.2a.1
A/Ukraine/667/2022	M.....										5a.2a.1
A/Skovde/2/2022	M.....										5a.2a.1
A/Norway/31694/2022	M.....										5a.2a.1
A/Rio_de_Janeiro/26715/2022	M.....										5a.2a.1
A/Bulgaria/180/2023	M.....										5a.2a.1
A/Uppsala/5/2022	M.....										5a.2a.1

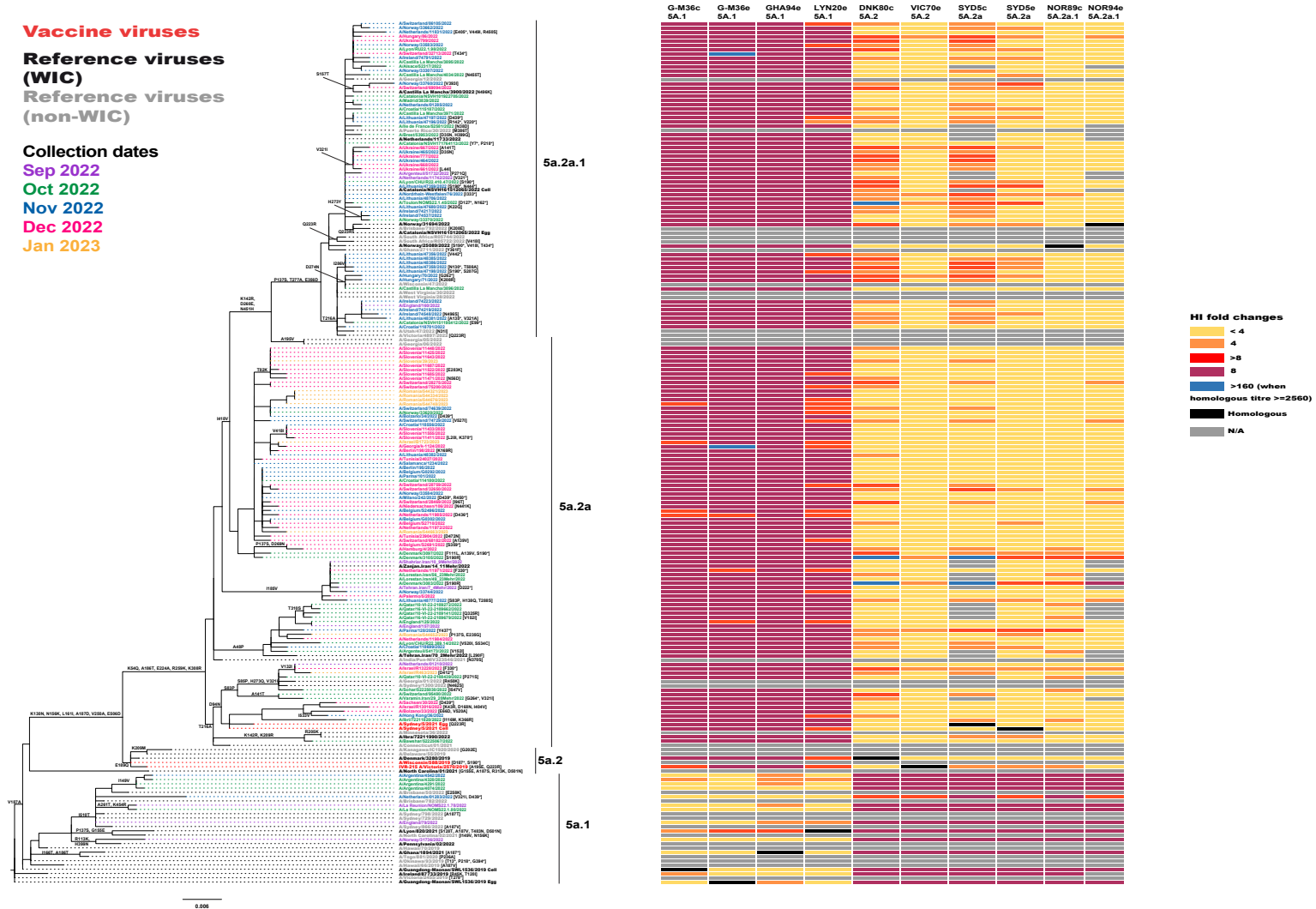
Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Structure

Figure S1: Pymol structure showing the HA1 amino acid substitutions compared with A/Wisconsin/588/2019 that define H1N1 6B.1A.5a2 (green) along with substitutions in emerging subgroups (black) [PDB Entry 1RUZ].



Genotype to Phenotype heatmap: H1N1



Antigenic analyses: H1N1

Haemagglutination inhibition tables: H1N1

Table H1-1. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2022-09-21)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre											
					Post-infection ferret antisera											
					NEW											
					A/ire 87733/19 Egg	A/G-M SWL1536/19 MDCK	A/G-M SWL1536/19 Egg	A/Ghana 1894/21 Egg	A/Ghana 1894/21 Egg	A/Lyon 820/21 Egg	A/Denmark 3280/19 MDCK	IVR-215 A/Vic/2570/19 Egg	A/Sydney 5/21 Egg	A/Sydney 5/21 Egg	A/Sydney 5/21 Egg	A/Sydney 5/21 Egg
		Passage history			St Jude's F18/20	F09/20	F12/20	F02/22	F35/22	F06/22	F28/20	F37/21	F04/22	F34/22		
		Genetic group			6B.1A.5a.1	5a.1	5a.1	5a.1	5a.1	5a.1	5a.2	5a.2	5a.2a	5a.2a		
REFERENCE VIRUSES																
A/Ireland/87733/2019		6B.1A.5a.1	2019-11-03	E4	1280	2560	2560	1280	1280	320	40	80	80	40		
A/Guangdong-Maonan/SWL1536/2019		5a.1	2019-06-17	C2/MDCK1	1280	2560	2560	1280	1280	320	80	80	80	40		
A/Guangdong-Maonan/SWL1536/2019		5a.1	2019-06-17	E3/E2	640	1280	1280	640	640	320	40	80	40	40		
A/Ghana/1894/2021		5a.1	2021-07-21	E2/E1	640	1280	2560	1280	1280	320	80	160	80	80		
A/Lyon/820/2021		5a.1	2021-11-16	E1/E2	160	320	320	160	160	320	40	40	40	<40		
A/Denmark/3280/2019	K130N, N156K, L161I, V250A, E189Q	5a.2	2019-11-10	MDCK4/MDCK5	40	80	80	<40	40	40	1280	1280	1280	640		
IVR-215 (A/Victoria/2570/2019)	K130N, N156K, L161I, V250A, E189Q (A195E, Q223R)	5a.2	2018-11-22	E4/D7/E2	40	160	80	40	40	80	640	1280	1280	320		
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A (Q223R)	5a.2a	2021-10-16	E3/E1	80	80	80	40	40	40	1280	1280	1280	1280		
TEST VIRUSES																
A/Lorraine/28657/2022		6B.1A.5a.1	2022-04-11	MDCK3	320	640	640	640	320	160	40	40	40	<40		
A/Norway/23877/2022		6B.1A.5a.1	2022-06-04	MDCK1	640	1280	1280	1280	40	160	40	40	40	640		
A/Norway/25093/2022		6B.1A.5a.2	2022-06-15	MDCK1	<40	40	<40	<40	<40	<40	640	1280	640	640		
A/Norway/25089/2022		6B.1A.5a.2	2022-06-15	MDCK1	<40	<40	<40	<40	<40	<40	640	1280	640	640		
A/Norway/29426/2022		6B.1A.5a.2	2022-08-02	MDCK1	<40	80	40	<40	<40	<40	2560	2560	2560	2560		
< relates to the lowest dilution of antiserum used																
ND = Not Done																
Vaccine NH 2020-21																
Vaccine SH 2021																
Vaccine NH 2021-22																
Vaccine SH 2022																
Vaccine NH 2022-23																

< 4-fold
4-fold
8-fold
> 8-fold
not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table H1-2. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2022-10-28)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre											
					Post-infection ferret antisera											
					A/Ire 8773/19 Egg	A/G-M SWL1536/19 MDCCK	A/G-M SWL1536/19 Egg	A/Ghana 1894/21 Egg	A/Lyon 820/21 Egg	A/Denmark 3280/19 MDCCK	IVR-215 A/Vic/2570/19 Egg	A/Sydney 5/21 MDCCK	A/Sydney 5/21 Egg	NEW A/Sydney 5/21 Egg	NEW A/Norway 2508/22 MDCCK	
					Ferret number	F09/20	F12/20	F02/22	F06/22	F28/20	F37/21	CDC Ferret A96111	F04/22	F40/22	F38/22	
					Genetic group	6B.1A.5a.1	5a.1	5a.1	5a.1	5a.1	5a.2	5a.2	5a.2a	5a.2a	5a.2a	5a.2a.1
REFERENCE VIRUSES																
A/Ireland/8773/2019		6B.1a.5a.1	2019-11-03	E4	640	5120	1280	1280	320	40	80	<40	40	40	40	40
A/Guangdong-Maonan/SWL1536/2019		5a.1	2019-06-17	C2/MDCCK1	640	1280	2560	1280	320	40	80	<40	40	<40	40	40
A/Guangdong-Maonan/SWL1536/2019		5a.1	2019-06-17	E3/E2	320	640	1280	320	160	<40	40	<40	<40	40	<40	<40
A/Ghana/1894/2021		5a.1	2021-07-21	E2/E1	640	2560	2560	1280	320	80	160	40	80	40	40	<40
A/Lyon/820/2021		5a.1	2021-11-16	E1/E2	160	320	320	160	320	40	80	<40	40	40	40	40
A/Denmark/3280/2019	K130N, N156K, L161I, V250A	5a.2	2019-11-10	MDCCK4/MDCCK5	40	80	80	40	40	1280	2560	320	1280	640	1280	1280
IVR-215 (A/Victoria/2570/2019)	K130N, N156K, L161I, V250A (A195E, Q223R)	5a.2	2018-11-22	E4/D7/E2	40	160	80	40	80	640	1280	320	640	640	640	640
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a	2021-10-16	MDCCK3/MDCCK1	<40	<40	<40	<40	<40	640	1280	320	640	640	640	640
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A (Q223R)	5a.2a	2021-10-16	E3/E2	<40	40	40	<40	<40	640	640	160	640	640	640	640
A/Norway/2508/2022	P137S, K142R, D260E, T277A	5a.2a.1	2022-06-15	MDCCK2	<40	<40	40	<40	<40	640	1280	320	1280	640	1280	1280
TEST VIRUSES																
A/Catalonia/NSVH101883797/2022		6B.1A.5a.2	2022-07-31	SIAT1/MDCCK1	<40	<40	40	<40	<40	640	640	320	640	640	1280	1280
< relates to the lowest dilution of antiserum used																
ND = Not Done																
< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)																

Table H1-3. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2022-11-25)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre							
					Post-infection ferret antisera							
					A/G-M SWL1536/19 MDCK	A/G-M SWL1536/19 Egg	A/Ghana 1894/21 Egg	A/Lyon 820/21 Egg	A/Denmark 3280/19 MDCK	IVR-215 A/Vic/2570/19 Egg	A/Sydney 5/21 Egg	A/Norway 25089/22 MDCK
					F09/20	F12/20	F02/22	F06/22	F28/20	F37/21	F04/22	F38/22
		Genetic group			5a.1	5a.1	5a.1	5a.1	5a.2	5a.2	5a.2a	5a.2a.1
REFERENCE VIRUSES												
A/Guangdong-Maonan/SWL1536/2019		5a.1	2019-06-17	C2/MDCK1	1280	2560	1280	320	40	80	40	40
A/Guangdong-Maonan/SWL1536/2019		5a.1	2019-06-17	E3/E2	1280	1280	640	320	40	80	40	<40
A/Ghana/1894/2021		5a.1	2021-07-21	E2/E1	2560	2560	2560	320	80	160	80	<40
A/Lyon/820/2021		5a.1	2021-11-16	E1/E2	320	320	160	640	40	40	40	<40
A/Denmark/3280/2019	K130N, N156K, L161I, V250A, E189Q	5a.2	2019-11-10	MDCK4/MDCK5	80	80	<40	80	2560	2560	1280	1280
IVR-215 (A/Victoria/2570/2019)	K130N, N156K, L161I, V250A, E189Q (A195E, Q223R)	5a.2	2018-11-22	E4/D7/E2	160	80	40	80	1280	2560	1280	1280
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a	2021-10-16	MDCK3/MDCK1	<40	<40	<40	<40	640	640	640	320
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A (Q223R)	5a.2a	2021-10-16	E3/E2	80	40	<40	40	640	1280	1280	640
A/Norway/25089/2022	P137S, K142R, D260E, T277A	5a.2a.1	2022-06-15	MDCK2	<40	40	<40	<40	1280	1280	1280	2560
TEST VIRUSES												
A/Netherlands/11714/2022		6B.1A.5a.2	2022-08-03	MDCK-MIX1/MDCK1	<40	<40	<40	<40	640	1280	640	1280
A/Netherlands/11699/2022		6B.1A.5a.2	2022-08-11	MDCK-MIX1/MDCK1	<40	<40	<40	<40	640	1280	1280	1280
A/Netherlands/11711/2022		6B.1A.5a.2	2022-08-17	MDCK-MIX1/MDCK1	<40	40	<40	<40	1280	1280	1280	1280
A/Netherlands/11733/2022		5a.2a.1	2022-09-06	MDCK-MIX1/MDCK1	<40	40	<40	<40	1280	1280	1280	1280
A/Netherlands/11742/2022		5a.2a.1	2022-09-20	MDCK-MIX1/MDCK1	<40	<40	<40	<40	640	1280	640	1280
< relates to the lowest dilution of antiserum used					Vaccine			Vaccine			Vaccine	
ND = Not Done					NH 2020-21			SH 2021			SH 2023	
								NH 2021-22			NH 2022-23	

Table H1-4. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2022-12-07)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/G-M SWL1536/19 MDCK	A/G-M SWL1536/19 Egg	A/Ghana 1894/21 Egg	A/Lyon 820/21 Egg	A/Denmark 3280/19 MDCK	IVR-215 A/Vic/2570/19 Egg	A/Sydney 5/21 Egg	A/Norway 25089/22 MDCK		
					F09/20	F12/20	F02/22	F06/22	F28/20	F37/21	F04/22	F38/22		
		Genetic group			5a.1	5a.1	5a.1	5a.1	5a.2	5a.2	5a.2a	5a.2a.1		
REFERENCE VIRUSES														
A/Guangdong-Maonan/SWL1536/2019		5a.1	2019-06-17	C2/MDCK1	1280	2560	1280	320	40	80	40	<40		
A/Guangdong-Maonan/SWL1536/2019		5a.1	2019-06-17	E3/E2	640	1280	640	160	40	40	40	<40		
A/Ghana/1894/2021		5a.1	2021-07-21	E2/E1	2560	2560	2560	320	80	160	40	<40		
A/Lyon/820/2021		5a.1	2021-11-16	E1/E2	320	160	160	320	40	40	40	<40		
A/Denmark/3280/2019	K130N, N156K, L161I, V250A, E189Q	5a.2	2019-11-10	MDCK4/MDCK5	80	80	<40	40	2560	2560	2560	1280		
IVR-215 (A/Victoria/2570/2019)	K130N, N156K, L161I, V250A, E189Q (A195E, Q223R)	5a.2	2018-11-22	E4/D7/E2	80	80	<40	40	640	1280	640	1280		
A/Sydney/5/2021		5a.2a	2022-10-31	E3/E2	80	40	<40	40	1280	1280	1280	640		
A/Norway/25089/2022	P137S, K142R, D260E, T277A	5a.2a.1	2022-06-15	MDCK2	<40	40	<40	<40	2560	2560	2560	2560		
TEST VIRUSES														
A/Shiraz.Iran/49_4Mordad/2022		6B.1A.5a.2	2022-07-20	SIAT2/MDCK1	<40	<40	<40	<40	640	640	640	640		
A/Shiraz.Iran/45_4Mordad/2022		6B.1A.5a.2	2022-07-21	SIAT2/MDCK2	40	<40	<40	<40	1280	1280	2560	640		
A/South Africa/R05848/2022		6B.1A.5a.2	2022-05-19	MDCK1/MDCK1	<40	<40	<40	<40	1280	1280	1280	1280		
A/Netherlands/11719/2022		6B.1A.5a.2	2022-08-28	MDCK1	<40	<40	<40	<40	640	640	640	1280		
A/Tehran.Iran/70_2Mehr/2022		5a.2a	2022-09-22	SIAT1/MDCK2	40	40	<40	<40	1280	2560	2560	1280		
A/Tehran.Iran/7_4Mehr/2022		5a.2a	2022-09-22	SIAT2/MDCK1	40	40	<40	<40	1280	2560	2560	1280		
A/Shahriar.Iran/10_9Mehr/2022		5a.2a	2022-09-26	SIAT1/MDCK1	<40	<40	<40	<40	1280	1280	1280	1280		
A/Zanjan.Iran/14_11Mehr/2022		5a.2a	2022-10-01	SIAT1/MDCK1	40	40	<40	<40	1280	2560	2560	1280		
A/Varamin.Iran/29_20Mehr/2022		5a.2a	2022-10-09	SIAT1/MDCK1	<40	<40	<40	<40	640	640	1280	1280		
A/Lorestan.Iran/56_23Mehr/2022		5a.2a	2022-10-10	SIAT1/MDCK1	<40	<40	<40	<40	640	1280	1280	640		
A/Lorestan.Iran/49_23Mehr/2022		5a.2a	2022-10-10	SIAT1/MDCK1	<40	<40	<40	<40	1280	1280	1280	1280		
A/Qatar/16-VI-22-2109662/2022		5a.2a	2022-10-24	MDCK1	40	40	<40	<40	1280	2560	1280	1280		
A/Qatar/10-VI-22-2109272/2022		5a.2a	2022-10-24	MDCK1	<40	40	<40	<40	1280	1280	1280	640		
A/Qatar/10-VI-22-2109141/2022		5a.2a	2022-10-24	MDCK1	40	40	<40	<40	1280	1280	1280	1280		
A/Qatar/10-VI-22-2108439/2022		5a.2a	2022-10-24	MDCK1	<40	<40	<40	<40	1280	1280	1280	640		
A/Catalonia/NSVH161512065/2022		5a.2a.1	2022-09-14	MDCK1	<40	<40	<40	<40	640	640	640	1280		
< relates to the lowest dilution of antiserum used					Vaccine NH 2020-21			Vaccine SH 2021 NH 2021-22 SH 2022 NH 2022-23			Vaccine SH 2023			
ND = Not Done														
					< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)									

Table H1-5. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2022-12-14)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre							
				Post-infection ferret antisera							
				A/G-M SWL1536/19 MDCK F09/20	A/G-M SWL1536/19 Egg F12/20	A/Ghana 1894/21 Egg F02/22	A/Lyon 820/21 Egg F06/22	A/Denmark 3280/19 MDCK F28/20	I/VR-215 A/Vic/2570/19 Egg F37/21	A/Sydney 5/21 Egg F04/22	A/Norway 25089/22 MDCK F38/22
	Passage history			5a.1	5a.1	5a.1	5a.1	5a.2	5a.2	5a.2a	5a.2a.1
REFERENCE VIRUSES											
A/Guangdong-Maonan/SWL1536/2019	5a.1	2019-06-17	C2/MDCK1	2560	2560	1280	320	40	80	40	<40
A/Guangdong-Maonan/SWL1536/2019	5a.1	2019-06-17	E3/E2	1280	1280	640	160	<40	40	<40	<40
A/Ghana/1894/2021	5a.1	2021-07-21	E2/E1	2560	2560	1280	320	80	80	40	<40
A/Lyon/820/2021	5a.1	2021-11-16	E1/E2	640	320	160	640	40	80	40	40
A/Denmark/3280/2019	5a.2	2019-11-10	MDCK4/MDCK5	80	80	40	80	1280	1280	1280	1280
I/VR-215 (A/Victoria/2570/2019)	5a.2	2018-11-22	E4/D7/E2	160	80	40	80	1280	1280	1280	1280
A/Sydney/5/2021	5a.2a	2021-10-16	MDCK3/MDCK3	40	40	<40	40	2560	2560	2560	1280
A/Sydney/5/2021	5a.2a	2021-10-16	E3/E2	80	40	<40	40	1280	640	1280	640
A/Norway/25089/2022	5a.2a.1	2022-06-15	MDCK2	40	40	<40	<40	2560	1280	1280	2560
TEST VIRUSES											
A/Paris/54771/2022	6B.1A.5a.2	2022-07-27	MDCK1/MDCK1	<40	<40	<40	<40	640	640	640	640
A/Argenteuil/54173/2022	5a.2a	2022-10-15	MDCK1/MDCK1	40	40	<40	<40	640	640	640	640
A/Qatar/16-VI-22-109679/2022	5a.2a	2022-10-24	MDCK2	40	40	<40	<40	640	640	640	640
A/Argenteuil/51732/2022	5a.2a.1	2022-09-27	MDCK1/MDCK1	<40	<40	<40	<40	640	640	640	1280
A/Alsace/52317/2022	5a.2a.1	2022-10-04	MDCK1/MDCK1	<40	40	<40	<40	1280	1280	1280	1280
A/Ile_de_France/52581/2022	5a.2a.1	2022-10-10	MDCK1/MDCK1	40	40	<40	<40	640	1280	640	1280
A/Brest/53953/2022	5a.2a.1	2022-10-13	MDCK2/MDCK1	<40	40	<40	<40	640	640	640	1280
< relates to the lowest dilution of antiserum used ND = Not Done				Vaccine NH 2020-21				Vaccine SH 2021 NH 2021-22 SH 2022 NH 2022-23	Vaccine SH 2023		

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table H1-6. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2022-12-21)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre										
					Post-infection ferret antisera										
					NEW					NEW					
					A/G-M SWL1536/19 MDCK	A/G-M SWL1536/19 Egg	A/Ghana 1894/21 Egg	A/Lyon 820/21 Egg	A/Denmark 3280/19 MDCK	IVR-215 A/Vic/2570/19 Egg	A/Sydney 5/21 MDCK	A/Sydney 5/21 MDCK	A/Sydney 5/21 Egg	A/Norway 25089/22 MDCK	
Ferret number	F09/20	F12/20	F02/22	F06/22	F28/20	F37/21	F46/22	F47/22	F04/22	F38/22					
Genetic group	5a.1	5a.1	5a.1	5a.1	5a.2	5a.2	5a.2a	5a.2a	5a.2a	5a.2a	5a.2a.1				
REFERENCE VIRUSES															
A/Guangdong-Maonan/SWL1536/2019		5a.1	2019-06-17	C2/MDCK1	1280	2560	1280	160	40	80	40	40	<40	<40	
A/Guangdong-Maonan/SWL1536/2019		5a.1	2019-06-17	E3/E2	1280	1280	320	160	40	40	<40	40	<40	<40	
A/Ghana/1894/2021		5a.1	2021-07-21	E2/E1	2560	2560	1280	320	80	80	40	80	40	<40	
A/Lyon/820/2021		5a.1	2021-11-16	E1/E2	320	160	160	320	40	40	40	<40	<40	<40	
A/Denmark/3280/2019	K130N, N156K, L161I, V250A, E189Q	5a.2	2019-11-10	MDCK4/MDCK5	80	40	<40	40	1280	1280	1280	1280	1280	1280	
IVR-215 (A/Victoria/2570/2019)	K130N, N156K, L161I, V250A, E189Q [A195E, Q223R]	5a.2	2018-11-22	E4/D7/E2	160	40	40	40	640	1280	640	1280	640	640	
A/Sydney/5/2021	K540, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a	2021-10-16	MDCK3/MDCK3	40	40	<40	<40	2560	2560	5120	2560	2560	2560	
A/Sydney/5/2021	K540, A186T, E224A, R259K, K308R, D94N, T216A (Q223R)	5a.2a	2021-10-16	E3/E2	80	40	<40	40	1280	1280	1280	2560	2560	1280	
A/Norway/25089/2022	P137S, K142R, D260E, T277A	5a.2a.1	2022-06-15	MDCK2	40	40	<40	<40	1280	2560	1280	2560	2560	2560	
A/Norway/31694/2022	P137S, K142R, D260E, T277A (Q223R)	5a.2a.1	2022-09-24	E2 (Am1A1) 10 ⁻⁵	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	
TEST VIRUSES															
A/Norway/30508/2022		6B.1A.5a.2	2022-08-30	MDCK1	<40	<40	<40	<40	320	640	640	1280	1280	640	
A/Norway/31736/2022		5a.1	2022-09-16	MDCK1	2560	2560	1280	320	40	80	40	80	40	40	
A/Ibri/72211820/2022		5a.2a	2022-10-23	MDCK1	<40	<40	<40	<40	1280	1280	640	2560	1280	640	
A/Ibri/72211990/2022		5a.2a	2022-10-26	MDCK1	<40	<40	<40	<40	640	1280	640	640	640	1280	
A/Sohari/52225030/2022		5a.2a	2022-10-27	MDCK3	<40	<40	<40	<40	640	1280	1280	1280	1280	640	
A/Bawshari/52225067/2022		5a.2a	2022-10-27	MDCK2	80	<40	<40	<40	2560	2560	2560	2560	1280	2560	
A/Salamancal/1234/2022		5a.2a	2022-11-08	MDCK2	40	<40	<40	<40	2560	2560	2560	>5120	2560	2560	
A/Madrid/3839/2022		5a.2a.1	2022-10-03	MDCK1	40	<40	<40	<40	1280	1280	1280	2560	1280	2560	
A/Castilla La Mancha/3971/2022		5a.2a.1	2022-10-05	MDCK1	40	<40	<40	<40	1280	1280	1280	1280	1280	2560	
A/Castilla La Mancha/3896/2022		5a.2a.1	2022-10-06	MDCK1	40	<40	<40	<40	1280	2560	1280	2560	2560	>5120	
A/Castilla La Mancha/3895/2022		5a.2a.1	2022-10-06	MDCK1	40	<40	<40	<40	1280	1280	1280	1280	1280	1280	
A/Castilla La Mancha/3900/2022		5a.2a.1	2022-10-07	MDCK1	40	<40	<40	<40	1280	1280	1280	1280	1280	1280	
A/Castilla La Mancha/4034/2022		5a.2a.1	2022-10-15	MDCK1	40	<40	<40	<40	1280	1280	1280	1280	640	1280	
< relates to the lowest dilution of antiserum used										Vaccine		Vaccine		Vaccine	
ND = Not Done										NH 2020-21		NH 2021-22		NH 2022-23	

Table H1-7. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2023-01-18)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/G-M SWL1536/19 MDCK	A/G-M SWL1536/19 Egg	A/Ghana 1894/21 Egg	A/Lyon 820/21 Egg	A/Denmark 3280/19 MDCK	IVR-215 A/Vic/2570/19 Egg	A/Sydney 5/21 MDCK	A/Sydney 29089/22 Egg	A/Norway 25089/22 MDCK	
					F09/20	F12/20	F02/22	F06/22	F28/20	F37/21	F46/22	F04/22	F38/22	
		Genetic group			5a.1	5a.1	5a.1	5a.1	5a.2	5a.2	5a.2a	5a.2a	5a.2a.1	
REFERENCE VIRUSES														
A/Guangdong-Maonan/SWL1536/2019		5a.1	2019-06-17	C2/MDCK1	2560	2560	2560	320	40	80	40	40	40	
A/Guangdong-Maonan/SWL1536/2019		5a.1	2019-06-17	E3/E2	1280	1280	640	160	40	80	<40	40	<40	
A/Ghana/1894/2021		5a.1	2021-07-21	E2/E1	2560	2560	2560	320	80	160	80	80	40	
A/Lyon/820/2021		5a.1	2021-11-16	E1/E2	320	160	320	320	40	80	40	40	40	
A/Denmark/3280/2019	K130N, N156K, L161L, V250A, E189Q	5a.2	2019-11-10	MDCK4/MDCK5	80	<40	40	<40	2560	1280	1280	2560	1280	
IVR-215 (A/Victoria/2570/2019)	K130N, N156K, L161L, V250A, E189Q (A195E, Q223R)	5a.2	2018-11-22	E4/D7/E2	80	80	40	80	640	1280	640	640	640	
A/Sydney/5/2021 clone 3.4.1	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a	2021-10-16	MDCK3/MDCK3	80	<40	40	<40	1280	2560	2560	2560	2560	
A/Sydney/5/2021 pooled clones 10-10	K54Q, A186T, E224A, R259K, K308R, D94N, T216A (Q223R)	5a.2a	2021-10-16	E3/E2	80	40	40	40	1280	1280	1280	2560	640	
A/Norway/25089/2022	P137S, K142R, D260E, T277A	5a.2a.1	2022-06-15	MDCK3	<40	<40	<40	<40	640	1280	640	1280	1280	
A/Norway/31694/2022	P137S, K142R, D260E, T277A (Q223R)	5a.2a.1	2022-09-24	E3/E1 10 ³	<40	<40	<40	<40	320	640	640	640	640	
TEST VIRUSES														
A/La Reunion/NOMS22.1.78/2022		5a.1	2022-09-19	MDCK2/MDCK1	1280	1280	640	160	40	80	40	80	<40	
A/La Reunion/NOMS22.1.80/2022		5a.1	2022-10-11	MDCK2/MDCK1	1280	1280	1280	160	40	80	40	80	<40	
A/Argentina/4074/2022		5a.1	2022-10-17	MDCK1	1280	1280	1280	160	40	80	40	80	<40	
A/Argentina/4291/2022		5a.1	2022-10-22	MDCK1	1280	640	640	160	40	80	40	80	<40	
A/Argentina/4320/2022		5a.1	2022-10-22	MDCK1	640	640	640	80	40	40	40	80	<40	
A/Argentina/4542/2022		5a.1	2022-11-04	MDCK1	1280	640	640	160	40	40	<40	40	<40	
A/Georgia/1/2022		5a.2a	not known	MDCK1	<40	<40	<40	<40	320	640	640	1280	640	
A/Denmark/3083/2022		5a.2a	2022-10-02	MDCK4/MDCK1	<40	<40	<40	<40	160	320	160	320	160	
A/Lyon/CHU/R22.389.14/2022		5a.2a	2022-10-14	MDCK2/MDCK2	40	<40	<40	<40	640	1280	640	1280	640	
A/Denmark/3097/2022		5a.2a	2022-10-18	MDCK3/MDCK1	40	<40	<40	<40	320	1280	640	640	320	
A/Denmark/3105/2022		5a.2a	2022-10-26	MDCK4/MDCK1	<40	<40	<40	<40	160	320	160	320	160	
A/Lithuania/48382/2022		5a.2a	2022-11-20	MDCK1	40	<40	40	<40	640	1280	1280	1280	640	
A/Lithuania/48777/2022		5a.2a	2022-11-25	MDCK1	<40	<40	<40	<40	640	640	640	640	640	
A/HongKong/36/2022		5a.2a	2022-11-28	MDCK1/MDCK1	80	40	<40	40	640	2560	1280	1280	640	
A/Georgia/5/2022		5a.2a.1	not known	MDCK1	40	<40	<40	<40	1280	2560	1280	2560	2560	
A/Lyon/RU22.1.98/2022		5a.2a.1	2022-10-10	MDCK2/MDCK1	<40	<40	<40	<40	320	640	640	640	640	
A/Toulon/NOMS22.1.45/2022		5a.2a.1	2022-10-14	MDCK2/MDCK2	40	<40	<40	<40	160	320	320	320	320	
A/Lyon/CHU/R22.410.47/2022		5a.2a.1	2022-10-31	MDCK2/MDCK2	<40	<40	<40	<40	320	640	640	640	640	
A/Lithuania/47198/2022		5a.2a.1	2022-11-02	MDCK2/MDCK1	40	<40	40	640	640	1280	640	1280	1280	
A/Lithuania/47197/2022		5a.2a.1	2022-11-02	MDCK2/MDCK1	40	<40	<40	40	640	640	640	640	1280	
A/Lithuania/47196/2022		5a.2a.1	2022-11-02	MDCK2/MDCK1	40	<40	80	640	640	640	640	1280	640	
A/Lithuania/47358/2022		5a.2a.1	2022-11-04	MDCK2/MDCK1	40	<40	<40	<40	320	640	320	640	640	
A/Lithuania/47356/2022		5a.2a.1	2022-11-05	MDCK2/MDCK1	40	<40	<40	40	640	1280	1280	1280	1280	
A/Lithuania/47359/2022		5a.2a.1	2022-11-06	MDCK2/MDCK1	<40	<40	<40	<40	320	640	320	320	640	
A/Lithuania/47680/2022		5a.2a.1	2022-11-10	MDCK2/MDCK1	40	<40	<40	<40	640	1280	640	1280	1280	
A/Lithuania/48385/2022		5a.2a.1	2022-11-20	MDCK1	<40	<40	<40	<40	320	640	640	640	640	
A/Lithuania/48381/2022		5a.2a.1	2022-11-20	MDCK1	<40	<40	<40	<40	640	1280	640	1280	1280	
A/Lithuania/48386/2022		5a.2a.1	2022-11-21	MDCK1	<40	<40	<40	<40	640	640	320	640	1280	
A/Lithuania/48706/2022		5a.2a.1	2022-11-23	MDCK1	40	<40	<40	<40	640	1280	640	1280	1280	
A/Ireland/74219/2022		5a.2a.1	2022-11-25	MDCK1	40	<40	<40	<40	1280	1280	1280	2560	2560	
A/Ireland/74223/2022		5a.2a.1	2022-11-27	MDCK1	<40	<40	<40	<40	640	1280	640	1280	1280	
A/Ireland/74217/2022		5a.2a.1	2022-11-27	MDCK1	40	<40	<40	<40	640	1280	640	1280	1280	
A/Ireland/74548/2022		5a.2a.1	2022-11-28	MDCK1	<40	<40	<40	<40	640	640	640	640	1280	
A/Ireland/74537/2022		5a.2a.1	2022-11-28	MDCK1	40	<40	<40	<40	1280	1280	640	1280	1280	
A/Ireland/74791/2022		5a.2a.1	2022-11-29	MDCK1	40	<40	<40	<40	640	1280	640	640	1280	

< relates to the lowest dilution of antiserum used

ND = Not Done

Vaccine
NH 2020-21Vaccine
SH 2021NH 2021-22
SH 2022

NH 2022-23

Vaccine
SH 2023

 < 4-fold
 4-fold
 8-fold
 > 8-fold
 not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table H1-8. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2023-01-24)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/G-M SWL1536/19 MDCK	A/G-M SWL1536/19 Egg	A/Ghana 1894/21 Egg	A/Lyon 820/21 Egg	A/Denmark 3280/19 MDCK	IVR-215 A/Vic/2570/19 Egg	A/Sydney 5/21 MDCK	A/Sydney 5/21 Egg	A/Norway 25089/22 MDCK	
				Passage history	F09/20	F12/20	F02/22	F06/22	F28/20	F37/21	F46/22	F04/22	F38/22
				Ferret number	5a.1	5a.1	5a.1	5a.1	5a.2	5a.2	5a.2a	5a.2a	5a.2a.1
Genetic group													
REFERENCE VIRUSES													
A/Guangdong-Maonan/SWL1536/2019		5a.1	2019-06-17	C2/MDCK1	2560	2560	1280	320	40	40	<40	40	<40
A/Guangdong-Maonan/SWL1536/2019		5a.1	2019-06-17	E3/E2	1280	1280	640	320	<40	80	<40	40	<40
A/Ghana/1894/2021		5a.1	2021-07-21	E2/E1	1280	2560	1280	320	40	80	40	40	<40
A/Lyon/820/2021		5a.1	2021-11-16	E1/E2	320	320	160	320	40	40	40	40	<40
A/Denmark/3280/2019	K130N, N156K, L161I, V250A, E189Q	5a.2	2019-11-10	MDCK4/MDCK5	160	40	<40	40	2560	2560	1280	2560	1280
IVR-215 (A/Victoria/2570/2019)	K130N, N156K, L161I, V250A, E189Q (A195E, Q223R)	5a.2	2018-11-22	E4/D7/E2	80	80	40	80	640	1280	640	640	320
A/Sydney/5/2021 clone 3.4.1	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a	2021-10-16	MDCK3/MDCK3	40	40	40	40	1280	2560	2560	2560	1280
A/Sydney/5/2021 pooled clones 10-10	K54Q, A186T, E224A, R259K, K308R, D94N, T216A (Q223R)	5a.2a	2022-10-31	E3/E2	80	40	40	40	640	1280	1280	1280	640
A/Norway/25089/2022	P137S, K142R, D260E, T277A	5a.2a.1	2022-06-15	MDCK3	<40	<40	<40	<40	320	640	640	640	640
A/Norway/31694/2022	P137S, K142R, D260E, T277A (Q223R)	5a.2a.1	2022-09-24	E3/E1 10-3	<40	40	<40	<40	320	640	320	320	320
TEST VIRUSES													
A/Switzerland/95480/2022		5a.2a	2022-10-13	MDCK1/MDCK1	<40	40	<40	<40	320	640	640	1280	640
A/Switzerland/74729/2022		5a.2a	2022-11-23	MDCK1/MDCK1	40	40	<40	40	640	640	640	1280	320
A/Switzerland/74639/2022		5a.2a	2022-11-24	MDCK1/MDCK1	40	40	<40	40	640	640	640	640	640
A/Switzerland/68182/2022		5a.2a	2022-12-13	MDCK1	40	40	<40	40	1280	1280	1280	1280	1280
A/Switzerland/28759/2022		5a.2a	2022-12-19	MDCK1	40	40	<40	40	320	640	640	640	640
A/Switzerland/28275/2022		5a.2a	2022-12-19	MDCK1	40	40	<40	40	320	640	640	640	320
A/Switzerland/28469/2022		5a.2a	2022-12-20	MDCK1	<40	40	<40	40	320	640	640	640	320
A/Switzerland/75200/2022		5a.2a	2022-12-23	MDCK1	40	40	<40	40	640	1280	1280	1280	1280
A/Ukraine/465/2022		5a.2a.1	2022-11-11	MDCK1	<40	<40	<40	<40	640	640	640	640	640
A/Ukraine/464/2022		5a.2a.1	2022-11-14	MDCK1	<40	<40	<40	<40	320	640	320	640	640
A/Switzerland/86105/2022		5a.2a.1	2022-11-15	MDCK1/MDCK1	<40	40	<40	40	640	640	640	640	1280
A/Hungary/70/2022		5a.2a.1	2022-11-21	MDCK1/MDCK1	<40	<40	<40	<40	320	320	320	320	320
A/Hungary/71/2022		5a.2a.1	2022-11-29	MDCK1/MDCK1	<40	<40	<40	<40	320	640	640	640	640
A/Ukraine/668/2022		5a.2a.1	2022-12-06	MDCK1	<40	40	<40	<40	640	1280	640	640	1280
A/Hungary/86/2022		5a.2a.1	2022-12-07	MDCK1/MDCK1	<40	<40	<40	<40	320	320	320	320	320
A/Ukraine/667/2022		5a.2a.1	2022-12-11	MDCK1	<40	<40	<40	<40	320	320	320	320	640
A/Ukraine/661/2022		5a.2a.1	2022-12-14	MDCK1	<40	40	<40	<40	640	640	640	640	1280
A/Switzerland/69094/2022		5a.2a.1	2022-12-14	MDCK1	<40	<40	<40	<40	320	640	640	320	640
A/Ukraine/777/2022		5a.2a.1	2022-12-17	MDCK1	<40	40	<40	<40	320	640	320	640	640
A/Ukraine/799/2022		5a.2a.1	2022-12-26	MDCK1	<40	<40	<40	<40	320	640	320	640	640
< relates to the lowest dilution of antiserum used					Vaccine NH 2020-21					Vaccine SH 2021		Vaccine SH 2023	
ND = Not Done										NH 2021-22			
										SH 2022			
										NH 2022-23			

< 4-fold 4-fold 8-fold > 8-fold * not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table H1-9a. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2023-01-31)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre											
					Post-infection ferret antisera											
					A/G-M SWL1536/19	A/G-M SWL1536/19	A/Ghana 1894/21	A/Lyon 820/21	A/Denmark 3280/19	IVR-215 A/Vic/2570/19	A/Sydney 5/21	A/Sydney 5/21	A/Norway 25089/22			
					MDCK	Egg	Egg	Egg	MDCK	Egg	MDCK	Egg	MDCK			
					F09/20	F12/20	F02/22	F06/22	F28/20	F37/21	F46/22	F04/22	F38/22			
Passage history		Collection date		Passage history		Collection date		Passage history		Collection date		Passage history		Collection date		
Ferret number		Genetic group		Passage history		Collection date		Passage history		Collection date		Passage history		Collection date		
Genetic group		Passage history		Collection date		Passage history		Collection date		Passage history		Collection date		Passage history		
REFERENCE VIRUSES																
A/Guangdong-Maonan/SWL1536/2019		5a.1	2019-06-17	C2/MDCK1	1280	1280	1280	320	40	40	40	40	40	40	40	40
A/Guangdong-Maonan/SWL1536/2019		5a.1	2019-06-17	E3/E2	1280	2560	1280	320	40	80	40	40	40	40	40	40
A/Ghana/1894/2021		5a.1	2021-07-21	E2/E1	2560	2560	1280	320	40	80	40	80	40	80	40	80
A/Lyon/820/2021		5a.1	2021-11-16	E1/E2	640	320	320	640	40	80	40	80	40	80	40	80
A/Denmark/3280/2019	K130N, N156K, L161I, V250A, E189Q	5a.2	2019-11-10	MDCK4/MDCK5	80	40	40	80	2560	2560	1280	2560	1280	2560	1280	2560
IVR-215 (A/Victoria/2570/2019)	K130N, N156K, L161I, V250A, E189Q (A195E, Q223R)	5a.2	2018-11-22	E4/D7/E2	160	160	80	80	640	1280	640	1280	640	1280	640	1280
A/Sydney/5/2021 clones 3.4.1	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a	2021-10-16	MDCK3/MDCK3	80	80	40	80	1280	2560	2560	2560	1280	2560	1280	2560
A/Sydney/5/2021 pooled clones 10-10	K54Q, A186T, E224A, R259K, K308R, D94N, T216A (Q223R)	5a.2a	2022-10-31	E3/E2	160	80	40	80	1280	1280	1280	2560	1280	2560	1280	2560
A/Norway/25089/2022	P137S, K142R, D260E, T277A	5a.2a.1	2022-06-15	MDCK3	40	40	<40	40	640	1280	1280	1280	1280	1280	1280	1280
A/Norway/31694/2022	P137S, K142R, D260E, T277A (Q223R)	5a.2a.1	2022-09-24	E3/E1 10 ⁻³	40	40	<40	40	640	640	640	320	640	640	320	640
TEST VIRUSES																
A/Croatia/114180/2022		5a.2a	2022-10-22	MDCKx/MDCK1	80	80	40	40	1280	2560	1280	2560	1280	2560	1280	2560
A/Norway/33620/2022		5a.2a	2022-10-31	MDCK1	80	80	40	80	1280	2560	2560	2560	2560	2560	2560	2560
A/Norway/33584/2022		5a.2a	2022-11-07	MDCK1	40	80	40	40	1280	1280	1280	1280	1280	1280	1280	1280
A/Norway/33744/2022		5a.2a	2022-11-09	MDCK1	80	80	40	80	1280	2560	2560	2560	1280	2560	1280	2560
A/Belgium/G0292/2022		5a.2a	2022-11-21	C1/MDCK1	80	80	40	40	1280	1280	1280	2560	1280	2560	1280	2560
A/Croatia/118556/2022		5a.2a	2022-11-22	MDCKx/MDCK1	80	80	40	40	1280	2560	2560	2560	1280	2560	1280	2560
A/Croatia/118699/2022		5a.2a	2022-11-23	MDCKx/MDCK1	40	40	<40	<40	640	640	640	640	640	640	640	640
A/Belgium/G0302/2022		5a.2a	2022-11-25	C1/MDCK1	80	80	40	40	1280	2560	1280	1280	1280	1280	1280	1280
A/Belgium/S2496/2022		5a.2a	2022-11-28	C1/MDCK1	160	80	40	80	1280	2560	2560	2560	1280	2560	1280	2560
A/Belgium/S2710/2022		5a.2a	2022-12-01	C1/MDCK1	40	40	<40	40	640	1280	1280	640	1280	1280	640	1280
A/Switzerland/32650/2022		5a.2a	2022-12-09	MDCK2	<40	40	<40	<40	640	640	320	640	640	640	320	640
A/Belgium/S2691/2022		5a.2a	2022-12-11	C1/MDCK1	40	40	<40	40	1280	1280	1280	1280	1280	1280	1280	1280
A/Catalonia/NSVH101922705/2022		5a.2a.1	2022-10-02	SIAT1/MDCK1	80	40	40	40	1280	2560	1280	2560	2560	2560	2560	2560
A/Catalonia/NSVH171764113/2022		5a.2a.1	2022-10-04	SIAT1/MDCK1	40	40	<40	<40	1280	1280	1280	2560	1280	2560	1280	2560
A/Catalonia/NSVH151195412/2022		5a.2a.1	2022-10-18	SIAT1/MDCK1	<40	<40	<40	<40	640	1280	640	1280	640	1280	640	1280
A/Croatia/115187/2022		5a.2a.1	2022-10-27	MDCKx/MDCK2	40	40	<40	<40	640	1280	640	640	1280	640	1280	640
A/Norway/33370/2022		5a.2a.1	2022-10-28	MDCK1	40	80	<40	40	1280	2560	1280	1280	1280	1280	1280	1280
A/Norway/33307/2022		5a.2a.1	2022-11-02	MDCK1	40	80	<40	40	1280	1280	640	1280	1280	1280	1280	1280
A/Norway/33583/2022		5a.2a.1	2022-11-07	MDCK1	40	40	<40	80	640	1280	640	1280	1280	1280	1280	1280
A/Norway/33662/2022		5a.2a.1	2022-11-09	MDCK1	40	40	<40	40	1280	1280	640	1280	1280	1280	1280	1280
A/Norway/33760/2022		5a.2a.1	2022-11-11	MDCK1	40	40	<40	40	640	640	640	320	640	640	320	640
A/Croatia/118701/2022		5a.2a.1	2022-11-21	MDCKx/MDCK2	40	40	<40	<40	1280	1280	1280	1280	1280	1280	1280	1280
A/Switzerland/32713/2022		5a.2a.1	2022-12-09	MDCK2	40	160	<40	40	1280	2560	1280	1280	1280	1280	1280	1280

< relates to the lowest dilution of antiserum used
ND = Not Done

Vaccine
NH 2020-21

Vaccine
SH 2021
NH 2021-22
SH 2022
NH 2022-23

Vaccine
SH 2023

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table H1-9b. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2023-01-31)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/G-M SWL1536/19	A/G-M SWL1536/19	A/Ghana 1894/21	A/Lyon 820/21	A/Denmark 3280/19	IVR-215 A/Vic/2570/19	A/Sydney 5/21	A/Sydney 5/21	A/Norway 25089/22	
					MDCK	Egg	Egg	Egg	MDCK	Egg	MDCK	Egg	MDCK	
					F09/20	F12/20	F02/22	F06/22	F28/20	F37/21	F46/22	F04/22	F38/22	

Table H1-10. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2023-02-07)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/G-M SWL1536/19 MDCK	A/G-M SWL1536/19 Egg	A/Ghana 1894/21 Egg	A/Lyon 820/21 Egg	A/Denmark 3280/19 MDCK	IVR-215 A/Vic/2570/19 Egg	A/Sydney 5/21 MDCK	A/Sydney 5/21 Egg	A/Norway 25089/22 MDCK	A/Norway/31694/2022 Egg
					F09/20	F12/20	F02/22	F06/22	F28/20	F37/21	F46/22	F04/22	F38/22	F48/22
		Genetic group			Sa.1	Sa.1	Sa.1	Sa.1	Sa.2	Sa.2	Sa.2a	Sa.2a	Sa.2a.1	Sa.2a.1
REFERENCE VIRUSES														
A/Guangdong-Maonan/SWL1536/2019		Sa.1	2019-06-17	C2/MDCK1	1280	1280	1280	320	40	80	40	40	<40	<40
A/Guangdong-Maonan/SWL1536/2019		Sa.1	2019-06-17	E3/E2	640	1280	640	160	<40	80	40	40	<40	<40
A/Ghana/1894/2021		Sa.1	2021-07-21	E2/E1	1280	2560	2560	320	80	160	80	80	40	<40
A/Lyon/820/2021		Sa.1	2021-11-16	E1/E2	640	320	320	640	40	80	40	40	40	<40
A/Denmark/3280/2019	K130N, N156K, L161L, V250A, E189Q	Sa.2	2019-11-10	MDCK4/MDCK5	80	40	<40	40	1280	2560	2560	2560	1280	640
IVR-215 (A/Victoria/2570/2019)	K130N, N156K, L161L, V250A, E189Q (A195E, Q223R)	Sa.2	2018-11-22	E4/D7/E2	80	80	40	80	640	1280	640	640	640	160
A/Sydney/5/2021 clone 3.4.1	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	Sa.2a	2021-10-16	MDCK3/MDCK3	80	80	40	80	1280	2560	2560	2560	1280	640
A/Sydney/5/2021 pooled clones 10-10	K54Q, A186T, E224A, R259K, K308R, D94N, T216A (Q223R)	Sa.2a	2022-10-31	E3/E2	80	40	40	40	640	1280	1280	1280	1280	320
A/Norway/25089/2022	P137S, K142R, D260E, T277A	Sa.2a.1	2022-06-15	MDCK3	40	40	<40	<40	1280	1280	1280	1280	1280	640
A/Norway/31694/2022	P137S, K142R, D260E, T277A (Q223R)	Sa.2a.1	2022-09-24	E3/E1 10-3	<40	40	<40	40	320	640	320	320	320	320
TEST VIRUSES														
A/England/79/2022		Sa.1	2022-09-15	SIAT1/MDCK1	640	640	640	160	40	80	40	40	40	<40
A/Netherlands/01283/2022		Sa.1	2022-11-15	hCK1/MDCK1	1280	1280	1280	640	160	160	80	160	80	40
A/Netherlands/01210/2022		Sa.2a	2022-09-02	hCK1/MDCK1	<40	40	<40	<40	640	1280	1280	1280	1280	320
A/England/157/2022		Sa.2a	2022-09-19	SIAT1/MDCK1	80	80	40	40	640	2560	1280	1280	1280	320
A/England/125/2022		Sa.2a	2022-10-24	SIAT1/MDCK1	80	160	80	80	2560	2560	2560	2560	1280	640
A/Milano/242/2022		Sa.2a	2022-11-14	MDCK2/MDCK1	40	80	40	40	640	1280	1280	1280	640	320
A/Parma/128/2022		Sa.2a	2022-11-17	MDCK2/MDCK1	<40	<40	<40	<40	320	320	320	160	160	160
A/Bolzano/34/2022		Sa.2a	2022-11-23	MDCK3/MDCK1	80	80	40	40	1280	2560	1280	2560	1280	320
A/Parma/101/2022		Sa.2a	2022-11-25	MDCK3/MDCK1	40	80	40	40	1280	1280	1280	1280	640	640
A/Berlin/195/2022		Sa.2a	2022-11-29	P2/MDCK1	40	40	<40	40	640	1280	1280	1280	640	320
A/Niedersachsen/106/2022		Sa.2a	2022-12-01	P1/MDCK1	80	80	40	40	640	1280	1280	1280	640	320
A/Palermo/5/2022		Sa.2a	2022-12-01	MDCK3/MDCK1	80	80	40	40	1280	1280	1280	2560	1280	320
A/Hamburg/4/2022		Sa.2a	2022-12-02	P1/MDCK1	40	80	<40	40	320	640	640	320	320	320
A/Berlin/198/2022		Sa.2a	2022-12-02	P1/MDCK1	40	40	<40	<40	1280	2560	1280	2560	1280	640
A/Bolzano/33/2022		Sa.2a	2022-12-05	MDCK3/MDCK1	40	40	<40	<40	640	1280	1280	1280	640	320
A/Sachsen/30/2022		Sa.2a	2022-12-12	P1/MDCK1	40	40	<40	<40	640	1280	1280	1280	640	320
A/Netherlands/11985/2022		Sa.2a	2022-12-20	MDCK-MIX2/MDCK1	160	160	80	80	2560	>5120	2560	>5120	2560	640
A/Netherlands/11972/2022		Sa.2a	2022-12-20	MDCK-MIX2/MDCK1	40	40	<40	40	1280	1280	1280	1280	1280	640
A/Netherlands/11971/2022		Sa.2a	2022-12-20	MDCK-MIX2/MDCK1	80	160	80	80	2560	2560	2560	2560	2560	1280
A/Netherlands/11984/2022		Sa.2a	2022-12-22	MDCK-MIX2/MDCK1	80	80	40	40	1280	1280	1280	1280	640	320
A/Tunisia/23904/2022		Sa.2a	2022-12-26	MDCK1	40	40	<40	40	640	1280	1280	1280	640	320
A/Tunisia/24027/2022		Sa.2a	2022-12-26	MDCK2	40	80	40	40	1280	1280	1280	2560	1280	640
A/England/160/2022		Sa.2a.1	2022-09-12	SIAT1/MDCK1	40	40	<40	<40	640	1280	640	1280	1280	640
A/Netherlands/01285/2022		Sa.2a.1	2022-11-10	hCK1/MDCK1	40	40	<40	<40	640	640	640	640	640	640
A/Netherlands/11831/2022		Sa.2a.1	2022-11-16	MDCK-MIX2/MDCK1	40	40	<40	<40	640	1280	1280	1280	1280	1280
A/Nordrhein-Westfalen/76/2022		Sa.2a.1	2022-11-22	P1/MDCK1	<40	<40	<40	<40	320	320	320	320	320	320

< relates to the lowest dilution of antiserum used
ND = Not Done

Vaccine
NH 2020-21

Vaccine
SH 2021
NH 2021-22
SH 2022

Vaccine
SH 2023

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table H1-11. Antigenic analyses of influenza A(H1N1)pdm09 viruses - Summary

Viruses	Passage history Ferret number Genetic group	Haemagglutination inhibition titre									
		Post-infection ferret antisera									
		A/G-M SWL1536/19	A/G-M SWL1536/19	A/Ghana 1894/21	A/Lyon 820/21	A/Denmark 3280/19	IVR-215 A/Vic/2570/19	A/Sydney 5/21	A/Sydney 5/21	A/Norway 25089/22	A/Norway/ 31694/22
		MDCK F09/20	Egg F12/20	Egg F02/22	Egg F06/22	MDCK F28/20	Egg F37/21	MDCK F46/22	Egg F04/22	MDCK F38/22	Egg F48/22
		5a.1	5a.1	5a.1	5a.1	5a.2	5a.2	5a.2a	5a.2a	5a.2a.1	5a.2a.1
REFERENCE VIRUSES											
A/Guangdong-Maonan/SWL1536/2019	5a.1	1280	1280	1280	320	40	80	40	40	<40	<40
A/Guangdong-Maonan/SWL1536/2019	5a.1	640	1280	640	160	<40	80	40	40	<40	<40
A/Ghana/1894/2021	5a.1	1280	2560	2560	320	80	160	80	80	40	<40
A/Lyon/820/2021	5a.1	640	320	320	640	40	80	40	40	40	<40
A/Denmark/3280/2019	5a.2	80	40	<40	40	1280	2560	2560	2560	1280	640
IVR-215 (A/Victoria/2570/2019)	5a.2	80	80	40	80	640	1280	640	640	640	160
A/Sydney/5/2021 clone 3.4.1	5a.2a	80	80	40	80	1280	2560	2560	2560	1280	640
A/Sydney/5/2021 pooled clones 10-10	5a.2a	80	40	40	40	640	1280	1280	1280	1280	320
A/Norway/25089/2022	5a.2a.1	40	40	<40	<40	1280	1280	1280	1280	1280	640
A/Norway/31694/2022	5a.2a.1	<40	40	<40	40	320	640	320	320	320	320
TEST VIRUSES											
Number tested	5a.1	9	9	9	9	9	9	9	9	9	9
No with titre reduction ≤2-fold		8	9	4	7						
%		88.9	100	44.4	77.8						
No with titre reduction =4-fold		1		5	2						
%		11.1		55.6	22.2						
No with titre reduction ≥8-fold						9	9	9	9	9	9
%						100	100	100	100	100	100
Number tested	5a.2a	88	88	88	88	88	88	75	88	88	75
No with titre reduction ≤2-fold						60	84	54	77	75	66
%						68.2	95.5	72.0	87.5	85.2	88.0
No with titre reduction =4-fold					1	19	4	17	8	10	7
%					1.1	21.6	4.5	22.7	9.1	11.4	9.3
No with titre reduction ≥8-fold		88	88	88	87	9		4	3	3	2
%		100	100	100	98.9	10.2		5.3	3.4	3.4	2.7
Number tested	5a.2a.1	61	61	61	61	61	61	54	61	61	54
No with titre reduction ≤2-fold						24	56	15	42	60	53
%						39.3	91.8	27.8	68.9	98.4	98.1
No with titre reduction =4-fold					1	23	5	28	16	1	1
%					1.6	37.7	8.2	51.9	26.2	1.6	1.9
No with titre reduction ≥8-fold		61	61	61	60	14		11	3		
%		100	100	100	98.4	23.0		20.3	4.9		
		Vaccine NH 2020-21				Vaccine SH 2021 NH 2021-22 SH 2022 NH 2022-23			Vaccine SH 2023		

Reference virus results are taken from an individual table as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

Panel 2

Public Health Wales panel:

Serum panel information:

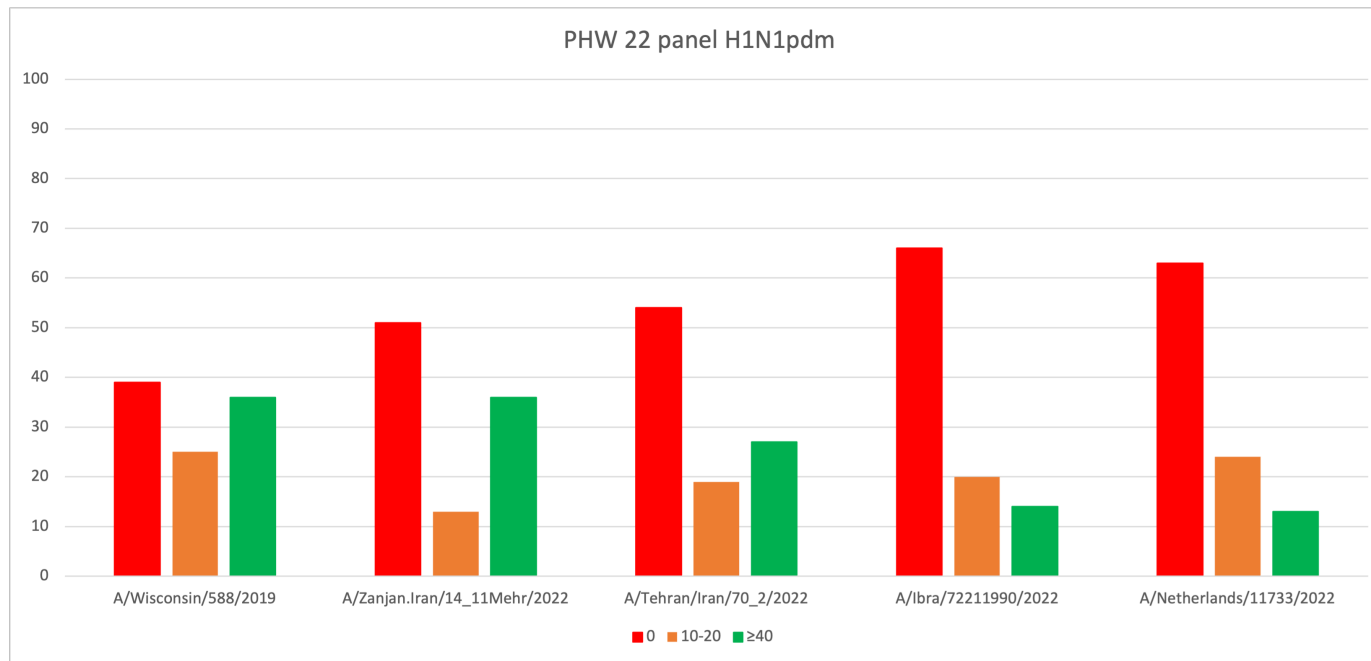
Total number of samples: 100

Collection dates: 16th February 2022 to 7th June 2022

Age range: 16 – 64 years

Average age: 36.5 years

Median age: 35.5 years



Reference reagents and isolates

A/H1N1: Egg isolates

Virus	Shipping status	Genetic group/clade
A/Lyon/820/2021	CBER-49B, NIB-132	6B.1A.5a.1 (128T, P137S, G155E)
A/Norway/31694/2022	sent out	6B.1A.5a.2a.1 (P137S, K142R)
A/Catalonia/NSVH161512065/2022	available to send out	6B.1A.5a.2a.1 (P137S, K142R)
A/Ghana/1894/2021	Reference only	6B.1A.5a.1 (I166T, A186T)

A/H1N1: HI reagents and references

Virus	Genetic group	Virus passage	Ferret ID
A/Guangdong-Maonan/SWL1536/2019	6B.1A.5a.1	C2/MDCK1	F09/20
A/Guangdong-Maonan/SWL1536/2019	6B.1A.5a.1	E3/E2	F12/20
A/Ghana/1894/2021	6B.1A.5a.1	E2/E1	F02/22
A/Lyon/820/2021	6B.1A.5a.1	E1/E2	F06/22
A/Denmark/3280/2019	6B.1A.5a.2	MDCK4/MDCK5	F28/20
A/Norway/25089/2022	6B.1A.5a.2	MDCK2	F38/22
A/Sydney/5/2021	6B.1A.5a.2	MDCK3/MDCK1	F40/22
A/Sydney/5/2021	6B.1A.5a.2	E3/E2	F04/22
IVR-215 (A/Victoria/2570/2019)	6B.1A.5a.2	E4/D7/E2	F37/21

Summary: H1N1

Genetic analyses

Clade 6B.1A.5a.1 viruses were detected since 1st September in Australia, Tasmania, La Reunion, Madagascar, Kenya, Mozambique, Uganda, the Netherlands, France, Norway, Argentina, Bolivia, Uruguay, Brazil, Canada, USA.

Signature amino acid substitutions: D187A, Q189E.

Vaccine virus: A/Guangdong-Maonan/SWL1536/2019 (egg) and A/Hawaii/70/2019 (cell)

Clade 6B.1A.5a.2 viruses predominated since 1st September in Europe, the Eastern Mediterranean, North Africa, Central and South Asia, North and Central America and parts of South East Asia.

Clade 5a.2.

Signature amino acid substitutions: N156K, L161I, V250A.

Vaccine virus: A/Victoria/2570/2019 (egg) and A/Wisconsin/588/2019 (cell)

Clade 5a.2a.

Signature amino acid substitutions: K54Q, A186T, Q189E, E224A, R259K, K308R Vaccine viruses: A/Sydney/5/2021 (cell and egg) which also contain D94N and T216A

5a.2a viruses predominated in Australasia, Asia, Africa, the Middle East and in some European countries, particularly towards the South and East of the region, though also in Scandinavia, the Netherlands and France.

Clade 5a.2a.1.

Signature amino acid substitutions: P137S, K142R, D260E

Clade 5a.2a.1 viruses predominated in the more northern and western European countries and also in North and Central America.

From the HA-NA tanglegrams we observe limited reassortment particularly for a small 5a.2a subclade, characterised by an HA with A195V substitution. Viruses from this clade link with a NA segment associated with 5a.1 viruses.

Antigenic and human serologic analyses

The small number of 5a.1 viruses analysed showed good recognition by post-infection ferret antisera raised against A/Guangdong Maonan/SWL1536/2021 and poor recognition by the antisera raised against 5a.2, 5a.2a and 5a.2a.1 reference viruses.

More recently-characterised 5a.2a.1 viruses showed variable recognition by post-infection ferret antisera raised against cell culture- (C) and egg-propagated (E) A/Sydney/5/2021 - the SH 2023 recommended vaccine virus from the 5a.2a clade, with 28%:16% (C:E) showing 4-fold loss in cross-reactivity against the ferret antisera raised against the SH 2023 viruses.

95% of viruses tested were well-recognised by the antisera raised against the 5a.2a.1 viruses A/Norway/25089/2022 (Cell) or the A/Norway/31694/2022 (Egg)

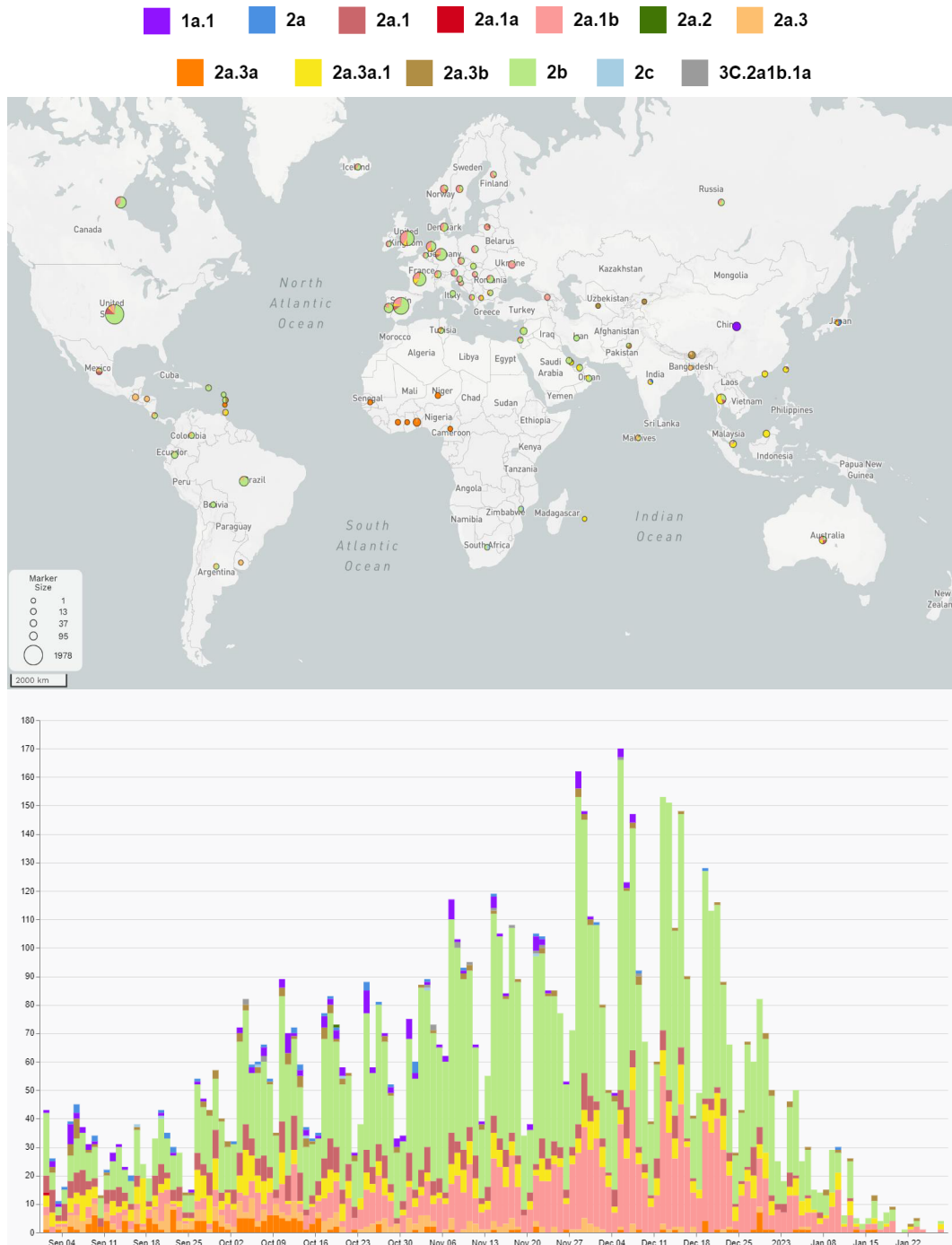
All viruses tested show a reduction in % GMT relative to A/Wisconsin/588/2019. Notably, all except A/Zanjan.Iran/14_11Mehr/2022 (5a.2a), i.e. A/Tehran/Iran/70_2/2022 (5a.2a), A/Ibra/72211990/2022 (5a.2a), A/Netherlands/11733/2022 (5a.2a.1), and A/Ghana/2711/2022 (5a.2a.1), showed greater than 50% reductions in % GMT relative to A/Wisconsin/588/2019.

All tested viruses were susceptible to neuraminidase inhibitors (NAI) by phenotypic testing and no markers for reduced inhibition by NAIs were identified in NA sequences derived from clinical specimens

Influenza A H3N2

Microreact: H3N2 by clade

MicroReact map for influenza viruses with collection dates after 31st August 2022 through to 7th February 2023 as deposited in GISAID, coloured by H3 clade. Geographic markers scaled to detection proportions. Full length HA.



Genetic analyses: H3N2

H3N2 subclades of 3C.2a1b.2a. Clade 1: Signature amino acid substitutions: G186S, S198P

Vaccine virus: A/Tasmania/503/2020

Clade 1a: Clade 1 with additional substitution: K171N

Vaccine virus: A/Cambodia/e0826360/2020

Clade 1a.1: Clade 1a with additional substitution: I48T comment: circulating almost exclusively in China

Clade 2: Signature amino acid substitutions: Y159N, T160I, L164Q, G186D, D190N comment: descendants predominate outside of China

Clade 2a: Signature amino acid substitution: Clade 2 + H156S contains subgroups with diversity at D53N/G or S

Vaccine virus: A/Darwin/6/2021

Clade 2a.1: 2a. with additional substitutions: D53G, D104G, K276R

Clade 2a.1a: 2a.1 with additional substitutions: L157I, K220R

Clade 2a.1b: 2a.1 with additional substitutions: I140K, R299K

Clade 2a.2: 2a. with additional substitutions: D53G, R201L, S219Y

Clade 2a.3: 2a. with additional substitutions: D53N, N96S, I192F

Clade 2a.3a: 2a.3 with additional substitution: E50K

Clade 2a.3a1: additional substitutions: 140K

Clade 2a.3b: 2a.3a with additional substitution: I140M

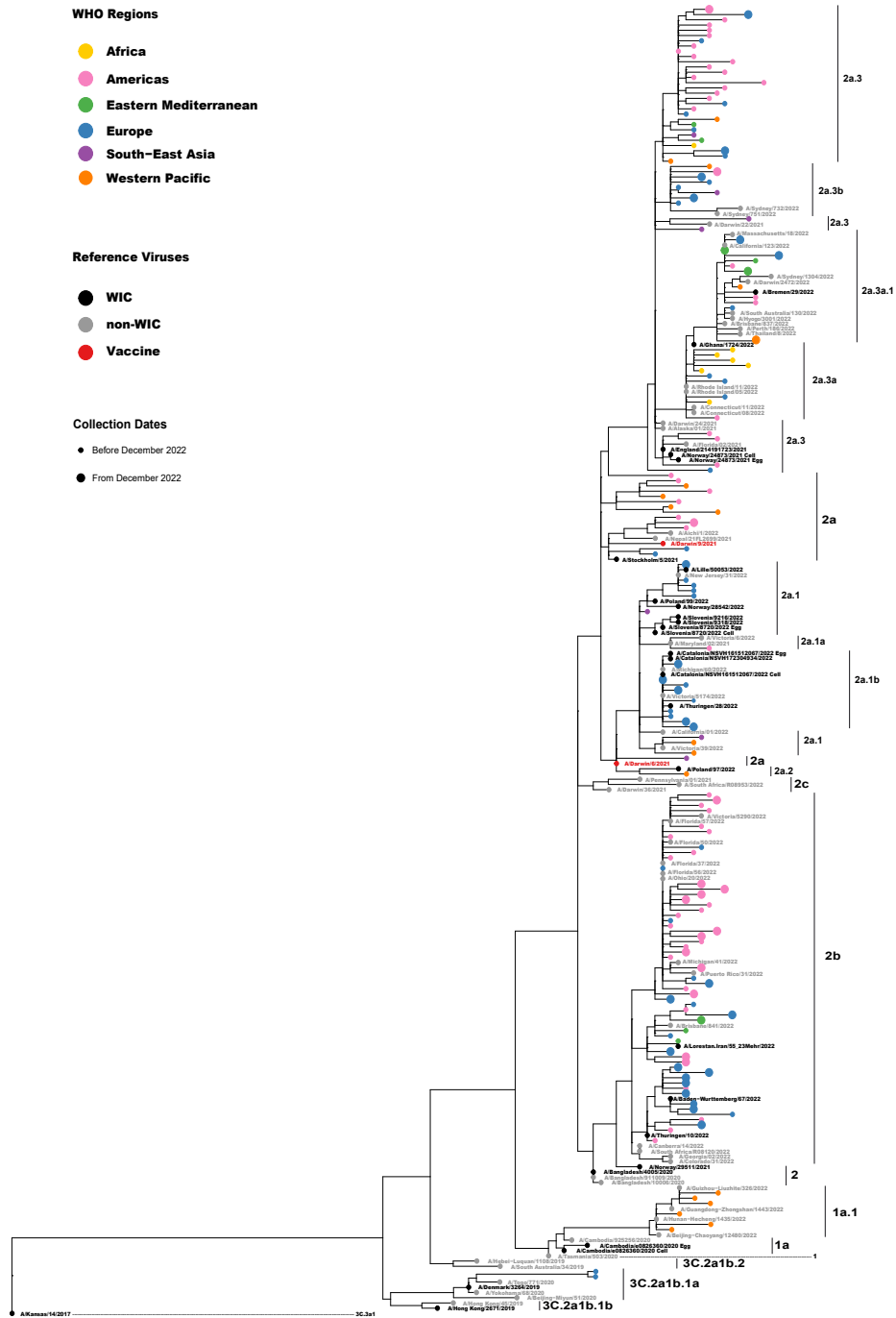
Clade 2b: Clade 2 with additional substitutions: E50K, F79V, I140K comment: has grown recently

Clade 2c: Clade 2 with additional substitutions: S205F, A212T comment: few recent viruses

Clade 2d: Clade 2 with additional substitutions: G62R, H156Q, S199P comment: few recent viruses

Maximum likelihood phylogenetic tree: H3N2

Maximum likelihood phylogenetic trees inferred from full-length HA sequence data obtained from GISAID for viruses with collection dates after 31st August 2022 and submitted to GISAID up to 7th February 2023, manually curated and then downsampled using Treemmer to retain a representative tree topology of 250 sequences.



Vaccine viruses
Reference viruses (WIC)
Reference viruses (non-WIC)

Collection dates

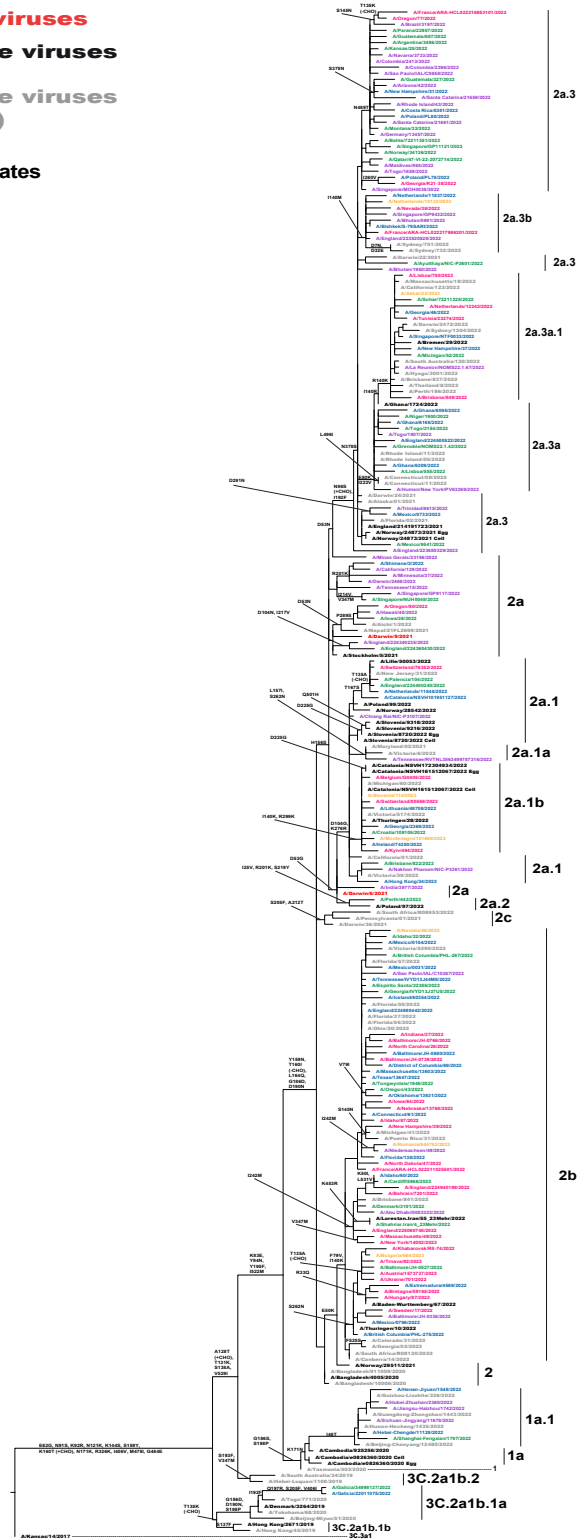
Sep 2022

Oct 2022

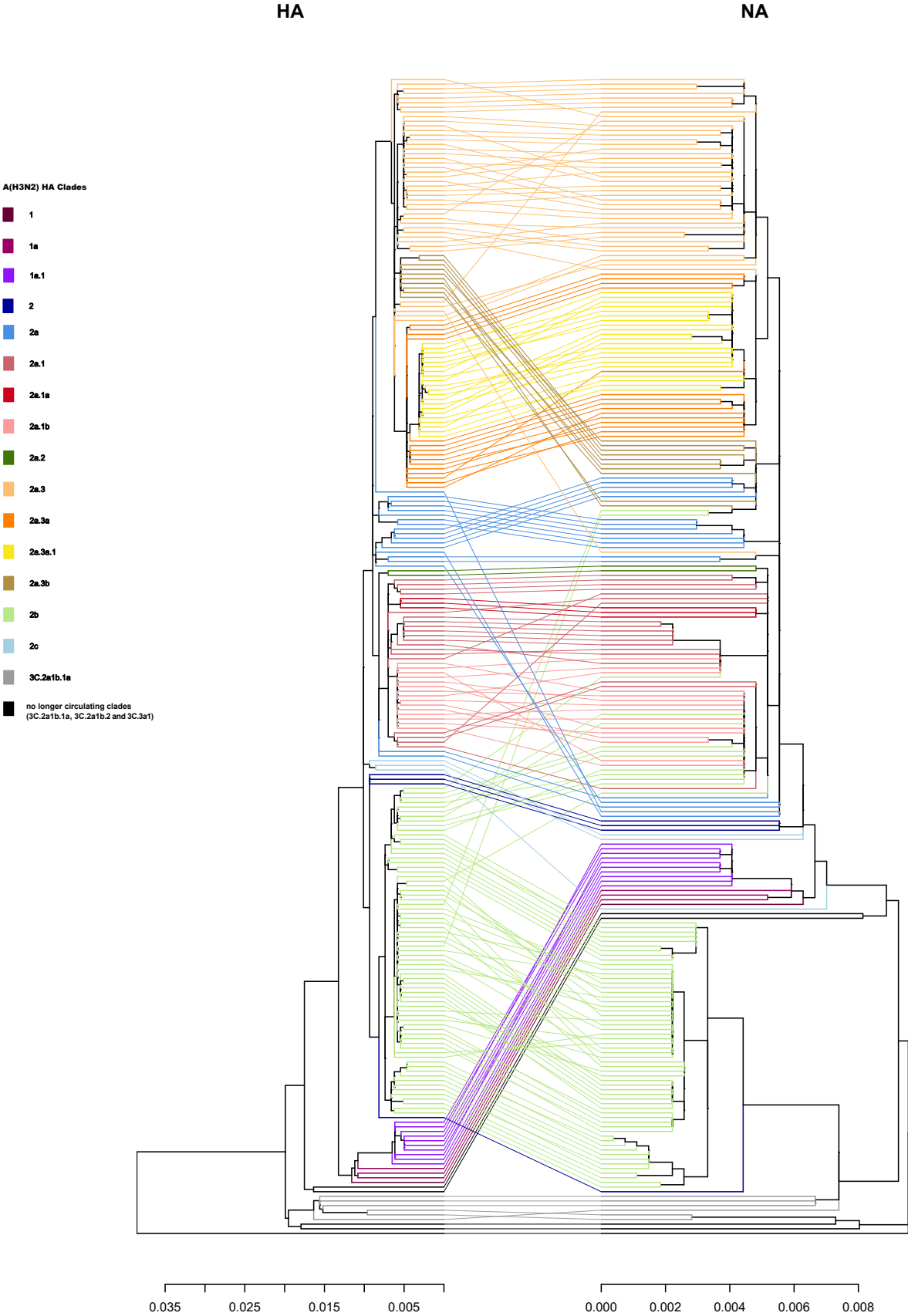
Nov 2022

Dec 2022

Jan 2023

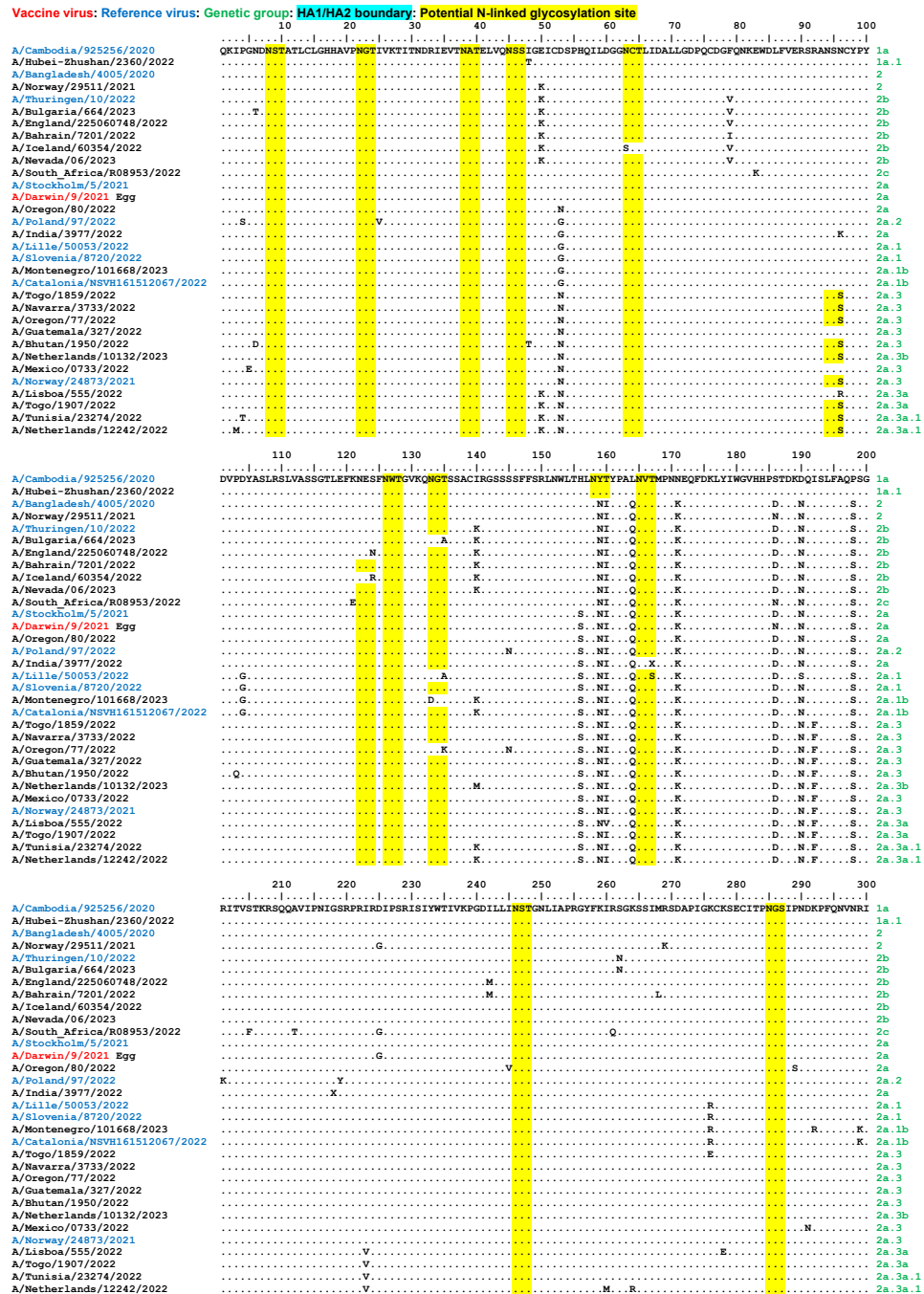


HA-NA tanglegrams



Protein alignments HA

Figure H3-P1. HA protein alignment for A(H3N2) viruses

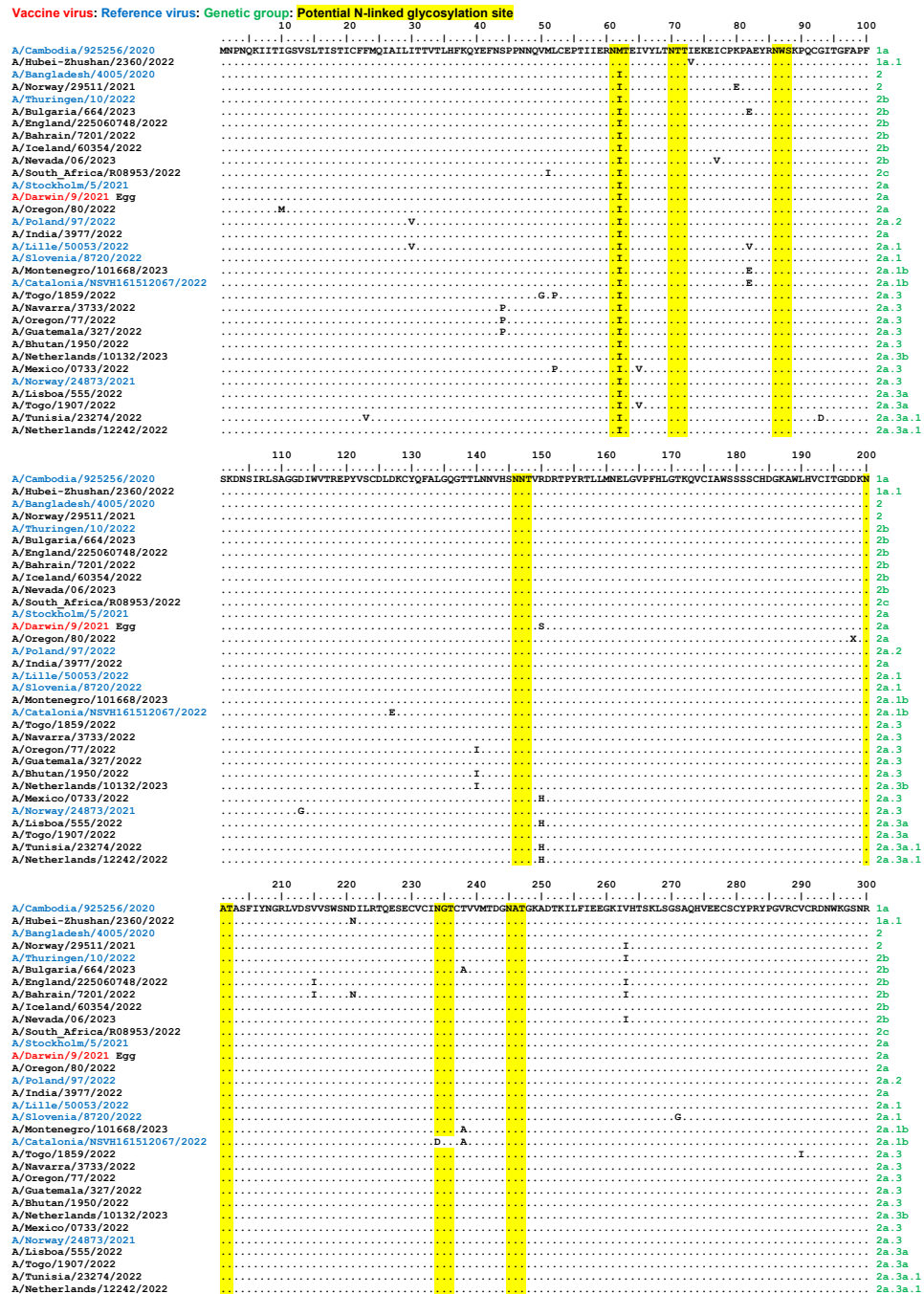


	310	320	330	340	350	360	370	380	390	400
A/Cambodia/925256/2020	TYGACPRYVKQSTIKLATGMGNVPEKQTR	GI	FGA	IAGFT	ENGWEGMVDGWYGRH	QNS	EGRGQAADLKSTQA	AI	DQINGKINRLIGKTKNEKFHQIEKFS	1a
A/Hubei-Zhushan/2360/2022										1a.1
A/Bangladesh/4005/2020										2
A/Norway/29511/2021										2
A/Thuringen/10/2022										2b
A/Bulgaria/664/2023										2b
A/England/225060748/2022										2b
A/Bahrain/7201/2022										2b
A/Iceland/60354/2022										2b
A/Nevada/06/2023										2b
A/South_Africa/R08953/2022										2c
A/Stockholm/5/2021						M				2a
A/Darwin/9/2021 Egg										2a
A/Oregon/80/2022						M				2a
A/Poland/97/2022										2a.2
A/India/3977/2022										2a
A/Lille/50053/2022										2a.1
A/Slovenia/8720/2022										2a.1
A/Montenegro/101668/2023										2a.1b
A/Catalonia/NSVH161512067/2022										2a.1b
A/Togo/1859/2022								S		2a.3
A/Navarra/3733/2022								S		2a.3
A/Oregon/77/2022								S		2a.3
A/Guatemala/327/2022										2a.3
A/Bhutan/1950/2022								S		2a.3
A/Netherlands/10132/2023								S		2a.3b
A/Mexico/0733/2022								S		2a.3
A/Norway/24873/2021								S		2a.3
A/Lisboa/555/2022								S		2a.3a
A/Togo/1907/2022								S		2a.3a
A/Tunisia/23274/2022								S		2a.3a.1
A/Netherlands/12242/2022								S		2a.3a.1
	410	420	430	440	450	460	470	480	490	500
A/Cambodia/925256/2020	EV	EG	R	VQD	L	E	K	V	E	D
A/Hubei-Zhushan/2360/2022										1a.1
A/Bangladesh/4005/2020										2
A/Norway/29511/2021										2
A/Thuringen/10/2022										2b
A/Bulgaria/664/2023										2b
A/England/225060748/2022										2b
A/Bahrain/7201/2022										2b
A/Iceland/60354/2022										2b
A/Nevada/06/2023										2b
A/South_Africa/R08953/2022										2c
A/Stockholm/5/2021										2a
A/Darwin/9/2021 Egg										2a
A/Oregon/80/2022										2a
A/Poland/97/2022										2a.2
A/India/3977/2022										2a
A/Lille/50053/2022										2a.1
A/Slovenia/8720/2022										2a.1
A/Montenegro/101668/2023										2a.1b
A/Catalonia/NSVH161512067/2022										2a.1b
A/Togo/1859/2022										2a.3
A/Navarra/3733/2022										2a.3
A/Oregon/77/2022										2a.3
A/Guatemala/327/2022										2a.3
A/Bhutan/1950/2022										2a.3
A/Netherlands/10132/2023										2a.3b
A/Mexico/0733/2022										2a.3
A/Norway/24873/2021										2a.3
A/Lisboa/555/2022										2a.3a
A/Togo/1907/2022										2a.3a
A/Tunisia/23274/2022										2a.3a.1
A/Netherlands/12242/2022										2a.3a.1
	510	520	530	540	550					
A/Cambodia/925256/2020	Q	I	K	G	V	E	L	K	S	G
A/Hubei-Zhushan/2360/2022										1a.1
A/Bangladesh/4005/2020										2
A/Norway/29511/2021										2
A/Thuringen/10/2022										2b
A/Bulgaria/664/2023										2b
A/England/225060748/2022										2b
A/Bahrain/7201/2022										2b
A/Iceland/60354/2022										2b
A/Nevada/06/2023										2b
A/South_Africa/R08953/2022										2c
A/Stockholm/5/2021										2a
A/Darwin/9/2021 Egg										2a
A/Oregon/80/2022										2a
A/Poland/97/2022										2a.2
A/India/3977/2022										2a
A/Lille/50053/2022										2a.1
A/Slovenia/8720/2022										2a.1
A/Montenegro/101668/2023										2a.1b
A/Catalonia/NSVH161512067/2022										2a.1b
A/Togo/1859/2022										2a.3
A/Navarra/3733/2022										2a.3
A/Oregon/77/2022										2a.3
A/Guatemala/327/2022										2a.3
A/Bhutan/1950/2022										2a.3
A/Netherlands/10132/2023										2a.3b
A/Mexico/0733/2022										2a.3
A/Norway/24873/2021										2a.3
A/Lisboa/555/2022										2a.3a
A/Togo/1907/2022										2a.3a
A/Tunisia/23274/2022										2a.3a.1
A/Netherlands/12242/2022										2a.3a.1

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Protein alignments NA

Figure H3-P2. NA protein alignment for A(H3N2) viruses



	310	320	330	340	350	360	370	380	390	400	
A/Cambodia/925256/2020	PIIDINIKHSIVSRVYVCSGLVGDTPPKSDSSSSSHCLNPNNKGDHGVKGWAFDDGNDVVMGRTINETSRLGYYETFKVVEGWSNPKSLQINRQVIVDR										1a
A/Hubei-Zhushan/2360/2022											1a.1
A/Bangladesh/4005/2020											2
A/Norway/29511/2021		S			S						2
A/Thuringen/10/2022											2b
A/Bulgaria/664/2023					G						2b
A/England/225060748/2022		S			S		T		H		2b
A/Bahrain/7201/2022		S			S		T		H		2b
A/Iceland/60354/2022				R		S					2b
A/Nevada/06/2023		S			S						2b
A/South_Africa/R08953/2022				N		G					G. 2c
A/Stockholm/5/2021											2a
A/Darwin/9/2021 Egg											2a
A/Oregon/80/2022					N						2a
A/Poland/97/2022					G						2a.2
A/India/3977/2022					G					K	2a
A/Lille/50053/2022					G						2a.1
A/Slovenia/8720/2022					G						2a.1
A/Montenegro/101668/2023			X		G						2a.1b
A/Catalonia/NSVH161512067/2022											2a.1b
A/Togo/1859/2022											G. 2a.3
A/Navarra/3733/2022			N								2a.3
A/Oregon/77/2022			N.R								G. 2a.3
A/Guatemala/327/2022			N								2a.3
A/Bhutan/1950/2022											2a.3
A/Netherlands/10132/2023			N								2a.3b
A/Mexico/0733/2022			N.G		N						2a.3
A/Norway/24873/2021			N								2a.3
A/Lisboa/555/2022											2a.3a
A/Togo/1907/2022											2a.3a
A/Tunisia/23274/2022											2a.3a.1
A/Netherlands/12242/2022											2a.3a.1

	410	420	430	440	450	460	
A/Cambodia/925256/2020	GDRSGYSICISFVEGKSCINRCFYVELIRGRKEETEVLWTSNSIVVFCGTSGTGTGSGWPDGANLGLMHI						1a
A/Hubei-Zhushan/2360/2022							1a.1
A/Bangladesh/4005/2020							2
A/Norway/29511/2021					D.N		2
A/Thuringen/10/2022					D.N		2b
A/Bulgaria/664/2023							2b
A/England/225060748/2022					D.N		2b
A/Bahrain/7201/2022					D.N		2b
A/Iceland/60354/2022						T	2b
A/Nevada/06/2023					D.N	V	2b
A/South_Africa/R08953/2022							2c
A/Stockholm/5/2021							2a
A/Darwin/9/2021 Egg							2a
A/Oregon/80/2022						T	2a
A/Poland/97/2022							2a.2
A/India/3977/2022						T	2a
A/Lille/50053/2022							2a.1
A/Slovenia/8720/2022					N		2a.1
A/Montenegro/101668/2023							2a.1b
A/Catalonia/NSVH161512067/2022							2a.1b
A/Togo/1859/2022						T	2a.3
A/Navarra/3733/2022							2a.3
A/Oregon/77/2022							2a.3
A/Guatemala/327/2022							2a.3
A/Bhutan/1950/2022							2a.3
A/Netherlands/10132/2023						T	2a.3b
A/Mexico/0733/2022							2a.3
A/Norway/24873/2021							2a.3
A/Lisboa/555/2022						T	2a.3a
A/Togo/1907/2022					D		T 2a.3a
A/Tunisia/23274/2022							T 2a.3a.1
A/Netherlands/12242/2022							T 2a.3a.1

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Structure

Figure S2. HA1 amino acid substitutions from A/Bangladesh/4005/2020 (red) defining clade 2 along with substitutions within the three main subgroups: D53G, H156S, (D104G and K276R in blue), D53N, H156S, (N96S and I192F in orange) and E50K, I140K, S262N, (F79V and T135A in green). Labels in black represent amino acid substitutions common to more than one group. [PDB Entry 2YP2]

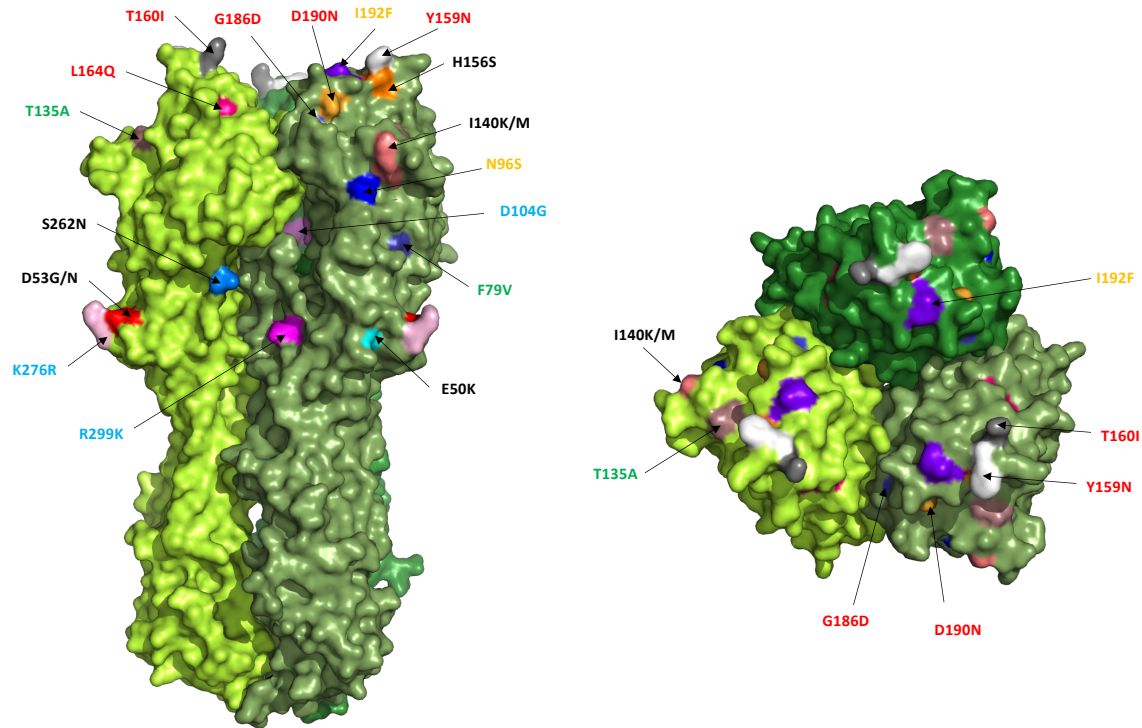
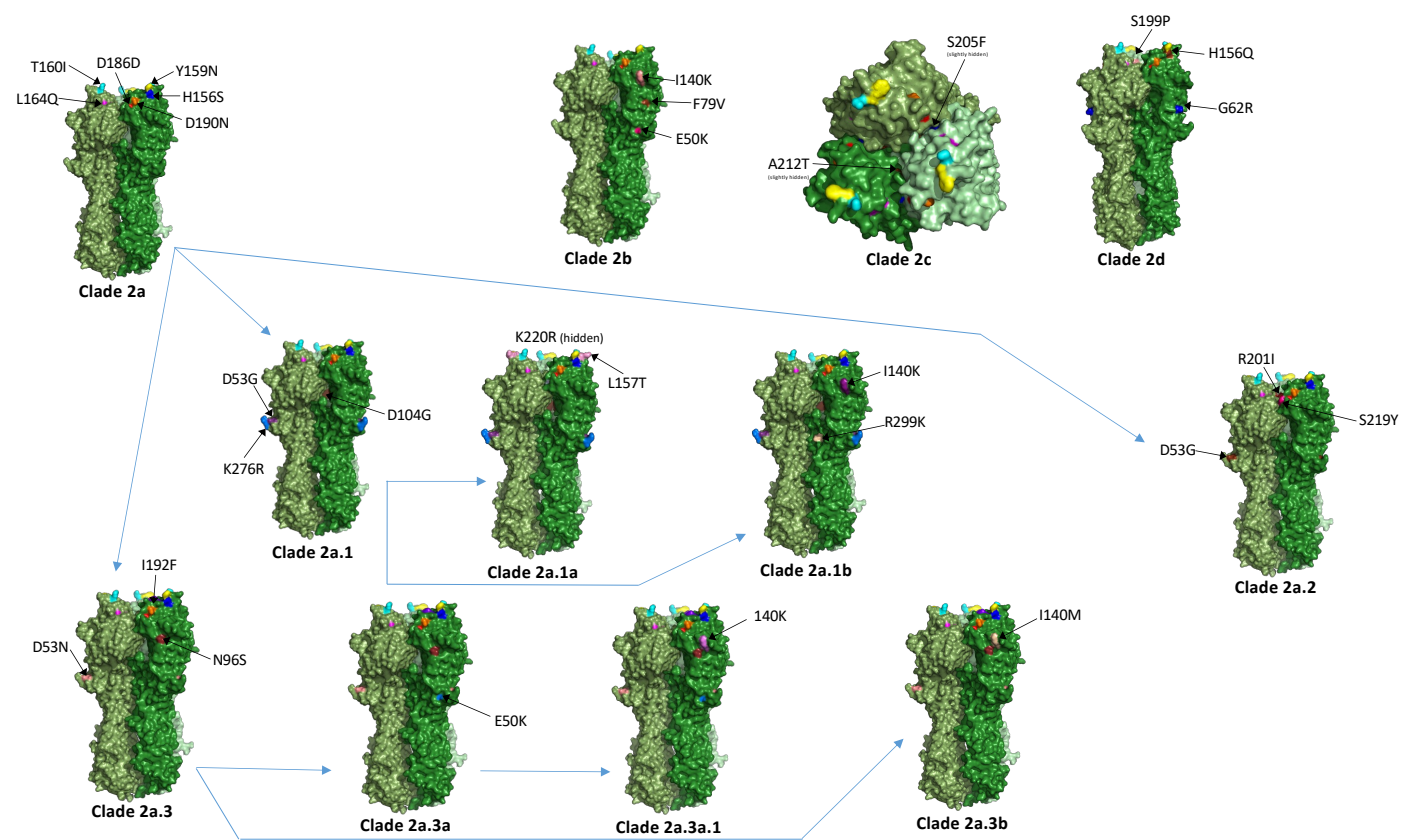
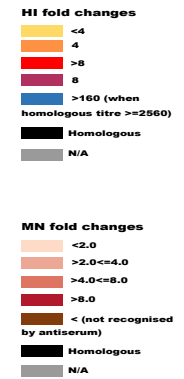
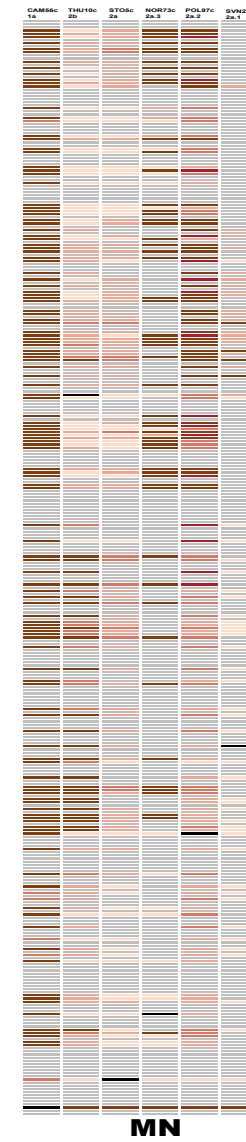
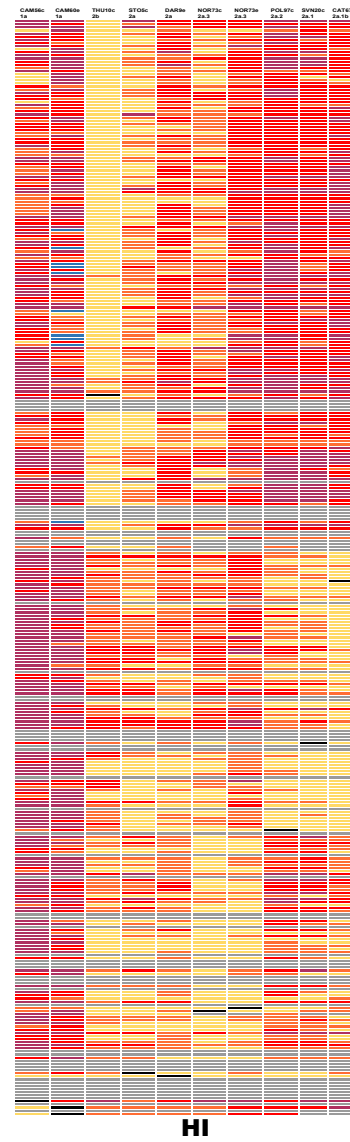
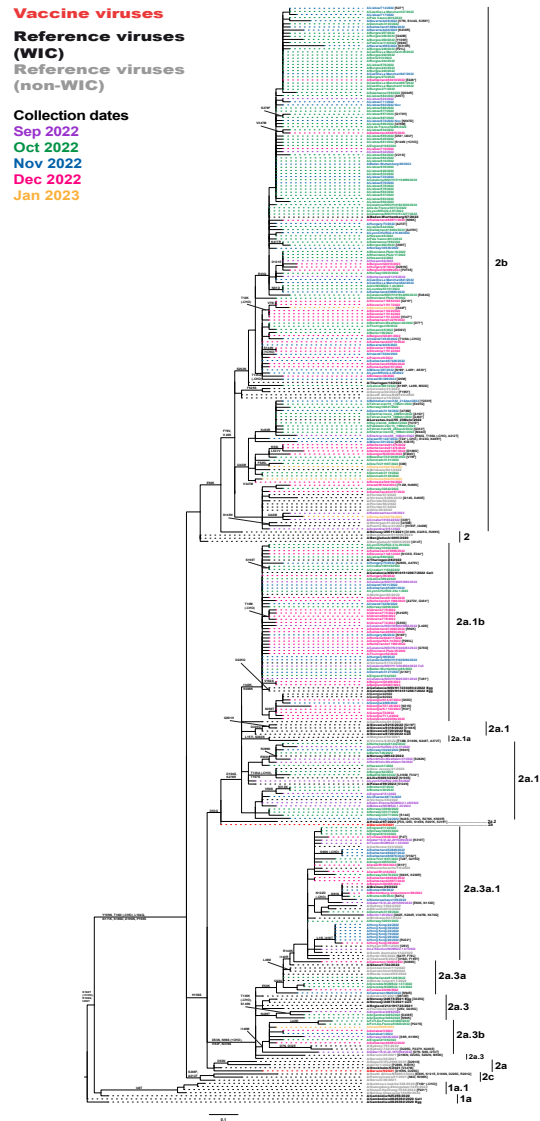


Figure S3. HA1 amino acid substitutions defining updated clade nomenclature for the H3N2 subclades of 3C.2a1b.2a - Clade 2. [PDB Entry 2YP2]



Genotype to Phenotype heatmap: H3N2



Antigenic analyses: H3N2

Haemagglutination inhibition tables: H3N2

Table H3-1. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-09-22

Haemagglutination inhibition titre														
Viruses		Collection date	Passage history	Post-infection ferret antisera										
				NEW										
				A/Denmark	A/HK	A/Camb	A/Camb	A/Bangladesh	A/Thuringen	A/Stockholm	A/Darwin	A/Eng	A/Kansas	
				3264/19	2671/19	925256/20	e0826360/20	4005/20	10/22	5/21	9/21	214191723/21	14/17	
HA1 substitutions compared to A/Bangladesh/4005/2020		Passage history	SIAT	Cell	SIAT	Egg	SIAT	SIAT	SIAT	SIAT	Egg	SIAT	SIAT	
Ferret number		F19/20	St Jude's F21/20	F03/21	F10/21	F07/21	F36/22	F35/21	F39/21	F07/22	F17/19			
Genetic group		3C.2a1b.1a	3C.2a1b.1b	1a	1a	2	2b	2a	2a	2a.3	3C.3a.1			
REFERENCE VIRUSES														
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT5	320	320	640	320	320	ND	160	320	40	160	
A/Hong Kong/2671/2019	3C.2a1b.1b	2019-06-17	MDCK1/SIAT5	320	320	640	160	320	<40	160	320	40	160	
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT5	160	160	1280	320	320	ND	160	640	80	160	
A/Cambodia/e0826360/2020	3C.2a1b.2a.1	2020-07-16	E5/E2	80	<40	80	1280	160	80	160	320	320	80	
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT3	160	40	160	160	640	320	640	640	640	160	
A/Thuringen/10/2022	E50K, F79V, I140K, S262N	2022-04-01	P1/SIAT2	80	<40	160	160	320	320	640	320	640	<40	
A/Stockholm/5/2021	H156S	2021-04-16	SIAT0/SIAT3	80	<40	80	160	320	160	640	640	320	40	
A/Darwin/9/2021	H156S (D186N, D225G)	2021-04-17	E3/E2	160	<40	80	640	320	320	640	1280	640	80	
A/England/214191723/2021	D53N, N96S(+cho), H156S, I192F	2021-10-12	MDCK1/SIAT3	80	<40	80	80	320	160	640	640	640	<40	
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2	40	40	160	160	80	40	160	320	80	640	
TEST VIRUSES														
A/Belgium/G0096/2022	3C.2a1b.2a.2	2022-03-07	SIAT1/SIAT1	40	<40	80	80	160	160	320	320	640	<40	
A/Belgium/S0822/2022	3C.2a1b.2a.2	2022-03-18	SIAT1/SIAT1	40	<40	80	160	160	160	320	320	640	40	
A/Belgium/S0826/2022	3C.2a1b.2a.2	2022-03-19	SIAT1/SIAT1	80	<40	80	160	320	160	320	640	640	40	
A/Belgium/G0153/2022	3C.2a1b.2a.2	2022-03-23	SIAT1/SIAT1	40	<40	80	160	320	160	320	640	640	40	
A/Poland/52/2022	3C.2a1b.2a.2	2022-03-30	SIAT2	80	<40	40	80	320	80	640	640	320	<40	
A/Poland/58/2022	3C.2a1b.2a.2	2022-04-07	SIAT1	80	<40	40	80	160	80	640	640	320	<40	
A/Poland/65/2022	3C.2a1b.2a.2	2022-04-14	SIAT1	80	<40	80	80	320	160	640	640	640	<40	
A/Argentina/2745/2022	3C.2a1b.2a.2	2022-04-21	SIAT1	<40	<40	<40	80	160	40	320	640	640	<40	
A/Togo/1112/2022	3C.2a1b.2a.2	2022-05-03	SIAT1	80	<40	40	160	320	160	640	640	640	40	
A/Argentina/2784/2022	3C.2a1b.2a.2	2022-05-16	SIAT1	80	<40	80	160	320	160	320	640	640	40	
A/Argentina/2863/2022	3C.2a1b.2a.2	2022-05-20	SIAT1	40	<40	80	80	160	80	320	640	640	<40	
A/Argentina/2822/2022	3C.2a1b.2a.2	2022-05-24	SIAT1	40	<40	80	80	320	80	320	640	640	40	
A/Ghana/341/2022	3C.2a1b.2a.2	2022-05-24	MDCK2/SIAT1	160	<40	80	160	320	160	640	640	640	40	
A/Ghana/3122/2022	3C.2a1b.2a.2	2022-06-02	MDCK1/SIAT1	80	<40	80	160	320	160	640	640	640	40	
A/Ghana/330/2022	3C.2a1b.2a.2	2022-06-02	MDCK2/SIAT1	80	<40	80	160	320	160	640	640	640	40	
A/Argentina/2894/2022	3C.2a1b.2a.2	2022-06-03	SIAT1	40	<40	160	80	160	160	640	640	640	40	
A/Ghana/3353/2022	3C.2a1b.2a.2	2022-06-03	SIAT1	80	<40	160	160	320	160	640	640	1280	40	
A/Argentina/2959/2022	3C.2a1b.2a.2	2022-06-06	SIAT1	40	<40	80	80	160	80	320	640	640	40	
A/Ghana/365/2022	3C.2a1b.2a.2	2022-06-06	SIAT1	160	<40	80	320	320	160	640	640	1280	40	
A/Argentina/2953/2022	3C.2a1b.2a.2	2022-06-08	SIAT1	40	<40	80	80	160	80	320	320	640	<40	
A/Ghana/390/2022	3C.2a1b.2a.2	2022-06-08	MDCK1/SIAT1	160	<40	160	320	640	320	1280	1280	2560	80	
A/Argentina/3023/2022	3C.2a1b.2a.2	2022-06-09	SIAT1	40	<40	80	80	160	80	320	320	640	<40	
A/Argentina/2973/2022	3C.2a1b.2a.2	2022-06-15	SIAT1	80	<40	80	80	320	80	640	640	640	40	
A/Norway/27853/2022	3C.2a1b.2a.2	2022-07-18	SIAT2	80	<40	160	320	320	160	640	640	640	40	
A/Norway/28394/2022	3C.2a1b.2a.2	2022-07-20	SIAT2	80	<40	40	80	320	80	640	640	640	<40	
A/Norway/28359/2022	3C.2a1b.2a.2	2022-07-24	SIAT3	80	<40	80	80	320	160	640	640	640	40	
A/Norway/29040/2022	3C.2a1b.2a.2	2022-07-27	SIAT2	80	<40	160	160	320	160	640	1280	640	40	
< relates to the lowest dilution of antiserum used ND = Not Done				Vaccine NH 2021-22					Vaccine SH 2022 NH 2022-23 SH 2023					

Table H3-2. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-09-29

Viruses	HA1 substitutions compared to A/Bangladesh/4005/2020	Passage history Ferret number Genetic group	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Denmark 3264/19 SIAT F19/20	A/HK 2671/19 Cell St Jude F21/20	A/Camb 925256/20 SIAT F03/21	A/Camb e0826360/20 Egg F10/21	A/Bangladesh 4005/20 SIAT F07/21	A/Stockholm 5/21 SIAT F35/21	A/Darwin 9/21 Egg F39/21	A/Eng 214191723/21 SIAT F07/22	A/Kansas 14/17 SIAT F17/19	
					3C.2a1b.1a	3C.2a1b.1b	1a	1a	2	2a	2a	2a.3	3C.3a.1	
					REFERENCE VIRUSES									
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT3/SIAT5	320	320	640	320	320	160	320	40	160	
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT5	320	160	640	160	320	160	320	40	160	
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT5	160	160	640	320	320	160	640	80	160	
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	E5/E2	80	<40	80	2560	160	160	320	320	80	
A/Bangladesh/4005/2020		3C.2a1b.2a.2	2020-10-04	SIAT3	160	40	160	160	640	640	640	640	160	
A/Stockholm/5/2021	H156S	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3	80	<40	80	160	320	640	640	320	40	
A/Darwin/9/2021	H156S (D186N, D225G)	3C.2a1b.2a.2	2021-04-17	E3/E2	160	<40	80	640	320	640	1280	640	80	
A/England/214191723/2021	D53N, N96S(+cho), H156S, I192F	3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT2	80	<40	80	80	320	640	640	1280	<40	
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	40	40	160	160	80	160	320	80	640	
TEST VIRUSES														
A/Croatia/77367/2022		3C.2a1b.2a.2	2022-05-07	SIAT1	40	<40	40	40	160	160	320	320	<40	
A/Argentina/2995/2022		3C.2a1b.2a.2	2022-06-21	SIAT1	40	<40	40	80	160	320	640	640	<40	
A/Ghana/392/2022		3C.2a1b.2a.2	2022-06-03	SIAT3	40	<40	40	40	160	320	320	640	<40	
< relates to the lowest dilution of antiserum used														
ND = Not Done														
Vaccine NH 2021-22														
Vaccine SH 2022 NH 2022-23 SH 2023														
< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)														

Table H3-3. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-10-14

Viruses	HA1 substitutions compared to A/Bangladesh/4005/2020	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Denmark 3264/19 SIAT	A/HK 2671/19 Cell	A/Camb 925256/20 SIAT	A/Camb e0826360/20 Egg	A/Bangladesh 4005/20 SIAT	A/Stockholm 5/21 SIAT	A/Darwin 9/21 Egg	A/Eng 214191723/21 SIAT	A/Kansas 14/17 SIAT	
					F19/20	St Jude's F21/20	F03/21	F10/21	F07/21	F35/21	F39/21	F07/22	F17/19	
		Genetic group			3C.2a1b.1a	3C.2a1b.1b	1a	1a	2	2a	2a	2a.3	3C.3a.1	
REFERENCE VIRUSES														
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT3/SIAT5	320	320	640	160	160	160	160	40	80	
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT5	160	160	640	80	80	80	160	<40	40	
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT5	80	160	640	160	160	160	160	<40	80	
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	E5/E2	80	<40	80	640	160	160	320	160	40	
A/Bangladesh/4005/2020		3C.2a1b.2a.2	2020-10-04	SIAT3	160	40	160	160	320	320	320	320	160	
A/Stockholm/5/2021	H156S	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3	80	<40	80	80	320	320	320	320	<40	
A/Darwin/9/2021	H156S (D186N, D225G)	3C.2a1b.2a.2	2021-04-17	E3/E4	80	40	80	320	320	640	640	40	80	
A/England/214191723/2021	D53N, N96S(+cho), H156S, I192F	3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT3	40	<40	40	80	160	320	320	640	<40	
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	40	<40	80	40	40	80	160	40	320	
TEST VIRUSES														
A/Pays de Loire/21194/2022		3C.2a1b.2a.2	2022-03-14	C1/SIAT1	80	<40	40	80	160	320	320	320	40	
A/Coquelles/21188/2022		3C.2a1b.2a.2	2022-03-15	C1/SIAT1	40	<40	40	40	160	320	320	160	<40	
A/Lorraine/20876/2022		3C.2a1b.2a.2	2022-03-16	C1/SIAT1	40	<40	40	40	160	320	320	160	<40	
A/Nord Pas de Calais/27131/2022		3C.2a1b.2a.2	2022-04-05	C1/SIAT1	40	<40	40	40	160	320	320	320	<40	
A/Bretagne/32468/2022		3C.2a1b.2a.2	2022-04-26	C1/SIAT1	40	<40	40	40	80	160	160	160	<40	
A/Champagne Ardenne/32870/2022		3C.2a1b.2a.2	2022-04-29	C1/SIAT1	80	<40	40	80	80	160	320	320	<40	

< relates to the lowest dilution of antiserum used

ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23
SH 2023

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table H3-4. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-10-28

Viruses	HA1 substitutions additional to those defining subclades of 3C.2a1b.2a	Passage history Ferret number Genetic group	Collection date	Passage history	Haemagglutination inhibition titre										
					Post-infection ferret antisera										
					A/Denmark 3264/19 SIAT F19/20	A/HK 2671/19 Cell St. Jude's F21/20	A/Camb 925256/20 SIAT F03/21	A/Camb e0826360/20 Egg F10/21	A/Bangladesh 4005/20 SIAT F07/21	A/Thuringen 10/22 SIAT F36/22	A/Stockholm 5/21 SIAT F35/21	A/Darwin 9/21 Egg F39/21	A/Eng 214191723/21 SIAT F07/22	A/Poland 97/22 SIAT F39/22	A/Kansas 14/17 SIAT F17/19
					3C.2a1b.1a	3C.2a1b.1b	1a	1a	2	2b	2a	2a	2a.3	2a.2	3C.3a.1
REFERENCE VIRUSES															
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT3/SIAT5	320	320	640	160	160	ND	160	160	40	ND	80
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT5	320	320	640	80	160	40	80	160	40	160	80
A/Cambodia/925256/2020		1a	2020-09-25	SIAT5	80	160	640	160	160	ND	160	160	<40	ND	80
A/Cambodia/e0826360/2020		1a	2020-07-16	E5/E2	160	<40	160	1280	160	80	160	320	320	80	80
A/Bangladesh/4005/2020		2	2020-10-04	SIAT3	160	<40	160	160	640	320	320	160	320	320	80
A/Thuringen/10/2022		2b	2022-04-01	P1/SIAT2	160	<40	80	160	320	320	640	160	320	ND	80
A/Stockholm/5/2021		2a	2021-04-16	SIAT0/SIAT3	40	<40	40	40	160	160	320	320	320	640	<40
A/Darwin/9/2021	D186N, D225G (egg)	2a	2021-04-17	E3/E4	80	40	40	320	160	160	320	640	320	640	<40
A/England/214191723/2021		2a.3	2021-10-12	MDCK1/SIAT3	40	<40	40	40	80	160	320	320	640	320	<40
A/Poland/97/2022	P3S, I25V, S145N, R201K	2a.2	2022-05-09	SIAT2	80	<40	80	160	160	ND	640	640	640	1280	40
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	40	<40	80	40	40	40	40	80	40	80	320
TEST VIRUSES															
A/Catalonia/NSVH101892403/2022		3C.2a1b.2a.2	2022-08-13	SIAT1/SIAT1	80	<40	40	80	160	80	320	320	320	ND	<40
A/Catalonia/NSVH161512067/2022		2a.1b	2022-09-14	SIAT1/SIAT1	160	<40	80	160	160	160	640	640	640	ND	40
< relates to the lowest dilution of antiserum used ND = Not Done															
< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre) Vaccine NH 2021-22 Vaccine SH 2022 NH 2022-23 SH 2023															

Table H3-5. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-11-18

Viruses	HA1 substitutions additional to those defining subclades of 3C.2a1b.2a	Passage history Ferret number Genetic group	Collection date	Passage history	Haemagglutination inhibition titre																		
					Post-infection ferret antisera																		
					A/Camb 925256/20	A/Camb e0826360/20	A/Thuringen 10/22	A/Stockholm 5/21	A/Darwin 9/21	A/Norway 24873/21	A/Norway 24873/21	A/Ghana 17/24/22	A/Poland 97/22	A/Slov 8720/2022	A/Poland 99/22	A/Nor 28542/22	NSVH-2067/22	A/Catal NSVH-2067/22					
					SIAT F03/21 1a	Egg F10/21 1a	SIAT F36/22 2b	SIAT F35/21 2a	SIAT F39/21 2a	SIAT F10/22 2a.3	SIAT F11/22 2a.3	SIAT F43/22 2a.3a	SIAT F39/22 2a.2	SIAT F24/22 2a.1	SIAT F44/22 2a.1	SIAT F42/22 2a.1	SIAT F42/22 2a.1	SIAT F41/22 2a.1b					
REFERENCE VIRUSES																							
A/Cambodia/925256/2020		1a	2020-09-25	SIAT5	640	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND		
A/Cambodia/e0826360/2020		1a	2020-07-16	SIAT	320	2560	160	320	640	80	160	<40	160	160	320	80	40	40	40	40	40		
A/Thuringen/10/2022	S262N	2b	2022-04-01	P1/SIAT2	40	160	160	320	160	80	80	<40	80	160	80	80	80	80	80	80	80		
A/Stockholm/5/2021		2a	2021-04-16	SIAT0/SIAT3	160	160	160	1280	640	160	320	<40	640	320	1280	320	160	160	160	160	160		
A/Darwin/9/2021	D186N, D225G (egg)	2a	2021-04-17	E3/E4	160	640	320	1280	640	320	320	80	640	640	1280	640	320	320	320	320	320		
A/Norway/24873/2021		2a.3	2021-10-24	SIAT2	80	80	160	640	320	320	640	80	320	320	640	160	160	160	160	160	160		
A/Norway/24873/2021	D225G (egg)	2a.3	2021-10-24	E3 (Am2A1)	320	320	320	1280	1280	640	1280	80	640	1280	1280	640	320	320	320	320	320		
A/Ghana/1724/2022		2a.3a	2022-05-18	SIAT2	80	80	160	640	640	320	320	80	320	320	1280	320	320	320	320	320	320		
A/Poland/97/2022	P35, I25V, S145N, R201K	2a.2	2022-05-09	SIAT2	160	160	160	1280	640	320	640	80	1280	640	2560	1280	640	640	640	640	640		
A/Slovenia/8720/2022		2a.1	2022-02-10	SIAT1/MDCK1/SIAT2	80	80	160	640	640	320	320	40	640	640	2560	640	640	640	640	640	640		
A/Poland/99/2022	S124N	2a.1	2022-05-10	SIAT2	80	40	160	640	640	160	160	<40	640	640	1280	320	640	640	640	640	640		
A/Norway/28542/2022	E50K, R299K	2a.1	2022-07-12	SIAT2	40	80	160	1280	640	320	320	80	1280	640	2560	2560	640	640	640	640	640		
A/Catalonia/NSVH161512067/2022		2a.1b	2022-09-14	SIAT1/SIAT2	40	80	160	640	320	160	160	<40	640	640	1280	320	640	640	640	640	640		
TEST VIRUSES																							
A/Rheinland-Pfalz/16/2022	S262N	2b	2022-10-04	P1/SIAT1	160	160	640	640	320	320	160	<40	160	160	320	160	80	80	80	80	80		
A/Niedersachsen/49/2022	I242M	2b	2022-09-26	P1/SIAT1	160	160	320	640	320	160	320	40	160	160	160	160	80	80	80	80	80		
A/Hessen/46/2022	R33Q, S262N	2b	2022-10-17	P1/SIAT1	80	160	640	320	320	80	160	<40	80	80	160	40	80	80	80	80	80		
A/Baden-Wuerttemberg/87/2022	R33Q, S262N	2b	2022-10-17	P1/SIAT1	80	160	320	320	320	80	160	<40	80	80	80	80	80	80	80	80	80		
A/Hessen/40/2022	R33Q, D101E, S262N	2b	2022-09-29	P1/SIAT1	160	160	320	320	320	80	160	40	80	80	320	80	40	40	40	40	40		
A/Rheinland-Pfalz/17/2022	R33Q, D101E, S262N	2b	2022-10-04	P1/SIAT1	80	160	160	160	160	80	160	<40	80	160	80	80	80	80	80	80	80		
A/Hessen/42/2022	R33Q, D101E, S262N	2b	2022-10-05	P1/SIAT1	40	320	320	320	160	80	80	<40	80	160	160	80	80	80	80	80	80		
A/Rheinland-Pfalz/18/2022	R33Q, D101E, S262N	2b	2022-10-11	P1/SIAT1	80	160	160	320	160	80	160	<40	80	160	80	80	80	80	80	80	80		
A/Berlin/150/2022	T135A(-cho), S262N	2b	2022-10-07	P1/SIAT1	160	160	640	640	640	320	160	<40	160	160	160	80	80	80	80	80	80		
A/Hessen/45/2022	T135A(-cho), S262N	2b	2022-10-13	P1/SIAT1	320	320	640	640	320	320	40	160	160	160	160	160	80	80	80	80	80		
A/Thuringen/30/2022	T135A(-cho), S262N	2b	2022-10-17	P1/SIAT1	80	160	320	320	320	80	80	<40	80	80	80	80	80	80	80	80	80		
A/Nordrhein-Westfalen/40/2022	D77X, T135A(-cho), S262N	2b	2022-10-13	P1/SIAT1	80	160	320	320	320	160	80	<40	80	80	80	80	80	80	80	80	80		
A/Hessen/41/2022		2a.1	2022-10-06	P1/SIAT1	40	80	80	640	640	80	160	<40	320	640	1280	320	640	640	640	640	640		
A/Nordrhein-Westfalen/36/2022	E50K	2a.1	2022-09-26	P2/SIAT1	40	80	80	640	640	160	320	40	320	640	2560	1280	640	640	640	640	640		
A/Berlin/146/2022	E50K, R299K	2a.1	2022-10-12	P1/SIAT1	80	40	160	640	640	160	320	80	640	640	2560	2560	640	640	640	640	640		
A/Bremen/27/2022	S312N	2a.1	2022-10-04	P1/SIAT1	160	80	320	1280	1280	160	640	40	640	640	2560	640	1280	1280	1280	1280	1280		
A/Bremen/30/2022	S312N	2a.1	2022-10-12	P1/SIAT1	80	80	160	640	640	320	320	40	640	1280	2560	640	1280	1280	1280	1280	1280		
A/Nordrhein-Westfalen/37/2022	E50K, S262N, R299K	2a.1	2022-09-29	P1/SIAT1	40	40	160	640	640	160	320	40	640	640	2560	1280	640	640	640	640	640		
A/Baden-Wuerttemberg/66/2022		2a.1b	2022-10-10	P1/SIAT1	80	80	160	640	640	320	320	<40	640	640	2560	640	640	640	640	640	640		
A/Thuringen/28/2022		2a.1b	2022-10-17	P1/SIAT1	80	80	160	640	320	160	160	<40	320	640	1280	320	1280	1280	1280	1280	1280		
A/Berlin/145/2022	G62E, K264R	2a.3a.1	2022-09-15	P1/SIAT1	40	80	160	320	320	320	640	80	160	160	320	160	160	160	160	160	160		
A/Bremen/29/2022	S92N, N122D	2a.3a.1	2022-10-04	P1/SIAT1	80	320	320	640	640	320	1280	160	320	320	640	320	320	320	320	320	320		
A/Bremen/28/2022	S47L(-cho), S92N, N122D(-cho)	2a.3a.1	2022-10-04	P1/SIAT1	160	320	640	640	1280	160	320	160	320	320	1280	320	320	320	320	320	320		
< relates to the lowest dilution of antiserum used ND = Not Done					Vaccine NH 2021-22					Vaccine SH 2022 NH 2022-23 SH 2023													

< relates to the lowest dilution of antiserum used
ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23
SH 2023

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table H3-6. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-12-09

Viruses	HA1 substitutions additional to those defining subclades of 3C.2a1b.2a	Passage history Ferret number Genetic group	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Camb 925256/20 SIAT F03/21	A/Camb e0826360/20 Egg F10/21	A/Thuringen 10/22 SIAT F36/22	A/Stockholm 5/21 SIAT F35/21	A/Darwin 9/21 Egg F39/21	A/Norway 24873/21 SIAT F10/22	A/Norway 24873/21 Egg F11/22	A/Poland 97/22 SIAT F39/22	A/Slov 8720/2022 SIAT F24/22	A/Catal NSVH-2067/22 SIAT F41/22
					1a	1a	2b	2a	2a	2a.3	2a.3	2a.2	2a.1	2a.1b
REFERENCE VIRUSES														
A/Cambodia/925256/2020		1a	2020-09-25	SIAT5	320	160	<40	160	160	<40	<40	80	<40	<40
A/Cambodia/e0826360/2020		1a	2020-07-16	E5/E2	160	1280	80	160	320	40	80	80	80	<40
A/Thuringen/10/2022	S262N	2b	2022-04-01	P1/SIAT2	40	160	320	320	320	80	160	80	160	80
A/Stockholm/5/2021		2a	2021-04-16	SIAT0/SIAT3	80	160	80	640	640	80	160	320	320	160
A/Darwin/9/2021	D186N, D225G (egg)	2a	2021-04-17	E3/E4	160	640	320	1280	1280	320	320	640	640	320
A/Norway/24873/2021		2a.3	2021-10-24	SIAT3	80	80	160	640	320	160	320	320	160	160
A/Norway/24873/2021	D225G (egg)	2a.3	2021-10-24	E3/E1	80	320	320	1280	640	320	640	640	640	160
A/Poland/97/2022	P3S, I25V, S145N, R201K	2a.2	2022-05-09	SIAT2	80	160	80	1280	640	160	320	640	320	640
A/Slovenia/8720/2022		2a.1	2022-02-10	SIAT1/MDCK1/SIAT3	40	80	80	640	320	160	160	320	640	640
A/Catalonia/NSVH161512067/2022		2a.1b	2022-09-14	SIAT1/SIAT2	40	80	80	320	320	160	160	640	640	640
TEST VIRUSES														
A/Qatar/21-VI-22-1929386/2022		3C.2a1b.2a.2	2022-08-26	SIAT2	40	40	40	80	160	80	160	80	80	40
A/Qatar/21-VI-22-1929305/2022		3C.2a1b.2a.2	2022-08-26	SIAT2	40	80	80	160	320	160	320	160	160	80
A/Qatar/25-VI-22-1918966/2022		3C.2a1b.2a.2	2022-08-23	SIAT2	40	160	160	160	320	160	640	80	160	80
A/Behbahan/Iran/33X 21Aban/2022	Y233H, I242M	2b	2022-11-04	SIAT1	80	160	320	320	160	80	80	160	80	80
A/Lisboa/533/2022	R33Q, S262N	2b	2022-10-02	SIAT2/SIAT1	40	80	160	160	160	40	80	80	80	40
A/Lisboa/576/2022	R33Q, S262N	2b	2022-10-03	SIAT2/SIAT1	160	80	320	160	160	320	80	80	640	80
A/Lisboa/583/2022	R33Q, S262N	2b	2022-10-04	SIAT2/SIAT1	80	160	320	320	160	80	80	80	80	40
A/Lisboa/580/2022	R33Q, S262N	2b	2022-10-07	SIAT2/SIAT1	80	80	320	320	160	40	80	80	80	40
A/Lisboa/582/2022	R33Q, S262N	2b	2022-10-09	SIAT2/SIAT1	80	80	320	320	160	40	80	80	80	40
A/Lisboa/537/2022	R33Q, S262N	2b	2022-10-10	SIAT2/SIAT1	80	80	320	320	160	40	80	80	80	40
A/Lisboa/596/2022	R33Q, S262N	2b	2022-10-18	SIAT1/SIAT1	80	80	160	160	160	40	80	80	80	40
A/Lisboa/593/2022	R33Q, S262N	2b	2022-10-20	SIAT1/SIAT1	80	80	320	320	160	40	80	80	80	40
A/Lisboa/545/2022	R33Q, S262N	2b	2022-10-20	SIAT2/SIAT1	80	80	160	320	160	40	80	80	80	40
A/Lisboa/544/2022	R33Q, S262N	2b	2022-10-24	SIAT2/SIAT1	80	160	320	320	160	80	80	80	80	40
A/Lisboa/616/2022	R33Q, S262N	2b	2022-10-26	SIAT1/SIAT1	80	160	320	320	160	40	80	80	80	40
A/Lisboa/560/2022	R33Q, S262N	2b	2022-10-26	SIAT1/SIAT1	80	80	320	320	160	40	80	80	80	40
A/Lisboa/626/2022	R33Q, S262N	2b	2022-10-27	SIAT1/SIAT1	80	80	320	320	160	40	80	80	80	40
A/Lisboa/563/2022	R33Q, S262N	2b	2022-11-02	SIAT1/SIAT1	40	80	160	160	160	40	80	40	80	40
A/Lisboa/584/2022	R33Q, A93T, S262N	2b	2022-10-05	SIAT2/SIAT1	80	160	320	320	160	80	80	80	80	40
A/Lisboa/554/2022	R33Q, V213I, S262N	2b	2022-10-24	SIAT1/SIAT1	80	160	320	320	160	80	80	80	80	40
A/Lisboa/595/2022	R33Q, D53X, I202X, S262N	2b	2022-10-24	SIAT1/SIAT1	80	160	320	320	320	80	160	80	160	80
A/Catalonia/NSVH198253092/2022		2a.1b	2022-09-01	SIAT2	40	80	80	320	320	40	160	320	320	320
A/Lisboa/566/2022		2a.1b	2022-10-31	SIAT1/SIAT1	<40	40	40	160	160	40	80	160	160	160

< relates to the lowest dilution of antiserum used
ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23
SH 2023

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table H3-7. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-12-15

Viruses	HA1 substitutions additional to those defining subclades of 3C.2a1b.2a	Passage history Ferret number Genetic group	Collection date	Passage history	Haemagglutination inhibition titre											
					Post-infection ferret antisera											
					A/Camb 925256/20 SIAT F03/21	A/Camb e0826360/20 Egg F10/21	A/Thuringen 10/22 SIAT F36/22	A/Stockholm 5/21 SIAT F35/21	A/Darwin 9/21 Egg F39/21	A/Norway 24873/21 SIAT F10/22	A/Norway 24873/21 Egg F11/22	A/Poland 97/22 SIAT F39/22	A/Slov 8720/2022 SIAT F24/22	NSVH-2067/22 SIAT F41/22	A/Catal NSVH-2067/22 SIAT F41/22	
					1a	1a	2b	2a	2a	2a.3	2a.3	2a.2	2a.1	2a.1b		
REFERENCE VIRUSES																
A/Cambodia/925256/2020		1a	2020-09-25	SIAT5	320	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND
A/Cambodia/e0826360/2020		1a	2020-07-16	E5E2	80	640	40	160	320	40	80	40	80	40	80	<40
A/Thuringen/10/2022	S262N	2b	2022-04-01	P1/SIAT2	80	160	320	640	320	80	80	80	160	40	80	40
A/Stockholm/5/2021		2a	2021-04-16	SIAT0/SIAT3	80	80	80	640	640	80	160	320	320	320	320	80
A/Darwin/9/2021	D186N, D225G (egg)	2a	2021-04-17	E3/E4	80	320	160	640	640	160	320	320	320	320	320	160
A/Norway/24873/2021		2a.3	2021-10-24	SIAT3	80	80	80	320	320	160	320	160	160	80	80	80
A/Norway/24873/2021	D225G (egg)	2a.3	2021-10-24	E3/E1	40	320	160	640	640	160	640	320	640	80	80	80
A/Poland/97/2022	P3S, I25V, S145N, R201K	2a.2	2022-05-09	SIAT2	40	160	80	1280	640	160	320	640	640	320	320	320
A/Slovenia/8720/2022		2a.1	2022-02-10	SIAT1/MDCK1/SIAT3	40	80	80	640	640	80	160	640	640	320	320	320
A/Catalonia/NSVH161512067/2022		2a.1b	2022-09-14	SIAT1/SIAT2	40	80	80	320	320	80	160	320	320	320	320	320
TEST VIRUSES																
A/South Africa/R07061/2022		3C.2a1b.2a.2	2022-06-09	MDCK1/SIAT1	80	320	160	320	320	40	80	160	80	160	40	40
A/Catalonia/NSVH198242704/2022		3C.2a1b.2a.2	2022-05-05	SIAT1/SIAT2	80	80	160	320	640	160	320	320	160	160	160	160
A/Qatar/25-VI-22-1930279/2022		3C.2a1b.2a.2	2022-08-27	SIAT3	80	80	80	320	640	320	320	160	160	160	160	160
A/Qatar/10-VI-22-1933793/2022		3C.2a1b.2a.2	2022-08-28	SIAT1	80	80	160	320	640	160	320	320	320	320	320	160
A/Shahriar.Iran/34 19Mehr/2022	I242M	2b	2022-10-08	SIAT1/SIAT1	80	160	320	640	320	80	160	160	160	80	80	80
A/Tehran.Iran/18 18Mehr/2022	I242M	2b	2022-10-08	SIAT1/SIAT1	80	80	320	320	160	80	80	80	80	80	80	40
A/Tehran.Iran/16 17Mehr/2022	I242M	2b	2022-10-08	SIAT1/SIAT1	40	80	320	320	160	40	80	80	80	80	80	40
A/Pakdasht.Iran/13 19Mehr/2022	I242M	2b	2022-10-09	SIAT1/SIAT1	40	80	320	320	160	40	80	80	80	80	80	40
A/Lorestan.Iran/55 23Mehr/2022	I242M	2b	2022-10-10	SIAT1/SIAT1	80	80	320	320	160	40	80	80	80	80	80	40
A/Rey.Iran/44 23Mehr/2022	I242M	2b	2022-10-12	SIAT1/SIAT1	80	80	320	320	160	40	80	160	80	80	80	80
A/Tehran.Iran/29 2Aba/2022	I242M	2b	2022-10-23	SIAT1/SIAT1	80	160	320	320	160	80	160	160	160	160	80	80
A/Shahriar.Iran/4 23Mehr/2022	A163X, I242M	2b	2022-10-10	SIAT1/SIAT1	80	160	320	320	320	80	80	80	80	160	40	40
A/Shahriar.Iran/66 10Mehr/2022	R92G, T135A(-cho), A212T, I242M	2b	2022-09-29	SIAT1/SIAT1	40	160	160	160	160	40	80	80	80	80	<40	<40
A/Lisboa/532/2022	R33Q, S262N	2b	2022-09-17	SIAT2/SIAT1	40	80	160	320	160	40	80	80	80	80	80	40
A/Lisboa/524/2022	R33Q, S262N	2b	2022-09-25	SIAT2/SIAT2	40	80	160	320	160	40	80	80	80	80	80	40
A/Lisboa/534/2022	R33Q, S262N	2b	2022-10-02	SIAT2/SIAT1	40	80	320	160	160	40	80	80	80	80	80	40
A/Lisboa/578/2022	R33Q, S262N	2b	2022-10-04	SIAT2/SIAT1	80	80	320	320	320	80	80	80	80	80	80	40
A/Lisboa/577/2022	R33Q, S262N	2b	2022-10-04	SIAT2/SIAT1	40	80	160	160	160	40	80	80	80	80	80	40
A/Lisboa/535/2022	R33Q, S262N	2b	2022-10-04	SIAT2/SIAT1	40	80	320	320	320	80	80	80	80	80	80	40
A/Île de France/52489/2022	R33Q, S262N	2b	2022-10-07	MDCK2/SIAT1	40	80	160	320	160	40	80	80	80	80	80	40
A/Lisboa/579/2022	R33Q, S262N	2b	2022-10-09	SIAT2/SIAT1	80	160	320	320	320	80	80	80	80	80	80	40
A/Île de France/53172/2022	R33Q, S262N	2b	2022-10-11	MDCK1/SIAT1	40	80	320	320	160	40	80	80	80	80	80	40
A/Lisboa/572/2022	R33Q, S262N	2b	2022-11-04	SIAT1/SIAT1	80	160	320	320	320	80	80	80	80	80	80	40
A/Lisboa/570/2022	R33Q, S262N	2b	2022-11-04	SIAT1/SIAT1	80	160	320	320	160	40	80	80	80	80	80	40
A/Lourdes/53101/2022	R33Q, N81D, S262N	2b	2022-10-08	MDCK1/SIAT1	40	160	320	320	160	40	80	80	80	80	80	40
A/Lisboa/561/2022	R33Q, S144N(+cho), S262N	2b	2022-10-24	SIAT1/SIAT1	80	160	320	320	320	80	160	80	160	80	80	40
A/Lisboa/597/2022	R33Q, S262N, S279F	2b	2022-10-19	SIAT1/SIAT1	40	80	320	320	160	40	80	80	80	80	80	40
A/Lisboa/557/2022	R33Q, Q173H, S262N, S279F	2b	2022-10-20	SIAT1/SIAT1	40	80	320	320	320	40	80	80	80	80	80	40
A/Lille/50053/2022	T135A(-cho), T167S, N190S	2a.1	2022-09-06	MDCK1/SIAT1	80	160	40	320	320	40	160	320	320	320	160	160
A/Madrid/3803/2022	T135A(-cho), T167S, L316M	2a.1	2022-10-05	SIAT2/SIAT2	40	80	40	320	320	40	80	320	320	320	320	640
A/Catalonia/NSVH560025968/2022		2a.1b	2022-09-01	SIAT1	<40	40	80	160	160	40	80	160	320	320	320	160
A/Catalonia/NSVH198253201/2022		2a.1b	2022-09-02	SIAT1	<40	40	80	160	320	40	80	320	320	320	320	320
A/Catalonia/NSVH172304934/2022		2a.1b	2022-09-13	SIAT1	40	80	80	320	320	40	160	320	320	320	320	320
A/Catalonia/NSVH531046263/2022	G78D	2a.1b	2022-09-08	SIAT1	<40	40	80	160	160	40	80	160	320	320	160	160
A/Qatar/10-VI-22-2015495/2022	K310T	2a.3a.1	2022-09-20	SIAT1	40	80	160	320	320	160	320	160	160	80	80	80
A/Qatar/10-VI-22-2015935/2022	E83K, N122D(-cho), N133D(-cho)	2a.3a.1	2022-09-20	SIAT1	<40	80	80	160	160	160	320	80	80	80	80	80
A/Fort-De-France/53862/2022		2a.3b	2022-10-18	MDCK1/SIAT1	80	80	160	640	320	160	320	160	160	160	160	80
A/Qatar/16-VI-22-1972944/2022	D7N, N8S, E50, I214T	2a.3b	2022-09-09	SIAT1	40	40	80	160	320	160	320	160	160	160	80	80

< relates to the lowest dilution of antiserum used
ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23
SH 2023

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table H3-8. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-12-16

Viruses	HA1 substitutions additional to those defining subclades of 3C.2a1b.2a	Passage history Ferret number Genetic group	Collection date	Passage history	Haemagglutination inhibition titre										
					Post-infection ferret antisera										
					A/Camb 925256/20	A/Camb e0826360/20	A/Thuringen 10/22	A/Stockholm 5/21	A/Darwin 9/21	A/Norway 24873/21	A/Norway 24873/21	A/Poland 97/22	A/Slov 8720/2022	A/Catal NSVH-2067/22	
					SIAT F03/21	Egg F10/21	SIAT F36/22	SIAT F35/21	Egg F39/21	SIAT F10/22	Egg F11/22	SIAT F39/22	SIAT F24/22	SIAT F41/22	
					1a	1a	2b	2a	2a	2a.3	2a.3	2a.2	2a.1	2a.1b	
REFERENCE VIRUSES															
A/Cambodia/925256/2020		1a	2020-09-25	SIAT5	320	160	<40	160	160	<40	<40	80	<40	<40	
A/Cambodia/e0826360/2020		1a	2020-07-16	E5/E2	160	1280	80	160	320	40	80	80	80	<40	
A/Thuringen/10/2022	S262N	2b	2022-04-01	P1/SIAT2	40	160	320	320	320	80	160	80	160	80	
A/Stockholm/5/2021		2a	2021-04-16	SIAT0/SIAT3	80	160	80	640	640	80	160	320	320	160	
A/Darwin/9/2021	D186N, D225G (egg)	2a	2021-04-17	E3/E4	160	640	320	1280	1280	320	320	640	640	320	
A/Norway/24873/2021		2a.3	2021-10-24	SIAT3	80	80	160	640	320	160	320	320	160	160	
A/Norway/24873/2021	D225G (egg)	2a.3	2021-10-24	E3/E1	80	320	320	1280	640	320	640	640	640	160	
A/Poland/97/2022	P3S, I25V, S145N, R201K	2a.2	2022-05-09	SIAT2	80	160	80	1280	640	160	320	640	320	640	
A/Slovenia/8720/2022		2a.1	2022-02-10	SIAT1/MDCK1/SIAT3	40	80	80	640	320	160	160	320	640	640	
A/Catalonia/NSVH161512067/2022		2a.1b	2022-09-14	SIAT1/SIAT2	40	80	80	320	320	160	160	640	640	320	
TEST VIRUSES															
A/Bawshar/72212056/2022	F79, I242M	2b	2022-10-30	SIAT1	40	40	320	320	160	80	160	40	80	80	
A/Burgos/271/2022	R33Q, S262N	2b	2022-10-14	SIAT1	160	160	320	320	160	160	160	80	80	40	
A/Burgos/272/2022	R33Q, S262N	2b	2022-10-15	SIAT1	80	80	160	160	160	80	160	40	80	40	
A/Burgos/245/2022	R33Q, S262N	2b	2022-10-18	SIAT1	160	80	320	320	320	160	160	80	80	80	
A/Burgos/244/2022	R33Q, S262N	2b	2022-10-18	SIAT1	160	80	320	320	320	160	160	80	80	80	
A/Burgos/243/2022	R33Q, S262N	2b	2022-10-18	SIAT1	320	80	320	320	320	160	160	80	160	160	
A/Burgos/242/2022	R33Q, S262N	2b	2022-10-18	SIAT1	80	40	320	320	320	160	80	40	80	40	
A/Castilla La Mancha/4129/2022	R33Q, S262N	2b	2022-10-19	SIAT1	40	40	320	160	80	40	80	40	40	40	
A/Pais Vasco/4012/2022	R33Q, S262N	2b	2022-10-19	SIAT1	80	40	320	320	160	80	160	40	80	80	
A/Pais Vasco/4010/2022	R33Q, S262N	2b	2022-10-19	SIAT1	320	80	640	640	320	160	320	160	80	80	
A/Burgos/249/2022	P21L, R33Q, S262N	2b	2022-10-16	SIAT1	80	80	160	160	160	80	80	40	40	40	
A/Burgos/260/2022	R33Q, Y105H, S262N	2b	2022-10-14	SIAT1	40	80	160	160	80	80	80	40	40	40	
A/Burgos/246/2022	R33Q, I242M, S262N	2b	2022-10-17	SIAT1	160	80	320	320	160	160	160	80	80	80	
A/Burgos/262/2022	R33Q, S262N, I300T(+cho)	2b	2022-10-14	SIAT1	80	320	320	320	320	160	320	80	80	40	
A/Ibra/72211987/2022	I3M, I242M, L243X	2b	2022-10-26	SIAT1	80	40	320	320	160	80	160	40	40	40	
A/Aragon/4055/2022		2a.3a.1	2022-10-18	SIAT1	80	40	160	320	320	320	320	160	160	160	
A/Ibra/72211857/2022	T28X, G275D	2a.3a.1	2022-10-23	SIAT1	40	80	160	320	320	160	320	80	80	80	

< relates to the lowest dilution of antiserum used
ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23
SH 2023

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table H3-9. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2023-01-13

Viruses	HA1 substitutions additional to those defining subclades of 3C.2a1b.2a	Passage history Ferret number Genetic group	Collection date	Passage history	Haemagglutination inhibition titre										
					Post-infection ferret antisera										
					A/Camb 925256/20	A/Camb e0826360/20	A/Thuringen 10/22	A/Stockholm 9/21	A/Darwin 9/21	A/Norway 24873/21	A/Norway 24873/21	A/Poland 97/22	A/Slov 8720/2022	A/Catal NSVH-2067/22	
					SIAT F03/21	Egg F10/21	SIAT F36/22	SIAT F35/21	Egg F39/21	SIAT F10/22	Egg F11/22	SIAT F39/22	SIAT F24/22	SIAT F41/22	
					1a	1a	2b	2a	2a	2a.3	2a.3	2a.2	2a.1	2a.1b	
REFERENCE VIRUSES															
A/Cambodia/925256/2020		1a	2020-09-25	SIAT5	640	160	<40	160	320	<40	40	40	<40	<40	
A/Cambodia/e0826360/2020		1a	2020-07-16	ES/E2	80	640		160	160	40	80	40	160	<40	
A/Thuringen/10/2022	S262N	2b	2022-04-01	P1/SIAT2	80	160	320	320	160	80	80	80	160	40	
A/Stockholm/5/2021		2a	2021-04-16	SIAT0/SIAT3	80	80	80	640	640	80	160	320	320	80	
A/Darwin/9/2021	D186N, D225G (egg)	2a	2021-04-17	E3/E4	80	320	320	640	1280	160	320	320	640	160	
A/Norway/24873/2021		2a.3	2021-10-24	SIAT3	80	80	80	320	320	160	320	160	320	80	
A/Norway/24873/2021	D225G (egg)	2a.3	2021-10-24	E3/E1	80	160	160	640	640	320	640	320	640	160	
A/Poland/97/2022	P3S, I25V, S145N, R201K	2a.2	2022-05-09	SIAT2	160	160	80	1280	1280	320	640	1280	640	640	
A/Slovenia/87/2022		2a.1	2022-02-10	SIAT1	80	80	160	640	640	160	320	640	1280	640	
A/Catalonia/NSVH161512067/2022		2a.1b	2022-09-14	SIAT1/SIAT2	40	80	80	320	320	80	160	320	640	320	
TEST VIRUSES															
A/Toulon/NOMS22.1.29/2022		3C.2a1b.2a.2	2022-08-19	MDCK2/SIAT1	40	160	160	320	320	320	640	160	320	160	
A/Denmark/3110/2022	I242M	2b	2022-10-17	SIAT2/SIAT1	80	80	320	320	160	80	160	80	80	40	
A/Denmark/3100/2022	I242M	2b	2022-10-17	SIAT2/SIAT1	80	80	320	320	160	80	160	80	80	40	
A/Denmark/3101/2022	I242M	2b	2022-10-21	SIAT2/SIAT1	80	80	320	320	160	80	160	80	80	40	
A/Denmark/3136/2022	I242M	2b	2022-11-05	SIAT2/SIAT1	80	80	320	320	160	80	160	80	80	40	
A/Lyon/NRU22.2.97/2022	R33Q, S262N	2b	2022-10-24	MDCK2/SIAT1	80	80	320	320	160	80	160	80	160	40	
A/Denmark/3103/2022	R33Q, S262N	2b	2022-11-05	SIAT3/SIAT1	80	80	320	320	160	80	160	80	80	40	
A/Lyon/CHU/R22.415.69/2022	R33Q, S262N	2b	2022-11-05	MDCK2/SIAT1	80	80	320	320	160	80	160	80	160	40	
A/Aix/NOMS22.1.44/2022	R33Q, N81D, S262N	2b	2022-11-14	MDCK2/SIAT1	80	80	320	320	160	80	160	80	160	40	
A/Navarra/4439/2022	R33Q, F79I, S144G, S262N	2b	2022-11-13	SIAT2/SIAT1	80	80	320	320	160	40	80	80	160	40	
A/Navarra/4440/2022	R33Q, K238R, S262N	2b	2022-11-17	SIAT2/SIAT1	80	160	320	320	160	80	160	80	160	40	
A/Lyon/NRU22.4.39/2022	T135A(-cho), S262N	2b	2022-11-14	MDCK2/SIAT1	80	160	320	320	160	80	160	80	160	40	
A/Navarra/4438/2022	T135A(-cho), S262N	2b	2022-11-18	SIAT2/SIAT1	80	80	160	160	160	80	80	40	160	40	
A/Lyon/CHU/R22.355.53/2022		2a.1	2022-09-05	MDCK2/SIAT1	40	40	80	320	320	80	160	320	640	320	
A/Saint-Etienne/NOMS22.1.28/2022		2a.1	2022-09-20	MDCK2/SIAT1	40	40	80	320	640	80	160	640	640	320	
A/Mesles/NOMS22.1.23/2022		2a.1	2022-09-26	MDCK2/SIAT1	40	40	80	320	320	80	160	640	320	320	
A/Lyon/CHU/R22.372.37/2022	E50K, R299K	2a.1	2022-09-28	MDCK2/SIAT1	<40	40	40	160	320	80	160	320	640	320	
A/Lyon/CHU/R22.374.29/2022		2a.1b	2022-10-01	MDCK2/SIAT1	40	40	80	320	320	80	160	320	640	320	
A/Lyon/CHU/R22.384.1/2022		2a.1b	2022-10-10	MDCK2/SIAT1	40	80	80	320	320	80	160	320	640	320	
A/Denmark/3127/2022	A163X	2a.1b	2022-11-01	SIAT3/SIAT1	40	80	80	320	320	80	160	320	320	320	
A/Grenoble/NOMS22.1.42/2022		2a.3a	2022-10-10	MDCK2/SIAT1	80	80	160	320	320	320	640	160	640	160	
A/Grenoble/NOMS22.1.57/2022		2a.3a	2022-10-26	MDCK2/SIAT1	80	80	160	320	320	320	640	160	320	160	
A/Toulon/NOMS22.1.33/2022		2a.3a.1	2022-09-15	MDCK2/SIAT1	80	160	160	320	320	320	640	160	160	160	
A/La Reunion/NOMS22.1.67/2022		2a.3a.1	2022-09-28	MDCK2/SIAT1	80	80	160	320	320	320	640	160	320	160	
A/Denmark/3108/2022		2a.3a.1	2022-10-27	SIAT3/SIAT1	80	160	160	320	320	320	640	160	160	160	
A/Ashabat/1/2022		2a.3b	2022-11-29	MDCKx/SIAT1	80	80	160	320	320	320	640	320	320	160	
A/Ashabat/3/2022		2a.3b	2022-12-06	MDCKx/SIAT1	80	80	160	320	320	320	640	320	320	160	
A/Fort-De-France/53820/2022	P227S	2a.3b	2022-10-17	MDCK1/SIAT1	80	80	160	320	640	160	320	160	320	160	

< relates to the lowest dilution of antiserum used
ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23
SH 2023

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table H3-10. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2023-01-20

Viruses	HA1 substitutions additional to those defining subclades of 3C.2a1b.2a	Passage history Ferret number Genetic group	Collection date	Passage history	Haemagglutination inhibition titre										
					Post-infection ferret antisera										
					A/Camb 925256/20 SIAT F03/21	A/Camb e0826360/20 Egg F19/21	A/Thuringen 10/22 SIAT F36/22	A/Stockholm 5/21 SIAT F35/21	A/Darwin 9/21 Egg F39/21	A/Norway 24873/21 SIAT F10/22	A/Norway 24873/21 Egg F11/22	A/Poland 97/22 SIAT F39/22	A/Slov 8720/2022 SIAT F24/22	A/Catal NSVH-2067/22 SIAT F41/22	
					2a.1	2a.1	2a.2	2a.2	2a.2	2a.2	2a.2	2a.2	2a.2	2a.2	
REFERENCE VIRUSES															
A/Cambodia/925256/2020		1a	2020-09-25	SIAT5	1280	160	40	320	640	<40	80	80	<40	<40	
A/Cambodia/e0826360/2020		1a	2020-07-16	E5/E3	160	1280	80	160	320	40	80	40	160	<40	
A/Thuringen/10/2022	S262N	2b	2022-04-01	P1/SIAT2	80	320	320	640	320	160	160	160	320	80	
A/Stockholm/5/2021		2a	2021-04-16	SIAT0/SIAT3	80	160	80	640	640	160	320	320	640	160	
A/Darwin/9/2021	D186N, D225G (egg)	2a	2021-04-17	E3/E4	80	640	160	640	1280	160	320	320	640	160	
A/Norway/24873/2021		2a.3	2021-10-24	SIAT3	80	80	80	320	640	320	320	160	320	160	
A/Norway/24873/2021	D225G (egg)	2a.3	2021-10-24	E3/E1	80	320	160	640	640	320	640	1280	320	160	
A/Poland/97/2022	P3S, I25V, S145N, R201K	2a.2	2022-05-09	SIAT2	80	160	80	640	640	320	320	640	1280	320	
A/Slovenia/8720/2022		2a.1	2022-02-10	SIAT1/MDCK1/SIAT2	80	160	160	640	640	160	320	640	1280	640	
A/Catalonia/NSVH161512067/2022		2a.1b	2022-09-14	SIAT1/SIAT3	40	160	80	640	640	80	320	320	640	320	
TEST VIRUSES															
A/Dakar/27/2022		3C.2a1b.2a.2	2022-06-28	P1/SIAT1	40	160	160	320	640	160	640	160	320	160	
A/Argentina/3155/2022		3C.2a1b.2a.2	2022-08-31	SIAT2	40	160	80	160	320	160	320	160	160	80	
A/Argentina/3151/2022		2b	2022-09-05	SIAT1	80	320	320	320	320	160	160	160	320	80	
A/Switzerland/24157/2022		2b	2022-12-23	SIAT3/SIAT1	80	160	320	320	320	80	160	80	320	40	
A/Switzerland/18284/2022	R33Q, S262N	2b	2022-10-06	SIAT2/SIAT1	80	80	160	160	160	80	80	80	160	40	
A/Burkina Faso/257/2022	R33Q, S262N	2b	2022-10-13	SIAT1	80	160	160	160	160	80	80	80	160	40	
A/Serbia/213/2022	R33Q, S262N	2b	2022-10-13	SIAT1	80	160	320	320	320	80	160	160	320	40	
A/Salamanca/198/2022	R33Q, S262N	2b	2022-10-13	SIAT1	80	160	320	320	320	80	160	160	320	80	
A/Salamanca/196/2022	R33Q, S262N	2b	2022-10-13	SIAT1	80	160	320	320	320	80	320	80	320	80	
A/Switzerland/18694/2022	R33Q, S262N	2b	2022-11-18	MDCK1/SIAT1	80	160	320	320	320	80	160	80	320	40	
A/Polencia/118/2022	R33Q, N94S, S262N	2b	2022-10-13	SIAT1	80	160	320	320	320	80	160	80	320	40	
A/Switzerland/39985/2022	R33Q, N81D, A163X, S262N	2b	2022-11-21	MDCK1/SIAT1	80	80	320	320	320	80	80	80	320	40	
A/Switzerland/07426/2022	T135A(-cho), S262N	2b	2022-11-17	SIAT1/SIAT1	80	160	320	320	320	80	80	80	160	40	
A/Ireland/73304/2022	T135A(-cho), S262N	2b	2022-11-23	SIAT1	80	160	320	320	320	80	160	80	160	40	
A/Ireland/74535/2022	T128A(-cho), T135A(-cho), S262N	2b	2022-11-28	SIAT2	80	160	160	160	160	80	80	80	160	40	
A/Lithuania/48774/2022		2a.1	2022-11-24	SIAT1	40	40	80	320	640	160	160	320	640	320	
A/Burkina Faso/82/2022	T135A(-cho), T167S	2a.1	2022-10-13	SIAT1	40	160	40	320	320	80	80	320	640	320	
A/Hong Kong/34/2022	N45S(-cho), K276	2a.1	2022-11-26	MDCK1/SIAT1	80	80	80	640	640	160	320	640	640	320	
A/Switzerland/54601/2022		2a.1b	2022-11-01	MDCK1/SIAT1	<40	40	40	160	160	40	80	160	320	80	
A/Ireland/74250/2022		2a.1b	2022-11-27	SIAT2	40	80	80	320	320	80	160	320	640	320	
A/Ireland/74511/2022		2a.1b	2022-11-29	SIAT2	40	80	80	320	320	80	160	320	640	320	
A/Switzerland/76550/2022		2a.1b	2022-12-05	SIAT1/SIAT1	40	40	80	320	320	80	160	320	640	160	
A/Switzerland/01284/2022		2a.1b	2022-12-05	SIAT1/SIAT1	40	80	80	640	640	160	320	320	1280	320	
A/Switzerland/89053/2022		2a.1b	2022-12-06	SIAT1/SIAT1	40	80	80	320	320	80	160	320	640	320	
A/Switzerland/12082/2022	R92K	2a.1b	2022-12-08	SIAT1/SIAT1	<40	40	80	160	320	40	80	160	320	160	
A/Georgia/2369/2022	N290T	2a.1b	2022-11-24	SIAT1	40	160	80	320	640	80	160	320	640	320	
A/Georgia/8/2022	N290T	2a.1b	not known	SIAT1	40	160	80	640	640	160	320	640	640	320	
A/Georgia/4/2022	N290T	2a.1b	not known	SIAT1	40	80	160	640	640	80	320	640	1280	320	
A/Argentina/3486/2022	N94K, I223	2a.3a	2022-10-01	SIAT1	80	80	80	320	640	320	640	160	320	160	
A/Argentina/3498/2022	T135K(-cho), S145N, I223	2a.3a	2022-09-29	SIAT1	40	160	80	160	320	320	320	160	320	160	
A/Argentina/3485/2022	T135K(-cho), S145N, I223	2a.3a	2022-10-01	SIAT1	40	160	80	160	320	160	320	160	160	80	
A/Hong Kong/19/2022	L151	2a.3a.1	2022-11-09	MDCK1/SIAT1	80	80	160	640	640	320	640	160	640	160	
A/Hong Kong/22/2022	L151	2a.3a.1	2022-11-13	MDCK1/SIAT1	80	160	160	640	640	320	640	320	640	160	
A/Hong Kong/24/2022	L151	2a.3a.1	2022-11-15	MDCK1/SIAT1	80	320	320	320	1280	640	1280	320	640	320	
A/Hong Kong/28/2022	L151	2a.3a.1	2022-11-16	MDCK1/SIAT1	80	160	160	320	640	320	640	160	320	80	
A/Hong Kong/30/2022	L151	2a.3a.1	2022-11-20	MDCK1/SIAT1	40	80	160	160	320	160	320	80	160	80	
A/Hong Kong/38/2022	L151	2a.3a.1	2022-12-04	MDCK1/SIAT1	80	80	160	320	320	320	640	160	320	80	
A/Hong Kong/26/2022	L151, R222X	2a.3a.1	2022-11-16	MDCK1/SIAT1	80	160	160	320	640	320	640	160	320	160	
A/Switzerland/52849/2022	N91	2a.3a.1	2022-11-11	SIAT1/SIAT1	40	80	80	160	320	160	320	160	160	80	
A/Switzerland/86247/2022	N91	2a.3a.1	2022-11-14	MDCK1/SIAT1	40	80	160	320	320	160	320	80	160	80	
A/Switzerland/85876/2022	N91, V182X	2a.3a.1	2022-11-14	MDCK1/SIAT1	40	80	160	320	320	160	320	160	320	80	
A/Hong Kong/15/2022	seq pending	seq pending	2022-10-04	MDCK1/SIAT1	40	80	80	160	320	160	160	160	160	80	
A/Hong Kong/17/2022	seq pending	seq pending	2022-10-24	MDCK1/SIAT1	40	80	80	80	160	80	160	80	80	80	
A/Hong Kong/32/2022	seq pending	seq pending	2022-11-25	MDCK1/SIAT1	40	80	160	320	320	160	320	160	320	80	
< relates to the lowest dilution of antiserum used ND = Not Done					Vaccine NH 2021-22			Vaccine SH 2022 NH 2022-23 SH 2023							

< relates to the lowest dilution of antiserum used
ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23
SH 2023

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table H3-11a. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2023-01-27

Viruses	HA1 substitutions additional to those defining subclades of 3C.2a1b.2a	Passage history Ferret number Genetic group	Collection date	Passage history	Haemagglutination inhibition titre													
					Post-infection ferret antisera													
					A/Camb 925256/20 SIAT F03/21	A/Camb e0826360/20 Egg F10/21	A/Thuringen 10/22 SIAT F36/22	A/Stock 9/21 SIAT F35/21	A/Darwin 9/21 Egg F39/21	A/Norway 24873/21 SIAT F10/22	A/Norway 24873/21 Egg F11/22	A/Poland 97/22 SIAT F39/22	A/Slov 8720/2022 SIAT F24/22	A/Catal NSVH-2067/22 SIAT F41/22				
					1a	1a	2b	2a	2a	2a.3	2a.3	2a.2	2a.1	2a.1b				
					REFERENCE VIRUSES													
A/Cambodia/925256/2020		1a	2020-09-25	SIAT5	640	160	<40	160	320	40	40	80	<40	<40				
A/Cambodia/e0826360/2020		1a	2020-07-16	E5/E3	80	160	80	160	160	40	80	40	160	<40				
A/Thuringen/10/2022	S262N	2b	2022-04-01	P1/SIAT2	80	160	320	320	320	80	160	80	320	40				
A/Stockholm/5/2021		2a	2021-04-16	SIAT0/SIAT3	80	80	80	640	640	160	160	320	640	80				
A/Darwin/9/2021	D186N, D225G (egg)	2a	2021-04-17	E3/E4	160	640	160	640	1280	160	320	640	1280	160				
A/Norway/24873/2021		2a.3	2021-10-24	SIAT3	40	80	80	320	320	320	320	160	160	80				
A/Norway/24873/2021	D225G (egg)	2a.3	2021-10-24	E3/E1	80	320	160	320	640	320	640	320	640	160				
A/Poland/97/2022	P3S, I25V, S145N, R201K	2a.2	2022-05-09	SIAT2	40	80	80	320	640	160	320	640	640	320				
A/Slovenia/8720/2022		2a.1	2022-02-10	SIAT1/MDCK1/SIAT2	80	80	160	640	640	160	320	640	1280	640				
A/Catalonia/NSVH161512067/2022		2a.1b	2022-09-14	SIAT1/SIAT3	40	80	80	640	320	80	160	320	640	320				
TEST VIRUSES																		
A/Croatia/101024/2022	I242M	2b	2022-09-01	MDCKx/SIAT1	80	80	320	320	160	80	160	80	160	40				
A/Norway/32641/2022	I224M	2b	2022-10-17	SIAT1	40	80	320	320	160	80	80	80	160	40				
A/Castilla La Mancha/4112/2022	R33Q, S262N	2b	2022-10-15	SIAT1	80	160	320	320	320	40	80	80	160	40				
A/Castilla La Mancha/4057/2022	R33Q, S262N	2b	2022-10-17	SIAT1	80	80	320	160	160	40	80	40	160	40				
A/Castilla La Mancha/4107/2022	R33Q, S262N	2b	2022-10-17	SIAT2	40	80	320	160	160	40	80	80	160	40				
A/Norway/33623/2022	R33Q, S262N	2b	2022-10-31	SIAT1	80	160	320	320	320	80	160	80	320	40				
A/Norway/33536/2022	R33Q, S262N	2b	2022-11-01	SIAT1	40	80	160	160	160	40	80	40	160	<40				
A/Castilla La Mancha/4542/2022	R33Q, S262N	2b	2022-11-24	SIAT1	40	80	160	160	160	80	40	40	80	<40				
A/Castilla La Mancha/4541/2022	R33Q, S262N	2b	2022-11-24	SIAT1	40	80	160	160	160	80	80	80	160	40				
A/Castilla La Mancha/4547/2022	R33Q, S262N	2b	2022-11-26	SIAT1	80	160	320	320	160	80	80	80	160	40				
A/Switzerland/68410/2022	R33Q, S262N	2b	2022-12-13	SIAT1	40	80	160	160	160	40	80	40	160	40				
A/Switzerland/68875/2022	R33Q, S262N	2b	2022-12-13	SIAT1	40	80	160	160	160	80	80	80	160	40				
A/Switzerland/68871/2022	R33Q, N96K, S262N	2b	2022-12-13	SIAT1	40	160	160	160	160	80	80	80	160	40				
A/Galicia/3991/2022	R33Q, S199P, S262N	2b	2022-10-16	SIAT1	40	80	160	160	160	40	80	40	160	<40				
A/Hungary/73/2022	R33Q, S262N, A272T	2b	2022-11-28	MDCK1/SIAT1	80	80	160	160	160	80	80	40	160	40				
A/Hungary/87/2022	R33Q, E41X, D101E, S262N, D291N	2b	2022-12-07	MDCK1/SIAT1	40	80	160	160	160	80	80	40	160	40				
A/Switzerland/12279/2022	T135A(-cho), S262N	2b	2022-12-09	SIAT1	80	160	160	160	160	80	160	80	160	40				
A/Switzerland/26719/2022	T135A(-cho), S262N	2b	2022-12-19	SIAT1	80	160	320	160	160	80	160	80	160	40				
A/Switzerland/68854/2022	T135A(-cho), S262N	2b	2022-12-23	SIAT1	40	80	160	160	160	80	80	40	80	40				
A/Slovenia/11512/2022	T135A(-cho), S144N(+cho), S262N	2b	2022-12-19	MDCKx/SIAT1	40	160	160	160	160	80	80	80	160	40				
A/Slovenia/11696/2022	T135A(-cho), S144N(+cho), S262N	2b	2022-12-29	SIATx/SIAT1	40	160	160	320	160	80	80	80	160	<40				
A/Slovenia/11516/2022	T10K(-cho), T135A(-cho), S262N	2b	2022-12-19	SIATx/SIAT1	40	160	160	160	160	80	80	80	160	<40				
A/Slovenia/11514/2022	T10K(-cho), T135A(-cho), S262N	2b	2022-12-19	SIATx/SIAT1	80	160	160	320	160	80	160	80	160	40				
A/Slovenia/11622/2022	T10K(-cho), T135A(-cho), S262N	2b	2022-12-27	MDCKx/SIAT1	80	80	160	160	160	80	80	40	160	<40				
A/Slovenia/11517/2022	T10K(-cho), V79I, T135A(-cho), S262N	2b	2022-12-20	MDCKx/SIAT1	80	160	160	160	160	80	80	80	160	40				
A/Slovenia/42/2023	T10K(-cho), V79I, T135A(-cho), S262N	2b	2023-01-03	MDCKx/SIAT1	80	160	160	320	320	80	160	80	160	<40				
A/Slovenia/11665/2022	T10K(-cho), V79I, T135A(-cho), Q210*, S262N	2b	2022-12-28	SIATx/SIAT1	160	320	320	640	320	80	160	80	320	40				

< relates to the lowest dilution of antiserum used

ND = Not Done

Vaccine

NH 2021-22

Vaccine

SH 2022

NH 2022-23

SH 2023

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table H3-11b. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2023-01-27

Viruses	HA1 substitutions additional to those defining subclades of 3C.2a1b.2a	Passage history Ferret number Genetic group	Collection date	Passage history	Haemagglutination inhibition titre										
					Post-infection ferret antisera										
					A/Camb 925256/20 SIAT F03/21 1a	A/Camb e0826360/20 Egg F10/21 1a	A/Thuringen 10/22 SIAT F36/22 2b	A/Stock 9/21 SIAT F35/21 2a	A/Darwin 9/21 Egg F39/21 2a	A/Norway 24873/21 SIAT F10/22 2a.3	A/Norway 24873/21 Egg F11/22 2a.3	A/Poland 97/22 SIAT F39/22 2a.2	A/Slov 8720/2022 SIAT F24/22 2a.1	A/Catal NSVH-2067/22 SIAT F41/22 2a.1b	
REFERENCE VIRUSES															
A/Cambodia/925256/2020		1a	2020-09-25	SIAT5	640	160	<40	160	320	40	40	80	<40	<40	
A/Cambodia/e0826360/2020		1a	2020-07-16	E5/E3	80	640	80	160	160	40	80	40	160	<40	
A/Thuringen/10/2021	S262N	2b	2022-04-01	P1/SIAT2	80	160	320	320	320	80	160	80	320	40	
A/Stockholm/5/2021		2a	2021-04-16	SIAT0/SIAT3	80	80	80	640	640	160	160	320	640	80	
A/Darwin/9/2021	D186N, D225G (egg)	2a	2021-04-17	E3/E4	160	640	160	640	1280	160	320	640	1280	160	
A/Norway/24873/2021		2a.3	2021-10-24	SIAT3	40	80	80	320	320	320	320	160	160	80	
A/Norway/24873/2021	D225G (egg)	2a.3	2021-10-24	E3/E1	80	320	160	320	640	320	640	320	640	160	
A/Poland/97/2022	P3S, I25V, S145N, R201K	2a.2	2022-05-09	SIAT2	40	80	80	320	640	160	320	640	640	320	
A/Slovenia/8720/2022		2a.1	2022-02-10	SIAT1/MDCK1/SIAT2	80	80	160	640	640	160	320	640	1280	640	
A/Catalonia/NSVH161512067/2022		2a.1b	2022-09-14	SIAT1/SIAT3	40	80	80	640	320	80	160	320	640	320	
TEST VIRUSES															
A/Norway/33317/2022		2a.1	2022-10-24	SIAT1	40	80	80	320	640	80	160	320	640	320	
A/Norway/33060/2022		2a.1	2022-10-27	SIAT1	40	40	80	320	320	80	160	320	640	160	
A/Norway/33371/2022	S144I	2a.1	2022-10-27	SIAT1	40	40	40	320	320	80	80	320	640	320	
A/Norway/33404/2022	E50K, N94H, R299K	2a.1	2022-11-05	SIAT1	40	80	80	320	640	80	160	320	1280	320	
A/Galicia/3994/2022		2a.1b	2022-10-17	SIAT1	<40	40	80	160	160	40	80	160	320	160	
A/Norway/33402/2022		2a.1b	2022-10-31	SIAT1	<40	40	40	80	160	40	80	160	320	160	
A/Hungary/69/2022		2a.1b	2022-11-11	MDCK1/SIAT1	40	160	80	320	640	80	160	320	640	320	
A/Hungary/85/2022		2a.1b	2022-12-02	MDCK1/SIAT1	40	80	80	160	320	80	160	320	640	320	
A/Switzerland/28394/2022		2a.1b	2022-12-19	SIAT1	40	80	40	160	160	40	80	160	320	160	
A/Ukraine/775/2022	T10M (-cho)	2a.1b	2022-12-21	SIAT1	40	80	80	160	320	80	80	320	640	640	
A/Ukraine/773/2022	T10M (-cho)	2a.1b	2022-12-21	SIAT1	<40	40	40	160	160	40	80	160	320	160	
A/Ukraine/779/2022	T10M (-cho)	2a.1b	2022-12-22	SIAT1	40	80	80	160	320	80	80	320	320	320	
A/Ukraine/804/2022	T10M (-cho)	2a.1b	2022-12-26	SIAT1	<40	80	80	160	160	80	80	160	320	160	
A/Ukraine/774/2022	G53S	2a.1b	2022-12-21	SIAT1	<40	40	40	160	160	80	80	160	320	160	
A/Slovenia/11481/2022	N133D(-cho)	2a.1b	2022-12-19	SIATx/SIAT1	40	40	80	160	320	80	80	320	640	320	
A/Hungary/68/2022	N165X(-cho)	2a.1b	2022-11-15	MDCK1/SIAT1	40	160	80	160	320	80	160	320	640	160	
A/Croatia/111538/2022	S198T	2a.1b	2022-10-11	MDCKx/SIAT1	40	80	80	160	320	80	160	320	320	320	
A/Hungary/75/2022	N290S	2a.1b	2022-11-28	MDCK1/SIAT1	40	40	80	160	160	80	80	160	640	160	
A/Tunisia/23298/2022		2a.3a	2022-12-14	SIAT1	80	160	160	320	640	320	1280	320	640	80	
A/Cameroon/9645/2022	N94S(-cho), I223	2a.3a	2022-11-14	SIAT1	<40	40	80	80	160	160	320	80	160	40	
A/Norway/32883/2022		2a.3a.1	2022-10-27	SIAT1	40	160	160	320	320	320	640	160	320	80	
A/Switzerland/68546/2022		2a.3a.1	2022-12-13	SIAT1	40	80	80	160	160	160	320	80	160	80	
A/Switzerland/28577/2022		2a.3a.1	2022-12-19	SIAT1	<40	80	80	160	160	160	320	80	160	80	
A/Tunisia/23669/2022	P4T	2a.3a.1	2022-12-13	SIAT1	<40	160	80	160	320	160	640	80	160	80	
A/Norway/32603/2022	V309X	2a.3a.1	2022-10-18	SIAT1	40	80	80	160	160	320	320	80	160	80	
A/Norway/33475/2022	N63K(-cho), K238R	2a.3a.1	2022-10-27	SIAT1	40	80	80	160	160	160	320	80	160	40	
A/Switzerland/68912/2022	E50, I223	2a.3b	2022-12-14	SIAT1	40	80	80	160	320	160	640	160	320	80	
A/Norway/32026/2022	S9R, E50, N159K, I223	2a.3b	2022-09-29	SIAT1	<40	80	80	160	320	160	160	80	160	80	
A/Ireland/74222/2022		seq pending	2022-11-27	SIAT1	<40	80	80	160	320	80	80	160	320	320	
A/Hungary/88/2022		seq pending	2022-12-08	MDCK1/SIAT1	40	80	160	160	160	40	80	40	80	<40	
< relates to the lowest dilution of antiserum used ND = Not Done					Vaccine NH 2021-22			Vaccine SH 2022 NH 2022-23 SH 2023							

< relates to the lowest dilution of antiserum used
ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23
SH 2023

< 4-fold
 4-fold
 8-fold
 > 8-fold
 not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table H3-12a. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2023-02-02

Haemagglutination inhibition titre														
Viruses	HA1 substitutions additional to those defining subclades of 3C.2a1b.2a	Passage history Ferret number Genetic group	Collection date	Passage history	Post-infection ferret antisera									
					A/Camb 92S256/20	A/Camb e0826360/20	A/Thuringen 10/22	A/Stockholm 5/21	A/Darwin 9/21	A/Norway 24873/21	A/Norway 24873/21	A/Poland 97/22	A/Slov 8720/2022	A/Catal NSVH-2067/22
					SIAT F03/21	Egg F10/21	SIAT F36/22	SIAT F35/21	Egg F39/21	SIAT F10/22	Egg F11/22	SIAT F39/22	SIAT F24/22	SIAT F41/22
					1a	1a	2b	2a	2a	2a.3	2a.3	2a.2	2a.1	2a.1b
REFERENCE VIRUSES														
A/Cambodia/92S256/2020		1a	2020-09-25	SIAT5	640	80	<40	160	160	<40	<40	40	<40	<40
A/Cambodia/e0826360/2020		1a	2020-07-16	E5/E3	80	1280	40	160	320	40	80	40	160	<40
A/Thuringen/10/2022	S262N	2b	2022-04-01	P1/SIAT2	40	80	320	320	160	80	80	80	160	40
A/Stockholm/5/2021		2a	2021-04-16	SIAT9/SIAT3	40	80	80	640	640	80	160	320	320	80
A/Darwin/9/2021	D186N, D225G (egg)	2a	2021-04-17	E3/E4	80	320	160	320	640	160	320	320	640	160
A/Norway/24873/2021		2a.3	2021-10-24	SIAT3	40	40	80	160	320	320	320	80	160	80
A/Norway/24873/2021	D225G (egg)	2a.3	2021-10-24	E3/E1	80	320	160	320	640	320	640	320	640	160
A/Poland/97/2022	P3S, I25V, S145N, R201K	2a.2	2022-05-09	SIAT2	40	80	80	160	320	80	160	640	640	160
A/Slovenia/8720/2022		2a.1	2022-02-10	SIAT1/MDCK1/SIAT2	40	80	80	320	640	80	320	640	1280	640
A/Catalonia/NSVH161512067/2022		2a.1b	2022-09-14	SIAT1/SIAT3	40	40	40	320	320	80	160	320	640	320
TEST VIRUSES														
A/Norway/32842/2022		2b	2022-10-23	SIAT2	40	40	160	160	160	40	80	40	80	40
A/Israel/B1644/2022	T12M	2b	2022-12-18	P1/SIAT1	40	40	160	160	80	40	80	40	80	<40
A/Georgia/RUS200/2022	I242M	2b	2022-12-12	SIAT1	40	40	160	160	80	40	40	40	80	40
A/Romania/544194/2022	I242M	2b	2022-12-28	SIAT1/SIAT1	40	80	160	160	80	40	80	40	80	<40
A/Romania/544283/2023	I242M	2b	2023-01-02	SIAT1/SIAT1	<40	40	160	160	80	40	40	40	80	<40
A/Romania/544760/2023	I242M	2b	2023-01-10	SIAT1/SIAT1	<40	40	160	80	80	<40	40	40	80	<40
A/Romania/544753/2023	I242M	2b	2023-01-10	SIAT1/SIAT1	40	40	160	160	80	40	40	40	80	<40
A/Israel/R11447/2022	T24X, S124G(-cho), I242M	2b	2022-11-25	P1/SIAT1	40	40	160	160	160	40	40	80	80	40
A/Netherlands/01867/2022	K50I, D186G, I242M	2b	2022-12-21	Hck1/SIAT1	<40	40	80	80	80	<40	40	40	40	<40
A/Milano/331/2022	I25V, I242M	2b	2022-11-29	SIAT2/SIAT1	40	40	160	160	160	40	80	80	80	<40
A/Netherlands/01476/2022	K50I, I242M	2b	2022-12-07	Hck1/SIAT1	40	40	160	160	80	40	80	40	80	<40
A/Netherlands/01475/2022	K50I, I242M	2b	2022-12-07	Hck1/SIAT1	40	40	80	160	80	40	40	40	80	<40
A/Netherlands/01218/2022	R33Q, S262N	2b	2022-09-12	Hck1/SIAT1	<40	40	160	160	80	40	40	40	80	<40
A/Catalonia/NSVH101923535/2022	R33Q, S262N	2b	2022-10-03	SIAT1/SIAT1	40	40	160	160	80	40	40	40	80	<40
A/Catalonia/NSVH161512277/2022	R33Q, S262N	2b	2022-10-04	SIAT1/SIAT1	40	40	160	80	80	40	40	40	80	<40
A/Catalonia/NSVH15119499/2022	R33Q, S262N	2b	2022-10-04	SIAT1/SIAT1	<40	40	160	80	80	40	40	40	80	<40
A/Catalonia/NSVH101934050/2022	R33Q, S262N	2b	2022-10-19	SIAT1/SIAT1	<40	40	160	80	80	<40	40	40	80	<40
A/Navarra/4550/2022	R33Q, S262N	2b	2022-11-25	SIAT2	40	80	320	160	160	40	80	40	80	40
A/England/106/2022	R33Q, S262N	2b	2022-10-24	SIAT1/SIAT1	40	80	160	160	160	40	80	40	160	<40
A/Baden-Wuerttemberg/88/2022	R33Q, S262N	2b	2022-11-29	P2/SIAT1	40	80	160	160	80	40	80	40	80	<40
A/Lisboa/717/2022	R33Q, S262N	2b	2022-11-29	SIAT1/SIAT1	<40	80	160	<40	160	40	80	40	160	40
A/Lisboa/711/2022	R33Q, S262N	2b	2022-11-29	SIAT1/SIAT1	40	160	320	<40	160	80	160	80	160	40
A/Lisboa/723/2022	R33Q, S262N	2b	2022-11-30	SIAT1/SIAT1	<40	80	160	<40	80	40	80	40	80	<40
A/Lisboa/716/2022	R33Q, S262N	2b	2022-12-04	SIAT1/SIAT1	40	80	160	160	80	40	80	40	160	<40
A/Lisboa/714/2022	K27X, R33Q, S262N	2b	2022-11-29	SIAT1/SIAT1	<40	80	160	<40	80	40	80	40	80	<40
A/Belgium/G0010/2023	R33Q, D101E, S262N	2b	2022-12-19	C1/SIAT1	<40	40	80	160	80	40	40	40	80	<40
A/Belgium/S2699/2022	R33Q, D101E, S262N, P273S	2b	2022-12-18	C1/SIAT1	40	80	160	160	80	40	40	40	80	<40
A/Belgium/G0401/2022	T135A(-cho), S262N	2b	2022-12-19	C1/SIAT1	40	40	80	80	160	<40	<40	40	40	40
A/Romania/544197/2022	T135A(-cho), S262N	2b	2022-12-28	SIAT1/SIAT1	40	80	80	160	80	40	40	40	80	<40
A/Milano/397/2022	T135A(-cho), N190X, S262N	2b	2022-11-30	SIAT2/SIAT1	40	40	80	80	80	40	40	40	80	<40
A/Israel/B1398/2022	I25M, T135A(-cho), S262N	2b	2022-11-28	P1/SIAT1	40	80	80	160	80	40	40	40	80	<40
< relates to the lowest dilution of antiserum used ND = Not Done					Vaccine NH 2021-22			Vaccine SH 2022 NH 2022-23 SH 2023						

< relates to the lowest dilution of antiserum used
ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23
SH 2023

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table H3-12b. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2023-02-02

Viruses	HA1 substitutions additional to those defining subclades of 3C.2a1b.2a	Passage history Ferret number Genetic group	Collection date	Passage history	Haemagglutination inhibition titre											
					Post-infection ferret antisera											
					A/Camb 92S256/20 SIAT F03/21	A/Camb e0826360/20 Egg F10/21	A/Thuringen 10/22 SIAT F36/22	A/Stockholm 5/21 SIAT F35/21	A/Darwin 9/21 Egg F39/21	A/Norway 24873/21 SIAT F10/22	A/Norway 24873/21 Egg F11/22	A/Poland 97/22 SIAT F39/22	A/Slov 8720/2022 SIAT F24/22	A/Catal NSVH-2067/22 SIAT F41/22		
					1a	1a	2b	2a	2a	2a.3	2a.3	2a.2	2a.1	2a.1b		
					REFERENCE VIRUSES											
A/Cambodia/92S256/2020		1a	2020-09-25	SIAT5	640	80	<40	160	160	<40	<40	40	<40	<40		
A/Cambodia/e0826360/2020		1a	2020-07-16	ES/E3	80	1280	40	160	320	40	80	40	160	<40		
A/Thuringen/10/2022	S262N	2b	2022-04-01	P1/SIAT2	40	80	320	320	160	80	80	80	160	40		
A/Stockholm/5/2021		2a	2021-04-16	SIAT0/SIAT3	40	80	80	640	640	80	160	320	320	80		
A/Darwin/9/2021	D186N, D225G (egg)	2a	2021-04-17	E3/E4	80	320	160	320	640	160	320	320	640	160		
A/Norway/24873/2021		2a.3	2021-10-24	SIAT3	40	40	80	160	320	320	320	80	160	80		
A/Norway/24873/2021	D225G (egg)	2a.3	2021-10-24	E3/E1	80	320	160	320	640	320	640	320	640	160		
A/Poland/97/2022		2a.2	2022-05-09	SIAT2	40	80	80	160	320	80	160	640	640	160		
A/Slovenia/8720/2022	P3S, I25V, S145N, R201K	2a.1	2022-02-10	SIAT1/MDCK1/SIAT2	40	80	80	320	640	80	320	640	1280	640		
A/Catalonia/NSVH161512067/2022		2a.1b	2022-09-14	SIAT1/SIAT3	40	40	40	320	320	80	160	320	640	320		
TEST VIRUSES																
A/England/151/2022		2a.1	2022-09-14	MDCK1/SIAT1	<40	40	40	80	160	80	80	160	320	320		
A/Netherlands/01232/2022	E50K, R299K	2a.1	2022-10-05	Hck1/SIAT1	<40	40	40	160	320	80	160	320	640	160		
A/Norway/32860/2022		2a.1b	2022-10-21	SIAT2	<40	40	40	80	160	40	80	80	320	160		
A/England/104/2022		2a.1b	2022-10-24	SIAT1/SIAT1	<40	40	40	80	160	40	80	80	160	80		
A/Catalonia/NSVH151029260/2022		2a.1b	2022-11-08	SIAT1/SIAT1	<40	40	40	80	160	40	80	160	320	160		
A/Rheinland-Pfalz/35/2022		2a.1b	2022-12-01	P1/SIAT1	<40	40	40	80	320	40	160	160	320	320		
A/Thuringen/63/2022		2a.1b	2022-12-05	P1/SIAT1	<40	<40	40	80	80	40	80	160	320	80		
A/Romania/544311/2022		2a.1b	2022-12-30	SIAT1/SIAT1	<40	40	40	80	160	40	80	80	160	160		
A/Belgium/G0007/2023	V182A	2a.1b	2022-12-19	C1/SIAT1	<40	40	40	80	160	40	80	80	320	80		
A/Belgium/G0405/2022	V182A	2a.1b	2022-12-19	C1/SIAT1	<40	40	40	80	160	40	80	80	320	160		
A/Croatia/109105/2022	S198T	2a.1b	2022-10-03	MDCKx/SIAT2	<40	40	40	160	160	40	80	160	320	160		
A/Georgia/T2/2022	N290T	2a.1b	2022-12-05	SIAT1	<40	40	40	80	80	40	40	160	160	80		
A/Georgia/T11-4/2022	N290T	2a.1b	2022-12-12	SIAT1	<40	40	40	80	80	40	40	160	320	160		
A/Georgia/T27-25/2022	N31S, N290T	2a.1b	2022-12-05	SIAT2	<40	40	80	160	320	40	80	160	320	320		
A/Georgia/k-1153/2022	R33X, N290T	2a.1b	2022-12-19	SIAT1	<40	40	40	80	160	80	160	640	640	640		
A/Georgia/K18-57/2022	G53D, N290T	2a.1b	2022-12-12	SIAT1	40	40	40	80	160	40	40	80	80	40		
A/Georgia/K20-14/2022	P293L	2a.1b	2022-12-12	SIAT1	40	40	40	80	80	40	40	80	160	80		
A/Netherlands/01248/2022		2a.3a	2022-10-14	Hck1/SIAT1	40	80	80	160	160	320	80	160	80	80		
A/Cameroon/10483/2022		2a.3a	2022-12-08	SIAT2	40	80	80	80	160	160	80	80	80	80		
A/England/110/2022		2a.3a.1	2022-10-05	SIAT1/SIAT1	<40	80	80	80	160	160	160	80	80	80		
A/England/103/2022		2a.3a.1	2022-10-06	SIAT1/SIAT1	<40	80	80	80	160	320	80	160	160	40		
A/Israel/B1418/2022		2a.3a.1	2022-12-05	P1/SIAT1	<40	40	80	80	160	160	160	40	80	40		
A/Israel/B1503/2022		2a.3a.1	2022-12-13	P1/SIAT1	40	40	40	40	160	80	160	40	80	40		
A/Bremen/50/2022	S91N, N122D(-cho)	2a.3a.1	2022-11-24	P1/SIAT1	<40	40	40	40	80	80	160	40	80	40		
A/Niedersachsen/105/2022	S91N, N122D(-cho)	2a.3a.1	2022-11-29	P1/SIAT1	<40	80	80	80	160	160	320	80	160	40		
A/Mecklenburg-Vorpommern/88/2022	S91N, N122D(-cho)	2a.3a.1	2022-12-01	P1/SIAT1	<40	40	80	80	160	80	320	40	160	40		
A/Belgium/G0055/2023	S91N, N122D(-cho)	2a.3a.1	2022-12-21	C1/SIAT1	<40	40	80	80	160	80	160	40	80	40		
A/England/105/2022		2a.3b	2022-10-21	SIAT1/SIAT1	<40	40	40	80	160	80	160	80	160	40		
A/Israel/R599/2023		2a.3b	2023-01-08	P1/SIAT1	<40	40	40	80	160	40	160	40	80	<40		
< relates to the lowest dilution of antiserum used ND = Not Done					Vaccine NH 2021-22				Vaccine SH 2022 NH 2022-23 SH 2023							

< relates to the lowest dilution of antiserum used
ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23
SH 2023

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table H3-13. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2023-02-10

Haemagglutination inhibition titre																
Viruses	HA1 substitutions additional to those defining subclades of 3C.2a1b.2a	Passage history Ferret number Genetic group	Collection date	Passage history	Post-infection ferret antisera											
					NEW											
					A/Camb 92S256/20	A/Camb e0826360/20	A/Thuringen 10/22	A/Stockholm 5/21	A/Stockholm 5/21	A/Darwin 9/21	A/Norway 24873/21	A/Norway 24873/21	A/Poland 97/22	A/Slov 8720/2022	A/Lille 50053/2022	A/Catal NSVH-2067/22
					SIAT	Egg	SIAT	SIAT	SIAT	Egg	SIAT	Egg	SIAT	SIAT	SIAT	SIAT
					F03/21 1a	F10/21 1a	F36/22 2b	F35/21 2a	F01/23 2a	F39/21 2a	F10/22 2a.3	F11/22 2a.3	F39/22 2a.2	F24/22 2a.1	F02/23 2a.1	F41/22 2a.1b
REFERENCE VIRUSES																
A/Cambodia/92S256/2020		1a	2020-09-25	SIAT5	1280	80	<40	160	160	160	<40	<40	40	<40	80	<40
A/Cambodia/e0826360/2020		1a	2020-07-16	E5/E3	80	1280	40	160	80	320	40	80	40	160	160	<40
A/Thuringen/10/2022	S262N	2b	2022-04-01	P1/SIAT2	40	80	320	320	80	160	80	80	80	160	160	40
A/Stockholm/5/2021		2a	2021-04-16	SIAT0/SIAT3	40	80	80	640	640	80	160	320	320	320	320	80
A/Darwin/9/2021	D186N, D225G (egg)	2a	2021-04-17	E3/E4	80	320	160	320	320	640	160	320	320	640	320	160
A/Norway/24873/2021		2a.3	2021-10-24	SIAT3	40	40	80	160	160	320	320	320	80	160	160	80
A/Norway/24873/2021	D225G (egg)	2a.3	2021-10-24	E3/E1	80	320	160	320	640	320	1280	320	640	640	160	160
A/Poland/97/2022	P3S, I25V, S145N, R201K	2a.2	2022-05-09	SIAT2	40	80	80	160	160	320	80	160	1280	640	1280	160
A/Slovenia/8720/2022		2a.1	2022-02-10	SIAT1/MDCK1/SIAT2	40	80	80	320	640	80	320	640	640	640	640	640
A/Lille/50053/2022	T135A(-cho), T167S, N190S	2a.1	2022-09-06	MDCK1/SIAT3	ND	ND	ND	ND	320	ND	ND	ND	ND	2560	ND	ND
A/Catalonia/NSVH161512067/2022		2a.1b	2022-09-14	SIAT1/SIAT3	40	40	40	320	320	320	80	160	320	640	640	320
TEST VIRUSES																
A/Bolzano/46/2022	T135A(-cho), S262N	2b	2022-12-05	SIAT2/SIAT2	40	80	160	160	40	160	40	80	40	80	320	<40
A/Palermo/6/2022	T135A(-cho), S262N	2b	2022-12-06	SIAT3/SIAT2	40	80	160	160	40	80	40	40	40	80	320	<40
A/Netherlands/11980/2022		2a.1b	2022-12-22	MDCK-MIX2/SIAT1	<40	40	40	80	160	160	40	80	160	320	320	160
A/Netherlands/11968/2022	A272V	2a.1b	2022-12-20	MDCK-MIX2/SIAT1	<40	80	80	160	160	160	40	80	160	320	320	160
< relates to the lowest dilution of antiserum used ND = Not Done					Vaccine NH 2021-22					Vaccine SH 2022 NH 2022-23 SH 2023						
< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)																

Microneutralisation tables: H3N2

Table MN-1. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2022-12-19

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre Post-infection ferret antisera				HA1 substitutions for 3C.2a/3a viruses compared to A/Bangladesh/4005/2020 (3c.2a1b.2a.2 = 2)
				A/Cambodia 925256/2020	A/Thuringen 10/2022	A/Poland 97/2022	A/Slovenia 8720/2022	
				SIAT F03/21 1a Read	SIAT F36/22 2b Read	SIAT F39/22 2a.2 Read	SIAT F24/22 2a.1 Read	
REFERENCE VIRUSES								
A/Cambodia/925256/2020	1a	2020-09-25	SIAT3 10 ⁻³	287	<	<	<	N159Y, I160T(+cho), Q164L, K171N, D186S, N190D, S198P
A/Thuringen/10/2022	2b	2022-04-01	P1/SIAT2 10 ⁻³	<	254	61	120	E50K, E79V, I140K, S262N
A/Poland/97/2022	2a.2	2022-05-09	SIAT3 10 ⁻⁵	<	206	490	579	P4S, I25V, D53G, S145N, H156S, R201K, S219Y
A/Slovenia/8720/2022	2a.1	2022-02-10	SIAT1/MDCK1/SIAT3 10 ⁻⁵	<	43	124	243	D53G, D104G, H156S, K276R
TEST VIRUSES								
A/Rheinland-Pfalz/16/2022	2b	2022-10-04	P1/SIAT2	<	76	<	48	E50K, F79V, I140K, S262N
A/Niedersachsen/49/2022	2b	2022-09-26	P1/SIAT1	<	60	42	73	E50K, F79V, I140K, I242M
A/Hessen/46/2022	2b	2022-10-17	P1/SIAT1	<	91	<	59	R33Q, E50K, F79V, I140K, S262N
A/Baden-Wuerttemberg/67/2022	2b	2022-10-17	P1/SIAT1	<	106	<	67	R33Q, E50K, F79V, I140K, S262N
A/Hessen/40/2022	2b	2022-09-29	P1/SIAT1	<	123	<	69	R33Q, E50K, F79V, D101E, I140K, S262N
A/Hessen/45/2022	2b	2022-10-13	P1/SIAT1	<	51	<	46	E50K, F79V, T135A(-cho), I140K, S262N
A/Thuringen/30/2022	2b	2022-10-17	P1/SIAT1	<	82	<	59	E50K, F79V, T135A(-cho), I140K, S262N
A/Nordrhein-Westfalen/40/2022	2b	2022-10-13	P1/SIAT1	43	97	47	66	E50K, D77X, F79V, T135A(-cho), I140K, S262N
A/Hessen/41/2022	2a.1	2022-10-06	P1/SIAT1	<	<	134	294	D53G, D104G, H156S, K276R
A/Bremen/27/2022	2a.1	2022-10-04	P1/SIAT1	<	<	121	238	D53G, D104G, H156S, K276R, S312N
A/Bremen/30/2022	2a.1	2022-10-12	P1/SIAT1	<	<	127	278	D53G, D104G, H156S, K276R, S312N
A/Baden-Wuerttemberg/66/2022	2a.1b	2022-10-10	P1/SIAT1	<	41	122	244	D53G, D104G, I140K, H156S, K276R, R299K
A/Thuringen/28/2022	2a.1b	2022-10-17	P1/SIAT1	<	<	60	117	D53G, D104G, I140K, H156S, K276R, R299K
A/Berlin/145/2022	2a.3a.1	2022-09-15	P1/SIAT1	<	78	107	222	E50K, D53N, G62E, N96S(+cho), I140K, H156S, I192F, I223V, K264R
A/Bremen/29/2022	2a.3a.1	2022-10-04	P1/SIAT1	40	99	104	248	E50K, D53N, S91N, N96S(+cho), N122D(-cho), I140K, H156S, I192F, I223V

Table MN-2. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2023-01-12

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre Post-infection ferret antisera					HA1 substitutions for 3C.2a/3a viruses compared to A/Bangladesh/4005/2020 (3c.2a1b.2a.2 = 2)
				A/Cambodia 925256/2020 SIAT F03/21 1a	A/Thuringen 10/2022 SIAT F36/22 2b	A/Poland 97/2022 SIAT F39/22 2a.2	A/Stockholm 5/2021 SIAT F15/22 2a	A/Norway 24873/2021 SIAT F10/22 2a.3	
				Read	Read	Read	Read	Read	
REFERENCE VIRUSES									
A/Cambodia/925256/2020	1a	2020-09-25	SIAT3 10 ⁻³	287	<	<	<	<	N159Y, I160T(+cho), Q164L, K171N, D186S, N190D, S198P
A/Thuringen/10/2022	2b	2022-04-01	P1/SIAT2 10 ⁻³	<	254	61	219	79	E50K, E79V, I140K, S262N
A/Poland/97/2022	2a.2	2022-05-09	SIAT3 10 ⁻⁵	<	206	490	219	231	P4S, I25V, D53G, S145N, H156S, R201K, S219Y
A/Stockholm/5/2021	2a	2021-04-16	SIAT0/SIAT3 10 ⁻³	44	127	253	252	56	H156S
A/Norway/24873/2021	2a.3	2021-10-24	SIAT 10 ⁻⁵	<	136	195	205	117	D53N, N96S(+cho), H156S, I192F
TEST VIRUSES									
A/Lisboa/583/2022	2b	2022-10-04	SIAT2/SIAT1	<	245	<	226	<	R33Q, E50K, F79V, I140K, S262N
A/Lisboa/580/2022	2b	2022-10-07	SIAT2/SIAT1	48	290	74	271	56	R33Q, E50K, F79V, I140K, S262N
A/Lisboa/582/2022	2b	2022-10-09	SIAT2/SIAT1	<	248	56	233	<	R33Q, E50K, F79V, I140K, S262N
A/Lisboa/537/2022	2b	2022-10-10	SIAT2/SIAT1	<	381	133	284	77	R33Q, E50K, F79V, I140K, S262N
A/Lisboa/596/2022	2b	2022-10-18	SIAT1/SIAT1	<	220	75	177	48	R33Q, E50K, F79V, I140K, S262N
A/Lisboa/593/2022	2b	2022-10-20	SIAT1/SIAT1	<	282	80	278	<	R33Q, E50K, F79V, I140K, S262N
A/Lisboa/545/2022	2b	2022-10-20	SIAT2/SIAT1	<	387	90	377	<	R33Q, E50K, F79V, I140K, S262N
A/Lisboa/544/2022	2b	2022-10-24	SIAT2/SIAT1	<	299	120	247	<	R33Q, E50K, F79V, I140K, S262N
A/Lisboa/616/2022	2b	2022-10-26	SIAT1/SIAT1	<	299	64	255	61	R33Q, E50K, F79V, I140K, S262N
A/Lisboa/560/2022	2b	2022-10-26	SIAT1/SIAT1	<	252	65	284	<	R33Q, E50K, F79V, I140K, S262N
A/Lisboa/626/2022	2b	2022-10-27	SIAT1/SIAT1	<	310	141	367	58	R33Q, E50K, F79V, I140K, S262N
A/Lisboa/563/2022	2b	2022-11-02	SIAT1/SIAT1	<	256	60	267	43	R33Q, E50K, F79V, I140K, S262N
A/Lisboa/584/2022	2b	2022-10-05	SIAT2/SIAT1	<	335	55	375	50	R33Q, E50K, F79V, A93T, I140K, S262N
A/Lisboa/554/2022	2b	2022-10-24	SIAT1/SIAT1	<	348	129	320	69	R33Q, E50K, F79V, I140K, V213I, S262N
A/Lisboa/595/2022	2b	2022-10-24	SIAT1/SIAT1	<	310	114	244	70	R33Q, E50K, D53X, F79V, I140K, I202X, S262N

Table MN-3. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2023-01-16

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre Post-infection ferret antisera					HA1 substitutions for 3C.2a/3a viruses compared to A/Bangladesh/4005/2020 (3c.2a1b.2a.2 = 2)
				A/Cambodia 925256/2020 SIAT F03/21 1a Read	A/Thuringen 10/2022 SIAT F36/22 2b Read	A/Poland 97/2022 SIAT F39/22 2a.2 Read	A/Stockholm 5/2021 SIAT F15/22 2a Read	A/Norway 24873/2021 SIAT F10/22 2a.3 Read	
REFERENCE VIRUSES									
A/Cambodia/925256/2020	1a	2020-09-25	SIAT3 10 ⁻³	287	<	<	<	<	N159Y, I160T(+cho), Q164L, K171N, D186S, N190D, S198P
A/Thuringen/10/2022	2b	2022-04-01	P1/SIAT2 10 ⁻³	<	254	61	219	79	E50K, E79V, I140K, S262N
A/Poland/97/2022	2a.2	2022-05-09	SIAT3 10 ⁻⁵	<	206	490	219	231	P4S, I25V, D53G, S145N, H156S, R201K, S219Y
A/Stockholm/5/2021	2a	2021-04-16	SIAT0/SIAT3 10 ⁻³	44	127	253	252	56	H156S
A/Norway/24873/2021	2a.3	2021-10-24	SIAT 10 ⁻⁵	<	136	195	205	117	D53N, N96S(+cho), H156S, I192F
TEST VIRUSES									
A/Bawshar/72212056/2022	2b	2022-10-30	SIAT1	<	109	<	95	<	E50K, I140K, I242M
A/Shahriar.Iran/34_19Mehr/2022	2b	2022-10-08	SIAT1/SIAT1	<	270	71	152	<	E50K, F79V, I140K, I242M
A/Tehran.Iran/18_18Mehr/2022	2b	2022-10-08	SIAT1/SIAT1	<	281	96	237	61	E50K, F79V, I140K, I242M
A/Tehran.Iran/16_17Mehr/2022	2b	2022-10-08	SIAT1/SIAT1	<	151	53	124	<	E50K, F79V, I140K, I242M
A/Pakdash.Iran/13_19Mehr/2022	2b	2022-10-09	SIAT1/SIAT1	<	94	<	71	<	E50K, F79V, I140K, I242M
A/Lorestan.Iran/55_23Mehr/2022	2b	2022-10-10	SIAT1/SIAT1	<	68	<	56	<	E50K, F79V, I140K, I242M
A/Rey.Iran/44_23Mehr/2022	2b	2022-10-12	SIAT1/SIAT1	<	236	44	213	40	E50K, F79V, I140K, I242M
A/Tehran.Iran/29_2AbanA/2022	2b	2022-10-23	SIAT1/SIAT1	<	210	63	142	<	E50K, F79V, I140K, I242M
A/Shahriar.Iran/4_23Mehr/2022	2b	2022-10-10	SIAT1/SIAT1	<	170	41	133	<	E50K, F79V, I140K, A163X, I242M
A/Ibra/72211987/2022	2b	2022-10-26	SIAT1	<	192	40	136	<	I3M, E50K, F79V, I140K, I242M, L243X
A/Shahriar.Iran/66_10Mehr/2022	2b	2022-09-29	SIAT1/SIAT1	<	95	66	140	<	E50K, F79V, R92G, T135A(-cho), I140K, A212T, I242M
A/Ille de France/52489/2022	2b	2022-10-07	MDCK2/SIAT1	<	116	<	71	<	R33Q, E50K, F79V, I140K, S262N
A/Ille de France/53172/2022	2b	2022-10-11	MDCK1/SIAT1	<	126	<	83	<	R33Q, E50K, F79V, I140K, S262N
A/Pais Vasco/4012/2022	2b	2022-10-19	SIAT1	<	95	<	72	<	R33Q, E50K, F79V, I140K, S262N
A/Pais Vasco/4010/2022	2b	2022-10-19	SIAT1	<	152	<	111	<	R33Q, E50K, F79V, I140K, S262N
A/Lourdes/53101/2022	2b	2022-10-08	MDCK1/SIAT1	<	116	<	92	<	R33Q, E50K, F79V, N81D, I140K, S262N
A/Lille/50053/2022	2a.1	2022-09-06	MDCK1/SIAT1	<	<	61	60	<	D53G, D104G, T135A(-cho), H156S, T167S, N190S, K276R
A/Aragon/4055/2022	2a.3a.1	2022-10-18	SIAT1	<	53	68	117	80	E50K, D53N, N96S(+cho), I140K, H156S, I192F, I223V
A/Qatar/10-VI-22-2015935/2022	2a.3a.1	2022-09-20	SIAT1	<	145	119	181	149	E50K, D53N, E83K, N96SI(+cho), N122D(-cho), N133D(-cho), I140K, H156S, I192F, I223V
A/Fort-De-France/53862/2022	2a.3b	2022-10-18	MDCK1/SIAT1	<	45	61	68	<	D53N, N96SI(+cho), I140M, H156S, I192F

Table MN-4. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2023-01-23

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre Post-infection ferret antisera					HA1 substitutions for 3C.2a/3a viruses compared to A/Bangladesh/4005/2020 (3c.2a1b.2a.2 = 2)
				A/Cambodia 925256/2020 SIAT F03/21 1a Read	A/Thuringen 10/2022 SIAT F36/22 2b Read	A/Poland 97/2022 SIAT F39/22 2a.2 Read	A/Stockholm 5/2021 SIAT F15/22 2a Read	A/Norway 24873/2021 SIAT F10/22 2a.3 Read	
REFERENCE VIRUSES									
A/Cambodia/925256/2020	1a	2020-09-25	SIAT3 10 ⁻³	287	<	<	<	<	N159Y, I160T(+cho), Q164L, K171N, D186S, N190D, S198P
A/Thuringen/10/2022	2b	2022-04-01	P1/SIAT2 10 ⁻³	<	254	61	219	79	E50K, E79V, I140K, S262N
A/Poland/97/2022	2a.2	2022-05-09	SIAT3 10 ⁻⁵	<	206	490	219	231	P4S, I25V, D53G, S145N, H156S, R201K, S219Y
A/Stockholm/5/2021	2a	2021-04-16	SIAT0/SIAT3 10 ⁻³	44	127	253	252	56	H156S
A/Norway/24873/2021	2a.3	2021-10-24	SIAT 10 ⁻⁵	<	136	195	205	117	D53N, N96S(+cho), H156S, I192F
TEST VIRUSES									
A/Catalonia/NSVH101892403/2022	3C.2a1b.2a.2	2022-08-13	SIAT1/SIAT2	<	<	82	65	<	D53G, D104G, I140K, H156S, K276R, R299K
A/Burgos/245/2022	2b	2022-10-18	SIAT1	<	116	<	70	<	R33Q, E50K, F79V, I140K, S262N
A/Burgos/244/2022	2b	2022-10-18	SIAT1	<	143	<	96	<	R33Q, E50K, F79V, I140K, S262N
A/Burgos/243/2022	2b	2022-10-18	SIAT1	<	142	46	92	<	R33Q, E50K, F79V, I140K, S262N
A/Burgos/242/2022	2b	2022-10-18	SIAT1	<	122	<	71	<	R33Q, E50K, F79V, I140K, S262N
A/Lyon/NU22.2.97/2022	2b	2022-10-24	MDCK2/SIAT1	<	127	<	77	<	R33Q, E50K, F79V, I140K, S262N
A/Lyon/CHU/R22.415.69/2022	2b	2022-11-05	MDCK2/SIAT1	<	118	<	73	<	R33Q, E50K, F79V, I140K, S262N
A/Burgos/249/2022	2b	2022-10-16	SIAT1	<	112	<	68	<	P21L, R33Q, E50K, F79V, I140K, S262N
A/Burgos/260/2022	2b	2022-10-14	SIAT1	<	85	<	60	<	R33Q, E50K, F79V, Y105H, I140K, S262N
A/Navarra/4439/2022	2b	2022-11-13	SIAT2/SIAT1	<	83	<	96	<	R33Q, E50K, F79I, I140K, S144G, S262N
A/Navarra/4440/2022	2b	2022-11-17	SIAT2/SIAT1	<	126	<	86	<	R33Q, E50K, F79V, I140K, K238R, S262N
A/Burgos/246/2022	2b	2022-10-17	SIAT1	<	74	<	51	<	R33Q, E50K, F79V, I140K, I242M, S262N
A/Burgos/262/2022	2b	2022-10-14	SIAT1	<	95	<	68	<	R33Q, E50K, F79V, I140K, S262N, I300T(+cho)
A/Berlin/150/2022	2b	2022-10-07	P1/SIAT2	<	56	<	66	<	E50K, F79V, T135A(-cho), I140K, S262N
A/Lyon/NU22.4.39/2022	2b	2022-11-14	MDCK2/SIAT1	<	65	<	71	<	E50K, F79V, T135A(-cho), I140K, S262N
A/Navarra/4438/2022	2b	2022-11-18	SIAT2/SIAT1	<	51	<	57	<	E50K, F79I, T135A(-cho), I140K, S144G, S262N
A/Madrid/3803/2022	2a.1	2022-10-05	SIAT2	<	<	78	54	<	D53G, D104G, T135A(-cho), H156S, T167S, K276R, L316M
A/Catalonia/NSVH161512067/2022	2a.1b	2022-09-14	SIAT1/SIAT2	<	<	59	47	<	D53G, D104G, I140K, H156S, K276R, R299K
A/Lyon/CHU/R22.374.29/2022	2a.1b	2022-10-01	MDCK2/SIAT1	<	<	63	40	<	D53G, D104G, S115X, I140K, H156S, K276R, R299K
A/Grenoble/NOMS22.1.42/2022	2a.3a	2022-10-10	MDCK2/SIAT1	<	67	134	198	66	E50K, D53N, N96S(+cho), H156S, I192F, I223V
A/Grenoble/NOMS22.1.57/2022	2a.3a	2022-10-26	MDCK2/SIAT1	<	77	151	200	69	E50K, D53N, N96S(+cho), H156S, I192F, I223V
A/Bremen/28/2022	2a.3a.1	2022-10-04	P1/SIAT2	<	107	137	158	77	S47L(-cho), E50K, D53N, S92N, N96S(+cho), N122D(-cho), I140K, H156S, I192F, I223V
A/Fort-De-France/53820/2022	2a.3b	2022-10-17	MDCK1/SIAT1	<	68	113	128	59	D53N, N96S(+cho), I140M, H156S, I192F, P227S

Table MN-5. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2023-01-26

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre Post-infection ferret antisera					HA1 substitutions for 3C.2a/3a viruses compared to A/Bangladesh/4005/2020 (3c.2a1b.2a.2 = 2)
				A/Cambodia 925256/2020 SIAT F03/21 1a Read	A/Thuringen 10/2022 SIAT F36/22 2b Read	A/Poland 97/2022 SIAT F39/22 2a.2 Read	A/Stockholm 5/2021 SIAT F15/22 2a Read	A/Norway 24873/2021 SIAT F10/22 2a.3 Read	
REFERENCE VIRUSES									
A/Cambodia/925256/2020	1a	2020-09-25	SIAT3 10 ⁻³	287	<	<	<	<	N159Y, I160T(+cho), Q164L, K171N, D186S, N190D, S198P
A/Thuringen/10/2022	2b	2022-04-01	P1/SIAT2 10 ⁻³	<	254	61	219	79	E50K, E79V, I140K, S262N
A/Poland/97/2022	2a.2	2022-05-09	SIAT3 10 ⁻⁵	<	206	490	219	231	P4S, I25V, D53G, S145N, H156S, R201K, S219Y
A/Stockholm/5/2021	2a	2021-04-16	SIAT0/SIAT3 10 ⁻³	44	127	253	252	56	H156S
A/Norway/24873/2021	2a.3	2021-10-24	SIAT 10 ⁻⁵	<	136	195	205	117	D53N, N96S(+cho), H156S, I192F
TEST VIRUSES									
A/Denmark/3110/2022	2b	2022-10-17	SIAT2/SIAT1	<	87	<	80	<	E50K, F79V, I140K, I242M
A/Denmark/3100/2022	2b	2022-10-17	SIAT2/SIAT1	<	108	<	71	<	E50K, F79V, I140K, I242M
A/Denmark/3101/2022	2b	2022-10-21	SIAT2/SIAT1	<	124	<	85	<	E50K, F79V, I140K, I242M
A/Denmark/3136/2022	2b	2022-11-05	SIAT2/SIAT1	<	124	<	86	<	E50K, F79V, I140K, I242M
A/Denmark/3103/2022	2b	2022-10-25	SIAT3/SIAT1	<	152	42	106	<	R33Q, E50K, F79V, I140K, S262N
A/Aix/NOMS22.1.44/2022	2b	2022-10-14	MDCK2/SIAT1	<	130	<	100	<	R33Q, E50K, F79V, N81D, I140K, S262N
A/Hessen/45/2022	2b	2022-10-13	P1/SIAT1	<	68	<	81	<	E50K, F79V, T135A(-cho), I140K, S262N
A/Thuringen/30/2022	2b	2022-10-17	P1/SIAT1	<	76	<	100	<	E50K, F79V, T135A(-cho), I140K, S262N
A/Nordrhein-Westfalen/40/2022	2b	2022-10-13	P1/SIAT1	<	110	43	130	<	E50K, D77X, F79V, T135A(-cho), I140K, S262N
A/Lyon/CHU/R22.355.53/2022	2a.1	2022-09-05	MDCK2/SIAT1	<	<	89	62	<	D53G, D104G, H156S, K276R
A/Saint-Etienne/NOMS22.1.28/2022	2a.1	2022-09-20	MDCK2/SIAT1	<	<	133	71	<	D53G, D104G, H156S, K276R
A/Melesse/NOMS22.1.23/2022	2a.1	2022-09-26	MDCK2/SIAT1	<	<	125	85	<	D53G, D104G, H156S, K276R
A/Lyon/CHU/R22.372.37/2022	2a.1	2022-09-28	MDCK2/SIAT1	<	<	148	99	<	E50K, D53G, D104G, H156S, K276R, R299K
A/Catalonia/NSVH560025968/2022	2a.1b	2022-09-01	SIAT1	<	<	66	48	<	D53G, D104G, I140K, H156S, K276R, R299K
A/Lyon/CHU/R22.384.1/2022	2a.1b	2022-10-10	MDCK2/SIAT1	<	<	82	55	<	D53G, D104G, I140K, H156S, K276R, R299K
A/Denmark/3127/2022	2a.1b	2022-11-01	SIAT3/SIAT1	<	40	119	82	<	D53G, D104G, I140K, H156S, A163*, K276R, R299K
A/La Reunion/NOMS22.1.67/2022	2a.3a.1	2022-09-28	MDCK2/SIAT1	<	124	187	227	125	E50K, D53N, N96S(+cho), I140K, H156S, I192F, I223V
A/Ashabat/1/2022	2a.3b	2022-11-29	MDCKx/SIAT1	<	67	129	143	60	D53N, N96S(+cho), I140M, H156S, I192F
A/Ashabat/3/2022	2a.3b	2022-12-06	MDCKx/SIAT1	<	70	148	136	66	D53N, N96S(+cho), I140M, H156S, I192F

Table MN-6. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2023-02-06

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre Post-infection ferret antisera					HA1 substitutions for 3C.2a/3a viruses compared to A/Bangladesh/4005/2020 (3c.2a1b.2a.2 = 2)
				A/Cambodia 925256/2020 SIAT F03/21 1a Read	A/Thuringen 10/2022 SIAT F36/22 2b Read	A/Poland 97/2022 SIAT F39/22 2a.2 Read	A/Stockholm 5/2021 SIAT F15/22 2a Read	A/Slovenia 8720/2022 SIAT F24/22 2a.1 Read	
REFERENCE VIRUSES									
A/Cambodia/925256/2020	1a	2020-09-25	SIAT3 10 ⁻³	287	<	<	<	<	N159Y, I160T(+cho), Q164L, K171N, D186S, N190D, S198P
A/Thuringen/10/2022	2b	2022-04-01	P1/SIAT2 10 ⁻³	<	254	61	219	58	E50K, E79V, I140K, S262N
A/Poland/97/2022	2a.2	2022-05-09	SIAT3 10 ⁻⁵	<	206	490	219	258	P4S, I25V, D53G, S145N, H156S, R201K, S219Y
A/Stockholm/5/2021	2a	2021-04-16	SIAT0/SIAT3 10 ⁻³	44	127	253	252	102	H156S
A/Slovenia/8720/2022	2a.1	2022-02-10	MDCK1/SIAT2 10 ⁻³	<	<	126	77	147	D53G, D104G, H156S, K276R
TEST VIRUSES									
A/Argentina/3155/2022	3C.2a1b.2a.2	2022-08-31	SIAT2	<	63	66	72	96	D53N, N96S(+cho), T135K(-cho), S145N, H156S, I192F
A/Argentina/3151/2022	2b	2022-09-05	SIAT1	<	165	53	128	71	E50K, F79V, I140K
A/Ireland/73304/2022	2b	2022-11-23	SIAT1	<	74	<	103	42	E50K, F79V, T135A(-cho), I140K, S262N
A/Switzerland/68854/2022	2b	2022-12-23	SIAT1	<	63	<	76	<	E50K, F79V, T135A(-cho), I140K, S262N
A/Switzerland/28719/2022	2b	2022-12-19	SIAT1	<	82	<	99	42	E50K, F79V, T135A(-cho), I140K, S262N
A/Switzerland/12279/2022	2b	2022-12-09	SIAT1	<	62	<	72	42	E50K, F79V, T135A(-cho), I140K, S262N
A/Ireland/74535/2022	2b	2022-11-28	SIAT2	<	57	<	78	<	E50K, F79V, T128A(-cho), T135A(-cho), I140K, S262N
A/Norway/33060/2022	2a.1	2022-10-27	SIAT1	<	<	150	111	257	D53G, D104G, H156S, K276R
A/Lithuania/48774/2022	2a.1	2022-11-24	SIAT1	<	<	139	79	197	D53G, D104G, H156S, K276R
A/Norway/33371/2022	2a.1	2022-10-27	SIAT1	<	<	111	73	194	D53G, D104G, S144I, H156S, K276R
A/Hong Kong/34/2022	2a.1	2022-11-26	SIAT1/MDCK1	<	<	130	87	138	N45S(-cho), D53G, D104G, H156S
A/Ireland/74250/2022	2a.1b	2022-11-27	SIAT2	<	<	90	62	129	D53G, D104G, I140K, H156S, K276R, R299K
A/Ireland/74511/2022	2a.1b	2022-11-29	SIAT2	<	<	110	70	127	D53G, D104G, I140K, H156S, K276R, R299K
A/Switzerland/28394/2022	2a.1b	2022-12-19	SIAT1	<	<	112	71	143	D53G, D104G, I140K, H156S, K276R, R299K
A/Georgia/8/2022	2a.1b	unknown	SIAT1	<	45	121	80	203	D53G, D104G, I140K, H156S, K276R, N290T, R299K
A/Georgia/2369/2022	2a.1b	2022-11-24	SIAT1	<	<	103	69	129	D53G, D104G, I140K, H156S, K276R, N290T, R299K
A/Argentina/3486/2022	2a.3a	2022-10-01	SIAT1	<	<	111	124	131	D53N, N94K(-cho), N96S, H156S, I192F
A/Switzerland/28577/2022	2a.3a.1	2022-12-19	SIAT1	44	110	155	204	221	E50K, D53N, N96S(+cho), I140K, H156S, I192F, I223V
A/Norway/33475/2022	2a.3a.1	2022-10-27	SIAT1	<	96	144	176	198	E50K, D53N, N63K(-cho), N96S(+cho), I140K, H156S, I192F, I223V, K238R
A/Hong Kong/28/2022	2a.3a.1	2022-11-16	SIAT1/MDCK1	<	106	140	152	197	L15I, E50K, D53N, N96S(+cho), I140K, H156S, I192F, I223V
A/Hong Kong/30/2022	2a.3a.1	2022-11-20	SIAT1/MDCK1	43	128	158	207	230	L15I, E50K, D53N, N96S(+cho), I140K, H156S, I192F, I223V
A/Hong Kong/38/2022	2a.3a.1	2022-12-04	SIAT1/MDCK1	40	123	160	221	281	L15I, E50K, D53N, N96S(+cho), I140K, H156S, I192F, I223V

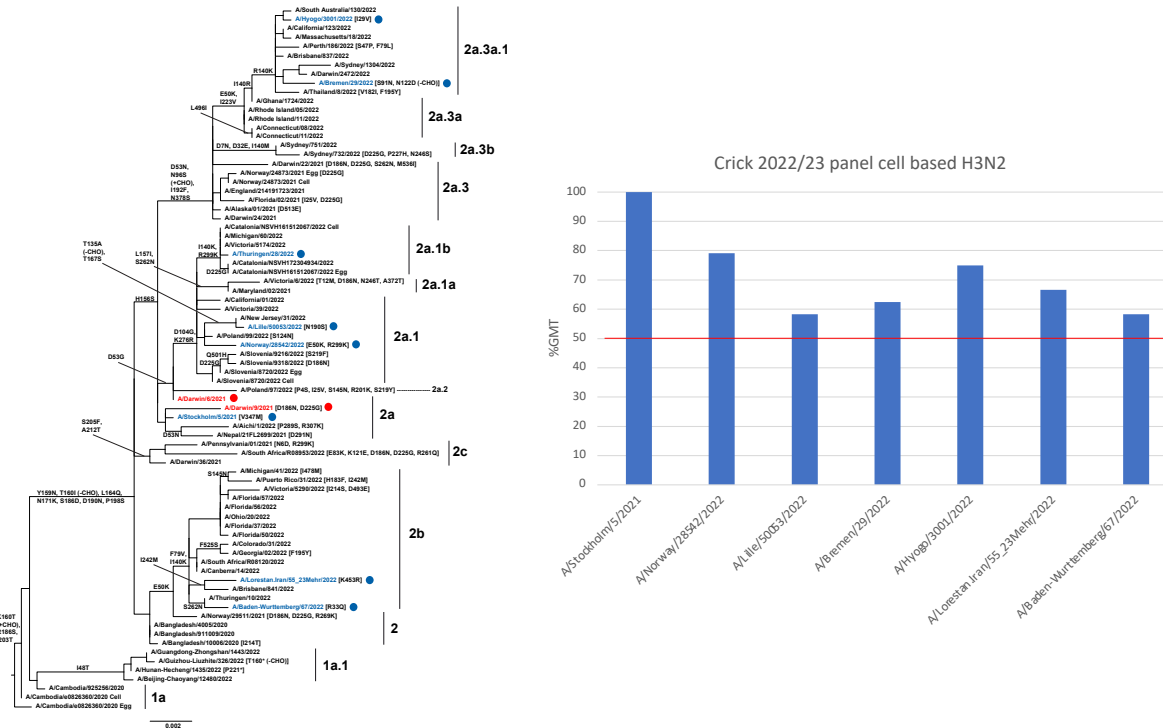
Table MN-7. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2023-02-09

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre Post-infection ferret antisera					HA1 substitutions for 3C.2a/3a viruses compared to A/Bangladesh/4005/2020 (3c.2a1b.2a.2 = 2)
				A/Cambodia 925256/2020	A/Thuringen 10/2022	A/Poland 97/2022	A/Stockholm 5/2021	A/Slovenia 8720/2022	
				SIAT F03/21 1a Read	SIAT F36/22 2b Read	SIAT F39/22 2a.2 Read	SIAT F15/22 2a Read	SIAT F24/22 2a.1 Read	
REFERENCE VIRUSES									
A/Cambodia/925256/2020	1a	2020-09-25	SIAT3 10 ⁻³	287	<	<	<	<	N159Y, I160T(+cho), Q164L, K171N, D186S, N190D, S198P
A/Thuringen/10/2022	2b	2022-04-01	P1/SIAT2 10 ⁻³	<	254	61	219	58	E50K, E79V, I140K, S262N
A/Poland/97/2022	2a.2	2022-05-09	SIAT3 10 ⁻⁵	<	206	490	219	258	P4S, I25V, D53G, S145N, H156S, R201K, S219Y
A/Stockholm/5/2021	2a	2021-04-16	SIAT0/SIAT3 10 ⁻³	44	127	253	252	102	H156S
A/Slovenia/8720/2022	2a.1	2022-02-10	MDCK1/SIAT2 10 ⁻³	<	<	126	77	147	D53G, D104G, H156S, K276R
TEST VIRUSES									
A/Galicia/3991/2022	2b	2022-10-16	SIAT1	<	242	68	160	110	E50K, F79V, I140K, S199P, S262N
A/Slovenia/11696/2022	2b	2022-12-29	SIATx/SIAT1	<	<	<	57	<	E50K, F79V, T135A(-cho), I140K, S144N(+cho), S262N
A/Norway/33623/2022	2b	2022-10-31	SIAT1	<	152	<	75	55	R33Q, E50K, F79V, I140K, S262N
A/Slovenia/11622/2022	2b	2022-12-27	MDCKx/SIAT1	<	55	<	58	<	T10K(-cho), E50K, F79V, T135A(-cho), I140K, S262N
A/Slovenia/42/2023	2b	2023-01-03	MDCKx/SIAT1	<	67	<	75	49	T10K(-cho), E50K, F79I, T135A(-cho), I140K, S262N
A/Slovenia/11665/2022	2b	2022-12-28	SIATx/SIAT1	<	65	<	85	47	T10K(-cho), E50K, F79I, T135A(-cho), I140K, Q210X, S262N
A/Norway/33536/2022	2b	2022-11-01	SIAT1	<	187	44	128	82	R33Q, E50K, F79V, I140K, S262N
A/Castilla La Mancha/4542/2022	2b	2022-11-24	SIAT1	<	156	<	103	57	R33Q, E50K, F79V, I140K, S262N
A/Castilla La Mancha/4541/2022	2b	2022-11-24	SIAT1	<	158	44	113	59	R33Q, E50K, F79V, I140K, S262N
A/Castilla La Mancha/4547/2022	2b	2022-11-26	SIAT1	<	145	<	102	57	R33Q, E50K, F79V, I140K, S262N
A/Hungary/73/2022	2b	2022-11-28	MDCK1/SIAT1	<	196	47	105	64	R33Q, E50K, F79V, I140K, S262N, A272T
A/Hungary/87/2022	2b	2022-12-07	MDCK1/SIAT1	<	152	52	99	61	R33Q, E41X, E50K, F79V, D101E, I140K, S262N, D291N
A/Norway/33404/2022	2a.1	2022-11-05	SIAT1	<	60	243	150	455	E50K, D53G, N94H, D104G, H156S, K276R, R299K
A/Galicia/3994/2022	2a.1b	2022-10-17	SIAT1	<	52	125	79	200	D53G, D104G, I140K, H156S, K276R, R299K
A/Norway/33402/2022	2a.1b	2022-10-31	SIAT1	<	<	74	50	105	D53G, D104G, I140K, H156S, K276R, R299K
A/Hungary/69/2022	2a.1b	2022-11-11	MDCK1/SIAT1	<	<	74	66	112	D53G, D104G, I140K, H156S, K276R, R299K
A/Ukraine/774/2022	2a.1b	2022-12-21	SIAT1	<	53	127	83	148	D53S, D104G, I140K, H156S, K276R, R299K
A/Hungary/68/2022	2a.1b	2022-11-15	MDCK1/SIAT1	<	44	115	79	150	D53G, D104G, I140K, H156S, N165X(-cho), K276R, R299K
A/Ukraine/775/2022	2a.1b	2022-12-21	SIAT1	<	54	133	89	188	T10M(-cho), D53G, D104G, I140K, H156S, K276R, R299K
A/Ukraine/773/2022	2a.1b	2022-12-21	SIAT1	<	48	122	76	165	T10M(-cho), D53G, D104G, I140K, H156S, K276R, R299K
A/Ukraine/779/2022	2a.1b	2022-12-22	SIAT1	<	56	141	94	223	T10M(-cho), D53G, D104G, I140K, H156S, K276R, R299K
A/Ukraine/804/2022	2a.1b	2022-12-26	SIAT1	<	<	98	57	126	T10M(-cho), D53G, D104G, I140K, H156S, K276R, R299K
A/Tunisia/23298/2022	2a.3a	2022-12-14	SIAT1	<	82	151	176	238	E50K, D53N, N96S(+cho), H156S, I192F, I223V
A/Tunisia/23669/2022	2a.3a.1	2022-12-13	SIAT1	<	109	147	178	244	P4T, E50K, D53N, N96S(+cho), I140K, H156S, I192F, I223V

Human Serology

Panel 1

A panel of pre- and post- vaccination serum samples was collected from healthy adult volunteers at the Francis Crick Institute who received cell-based quadrivalent vaccines in the course of the 2022-2023 vaccination campaign. Samples were tested by HI assay and post vaccination GMTs calculated. The bar chart shows %GMT for each virus shown as a percentage of the A/Stockholm/5/2021 virus result expressed as 100%. All viruses tested show a reduction in GMT, but none show a greater than 50% reductions in GMT. All viruses were grown in SIAT cells.



Panel 2

Public Health Wales panel:

Serum panel information:

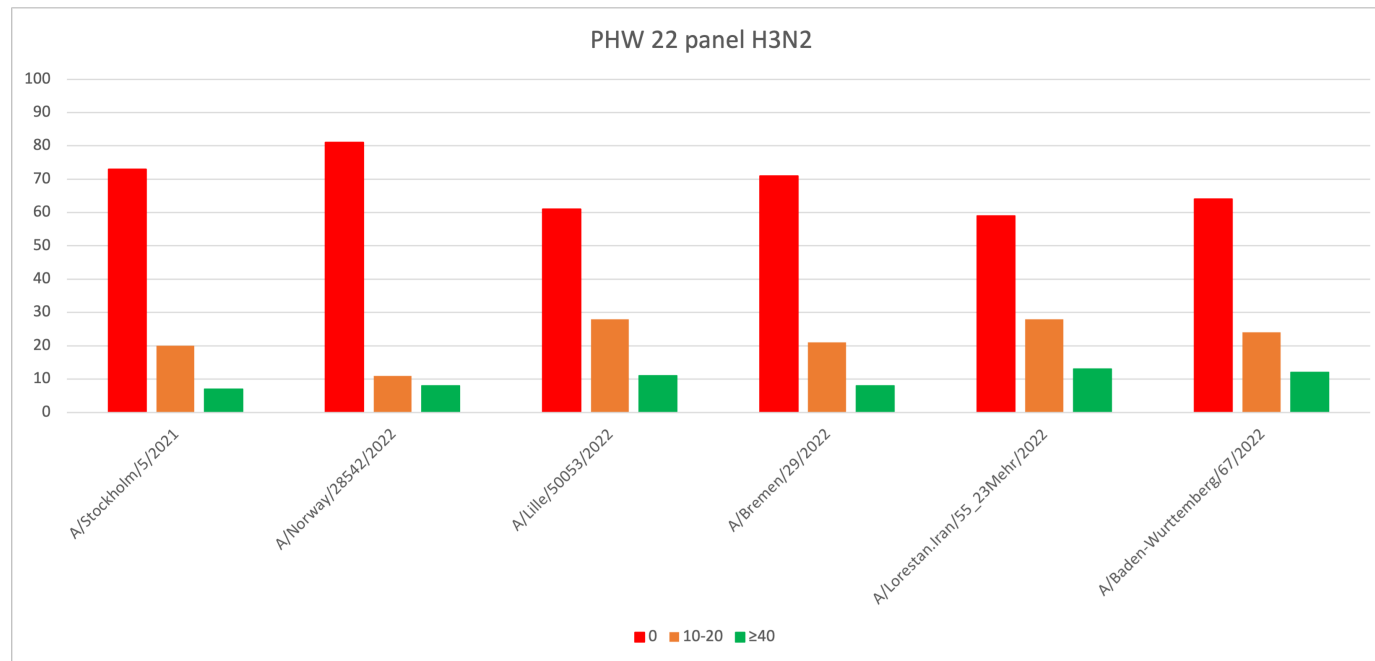
Total number of samples: 100

Collection dates: 16th February 2022 to 7th June 2022

Age range: 16 – 64 years

Average age: 36.5 years

Median age: 35.5 years



Reference reagents and isolates

A/H3N2: Egg isolates

Virus	Status	Genetic group/clade	Comment
A/Norway/29511/2021	San-017/017	3C.2a1b.2a.2 (E50K)	Failed 2-way
A/Norway/24873/2021	sent out	3C.2a1b.2a.2 (N96S, I192F, N378S)	
A/Slovenia/9216/2022	sent out	3C.2a1b.2a.2 (H156S, D53G, D104G)	Egg adaptation: (S219F, D225G)
A/Slovenia/9318/2022	sent out	3C.2a1b.2a.2 (H156S, D53G, D104G)	Egg adaptation: (D186N, D225G)
A/Catalonia/NSVH161512067/2022	sent out	3C.2a1b.2a.2 (I140K, R299K)	
A/Catalonia/NSVH172304934/2022	available to send out	3C.2a1b.2a.2 (I140K, R299K).	
A/Poland/97/2022	Reference only	3C.2a1b.2a.2 (S145N)	

A/H3N2: HI reagents and references

Virus	Genetic group	Virus passage	Ferret ID
A/Cambodia/925256/2020	3C.2a1b.2a.1	SIAT5	F03/21
A/Cambodia/e0826360/2020	3C.2a1b.2a.1	E5/E2	F10/21
A/Thuringen/10/2022	3C.2a1b.2a.2	P1/SIAT2	F36/22
A/Poland/97/2022	3C.2a1b.2a.2	SIAT2	F39/22
A/Stockholm/5/2021	3C.2a1b.2a.2	SIAT0/SIAT3	F35/21
A/Darwin/9/2021	3C.2a1b.2a.2	E3/E4	F39/21
A/Norway/24873/2021	3C.2a1b.2a.2	SIAT2	F10/22
A/Norway/24873/2021	3C.2a1b.2a.2	E3 (Am2A11)	F11/22
A/Slovenia/8720/2022	3C.2a1b.2a.2	SIAT1/MDCK1/SIAT2	F24/22
A/Slovenia/8720/2022	3C.2a1b.2a.2	E3(Am1A12)	F25/22
A/Catalonia/NSVH161512067/2022	3C.2a1b.2a.2	SIAT1/SIAT2	F41/22
A/Norway/28542/2022	3C.2a1b.2a.2	SIAT2	F42/22
A/Ghana/1724/2022	3C.2a1b.2a.2	SIAT2	F43/22
A/Poland/99/2022	3C.2a1b.2a.2	SIAT2	F44/22

A/H3N2: MN reagents and references

Virus	Genetic clade	Virus passage	Ferret ID
A/Cambodia/925256/2020	3C.2a1b.2a.1	S5 10-3	F03/21
A/Thuringen/10/2022	3C.2a1b.2a.2	P1/S2 10-3	F36/22
A/Poland/97/2022	3C.2a1b.2a.2	S3 10-5	F39/22
A/Stockholm/5/2021	3C.2a1b.2a.2	S0/S3 10-3	F35/21
A/Slovenia/8720/2022	3C.2a1b.2a.2	S1/M1/S3 10-5	F24/22
A/Norway/24873/2021	3C.2a1b.2a.2	SIAT3 10-5	F10/22

Summary: H3N2

Genetic analyses

Clade 3C.2a1b.2a.2 predominated since 1st September 2022 in all geographic regions where H3N2 circulated. Clade 3C.2a1b.1a.1 viruses were detected in China only.

Owing to substantial genetic diversification and co-circulation of multiple genetic groups within the 2a.2 clade, we use a new nomenclature designation.

Geographically, although we observed limited spatial segregation of some subclades; for example, the 2b subclade was predominantly detected in the Americas and Europe whereas the 2a.3a subclade was identified in Africa, many subclades co-circulated in multiple geographic regions.

From the HA-NA tanglegrams we note that viruses from clade 2b possess a NA segment that is closely related to 3C.2a1b.1a viruses, which may indicate a reassortment event. Some viruses within 2b subclade also show reassortment with NA associated with 2a.1b or 2a HA clades.

Antigenic analyses

For the 2a.2 viruses we observed that the post-infection antisera raised against new reference viruses discriminated antigenic properties among the different genetic subgroups (see heat map).

We observed the broadest cross-recognition of viruses in different genetic groups by an antiserum raised against a 2a virus A/Stockholm/5/2021 (SIAT) - an A/Darwin/6-like virus, compared with antisera raised against other reference or vaccine viruses. However we noted consistent 4-fold drop in cross-reactivity against the 2b subclade viruses represented in our dataset by A/Thuringen/10/2022, and in more recent viruses from the 2a.1b, 2a.3a, 2a.3a.1 and the 2a.3b subclades (4-fold).

Conversely, the post-infection ferret antiserum raised against A/Thuringen/10/2022 (2b) recognised well many of the viruses in the emerging genetic groups, with the exception of those in the 2a.1 group with a loss of glycosylation at position 133 due to amino acid substitution at position 135 (>4-fold), and limited loss in recognition (2-fold) of viruses in the 2a.1b group.

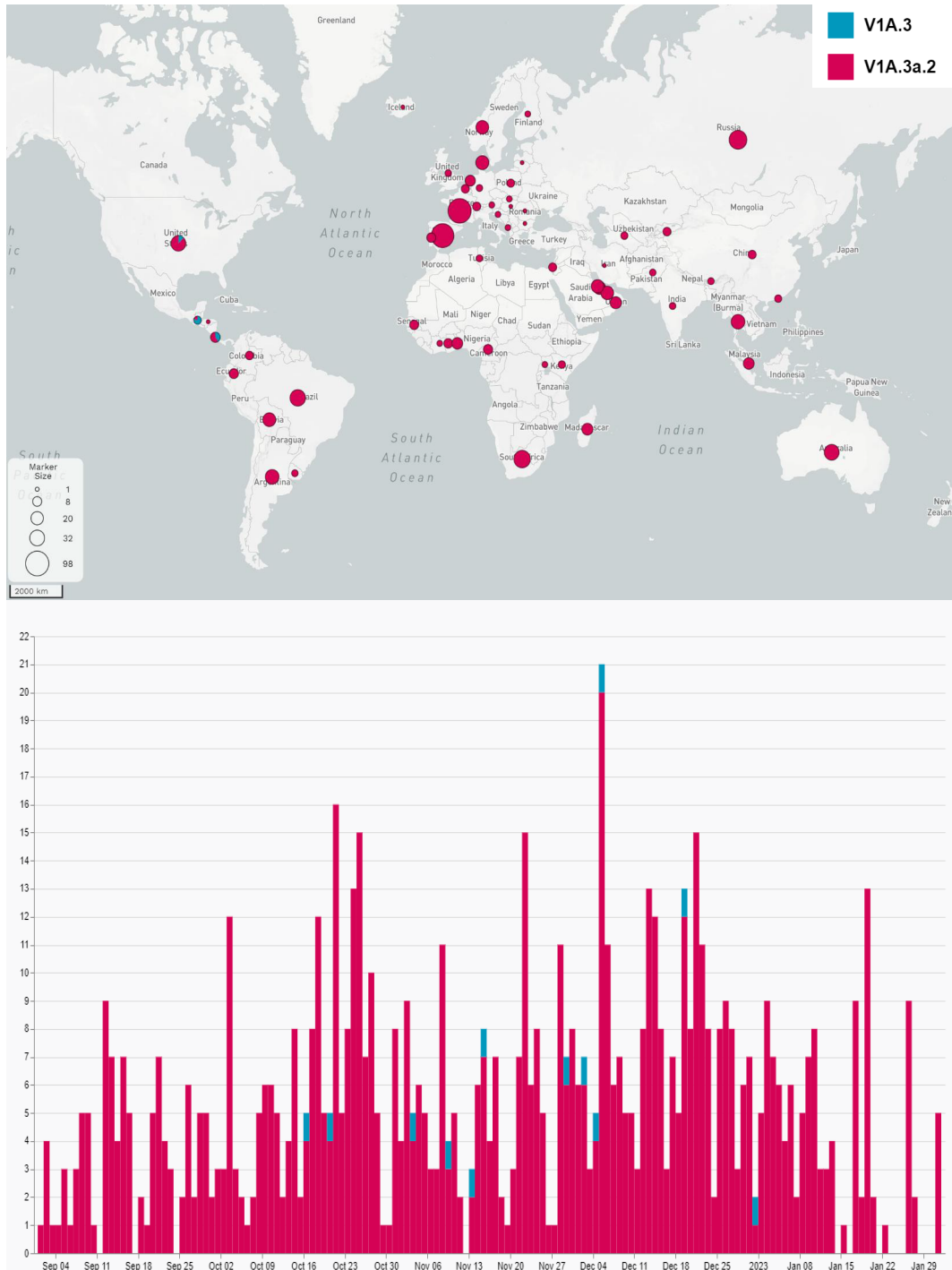
For other reference antisera we found that there was good recognition of viruses within-genetic subclade, e.g. A/Norway/24873/2021 for 2a.3, A/Poland/97/22 for 2a.2, A/Slovenia/8720/22 for the 2a.1 and A/Catalonia/NSVH-2067/22 for 2a.1b subclades but poor among viruses from different subclades, particularly in recognition of viruses from the 2b group represented by A/Thuringen/10/2022.

Microneutralisation (PRN) data. Broadest cross-neutralisation was seen for antisera raised against A/Stockholm/5/2021 (2a) and A/Thuringen/10/2022 (2b) for viruses in most genetic subclades. The 2b antiserum however, did not recognise most viruses in the 2a.1 and 2a.1b subclade whereas A/Stockholm/5/2021 did recognise well viruses from these subclades.

All tested viruses were susceptible to neuraminidase inhibitors (NAI) by phenotypic testing and no markers for reduced inhibition by NAIs were identified in NA sequences derived from clinical specimens

Influenza B

MicroReact map for influenza viruses with collection dates after 31st August 2022 through to 7th February 2023 as deposited in GISAID, coloured by B lineage. Geographic markers scaled to detection proportions. Full length HA.



Genetic analyses: B/Victoria

Maximum likelihood phylogenetic tree: B/Victoria

Maximum likelihood phylogenetic trees inferred from full-length HA sequence data obtained from GISAID for viruses with collection dates after 31st August 2022 and submitted to GISAID up to 7th February 2023, manually curated and then downsampled using Treemmer to retain a representative tree topology of 250 sequences.



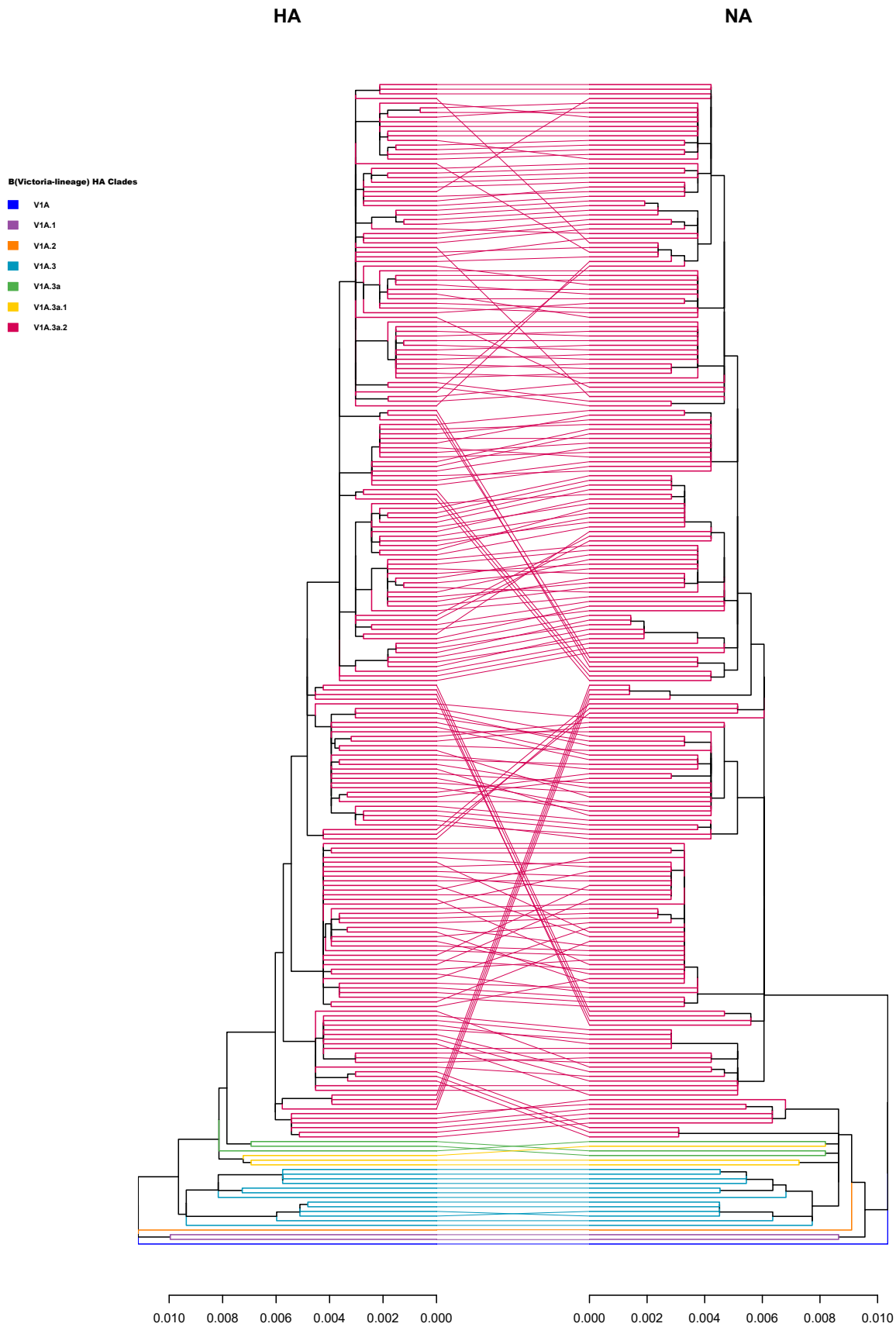
Vaccine viruses
Reference viruses (WIC)
 Reference viruses (non-WIC)

Collection dates
 Sep 2022
 Oct 2022
 Nov 2022
 Dec 2022
 Jan 2023



VIA.3a.2

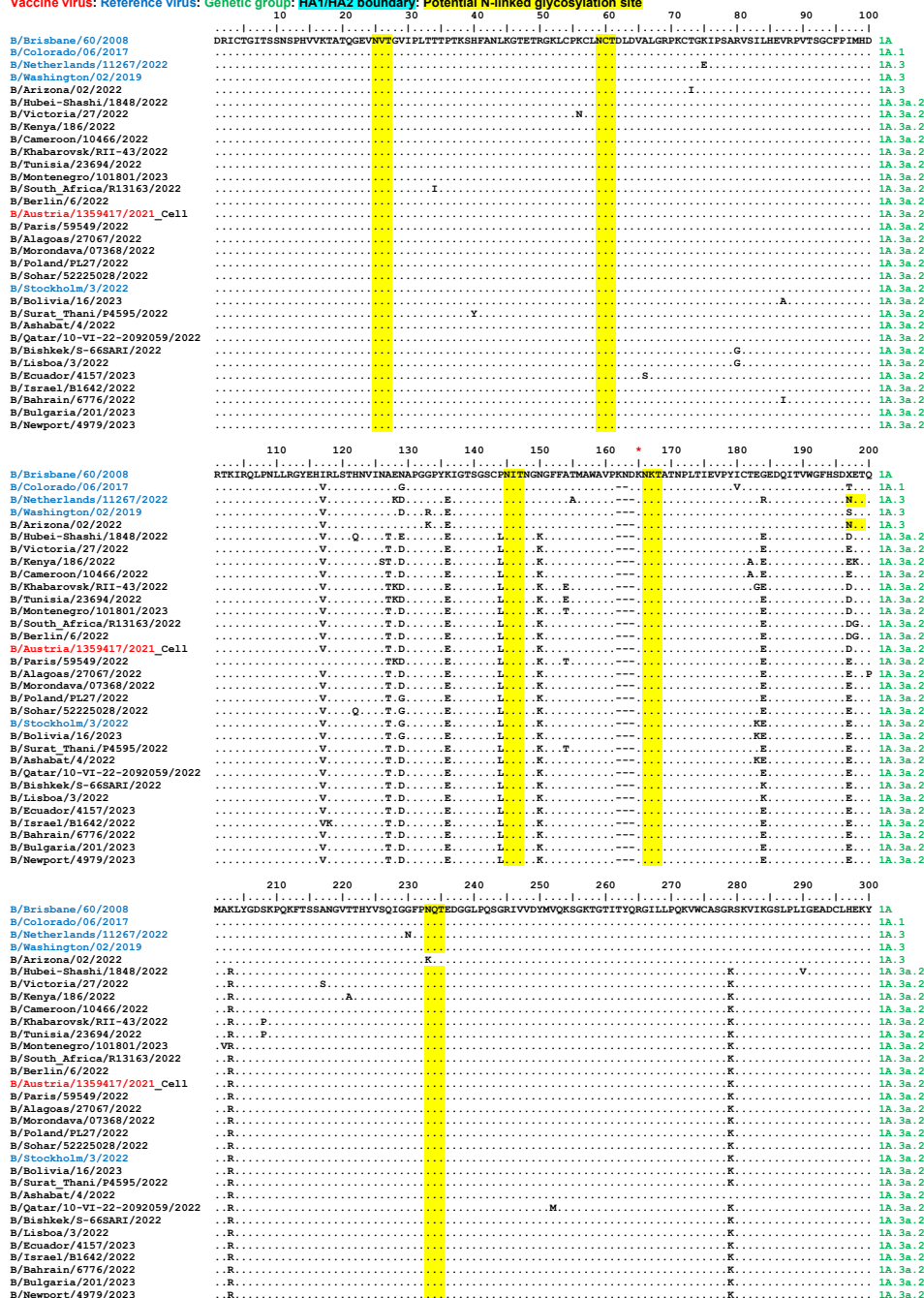
HA-NA tanglegrams



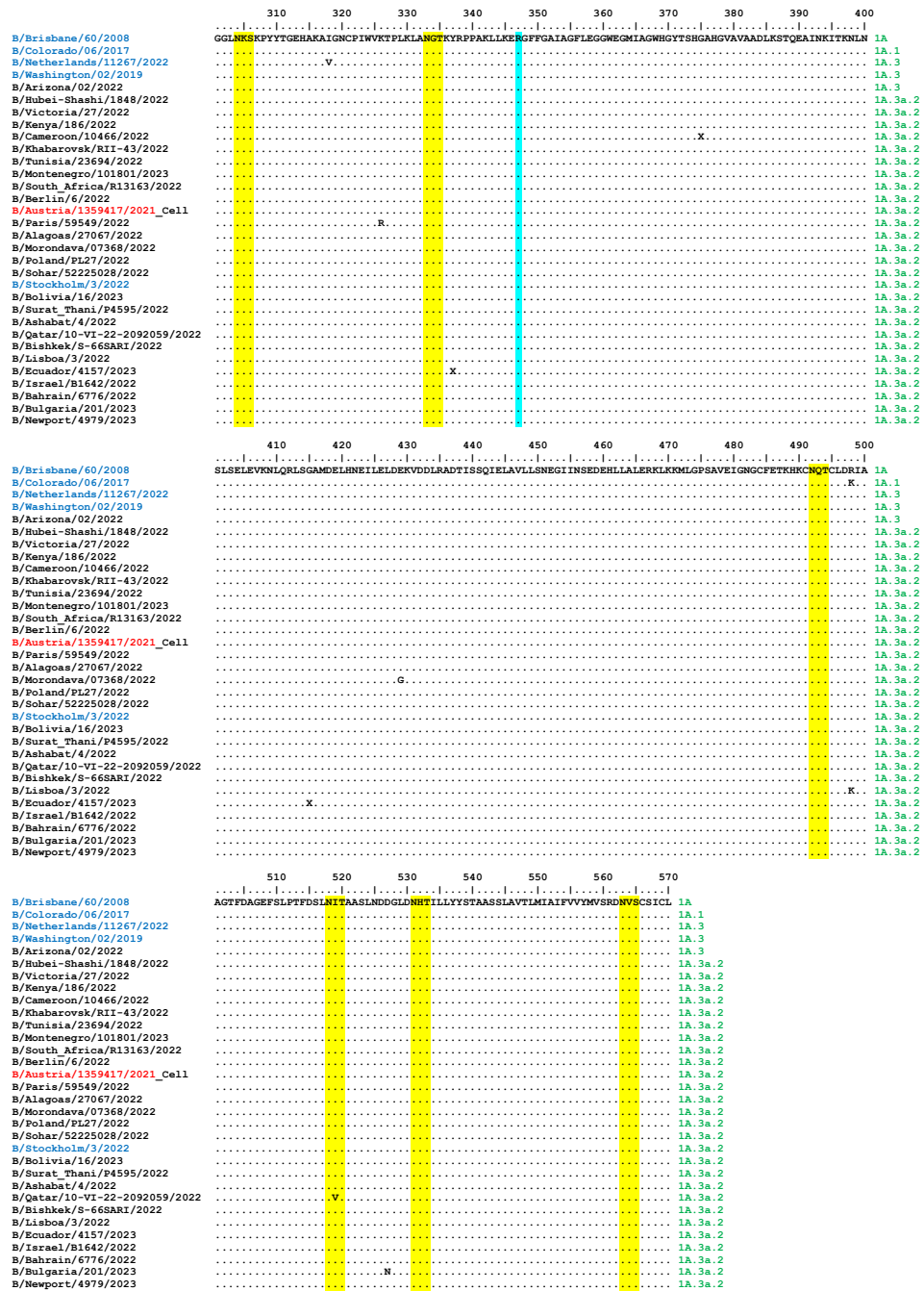
Protein alignments HA

Figure B-P1. HA protein alignment for B (Victoria-lineage) viruses

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

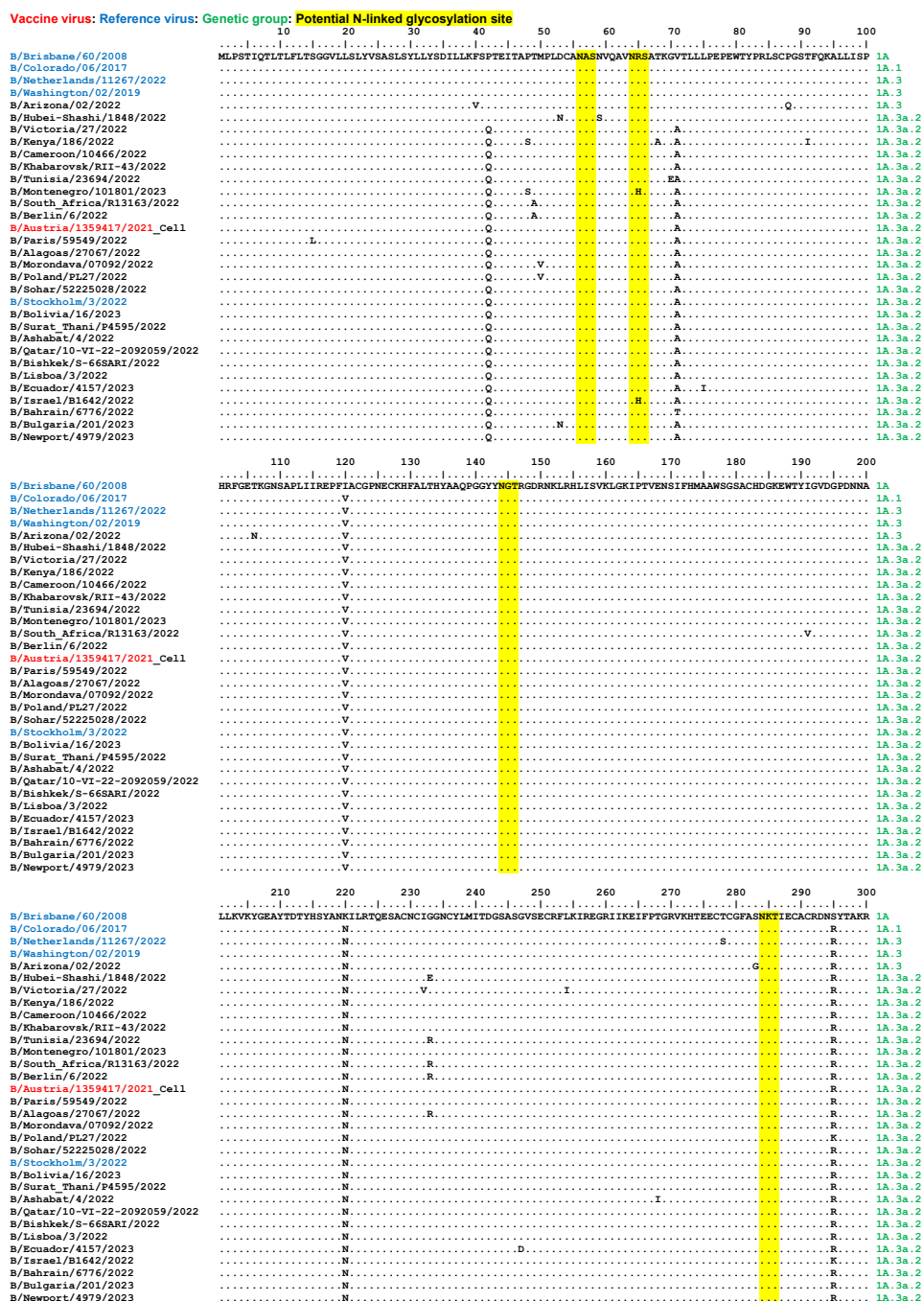


* Some viruses recently detected in Spain (Catalonia) also carry residue 165 deletions



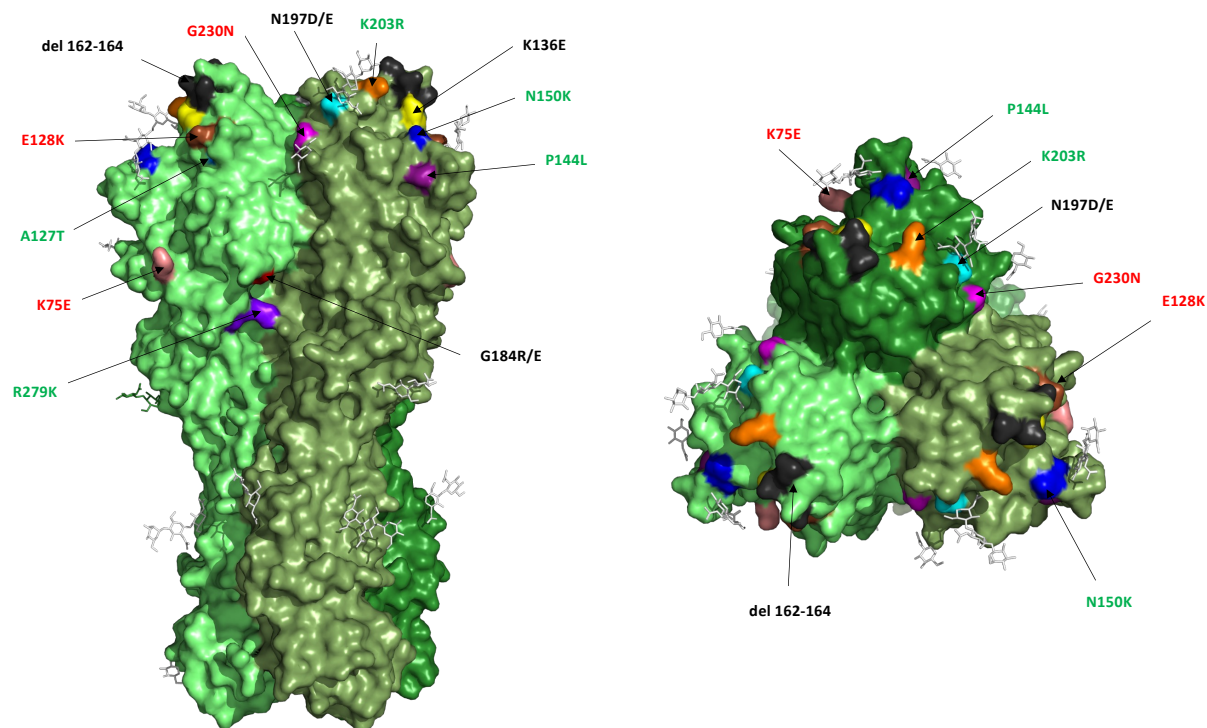
Protein alignments NA

Figure B-P2. NA protein alignment for B (Victoria-lineage) viruses



Structure

Figure S4. HA1 amino acid substitutions from A/Washington/02/2019 along with contemporary B-Victoria lineage substitutions in an emerging group in subclade V1A.3 (red) and subgroup V1A.3a.2 (green). Labels in black represent amino acid substitutions common to more than one group. [PDB Entry 4FQM]

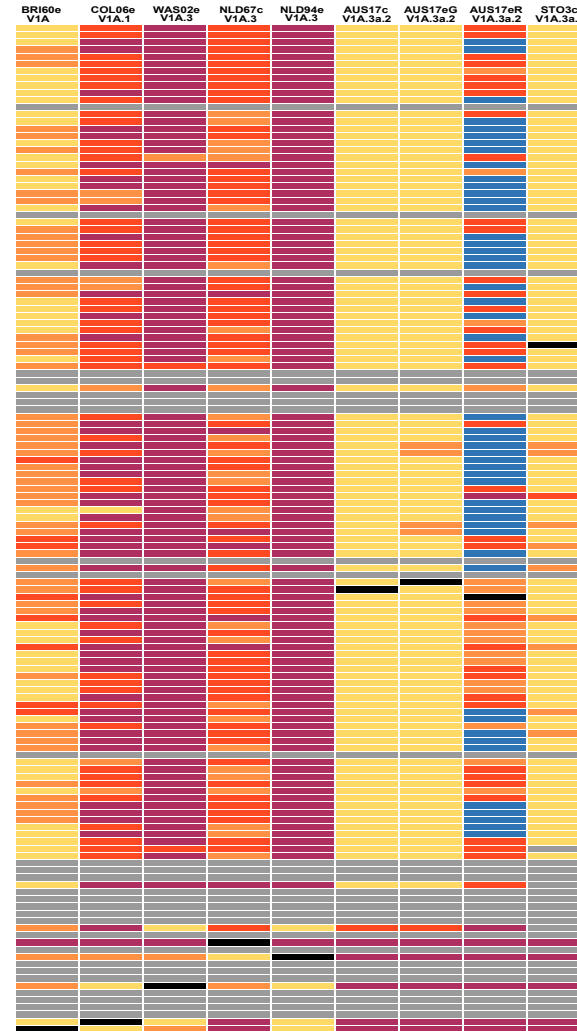
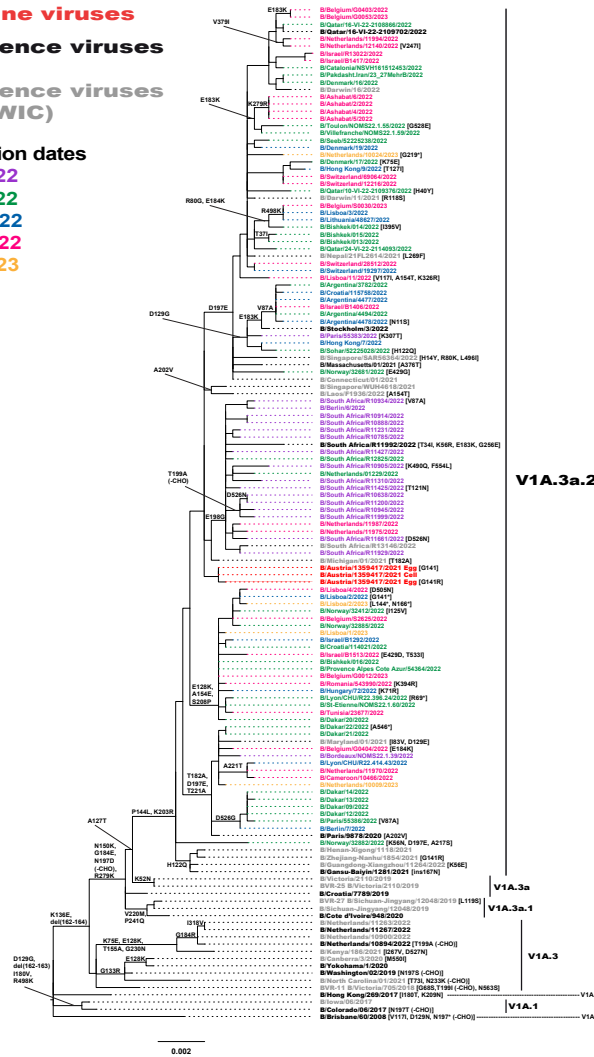


Genotype to Phenotype heatmap: B/Victoria

Vaccine viruses
Reference viruses (WIC)
Reference viruses (non-WIC)

Collection dates

Sep 2022
 Oct 2022
 Nov 2022
 Dec 2022
 Jan 2023



HI fold changes
 < 4
 4
 > 8
 8
 > 160 (when
 homologous titre >= 2560)
 Homologous
 N/A

Antigenic analyses: B/Victoria

Haemagglutination inhibition tables: B/Victoria

Table B-1. Antigenic analyses of influenza B viruses (Victoria lineage) 2022-09-21

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre											
					Post-infection ferret antiserum											
					NEW											
					B/Bris 60/08 Egg	B/Colorado 06/17 Egg	B/Wash'ton 02/19 Egg	B/Neth 11267/22 MDCK	B/Neth 10894/22 Egg	B/CIV 948/20 MDCK	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R	B/Paris 9878/20 MDCK	B/G-Baiyin 1281/21 MDCK	
					Sh 539, 540, 543, 544, 570, 571, 574 ¹	F44/18	F20/20	F29/22	F37/22	F08/21	NIB F01/21	F15/21	F44/21	F12/21	F08/22	
Genetic group	V1A	V1A.1	V1A.3	V1A.3	V1A.3	V1A.3a.1	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2					
REFERENCE VIRUSES																
B/Brisbane/60/2008		V1A	2008-08-04	E4/E4	1280	160	40	<40	80	<80	<40	<40	<40	<40	<40	
B/Colorado/06/2017		V1A.1	2017-02-05	E5/E2	1280	640	40	<40	160	<80	<40	<40	<40	<40	<40	
B/Washington/02/2019		V1A.3	2019-01-19	E3/E3	1280	160	80	40	320	<80	<40	<40	<40	<40	<40	
B/Netherlands/11267/2022	G184R	V1A.3	2022-04-14	MDCK-MIX/MDCK1	40	10	<10	320	40	<80	<40	<40	<40	<40	<40	
B/Netherlands/10894/2022	G184R, T199A (-cho), V318I	V1A.3	2022-04-02	E4	640	<40	<10	80	160	<40	<40	<40	<40	<40	<40	
B/Cote d'Ivoire/948/2020	V220M, P241Q	V1A.3a.1	2020-05-28	MDCK4	320	40	40	40	80	640	160	160	80	80	40	
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2	2021-01-09	SIAT1/MDCK4	640	40	<10	40	<40	160	1280	1280	320	320	320	
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141)	V1A.3a.2	2021-01-09	E3/E5	640	20	<10	40	<40	160	2560	1280	320	320	320	
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141R)	V1A.3a.2	2021-01-09	E3/E5	320	20	<10	40	<40	320	1280	2560	320	320	320	
B/Paris/9878/2020	T182A, D197E, T221A	V1A.3a.2	2020-11-20	MDCK2	640	80	10	40	<40	160	1280	1280	320	640	320	
B/Gansu-Baiyin/1281/2021	ins167N, H122Q	V1A.3a.2	2021-04-13	C1/C1/MDCK2	640	40	<10	<40	<40	160	1280	1280	320	320	320	
TEST VIRUSES																
B/Netherlands/10894/2022	G184R, T199A (-cho), V318I	V1A.3	2022-04-02	E3(Am1AL2)	640	40	20	40	ND	<80	<40	<40	<40	<40	<40	
B/Netherlands/11264/2022	G184R, N197S (-cho), V318I	V1A.3	2022-04-13	E4(Am1AL3)	640	160	80	80	ND	<80	<40	<40	<40	<40	<40	
B/Netherlands/11264/2022	G184R, N197S (-cho), V318I	V1A.3	2022-04-13	E4(Am2AL2)	1280	160	40	80	ND	<80	<40	<40	<40	<40	<40	
B/S. Petersburg/RII-04/2022		V1A.3a.2	2022-03-10	MDCK3/MDCK1	640	40	<10	40	ND	160	1280	1280	320	320	320	
B/S. Petersburg/RII-06/2022		V1A.3a.2	2022-04-27	MDCK3/MDCK1	640	40	<10	40	ND	160	1280	1280	320	320	320	

< relates to the lowest dilution of antiserum used

¹ hyperimmune sheep serum; ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23
SH 2023

< 4-fold
4-fold
8-fold
> 8-fold
not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table B-2. Antigenic analyses of influenza B viruses (Victoria lineage) 2022-11-29

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre								
					Post-infection ferret antiserum								
					B/Bris 60/08 Egg Sh 539, 540, 543, 544, 570, 571, 574 ¹	B/Colorado 06/17 Egg	B/Wash'ton 02/19 Egg	B/Neth 11267/22 MDCK	B/Neth 10894/22 Egg	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R	B/Stockholm 3/22 MDCK
						F44/18	F20/20	F29/22	F37/22	NIB F01/21	F15/21	F44/21	F28/22
					Genetic group	V1A	V1A.1	V1A.3	V1A.3	V1A.3	V1A.3a.2	V1A.3a.2	V1A.3a.2
REFERENCE VIRUSES													
B/Brisbane/60/2008		V1A	2008-08-04	E4/E4	1280	320	40	20	160	<40	<40	<40	<40
B/Colorado/06/2017		V1A.1	2017-02-05	E5/E2	640	320	40	20	320	<40	<40	<40	<40
B/Washington/02/2019		V1A.3	2019-01-19	E3/E3	640	<40	40	40	160	<40	<40	<40	<40
B/Netherlands/11267/2022	G184R	V1A.3	2022-04-14	MDCK-MIX/MDCK2	<40	<40	<10	320	40	<40	<40	<40	<40
B/Netherlands/10894/2022	G184R, T199A (-cho), V318I	V1A.3	2022-04-02	E4	160	<40	<10	80	320	<40	<40	<40	<40
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2	2021-01-09	SIAT1/MDCK4	160	<40	<10	80	10	640	640	320	640
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141)	V1A.3a.2	2021-01-09	E3/E5	640	<40	<10	80	<10	1280	1280	640	1280
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141R)	V1A.3a.2	2021-01-09	E3/E5	160	<40	<10	40	<10	1280	640	2560	640
B/Stockholm/3/2022	D197E, D129G, E183K	V1A.3a.2	2022-03-22	SIAT1/MDCK3	320	<40	<10	40	10	640	640	320	1280
TEST VIRUSES													
B/Netherlands/11591/2022	G184R, D129N, V318I	V1A.3	2022-05-17	MDCK-MIX1/MDCK1	<40	<40	<10	40	10	<40	<40	<40	<40
B/Netherlands/11678/2022	G184R, V318I	V1A.3	2022-06-30	MDCK-MIX1/MDCK2	<40	<40	<10	160	40	<40	<40	<40	<40
B/Berlin/6/2022		V1A.3a.2	2022-09-15	P1/MDCK1	320	<40	<10	40	10	1280	640	320	640
B/Pakdasht.Iran/23_27MehrB/2022		V1A.3a.2	2022-10-16	SIAT1/MDCK1	640	<40	<10	40	10	640	640	320	1280

< relates to the lowest dilution of antiserum used

¹ hyperimmune sheep serum; ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23
SH 2023

 < 4-fold  4-fold  8-fold  > 8-fold  < not recognised by the antiserum  ≥ 160 (when homologous titre ≥ 2560)  ≥ 160 (no homologous titre)

Table B-3. Antigenic analyses of influenza B viruses (Victoria lineage) 2022-12-06

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antiserum									
				B/Bris 60/08 Egg	B/Colorado 06/17 Egg	B/Wash'ton 02/19 Egg	B/Neth 11267/22 MDCK	B/Neth 10894/22 Egg	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R	B/Stock 3/22 MDCK	
				Sh 539, 540, 543, 544, 570, 571, 574 ¹	F44/18	F20/20	F29/22	F37/22	NIB F01/21	F15/21	F44/21	F28/22	
				V1A	V1A.1	V1A.3	V1A.3	V1A.3	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2	
REFERENCE VIRUSES													
B/Brisbane/60/2008		V1A	2008-08-04	E4/E4	2560	320	40	20	160	<40	<40	<40	<40
B/Colorado/06/2017		V1A.1	2017-02-05	E5/E2	1280	640	80	20	320	<40	<40	<40	<40
B/Washington/02/2019		V1A.3	2019-01-19	E3/E3	640	<10	80	40	160	<40	<40	<40	<40
B/Netherlands/11267/2022	G184R	V1A.3	2022-04-14	MDCK-MIX/MDCK2	<40	<10	<10	320	<40	<40	<40	<40	<40
B/Netherlands/10694/2022	G184R, T199A (-cho), V318I	V1A.3	2022-04-02	E4	320	<10	<10	80	320	<40	<40	<40	<40
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2	2021-01-09	SIAT1/MDCK4	320	<10	<10	80	10	1280	1280	640	640
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141)	V1A.3a.2	2021-01-09	E3/E5	640	<10	<10	80	<10	2560	1280	640	1280
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141R)	V1A.3a.2	2021-01-09	E3/E5	320	<10	<10	40	<10	2560	1280	5120	640
B/Stockholm/9/2022	D197E, D129G, E183K	V1A.3a.2	2022-03-22	SIAT1/MDCK3	640	<10	<10	40	10	1280	640	320	1280
TEST VIRUSES													
B/South Africa/R09303/2022		V1A.3a.2	2022-07-22	MDCK1/MDCK1	320	160	<10	40	10	1280	640	320	640
B/South Africa/R09302/2022		V1A.3a.2	2022-07-25	MDCK1/MDCK1	320	80	<10	80	10	640	640	320	640
B/South Africa/R09298/2022		V1A.3a.2	2022-07-25	MDCK1/MDCK1	320	80	<10	80	10	1280	640	160	640
B/South Africa/R09294/2022		V1A.3a.2	2022-07-25	MDCK1/MDCK1	320	80	<10	80	10	1280	640	320	640
B/South Africa/R09387/2022		V1A.3a.2	2022-08-10	MDCK1/MDCK1	320	160	<10	80	10	1280	640	320	640
B/South Africa/R09666/2022		V1A.3a.2	2022-08-15	MDCK1/MDCK1	320	80	<10	40	10	1280	640	320	640
B/South Africa/R09679/2022		V1A.3a.2	2022-08-16	MDCK1/MDCK1	320	80	<10	80	10	1280	640	320	640
B/Norway/30217/2022		V1A.3a.2	2022-08-18	MDCK1	640	<10	<10	40	10	2560	1280	320	640
B/Norway/30216/2022		V1A.3a.2	2022-08-20	MDCK1	640	<10	<10	40	10	2560	1280	640	640
B/South Africa/R10101/2022		V1A.3a.2	2022-08-24	MDCK1/MDCK1	640	80	<10	80	10	640	640	160	640
B/South Africa/R10090/2022		V1A.3a.2	2022-08-24	MDCK1/MDCK1	640	80	<10	80	20	1280	640	160	640
B/South Africa/R10431/2022		V1A.3a.2	2022-08-30	MDCK1/MDCK1	1280	160	<10	40	10	1280	640	320	640
B/South Africa/R10483/2022		V1A.3a.2	2022-08-31	MDCK1/MDCK1	1280	160	<10	40	10	1280	640	320	640
B/South Africa/R10905/2022		V1A.3a.2	2022-09-05	MDCK1/MDCK1	640	80	<10	80	10	1280	640	320	640
B/South Africa/R10638/2022		V1A.3a.2	2022-09-05	MDCK1/MDCK1	1280	320	<10	80	10	1280	1280	320	640
B/South Africa/R10945/2022		V1A.3a.2	2022-09-07	MDCK1/MDCK1	640	80	<10	40	10	640	320	160	320
B/South Africa/R10934/2022		V1A.3a.2	2022-09-07	MDCK1/MDCK1	640	80	<10	80	10	1280	640	320	640
B/South Africa/R10785/2022		V1A.3a.2	2022-09-07	MDCK1/MDCK1	640	80	<10	80	10	640	320	160	320
B/South Africa/R10888/2022		V1A.3a.2	2022-09-08	MDCK1/MDCK1	640	80	<10	80	10	640	640	320	640
B/South Africa/R10914/2022		V1A.3a.2	2022-09-09	MDCK1/MDCK1	640	<10	<10	80	<10	640	640	320	640
B/South Africa/R11200/2022		V1A.3a.2	2022-09-10	MDCK1/MDCK1	1280	<10	<10	80	20	640	640	320	640
B/South Africa/R11427/2022		V1A.3a.2	2022-09-12	MDCK1/MDCK1	640	<10	<10	40	10	1280	640	320	640
B/South Africa/R11425/2022		V1A.3a.2	2022-09-12	MDCK1/MDCK1	640	<10	<10	40	10	640	640	320	640
B/South Africa/R11231/2022		V1A.3a.2	2022-09-12	MDCK1/MDCK1	640	<10	<10	40	10	640	320	160	320
B/South Africa/R11310/2022		V1A.3a.2	2022-09-14	MDCK1/MDCK1	640	<10	<10	40	10	640	640	80	160
B/South Africa/R11661/2022		V1A.3a.2	2022-09-21	MDCK1/MDCK1	640	<10	<10	40	10	640	640	160	640
B/South Africa/R11929/2022		V1A.3a.2	2022-09-26	MDCK1/MDCK1	640	<10	<10	40	10	640	640	160	320
B/South Africa/R11999/2022		V1A.3a.2	2022-09-28	MDCK1/MDCK1	640	<10	<10	20	10	640	320	160	640
B/South Africa/R11992/2022		V1A.3a.2	2022-09-28	MDCK1/MDCK1	320	<10	<10	40	<10	1280	640	320	640
B/South Africa/R12825/2022		V1A.3a.2	2022-10-17	MDCK1/MDCK1	640	<10	<10	80	10	1280	640	320	640
B/Qatar/24-VI-22-2114093/2022		V1A.3a.2	2022-10-23	MDCK1	1280	<10	<10	80	10	1280	1280	320	1280
B/Qatar/16-VI-22-2109702/2022		V1A.3a.2	2022-10-24	MDCK1	640	<10	<10	40	10	1280	1280	320	1280
B/Qatar/10-VI-22-2109376/2022		V1A.3a.2	2022-10-24	MDCK1	1280	<10	<10	80	10	1280	640	320	1280
B/Qatar/16-VI-22-2108866/2022		V1A.3a.2	2022-10-24	MDCK1	1280	<10	<10	40	10	640	1280	320	1280

< relates to the lowest dilution of antiserum used

¹ hyperimmune sheep serum; ND = Not DoneVaccine
NH 2021-22Vaccine
SH 2022
NH 2022-23
SH 2023

< 4-fold
 4-fold
 8-fold
 > 8-fold
 not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table B-4. Antigenic analyses of influenza B viruses (Victoria lineage) 2023-01-17

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antiserum									
					B/Bris 60/08 Egg Sh 539, 540, 543, 544, 570, 571, 574 ¹	B/Colorado 06/17 Egg	B/Wash'ton 02/19 Egg	B/Neth 11267/22 MDCK	B/Neth 10894/22 Egg	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R	B/Stock 3/22 MDCK	
					Ferret number	F44/18	F20/20	F29/22	F37/22	NIB F01/21	F15/21	F44/21	F28/22	
		Genetic group			V1A	V1A.1	V1A.3	V1A.3	V1A.3	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2	
REFERENCE VIRUSES														
B/Brisbane/60/2008		V1A	2008-08-04	E4/E4	1280	160	40	10	80	<40	<40	<40	<40	
B/Colorado/06/2017		V1A.1	2017-02-05	E5/E2	640	640	80	10	160	<40	<40	<40	<40	
B/Washington/02/2019		V1A.3	2019-01-19	E3/E3	640	160	160	20	160	<40	<40	<40	<40	
B/Netherlands/11267/2022	G184R	V1A.3	2022-04-14	MDCK-MIX/MDCK2	<40	10	<10	160	20	<40	<40	<40	<40	
B/Netherlands/10894/2022	G184R, T199A (-cho), V318I	V1A.3	2022-04-02	E4(Am1AL3)	320	80	10	80	160	<40	<40	<40	<40	
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2	2021-01-09	SIAT1/MDCK4	320	40	<10	40	<10	1280	1280	320	640	
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141)	V1A.3a.2	2021-01-09	E3/E5	320	40	<10	40	<10	1280	1280	320	640	
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141R)	V1A.3a.2	2021-01-09	E3/E5	160	20	<10	40	<10	1280	640	5120	320	
B/Stockholm/3/2022	D197E, D129G, E183K	V1A.3a.2	2022-03-22	SIAT1/MDCK3	320	40	<10	40	10	1280	1280	320	1280	
TEST VIRUSES														
B/Bordeaux/NOMS22.1.39/2022		V1A.3a.2	2022-09-23	MDCK2/MDCK1	640	80	10	40	10	1280	1280	640	1280	
B/Argentina/3782/2022		V1A.3a.2	2022-10-13	MDCK1	640	80	<10	20	<10	1280	1280	320	1280	
B/Denmark/16/2022		V1A.3a.2	2022-10-14	MDCK2/MDCK1	640	80	<10	20	10	1280	1280	320	640	
B/Lyon/CHUR22.396.24/2022		V1A.3a.2	2022-10-20	MDCK2/MDCK1	160	20	<10	20	<10	1280	640	320	320	
B/Toulon/NOMS22.1.55/2022		V1A.3a.2	2022-10-21	MDCK2/MDCK1	640	80	<10	40	10	1280	1280	320	1280	
B/Dakar/20/2022		V1A.3a.2	2022-10-23	P1/MDCK1	320	40	<10	20	<10	2560	1280	320	320	
B/Dakar/12/2022		V1A.3a.2	2022-10-23	P2/MDCK1	640	80	<10	40	10	1280	1280	320	1280	
B/Dakar/09/2022		V1A.3a.2	2022-10-23	P1/MDCK1	320	40	<10	20	<10	640	1280	320	640	
B/Dakar/22/2022		V1A.3a.2	2022-10-24	P1/MDCK1	320	40	<10	20	<10	1280	1280	320	1280	
B/Dakar/21/2022		V1A.3a.2	2022-10-24	P2/MDCK1	320	40	<10	40	<10	1280	1280	320	640	
B/Argentina/4494/2022		V1A.3a.2	2022-10-25	MDCK1	1280	80	<10	40	<10	1280	1280	640	1280	
B/Dakar/14/2022		V1A.3a.2	2022-10-25	P2/MDCK1	320	40	<10	20	<10	1280	1280	320	1280	
B/Dakar/13/2022		V1A.3a.2	2022-10-25	P2/MDCK1	320	40	<10	20	<10	640	640	320	640	
B/Villefranche/NOMS22.1.59/2022		V1A.3a.2	2022-10-26	MDCK2/MDCK1	320	80	10	40	10	1280	1280	320	1280	
B/Denmark/17/2022		V1A.3a.2	2022-10-27	MDCK3/MDCK1	640	40	<10	20	10	1280	1280	320	640	
B/St-Etienne/NOMS22.1.60/2022		V1A.3a.2	2022-10-28	MDCK2/MDCK1	640	40	<10	40	<10	2560	1280	320	640	
B/Argentina/4478/2022		V1A.3a.2	2022-11-02	MDCK1	320	40	<10	20	<10	640	640	160	640	
B/Argentina/4477/2022		V1A.3a.2	2022-11-02	MDCK1	640	40	<10	20	<10	1280	640	320	640	
B/Lyon/CHUR22.414.43/2022		V1A.3a.2	2022-11-04	MDCK3/MDCK1	640	80	10	40	10	1280	1280	640	1280	
B/Denmark/19/2022		V1A.3a.2	2022-11-08	MDCK3/MDCK1	640	40	<10	10	<10	640	640	160	640	
B/Hong Kong/7/2022		V1A.3a.2	2022-11-09	MDCK1/MDCK1	640	80	<10	40	<10	1280	1280	320	1280	
B/Hong Kong/9/2022		V1A.3a.2	2022-11-21	MDCK1/MDCK1	640	40	<10	20	10	1280	640	320	640	
B/Lithuania/48627/2022		V1A.3a.2	2022-11-22	MDCK1	320	40	<10	20	<10	1280	1280	320	640	
B/Ashabat/2/2022		V1A.3a.2	2022-12-06	MDCKx/MDCK1	640	80	<10	40	10	1280	1280	320	1280	
B/Ashabat/5/2022		V1A.3a.2	2022-12-10	MDCKx/MDCK1	320	40	10	20	<10	640	640	320	640	
B/Ashabat/4/2022		V1A.3a.2	2022-12-10	MDCKx/MDCK1	320	40	<10	20	<10	640	1280	320	640	
B/Ashabat/6/2022		V1A.3a.2	2022-12-15	MDCKx/MDCK1	640	80	<10	40	10	1280	1280	640	1280	

< relates to the lowest dilution of antiserum used

¹ hyperimmune sheep serum; ND = Not DoneVaccine
NH 2021-22Vaccine
SH 2022
NH 2022-23
SH 2023

< 4-fold
 4-fold
 8-fold
 > 8-fold
 not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table B-5. Antigenic analyses of influenza B viruses (Victoria lineage) 2022-12-20

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antiserum									
					B/Bris 60/08 Egg	B/Colorado 06/17 Egg	B/Wash'ton 02/19 Egg	B/Neth 11267/22 MDCK	B/Neth 10894/22 Egg	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R	B/Stock 3/22 MDCK	
					Sh 539, 540, 543, 544, 570, 571, 574 ¹	F44/18	F20/20	F29/22	F37/22	NIB F01/21	F15/21	F44/21	F28/22	
Genetic group					V1A	V1A.1	V1A.3	V1A.3	V1A.3	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2	
REFERENCE VIRUSES														
B/Brisbane/60/2008		V1A	2008-08-04	E4/E4	1280	320	20	20	160	<40	<40	<40	<40	
B/Colorado/06/2017		V1A.1	2017-02-05	E5/E2	1280	640	40	20	320	<40	<40	<40	<40	
B/Washington/02/2019		V1A.3	2019-01-19	E3/E3	640	320	80	40	160	<40	<40	<40	<40	
B/Netherlands/11267/2022	G184R	V1A.3	2022-04-14	MDCK-MIX/MDCK2	<40	20	<10	320	40	<40	<40	<40	<40	
B/Netherlands/10894/2022	G184R, T199A (-cho), V318I	V1A.3	2022-04-02	E4(Am1AL3)	320	160	<10	80	320	<40	<40	<40	<40	
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2	2021-01-09	SIAT1/MDCK4	320	80	<10	80	10	1280	640	320	640	
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141)	V1A.3a.2	2021-01-09	E3/E5	640	80	<10	80	<10	2560	1280	640	640	
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141R)	V1A.3a.2	2021-01-09	E3/E5	160	40	<10	40	<10	1280	640	2560	640	
B/Stockholm/3/2022	D197E, D129G, E183K	V1A.3a.2	2022-03-22	SIAT1/MDCK3	320	40	<10	40	10	640	640	320	640	
TEST VIRUSES														
B/Netherlands/11473/2022	G184R, V318I	V1A.3	2022-04-25	MDCK3	640	320	<10	160	320	<40	<40	<40	<40	
B/Paris/55383/2022		V1A.3a.2	2022-09-21	MDCK1/MDCK1	320	80	<10	40	<10	1280	640	320	1280	
B/Paris/55386/2022		V1A.3a.2	2022-10-14	MDCK1/MDCK1	640	40	<10	80	10	640	640	160	320	
B/Provence_Alpes_Cote_Azur/54364/2022		V1A.3a.2	2022-10-25	MDCK1/MDCK1	640	80	<10	40	10	1280	1280	640	640	
B/Seeb/52225238/2022		V1A.3a.2	2022-10-27	MDCK1	640	80	20	80	10	640	1280	320	1280	
B/Sohar/52225028/2022		V1A.3a.2	2022-10-27	MDCK1	320	80	10	40	<10	640	640	320	640	
< relates to the lowest dilution of antiserum used							Vaccine NH 2021-22			Vaccine SH 2022 NH 2022-23 SH 2023				
¹ hyperimmune sheep serum; ND = Not Done														

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table B-6. Antigenic analyses of influenza B viruses (Victoria lineage) 2023-01-24

Viruses	Other information	Passage history	Collection date	Haemagglutination inhibition titre									
				Post-infection ferret antiserum									
				B/Bris 60/08 Egg	B/Colorado 06/17 Egg	B/Wash'ton 02/19 Egg	B/Neth 11267/22 MDCK	B/Neth 10894/22 Egg	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R	B/Stock 3/22 MDCK	
				Sh 539, 540, 543, 544, 570, 571, 574 ¹	F44/18	F20/20	F29/22	F37/22	NIB F01/21	F15/21	F44/21	F28/22	
				V1A	V1A.1	V1A.3	V1A.3	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2		
REFERENCE VIRUSES													
B/Brisbane/60/2008		V1A	2008-08-04	E4/E4	2560	320	80	<10	160	<40	<40	<40	<40
B/Colorado/06/2017		V1A.1	2017-02-05	E5/E2	1280	320	80	<10	160	<40	<40	<40	<40
B/Washington/02/2019		V1A.3	2019-01-19	E3/E3	320	160	160	20	80	<40	<40	<40	<40
B/Netherlands/11267/2022	G184R	V1A.3	2022-04-14	MDCK-MIX/MDCK2	<40	10	<10	160	40	<40	<40	<40	<40
B/Netherlands/10894/2022	G184R, T199A (-cho), V318I	V1A.3	2022-04-02	E4(Am1AL3)	320	80	20	40	160	<40	<40	<40	<40
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2	2021-01-09	SIAT1/MDCK4	320	40	<10	20	<10	1280	1280	320	640
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141)	V1A.3a.2	2021-01-09	E3/E5	640	20	<10	40	10	2560	1280	640	640
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141R)	V1A.3a.2	2021-01-09	E3/E5	320	40	<10	40	10	2560	1280	>5120	640
B/Stockholm/3/2022	D197E, D129G, E163K	V1A.3a.2	2022-03-22	SIAT1/MDCK2	320	40	<10	20	<10	1280	640	320	640
TEST VIRUSES													
B/Bishkek/013/2022		V1A.3a.2	2022-10-08	MDCK1/MDCK1	640	40	<10	20	<10	1280	640	320	640
B/Bishkek/014/2022		V1A.3a.2	2022-10-14	MDCK1/MDCK1	640	40	<10	20	<10	1280	1280	320	1280
B/Bishkek/015/2022		V1A.3a.2	2022-10-22	MDCK1/MDCK1	640	40	<10	20	<10	1280	1280	320	640
B/Bishkek/016/2022		V1A.3a.2	2022-10-23	MDCK1/MDCK1	640	40	<10	20	<10	1280	1280	640	640
B/Hungary/72/2022		V1A.3a.2	2022-11-22	MDCK1/MDCK1	320	40	<10	40	<10	1280	1280	640	640
B/Switzerland/19297/2022		V1A.3a.2	2022-11-28	MDCK1	640	80	<10	20	<10	1280	1280	320	640
B/Switzerland/12216/2022		V1A.3a.2	2022-12-09	MDCK1/MDCK1	640	80	<10	20	<10	1280	1280	320	640
B/Switzerland/69064/2022		V1A.3a.2	2022-12-13	MDCK1	640	80	<10	20	<10	1280	1280	320	640
B/Switzerland/28512/2022		V1A.3a.2	2022-12-19	MDCK1	640	80	<10	20	<10	1280	1280	640	640

< relates to the lowest dilution of antiserum used

¹ hyperimmune sheep serum; ND = Not DoneVaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23
SH 2023

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table B-7. Antigenic analyses of influenza B viruses (Victoria lineage) 2023-01-31

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antiserum									
					B/Bris 60/08 Egg	B/Colorado 06/17 Egg	B/Wash'ton 02/19 Egg	B/Neth 11267/22 MDCK	B/Neth 10894/22 Egg	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R	B/Stock 3/22 MDCK	
					Sh 539, 540, 543, 544, 570, 571, 574	F44/18	F20/20	F29/22	F37/22	NIB F01/21	F15/21	F44/21	F28/22	
		Genetic group			V1A	V1A.1	V1A.3	V1A.3	V1A.3	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2	
REFERENCE VIRUSES														
B/Brisbane/60/2008		V1A	2008-08-04	E4/E4	1280	160	40	<10	80	<40	<40	<40	<40	
B/Colorado/06/2017		V1A.1	2017-02-05	E5/E2	1280	320	80	10	80	<40	<40	<40	<40	
B/Washington/02/2019		V1A.3	2019-01-19	E3/E3	640	160	80	20	160	<40	<40	<40	<40	
B/Netherlands/11267/2022	G184R	V1A.3	2022-04-14	MDCK-MIX/MDCK2	40	10	<10	160	10	<40	<40	<40	<40	
B/Netherlands/10894/2022	G184R, T199A (-cho), V318I	V1A.3	2022-04-02	E4(Am1AL3)	320	40	40	40	80	<40	<40	<40	<40	
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2	2021-01-09	SIAT1/MDCK4	40	40	<10	40	<10	1280	1280	320	640	
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141)	V1A.3a.2	2021-01-09	E3/E5	320	20	<10	40	<10	1280	1280	320	640	
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141R)	V1A.3a.2	2021-01-09	E3/E5	320	40	10	40	<10	1280	1280	2560	640	
B/Stockholm/3/2022	D197E, D129G, E183K	V1A.3a.2	2022-03-22	SIAT1/MDCK2	640	40	<10	40	<10	1280	1280	320	1280	
TEST VIRUSES														
B/Norway/32412/2022		V1A.3a.2	2022-10-03	MDCK2	640	40	<10	40	<10	2560	1280	640	640	
B/Norway/32681/2022		V1A.3a.2	2022-10-14	MDCK1	1280	80	<10	40	<10	1280	1280	640	1280	
B/Catalonia/NSVH161512453/2022		V1A.3a.2	2022-10-18	SIAT1/MDCK1	640	40	<10	20	<10	1280	1280	320	1280	
B/Croatia/114021/2022		V1A.3a.2	2022-10-22	MDCKx/MDCK1	640	20	<10	20	<10	1280	1280	640	640	
B/Norway/32882/2022		V1A.3a.2	2022-10-25	MDCK1	640	40	<10	40	<10	1280	1280	320	1280	
B/Norway/32885/2022		V1A.3a.2	2022-10-29	MDCK1	640	40	<10	40	<10	2560	1280	640	640	
B/Croatia/115758/2022		V1A.3a.2	2022-11-02	MDCKx/MDCK2	640	40	<10	20	<10	1280	640	320	1280	
B/Israel/B1292/2022		V1A.3a.2	2022-11-20	P1/MDCK1	640	20	<10	20	<10	1280	1280	640	640	
B/Israel/B1406/2022		V1A.3a.2	2022-12-01	P1/MDCK1	640	40	<10	20	<10	1280	1280	640	1280	
B/Cameroon/10466/2022		V1A.3a.2	2022-12-05	MDCK1	640	80	<10	40	<10	1280	1280	640	1280	
B/Israel/B1417/2022		V1A.3a.2	2022-12-05	P2/MDCK1	640	40	<10	20	<10	640	640	320	1280	
B/Israel/B1513/2022		V1A.3a.2	2022-12-09	P1/MDCK1	640	20	<10	20	<10	1280	1280	320	640	
B/Belgium/G0012/2023		V1A.3a.2	2022-12-15	C1/MDCK1	640	40	<10	20	<10	2560	1280	640	640	
B/Belgium/S2625/2022		V1A.3a.2	2022-12-16	C1/MDCK1	640	20	<10	20	<10	1280	1280	640	640	
B/Belgium/G0053/2023		V1A.3a.2	2022-12-19	C1/MDCK1	640	40	<10	20	<10	1280	1280	320	1280	
B/Belgium/G0404/2022		V1A.3a.2	2022-12-19	C1/MDCK1	640	80	<10	20	<10	1280	1280	640	1280	
B/Belgium/G0403/2022		V1A.3a.2	2022-12-19	C1/MDCK1	640	40	<10	20	<10	640	640	320	640	
B/Belgium/S0030/2023		V1A.3a.2	2022-12-21	C1/MDCK1	640	40	<10	20	<10	1280	640	320	1280	
B/Israel/R13022/2022		V1A.3a.2	2022-12-25	P1/MDCK1	640	40	<10	20	<10	1280	1280	640	1280	
B/Romania/543990/2022		V1A.3a.2	2022-12-26	SIAT1/MDCK1	640	20	<10	20	<10	1280	640	320	640	
< relates to the lowest dilution of antiserum used							Vaccine NH 2021-22			Vaccine SH 2022				
1 hyperimmune sheep serum; ND = Not Done										NH 2022-23				
										SH 2023				

¹ hyperimmune sheep serum: ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23
SH 2023

 < 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table B-8. Antigenic analyses of influenza B viruses (Victoria lineage) 2023-02-07

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre								
					Post-infection ferret antiserum								
					B/Bris 60/08 Egg	B/Colorado 06/17 Egg	B/Wash'ton 02/19 Egg	B/Neth 11267/22 MDCK	B/Neth 10894/22 Egg	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R	B/Stock 3/22 MDCK
					Sh 539, 540, 543, 544, 570, 571, 574 ¹	F44/18	F20/20	F29/22	F37/22	NIB F01/21	F15/21	F44/21	F28/22
		Genetic group			V1A	V1A.1	V1A.3	V1A.3	V1A.3	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2
REFERENCE VIRUSES													
B/Brisbane/60/2008		V1A	2008-08-04	E4/E4	2560	160	40	<10	80	<40	<40	<40	<40
B/Colorado/06/2017		V1A.1	2017-02-05	E5/E2	1280	320	80	<10	160	<40	<40	<40	<40
B/Washington/02/2019		V1A.3	2019-01-19	E3/E3	640	160	160	40	160	<40	<40	<40	<40
B/Netherlands/11267/2022	G184R	V1A.3	2022-04-14	MDCK-MIX/MDCK2	40	20	<20	160	<10	<40	<40	<40	<40
B/Netherlands/10894/2022	G184R, T199A (-cho), V318I	V1A.3	2022-04-02	E4/E1	640	80	40	80	160	<40	<40	<40	<40
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2	2021-01-09	SIAT1/MDCK4	640	40	<20	20	<10	1280	1280	640	1280
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141)	V1A.3a.2	2021-01-09	E3/E5	640	40	<20	40	<10	2560	1280	640	640
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141R)	V1A.3a.2	2021-01-09	E3/E5	320	20	<20	20	<10	1280	640	2560	640
B/Stockholm/3/2022	D197E, D129G, E183K	V1A.3a.2	2022-03-22	SIAT1/MDCK2	640	40	<20	20	<10	1280	1280	320	1280
TEST VIRUSES													
B/Netherlands/01229/2022		V1A.3a.2	2022-10-02	hCK1/MDCK1	640	40	<20	20	<10	1280	640	320	640
B/Berlin/7/2022		V1A.3a.2	2022-11-09	P1/MDCK1	1280	40	<20	20	<10	1280	1280	320	1280
B/Lisboa/3/2022		V1A.3a.2	2022-11-19	MDCK1/MDCK1	640	40	<20	20	<10	640	640	320	640
B/Lisboa/2/2022		V1A.3a.2	2022-11-22	MDCK1/MDCK1	640	20	<20	20	<10	1280	1280	640	640
B/Lisboa/4/2022		V1A.3a.2	2022-12-13	MDCK1/MDCK1	640	40	<20	20	<10	2560	1280	640	640
B/Tunisia/23677/2022		V1A.3a.2	2022-12-20	MDCK1	640	40	<20	20	<10	2560	1280	640	640
B/Netherlands/11975/2022		V1A.3a.2	2022-12-20	MDCK-MIX2/MDCK1	320	20	<20	10	<10	1280	640	320	320
B/Netherlands/11970/2022		V1A.3a.2	2022-12-20	MDCK-MIX2/MDCK1	640	40	<20	20	<10	640	1280	320	640
B/Netherlands/11987/2022		V1A.3a.2	2022-12-21	MDCK-MIX2/MDCK1	320	20	<20	20	<10	1280	640	320	640
B/Netherlands/11994/2022		V1A.3a.2	2022-12-23	MDCK-MIX2/MDCK1	640	40	<20	20	<10	1280	1280	320	1280
B/Netherlands/12140/2022		V1A.3a.2	2022-12-27	MDCK-MIX2/MDCK1	640	40	<20	20	<10	1280	640	320	1280
B/Lisboa/11/2022		V1A.3a.2	2022-12-30	MDCK1/MDCK1	640	20	<20	10	<10	640	640	320	1280
B/Netherlands/10009/2023		V1A.3a.2	2023-01-02	MDCK-MIX2/MDCK1	640	40	<20	20	<10	1280	1280	320	1280
B/Lisboa/1/2023		V1A.3a.2	2023-01-04	MDCK1/MDCK1	320	20	<20	10	<10	1280	640	320	320
B/Netherlands/10024/2023		V1A.3a.2	2023-01-04	MDCK-MIX2/MDCK1	640	40	<20	20	<10	1280	1280	640	1280
B/Lisboa/2/2023		V1A.3a.2	2023-01-09	MDCK1/MDCK1	320	<10	<20	<10	<10	640	640	320	320

< relates to the lowest dilution of antiserum used

¹ hyperimmune sheep serum; ND = Not DoneVaccine
NH 2021-22Vaccine
SH 2022
NH 2022-23
SH 2023

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table B-9. Antigenic analyses of influenza B viruses (Victoria lineage) - Summary

Viruses		B/Colorado 06/17 Egg F44/18 Genetic group V1A.1	B/Wash'ton 02/19 Egg F20/20 V1A.3	B/Neth 11267/22 MDCK F29/22 V1A.3	B/Neth 10894/22 Egg F37/22 V1A.3	B/Austria 1359417/21 MDCK NIB F01/21 V1A.3a.2	B/Austria 1359417/21 Egg G141 F15/21 V1A.3a.2	B/Austria 1359417/21 Egg G141R F44/21 V1A.3a.2	B/Stock 3/22 MDCK F28/22 V1A.3a.2
REFERENCE VIRUSES									
B/Brisbane/60/2008	V1A	160	40	<10	80	<40	<40	<40	<40
B/Colorado/06/2017	V1A.1	320	80	<10	160	<40	<40	<40	<40
B/Washington/02/2019	V1A.3	160	160	40	160	<40	<40	<40	<40
B/Netherlands/11267/2022	V1A.3	20	<20	160	<10	<40	<40	<40	<40
B/Netherlands/10894/2022	V1A.3	80	40	80	160	<40	<40	<40	<40
B/Austria/1359417/2021	V1A.3a.2	40	<20	20	<10	1280	1280	640	1280
B/Austria/1359417/2021 Isolate 2	V1A.3a.2	40	<20	40	<10	2560	1280	640	640
B/Austria/1359417/2021 Isolate 2	V1A.3a.2	20	<20	20	<10	1280	640	2560	640
B/Stockholm/3/2022	V1A.3a.2	40	<20	20	<10	1280	1280	320	1280
TEST VIRUSES									
Number tested	V1A.3a.2	100	100	100	100	100	100	100*	100
No with titre reduction ≤2-fold		1				100	96		90
%		1.0				100	96.0		90.0
No with titre reduction =4-fold		7	1	28			4	16	9
%		7.0	1.0	28.0			4.0	16.0	9.0
No with titre reduction ≥8-fold		92	99	72	100			84	1
%		92.0	99.0	72.0	100			84.0	1.0
Vaccine NH 2021-22 Vaccine SH 2022 NH 2022-23 SH 2023									

* All but one test virus gave a titre of at least 160 with this antiserum.

Reference virus results are taken from an individual table as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

B/Victoria: Egg isolates

Type	Virus	Status	Genetic group/clade
B/Victoria	B/Netherlands/10894/2022	Sent out	V1A.3 (184R)

B/Victoria: Reagents and references

Virus	Genetic group	Virus passage	Ferret ID
B/Brisbane/60/2008	V1A	E4/E4	sheep pool
B/Colorado/06/2017	V1A.1	E5/E2	F44/18
B/Washington/02/2019	V1A.3	E3/E3	F20/20
B/Netherlands/11267/2022	V1A.3	MDCKMIX/MDCK2	F29/22
B/Netherlands/10894/2022	V1A.3	E4(Am1A13)	F37/22
B/Stockholm/3/2022	V1A.3a.2	SIAT1/MDCK3	F28/22
B/Austria/1359417/2021	V1A.3a.2	SIAT1/MDCK4	NIB F01/21
B/Austria/1359417/2021 G141	V1A.3a.2	E3/E5	F15/21
B/Austria/1359417/2021 G141R	V1A.3a.2	E3/E5	F44/21

Summary: B/Victoria

Genetic analyses

B/Victoria-lineage viruses of Clade V1A.3a.2 predominated since 1st September 2022, with a wide geographic distribution, but primarily in Europe, the Middle East, West and South Africa and several South American countries.

No Clade V1A.3 viruses were detected since 1st September 2022.

HA-NA tanglegrams showed limited reassortment and all within the V1A.3a.2 associated HA and NA clades.

Antigenic analyses

Despite accumulating limited amino acid substitutions within the 3a.2 clade , 96% of V1A.3a.2 viruses tested were recognised to within 2-fold and 4% within 4-fold reductions by antisera raised against B/Austria/1359417/2021 and -like viruses.

All tested viruses were susceptible to neuraminidase inhibitors (NAI) by phenotypic testing and no markers for reduced inhibition by NAIs were identified in NA sequences derived from clinical specimens

Susceptibility to antivirals

Antiviral (NAI) testing of isolates with collection dates after 2022-08-31

Month of Collection	Number of viruses tested for phenotype ¹						Totals
	A(H1N1)pdm09		A(H3N2)		Influenza B-Victoria		
	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	
2022							
September	10		27		19		56
October	44		109		35		188
November	50		55		13		118
December	44		68		25		137
2023							
January	10		5		4		19
Totals	158	0	264	0	96	0	518

¹ Phenotype: NI (normal inhibition), RI (reduced inhibition), HRI (highly reduced inhibition) as defined in WHO Wkly. Epidemiol. Rec. (2012) 87, 369-374

* As of 2023-02-15

Virus name	Subtype/ Lineage	Collection date	Fold difference		NA substitution
			Oseltamivir	Zanamivir	
Outliers	NONE				

In addition, NA gene sequencing was used to assess the presence of markers associated with RI or HRI by neuraminidase inhibitors (NAIs: oseltamivir, peramivir, laninamivir and zanamivir). At the WIC totals of 417 H1N1-, 713 H3N2- and 176 B/Victoria-associated NA gene sequences from individual viruses with collection dates after 31 August 2022 were determined, as of 15th February 2023. No sequences with mutations encoding amino acid markers associated with RI/HRI by NAIs were identified.

Based on whole genome sequencing at the WIC, full-length gene sequences were generated for totals of 269 H1N1-, 642 H3N2- and 167 B/Victoria-associated PA genes from individual viruses with collection dates after 31 August 2022 were determined, as of 15th February 2023. No sequences with mutations encoding amino acid markers associated with reduced susceptibility/resistance to Baloxavir marboxil were identified. For 14 B/Victoria lineage viruses the lack of reduced susceptibility was confirmed by phenotypic testing.

Annex

Tanglegrams with tip labels

Figure A1: H1N1

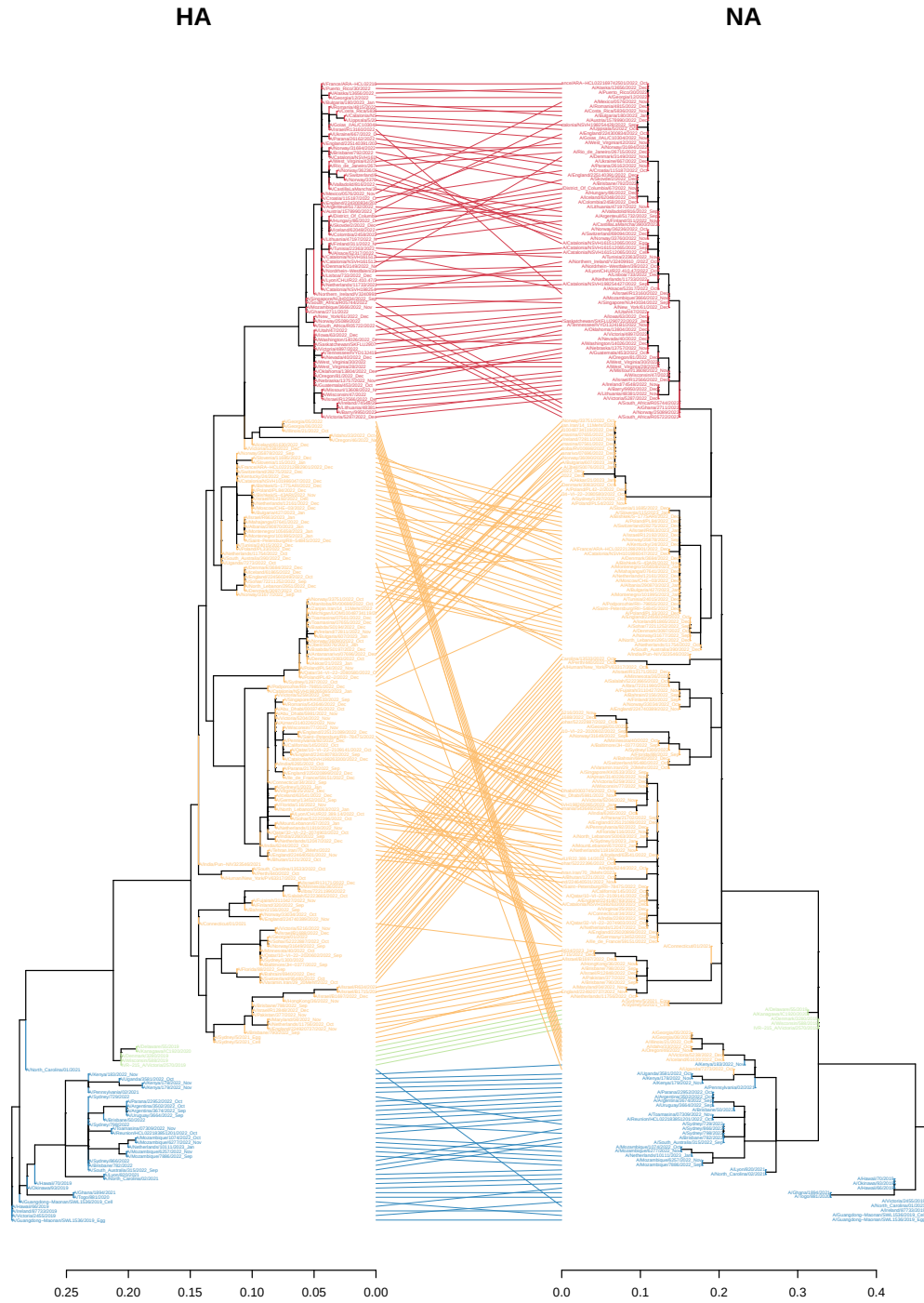


Figure A2: H3N2

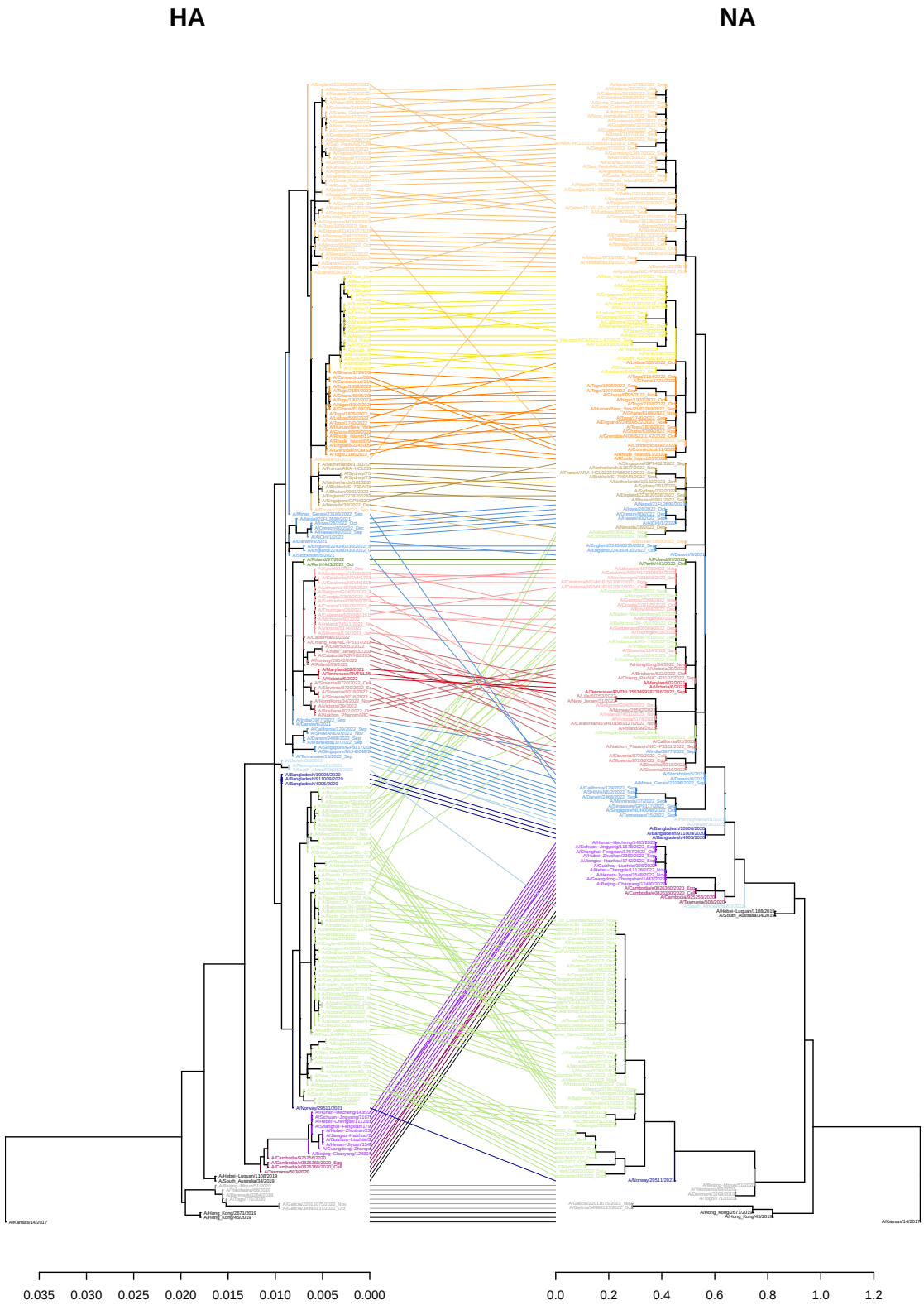


Figure A3: B

